# Power Systems

Detailed Solutions • 1991–2026

## GATE ELECTRICAL ENGINEERING

**(EE)**

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

# Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Power Systems (EE), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

## Meet your Authors


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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

## How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

**Q3.24.7**

| | |
|---|---|
| **3** | Unit (chapter) number — Unit III of this book |
| **24** | GATE exam year — 2024 (two digits; 98 = 1998) |
| **7** | Question number within that unit and year |

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

## What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

## How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

| | |
|---|---|
| **Questions** | the real total for this unit. |
| **MCQ, MSQ, NAT** | that same total, split by question type. |
| **1M, 2M** | how many of those questions carry 1 or 2 marks. |
| **Descriptive** | questions worth more than 2 marks, or whose marks value isn't recorded. |
| **Avg Weightage** | the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one. |
| **Range** | the lowest and highest that same per-sitting value has been, with the year each occurred. |

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

### Worked example

**Unit Snapshot***

| | | | | |
|---|---|---|---|---|
| **Stream:** EE | **Questions:** 45 | **MCQ:** 30 | **MSQ:** 3 | **NAT:** 12 |
| **1M:** 20 | **2M:** 22 | **Descriptive:** 3 | **Avg Weightage** ~4M | **Range:** 2M(2015)–6M(2021) |

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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# Preface

This volume presents GATE previous year questions arranged subject-wise and chapter-wise for streamlined practice and revision.

**Scope of this volume**

- Stream: EE
- Structure: Subject → Chapter → Year-descending questions
- Question types: MCQ, MSQ, NAT, and Common Data questions
- Solutions: not included in this volume

Access solutions, updates, and additional material online

SOLUTIONS

# Basics of Power System

## Topics

1. 3 phase systems
2. Single Line Diagrams and p.u. analysis
3. AC and DC Transmission Concepts

## Unit Snapshot*

| | | | | |
|---|---|---|---|---|
| **Stream:** EE | **Questions:** 28 | **MCQ:** 22 | **MSQ:** 1 | **NAT:** 5 |
| **1M:** 16 | **2M:** 11 | **Descriptive:** 1 | **Avg Weightage** ~2M | **Range:** 1M(2022)–2M(2024) |

**See preface for how Avg Weightage and Range are calculated.*

**Q1.24.1** MSQ | 2024 | 2M | Ans: B,D | ● ● ●

Automatic Generation Control (AGC) controls the real power balance in the power system and thereby regulates system frequency.
Automatic Voltage Regulator (AVR) controls the excitation of the synchronous generator and thereby regulates terminal voltage.

**Option A**
AGC affects the overall system frequency, which is a global quantity.
AVR primarily affects voltage locally around the generator bus.
Hence, Option A is incorrect.

**Option B**
AGC response has a global effect on frequency while AVR response has a local effect on voltage.
Hence, Option B is correct.

**Option C**
AGC does not regulate field current. Field current is controlled by AVR.
Hence, Option C is incorrect.

**Option D**
AGC regulates the generator mechanical power input through governor action, while AVR regulates excitation field current.
Hence, Option D is correct.
Therefore, the correct options are

B, D

**Key Points**

- AGC controls real power and frequency.
- AVR controls excitation and terminal voltage.
- Frequency is a global parameter in an interconnected power system.
- Voltage control is predominantly local.
- Mechanical power is controlled through the governor system, not AVR.
- AVR acts on excitation/field current, not turbine input.

**Q1.23.1** NAT | 2023 | 2M | Ans: 415 | ● ● ○

**Method 1**

The line reactance is

$$X_L = 2\pi f L = 2\pi(50)(31.83 \times 10^{-3}) = 10\ \Omega$$

Hence,

$$Z_{line} = j10\ \Omega$$

The capacitive reactance is

$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi(50)(212.21 \times 10^{-6})} = 15\ \Omega$$

Therefore,

$$Z_\Delta = 15 - j15\ \Omega$$

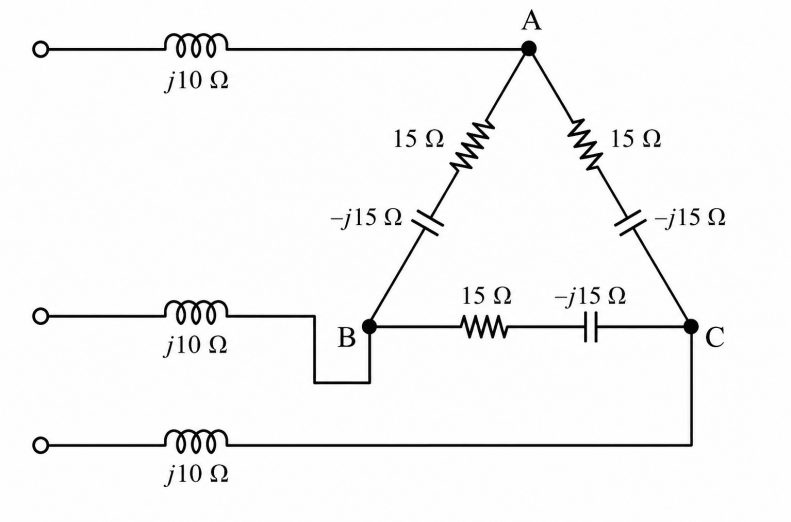
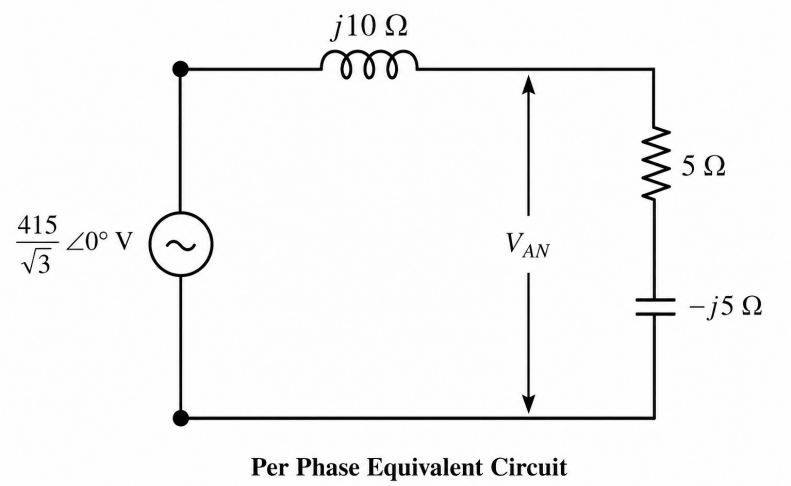



Fig. Solution Q1.23.1

Converting the balanced delta load into an equivalent star,

$$Z_Y = \frac{Z_\Delta}{3} = 5 - j5\ \Omega$$

The per-phase equivalent circuit is shown below.




**Per Phase Equivalent Circuit**

Fig. Solution Q1.23.1

The phase voltage across the equivalent load is

$$V_{AN} = \frac{5 - j5}{(5 - j5) + j10} \times \frac{415}{\sqrt{3}}$$

$$= \frac{5 - j5}{5 + j5} \times \frac{415}{\sqrt{3}}$$

Since

$$\left|\frac{5-j5}{5+j5}\right| = 1$$

$$|V_{AN}| = \frac{415}{\sqrt{3}}$$

Hence,

$$|V_{AB}| = \sqrt{3}\,|V_{AN}| = 415\ V$$

**Method 2**

Let the load phase voltage be $V_{AB}$.
For a balanced delta load,

$$I_{AB} = \frac{V_{AB}}{15-j15}$$

Applying KVL between source terminals A and B,

$$V_{AB,\text{source}} = V_{AB} + j10\,(I_A - I_B)$$

For a balanced delta,

$$I_A - I_B = 3I_{AB}$$

Hence,

$$V_{AB,\text{source}} = V_{AB} + 3j10\,I_{AB}$$

Substituting

$$I_{AB} = \frac{V_{AB}}{15-j15}$$

gives

$$V_{AB,\text{source}} = V_{AB}\left(1 + \frac{j30}{15-j15}\right)$$

Now,

$$\frac{j30}{15-j15} = -1 + j$$

Therefore,

$$1 + \frac{j30}{15-j15} = j$$

Hence,

$$V_{AB,\text{source}} = jV_{AB}$$

Taking magnitudes,

$$|V_{AB,\text{source}}| = |V_{AB}|$$

Since

$$|V_{AB,\text{source}}| = 415\ \text{V}$$

we obtain

$$|V_{AB}| = 415\ \text{V}$$

Therefore, the correct answer is

$$\boxed{V_{AB} = 415\ V}$$

**Q1.22.1** NAT | 2022 | 1M | Ans: 20 | ● ○ ○

Given:

$$V_{LL} = 100\sqrt{3}\ \text{V}$$

$$Z_1 = (18 + j24)\ \Omega$$

$$Z_2 = (6 + j8)\ \Omega$$

The inner star-connected load is first converted into an equivalent delta load.
For balanced star-to-delta conversion,

$$Z_\Delta = 3Z_2$$

Hence, the equivalent per-phase impedance becomes

$$Z_{ph} = Z_1 \parallel 3Z_2$$

$$Z_{ph} = \frac{Z_1(3Z_2)}{Z_1 + 3Z_2}$$

$$Z_{ph} = \frac{(18 + j24)(18 + j24)}{(18 + j24) + (18 + j24)}$$

$$Z_{ph} = 15\angle 53.1^\circ\ \Omega$$

After conversion, the network becomes a balanced delta load.

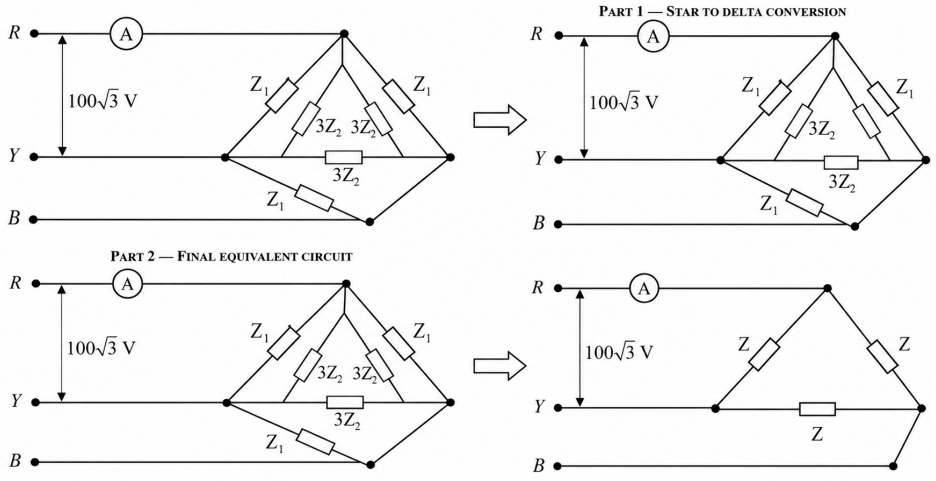



Fig. Solution Q1.22.1

For a delta connection,

$$V_{ph} = V_{LL}$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}}$$

$$I_{ph} = \frac{100\sqrt{3}}{15\angle 53.1^\circ}$$

$$|I_{ph}| = 11.547\ \text{A}$$

Line current is

$$I_L = \sqrt{3}\, I_{ph}$$

$$I_L = \sqrt{3} \times 11.547$$

$$I_L = 20\ \text{A}$$

Therefore, the ammeter reading is

$$\boxed{20}$$

**Key Points**

- Balanced star impedance converts to delta impedance as $Z_\Delta = 3Z_Y$.
- For balanced delta loads, $V_{ph} = V_{LL}$.
- Line current in a delta connection is $\sqrt{3}$ times the phase current.

**Q1.16.1** MCQ | 2016 | 2M | Ans: A | ● ○ ○

Let the right node be the reference node.

$$V_A = V_3$$

Applying KCL at node $A$ (taking currents entering the node as positive), the current source contributes $-i$ since its current leaves node $A$.

$$\frac{V_3 - V_1}{-j1} + \frac{V_3 - V_2}{j1} - i = 0$$

Hence,

$$i = \frac{V_3 - V_1}{-j1} + \frac{V_3 - V_2}{j1}$$

$$i = \frac{1}{j}(V_1 - V_2)$$

Substituting the given phasors,

$$i = \frac{1}{j}(100\angle 0^\circ - 100\angle -120^\circ)$$

Now,

$$100\angle 0^\circ - 100\angle -120^\circ = 173.2\angle 30^\circ$$

Therefore,

$$i = 173.2\angle 30^\circ \times \angle(-90^\circ)$$

$$i = 173.2\angle -60^\circ$$

Thus the correct option is

$$\boxed{\text{(A) } 173.2\angle -60^\circ}$$

**Key Points**

- Use nodal analysis by choosing the common right-side node as the reference node.
- Since currents entering node $A$ are taken as positive, the current source appears as $-i$ in the KCL equation because it leaves the node.
- Impedance of the capacitor is $-j1\ \Omega$ and that of the inductor is $j1\ \Omega$.
- Multiplication by $\frac{1}{j}$ rotates a phasor by $-90^\circ$.

**Q1.15.1** MCQ | 2015 | 1M | Ans: C | ● ○ ○

Base load plants are those generating stations which are operated continuously to meet the minimum constant demand of the power system.

- Wind farms are generally treated as base load plants because once available, they are operated continuously due to negligible fuel cost.
- Run-of-river hydro plants operate continuously based on river flow and are used as base load plants.
- Nuclear power plants are highly suitable for base load operation because they operate most efficiently at constant output.
- Diesel power plants are generally used as peak load or standby plants due to high operating cost.

Hence, the correct statements are:

$$\boxed{\text{(C) } P,\ Q,\text{ and } R}$$

**Q1.14.1** NAT | 2014 | 1M | Ans: 117 to 120 | ● ○ ○

Per-unit value is defined as

$$\text{p.u. value} = \frac{\text{Actual value}}{\text{Base value}}$$

Given,

$$S_{\text{base}} = 100\,\text{MVA}$$

$$I_{\text{base}} = 437.38\,\text{A}$$

For a three-phase system,

$$S_{\text{base}} = \sqrt{3} V_{LL,\text{base}} I_{\text{base}}$$

Hence,

$$V_{LL,\text{base}} = \frac{100 \times 10^6}{\sqrt{3} \times 437.38}$$

$$V_{LL,\text{base}} \approx 132\,\text{kV}$$

Given voltage in per unit is

$$V_{\text{pu}} = 0.9$$

Therefore,

$$V_{LL,\text{actual}} = 0.9 \times 132$$

$$V_{LL,\text{actual}} = 118.8\,\text{kV}$$

$$\boxed{118.8}$$

**Key Points**

- Per-unit relation:

$$\text{p.u. value} = \frac{\text{Actual value}}{\text{Base value}}$$

- Three-phase apparent power relation:

$$S = \sqrt{3} V_{LL} I_L$$

- Base quantities must belong to the same system type and units.

**Mistakes to be avoided**

- Do not omit the factor:

$$\sqrt{3}$$

in three-phase calculations.

- The given voltage is already in per-unit form; directly multiply with base voltage.
- Power factor is not required for calculating voltage in this problem.

**Q1.14.2** MCQ | 2014 | 1M | Ans: C | ● ○ ○

Given,

$$V_{AN} = 10\angle 15^\circ\ \text{V}$$

For a balanced three-phase system with phase sequence $ABC$,

$$V_{BN} = 10\angle(15^\circ - 120^\circ)$$

$$V_{BN} = 10\angle(-105^\circ)\ \text{V}$$

Similarly,

$$V_{CN} = 10\angle(-105^\circ - 120^\circ)$$

$$V_{CN} = 10\angle(-225^\circ)\ \text{V}$$

The required line voltage is

$$V_{BC} = V_{BN} - V_{CN}$$

Substituting,

$$V_{BC} = 10\angle(-105^\circ) - 10\angle(-225^\circ)$$

For a balanced three-phase system,

$$V_{LL} = \sqrt{3}\,V_{ph}\angle 30^\circ$$

Hence,

$$V_{BC} = 10\sqrt{3}\angle(-105^\circ + 30^\circ)$$

$$V_{BC} = 10\sqrt{3}\angle(-75^\circ)\ \text{V}$$

Therefore,

$$\boxed{\text{(C) } 10\sqrt{3}\angle(-75^\circ)\ \text{V}}$$

**Key Points**

- For phase sequence $ABC$, each phase lags the previous phase by:

$$120^\circ$$

**Q1.10.1** MCQ | 2010 | 2M | Ans: B | ● ○ ○

The per-unit reactance conversion formula is

$$X_{\text{new}} = X_{\text{old}}\left(\frac{\text{New MVA}}{\text{Old MVA}}\right)\left(\frac{\text{Old kV}}{\text{New kV}}\right)^2$$

For generators $G_1$ and $G_2$,

$$X_{G_1} = X_{G_2} = j0.09 \times \frac{200}{100} \times \left(\frac{25}{25}\right)^2 = j0.18$$

For transformer $T_1$,

$$X_{T_1} = j0.12 \times \frac{200}{90} \times \left(\frac{25}{25}\right)^2 = j0.27$$

Similarly,

$$X_{T_2} = j0.12 \times \frac{200}{90} \times \left(\frac{25}{25}\right)^2 = j0.27$$

For the transmission line,

$$Z_{\text{base}} = \frac{(220)^2}{200} = 242\ \Omega$$

$$X_{\text{Line}} = \frac{150}{242} = 0.62\ \text{pu}$$

Therefore,

$$X_{\text{Line}} = j0.62$$

Hence the impedance diagram becomes

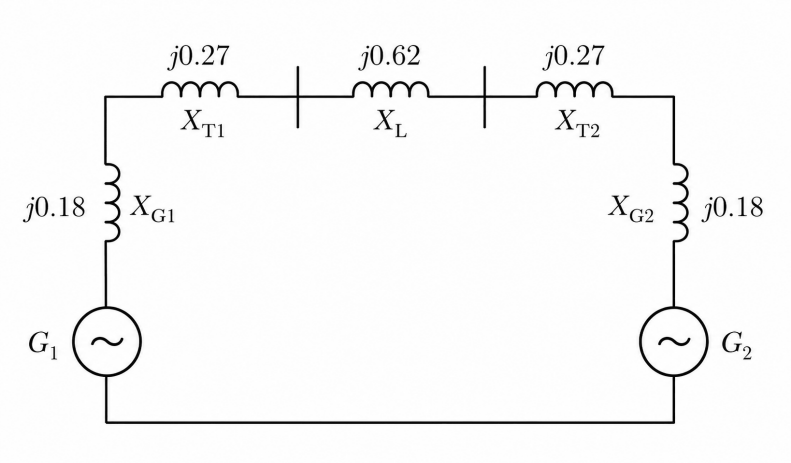



Fig. Solution Q1.10.1

Thus,

(B)

**Key Points**

- Per-unit quantities must be converted to a common MVA base.
- Base voltage changes across transformer sides according to transformer turns ratio.
- For transmission lines specified in ohms, first calculate base impedance.
- Base impedance: $Z_{\text{base}} = \dfrac{(kV_{\text{base}})^2}{MVA_{\text{base}}}$.

## Q1.09.1 MCQ 2009 1M Ans: A ●○○

Base-load power plants are generating stations that operate continuously to supply the minimum demand of the power system.

**Nuclear power plants**

- Have very high capital cost.
- Have very low running cost.
- Are most economical when operated continuously.
- Therefore, they are used as base-load stations.

**Run-of-river hydro plants**

- Generate power continuously depending on river flow.
- Are generally used to meet base-load demand.

**Pumped-storage plants**

- Are primarily used for peak-load operation and load balancing.
- Hence, they are not base-load plants.

**Diesel power plants**

- Have high operating cost.
- Are generally used for standby, emergency and peak-load duties.
- Hence, they are not base-load plants.

Therefore, the base-load power plants are

(A) (i) Nuclear and (ii) Run-of-river

**Q1.07.1** MCQ | 2007 | 1M | Ans: A

Take the generator-side voltage as the reference.

$$V_1 = 15\angle 0^\circ \text{ kV}$$

The 400 kV/15 kV transformer is a Δ-Y transformer with a phase shift of

$$+30^\circ$$

Therefore,

$$V_2 = 400\angle 30^\circ \text{ kV}$$

The autotransformer has zero phase shift.
Hence,

$$V_3 = 220\angle 30^\circ \text{ kV}$$

The upper transformer is a Y-Δ transformer with phase shift

$$-30^\circ$$

Thus the voltage at bus $A$ becomes

$$V_4 = 220\angle 0^\circ$$

The transformer between buses $A$ and $B$ must therefore introduce

$$\theta = 0^\circ$$

so that the phase relationship around the network remains consistent.

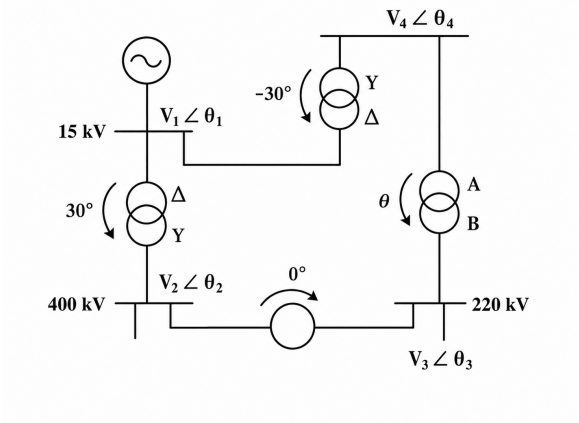



Fig. Solution Q1.07.1

Hence, the required transformer connection is

(A) Star–Star with $\theta = 0°$

## Q1.07.2 MCQ 2007 2M Ans: D ●○○

For a balanced star-connected system with phase sequence ABC,

$$V_{an} = V\angle 0°$$

$$V_{bn} = V\angle -120°$$

$$V_{cn} = V\angle 120°$$

Since the load is purely inductive, each phase current lags its corresponding phase voltage by 90°. Hence,

$$i_a = I\angle -90°$$

$$i_b = I\angle -210°$$

$$i_c = I\angle 30°$$

The matrix multiplication can be rewritten as

$$R = \begin{bmatrix} V_{an} & V_{bn} & V_{cn} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}}(i_b - i_c) \\ \frac{1}{\sqrt{3}}(i_c - i_a) \\ \frac{1}{\sqrt{3}}(i_a - i_b) \end{bmatrix}$$

Now,

$$i_b - i_c = -\sqrt{3}I\angle 180°$$

$$i_c - i_a = \sqrt{3}I\angle 60°$$

$$i_a - i_b = \sqrt{3} I \angle -60^\circ$$

Substituting,

$$R = V\angle 0^\circ \, (I\angle 180^\circ) + V\angle(-120^\circ) \, (I\angle 60^\circ) + V\angle 120^\circ \, (I\angle -60^\circ)$$

$$R = VI \, (1\angle 180^\circ + 1\angle(-60^\circ) + 1\angle 60^\circ)$$

Using phasor addition,

$$1\angle 180^\circ + 1\angle(-60^\circ) + 1\angle 60^\circ = -1 + \frac{1}{2} + \frac{1}{2} = 0$$

Therefore,

$$R = 0\ \Omega$$

Hence, the correct answer is

(D) 0

**Q1.06.1** MCQ | 2006 | 2M | Ans: C

The load is given as

0.72 pu

on the load-side transformer ratings as base values:

$$S_{\text{old}} = 10 \text{ MVA}, \qquad V_{\text{old}} = 69 \text{ kV}$$

The chosen system base values in the load circuit are

$$S_{\text{base}} = 10 \text{ MVA}, \qquad V_{\text{base}} = 69 \text{ kV}$$

These are the same as the load-side rating, so in the load circuit the load is still

0.72 pu

The base voltages in the other circuits follow from the transformer ratios:

$$V_{\text{base, line}} = 69 \times \frac{138}{69} = 138 \text{ kV}$$

$$V_{\text{base, generator}} = 138 \times \frac{13.8}{138} = 13.8 \text{ kV}$$

The base MVA (10 MVA) is the same throughout the system. Because the base voltages change in exactly the same ratio as the transformer turns ratios, a per-unit quantity (power or impedance) has the same value on either side of an ideal transformer:

$$Z_{\text{pu, generator side}} = \frac{Z_{\text{load}} \left(\frac{13.8}{69}\right)^2}{Z_{\text{base}} \left(\frac{13.8}{69}\right)^2} = Z_{\text{pu, load side}}$$

Hence, the value of the load in the generator circuit is

(C) 0.72 pu

**Key Points**

- Choose one base MVA for the whole system.
- Base voltages in different circuits are related by the transformer turns ratios.
- With such consistent bases, a per-unit value does not change when referred through a transformer.

**Q1.05.1** MCQ | 2005 | 1M | Ans: D | ● ○ ○

Given,

$$M_{old} = 20 \text{ pu}$$

$$X_{old} = 2 \text{ pu}$$

$$S_{old} = 500 \text{ MVA}$$

$$S_{new} = 100 \text{ MVA}$$

The reactance on the new base is

$$X_{new} = X_{old}\left(\frac{S_{new}}{S_{old}}\right)$$

$$= 2\left(\frac{100}{500}\right) = 0.4 \text{ pu}$$

The inertia constant on the new base is

$$M_{new}S_{new} = M_{old}S_{old}$$

$$M_{new} = \frac{20 \times 500}{100} = 100 \text{ pu}$$

Hence, the correct answer is

(D) $M_{new} = 100$ pu, $X_{new} = 0.4$ pu

**Key Points**

- Reactance changes directly with MVA base.
- Inertia constant changes inversely with MVA base.
- $M$ and $S$ remains invariant when changing base.

**Q1.04.1** MCQ 2004 1M Ans: C ● ○ ○

The rated voltage of a three-phase power system is specified in terms of the rms line-to-line voltage. For example, a 220 kV system refers to

$$V_{LL} = 220 \text{ kV (rms)}$$

Hence, the correct option is

(C) rms line-to-line voltage

**Q1.04.2** MCQ 2004 1M Ans: B ● ○ ○

From the delta connection shown,

$V_{RY}$

is represented by the side joining phases R and Y,

$V_{YB}$

by the side joining phases Y and B, and

$V_{BR}$

by the remaining side.
Traversing the phasors in the direction indicated by the delta voltage triangle gives the sequence

$$R \rightarrow B \rightarrow Y$$

Therefore, the phase sequence is

(B) RBY

**Key Points**

- Phase sequence is determined from the order in which phase voltages attain their positive maxima.
- The phase-voltage triangle can be used to identify the sequence.

**Q1.04.3** MCQ 2004 2M Ans: B ● ○ ○

The bus voltage is obtained from the connection of two sources through their reactances.

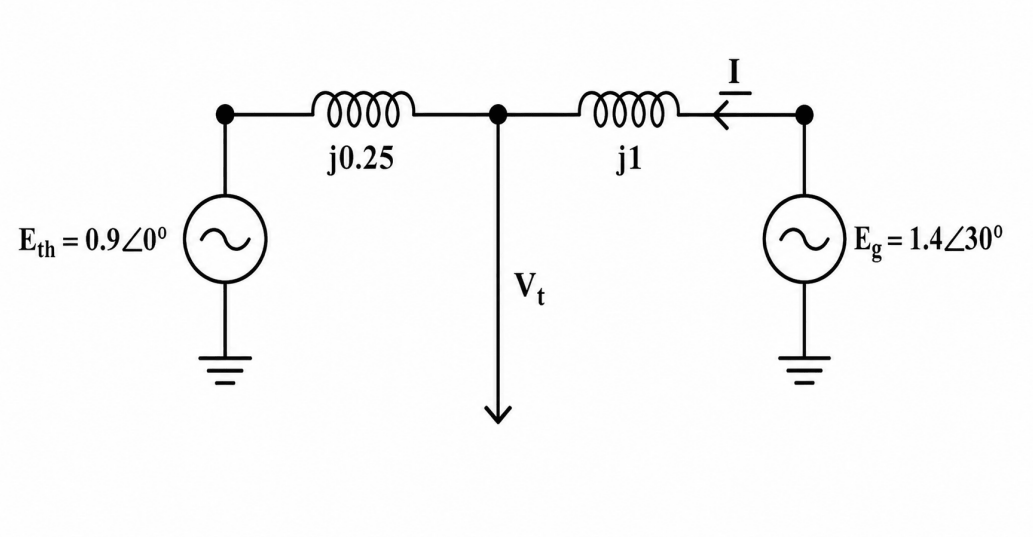



Fig. Solution Q1.04.3

Using nodal analysis,

$$V_t = \frac{\frac{E_g}{j1.0} + \frac{E_{th}}{j0.25}}{\frac{1}{j1.0} + \frac{1}{j0.25}}$$

Substituting,

$$E_g = 1.4\angle 30^\circ = 1.212 + j0.7$$

$$E_{th} = 0.9 + j0$$

Evaluating,

$$V_t = 0.970 + j0.078$$

$$|V_t| = 0.973$$

Hence, the correct answer is

(B) 0.973

## Q1.04.4 MCQ 2004 1M Ans: C ● ○ ○

In a thermal power plant, the feed water pump raises the pressure of the working fluid before it enters the boiler. The condenser decreases pressure, the superheater increases steam temperature, and the turbine extracts energy from the steam.

Therefore, pressure in the working fluid cycle is developed by the feed water pump.

(C) Feed water pump

**Q1.04.5** MCQ 2004 1M Ans: A ● ○ ○

Kaplan turbines are axial-flow reaction turbines specially designed for low-head and high-discharge applications.
They possess a high degree of reaction and adjustable runner blades, allowing efficient operation over a wide range of water flow conditions.
Therefore, the suitable turbine is

(A) Kaplan Turbine

**Q1.01.1** MCQ 2001 1M Ans: B ● ○ ○

For phase-$a$,

$$p_a = v_a i_a$$

$$= (V\cos\omega t)\,(I\cos(\omega t - \phi))$$

Using

$$\cos A\cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$$

$$p_a = \frac{VI}{2}[\cos\phi + \cos(2\omega t - \phi)]$$

Similarly, for phase-$b$,

$$p_b = \frac{VI}{2}[\cos\phi + \cos(2\omega t - 240^\circ - \phi)]$$

and for phase-$c$,

$$p_c = \frac{VI}{2}[\cos\phi + \cos(2\omega t - 120^\circ - \phi)]$$

Therefore, the total instantaneous three-phase power is

$$p = p_a + p_b + p_c$$

$$p = \frac{VI}{2}\Big[3\cos\phi + \cos(2\omega t - \phi) + \cos(2\omega t - 120^\circ - \phi) + \cos(2\omega t - 240^\circ - \phi)\Big]$$

For three balanced phasors separated by 120°,

$$\cos\theta + \cos(\theta - 120^\circ) + \cos(\theta - 240^\circ) = 0$$

Hence,

$$p = \frac{VI}{2}(3\cos\phi)$$

$$p = \frac{3}{2}VI\cos\phi$$

Thus, the instantaneous three-phase power is constant and equal to

$$\frac{3}{2}VI\cos\phi$$

Hence, the correct answer is

$$\boxed{\text{(B) a constant with a magnitude of } \frac{3}{2}VI\cos\phi}$$

**Q1.01.2** MCQ | 2001 | 2M | Ans: D | ● ○ ○

Given,

$$X_{d,\text{old}} = 0.4 \text{ pu}$$

on the base

$$S_{\text{old}} = 75 \text{ MVA}$$

and

$$V_{\text{old}} = 10 \text{ kV}$$

The new base values are

$$S_{\text{new}} = 100 \text{ MVA}$$

and

$$V_{\text{new}} = 11 \text{ kV}$$

The per-unit impedance conversion formula is

$$X_{\text{new}} = X_{\text{old}} \left(\frac{S_{\text{new}}}{S_{\text{old}}}\right)\left(\frac{V_{\text{old}}}{V_{\text{new}}}\right)^2$$

Substituting the given values,

$$X_d = 0.4 \left(\frac{100}{75}\right)\left(\frac{10}{11}\right)^2$$

$$X_d = 0.4 \times 1.3333 \times 0.8264$$

$$X_d = 0.4407$$

$$X_d \approx 0.44 \text{ pu}$$

Hence, the correct answer is

$$\boxed{\text{(D) } 0.44}$$

**Q1.00.1** MCQ 2000 1M Ans: A ● ○ ○

Given,

$$Z_{\text{pu,old}} = x$$

The per-unit impedance conversion formula is

$$Z_{\text{pu,new}} = Z_{\text{pu,old}} \left(\frac{V_{\text{old}}}{V_{\text{new}}}\right)^2 \left(\frac{S_{\text{new}}}{S_{\text{old}}}\right)$$

The new base voltage and base apparent power are both doubled.
Hence,

$$V_{\text{new}} = 2V_{\text{old}}$$

and

$$S_{\text{new}} = 2S_{\text{old}}$$

Substituting,

$$Z_{\text{pu,new}} = x \left(\frac{V_{\text{old}}}{2V_{\text{old}}}\right)^2 \left(\frac{2S_{\text{old}}}{S_{\text{old}}}\right)$$

$$Z_{\text{pu,new}} = x \left(\frac{1}{2}\right)^2 (2)$$

$$Z_{\text{pu,new}} = x \left(\frac{1}{4}\right) (2)$$

$$Z_{\text{pu,new}} = \frac{x}{2}$$

$$Z_{\text{pu,new}} = 0.5x$$

Hence, the correct answer is

$$\boxed{\text{(A) } 0.5x}$$

**Q1.00.2** NAT 2000 5M Ans: * ● ○ ○

For a balanced three-phase system, the phase currents of the $h^{\text{th}}$ harmonic are

$$i_{ah}(t) = I_h \cos(h\omega t)$$

$$i_{bh}(t) = I_h \cos(h\omega t - h \times 120^\circ)$$

$$i_{ch}(t) = I_h \cos(h\omega t + h \times 120^\circ)$$

The sequence of a harmonic depends on its order $h$.

$$h = 3k + 1 \quad \Rightarrow \quad \text{Positive sequence}$$

$$h = 3k - 1 \quad \Rightarrow \quad \text{Negative sequence}$$

$$h = 3k \quad \Rightarrow \quad \text{Zero sequence}$$

For the $5^{th}$ harmonic,

$$5 = 3(2) - 1$$

Hence, the $5^{th}$ harmonic belongs to the $3k - 1$ family and therefore forms a negative-sequence system.
For the $7^{th}$ harmonic,

$$7 = 3(2) + 1$$

Hence, the $7^{th}$ harmonic belongs to the $3k + 1$ family and therefore forms a positive-sequence system.

5th harmonic: Negative sequence

7th harmonic: Positive sequence

**Q1.00.3** MCQ | 2000 | 1M | Ans: D | ● ○ ○

An economizer is a heat exchanger installed in the path of boiler exhaust gases.
The feed water entering the boiler is preheated by recovering heat from the outgoing flue gases.
This improves the overall thermal efficiency of the power plant by reducing fuel consumption and utilizing waste heat.
Therefore, the feed water coming to the economizer is heated using

(D) flue gases

**Q1.99.1** MCQ | 1999 | 1M | Ans: B | ● ○ ○

The load carrying capability of a long AC transmission line is generally restricted by steady-state stability rather than thermal limits.
The steady-state stability limit is

$$P_{\max} = \frac{|V_s||V_r|}{Z_c \sin \beta \ell}$$

For long transmission lines, the stability limit is reached before the thermal limit.
Hence, the load carrying capability is limited by stability considerations.
Therefore, the correct answer is

(B) limited by stability considerations

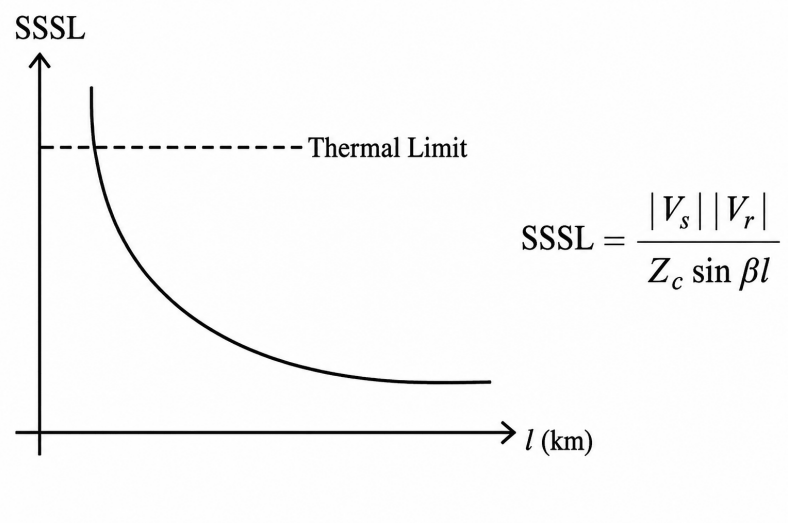



Fig. Solution Q1.99.1

**Q1.99.2** MCQ | 1999 | 2M | Ans: B

Load factor

$$= \frac{\text{Average Load}}{\text{Maximum Demand}}$$

Energy generated per day

$$= (2000 \times 12) + (1000 \times 12)$$

$$= 36000 \text{ kWh}$$

Average load

$$= \frac{36000}{24}$$

$$= 1500 \text{ kW}$$

Maximum demand

$$= 2000 \text{ kW}$$

Therefore,

$$\text{Load Factor} = \frac{1500}{2000}$$

$$= 0.75$$

Hence, the correct option is

(B) 0.75

**Q1.97.1** NAT | 1997 | 2M | Ans: 1 | ○ ○ ○

The base-conversion formula is

$$X_{pu,new} = X_{pu,old}\left(\frac{S_{new}}{S_{old}}\right)\left(\frac{V_{old}}{V_{new}}\right)^2$$

Given,

$$X_{pu,old} = 1.0$$

$$S_{old} = 200 \text{ MVA}$$

$$S_{new} = 100 \text{ MVA}$$

$$V_{old} = 10 \text{ kV}$$

$$V_{new} = 20 \text{ kV}$$

Substituting,

$$X_{pu,new} = 1.0\left(\frac{100}{200}\right)\left(\frac{10}{20}\right)^2$$

$$= 0.5 \times 0.25$$

$$= 0.125$$

Hence,

0.125

**Q1.96.1** MCQ | 1996 | 2M | Ans: C | ● ○ ○

Control rods regulate the neutron population inside a nuclear reactor.
The material used for control rods should have a high neutron-absorption cross section.
Boron satisfies this requirement and effectively absorbs excess neutrons.
Therefore the fission chain reaction can be controlled by inserting or withdrawing boron control rods.

(C) Boron

**Key Points**

- Boron and cadmium are common control-rod materials.
- Heavy water acts mainly as a moderator.
- Thorium is a fertile fuel material.

**Q1.95.1** MCQ | 1995 | 1M | Ans: D

The cost of electrical energy generation depends strongly on how effectively the installed generating capacity is utilized.
The load factor is defined as

$$\text{Load Factor} = \frac{\text{Average Load}}{\text{Maximum Demand}}$$

A high load factor indicates that the generating plant is utilized more uniformly throughout the day. This improves utilization of equipment and reduces the cost per unit of generated energy.
The diversity factor is defined as

$$\text{Diversity Factor} = \frac{\sum \text{Individual Maximum Demands}}{\text{Simultaneous Maximum Demand}}$$

A higher diversity factor means that all consumers do not reach their maximum demand at the same instant. Consequently, the installed generation capacity required is reduced.
Therefore,

- High load factor improves plant utilization.
- High diversity factor reduces installed capacity requirements.
- Both effects reduce the overall cost of generation.

Hence, for minimum generation cost,

(D) Load factor and diversity factor should be high

PrepFusion

SOLUTIONS

# 11

# Symmetrical Fault Analysis

## Topics

1. Types of Symmetrical Faults
2. Reactors in Power System
3. Transient Behaviour of Alternator during Fault

## Unit Snapshot*

| | | | | |
|---|---|---|---|---|
| **Stream:** EE | **Questions:** 53 | **MCQ:** 33 | **MSQ:** 1 | **NAT:** 19 |
| **1M:** 15 | **2M:** 34 | **Descriptive:** 4 | **Avg Weightage** ~3M | **Range:** 1M(2026)–4M(2018) |

**See preface for how Avg Weightage and Range are calculated.*

**Q2.26.1** MCQ | 2026 | 1M | Ans: D | ● ○ ○

The zero sequence component of a three-phase system is given by

$$\bar{V}_0 = \frac{1}{3}\left(\bar{V}_A + \bar{V}_B + \bar{V}_C\right)$$

For Case-1:

$$\bar{V}_{0,\text{Case-1}} = \frac{1}{3}\left(V_A\angle\theta_A + V_B\angle\theta_B + V_C\angle\theta_C\right)$$

In Case-2, each phasor undergoes the same phase shift $\Delta\theta$ while retaining the same magnitude.
Hence,

$$\bar{V}'_A = V_A\angle(\theta_A + \Delta\theta)$$

$$\bar{V}'_B = V_B\angle(\theta_B + \Delta\theta)$$

$$\bar{V}'_C = V_C\angle(\theta_C + \Delta\theta)$$

Therefore,

$$\begin{aligned}\bar{V}_{0,\text{Case-2}} &= \frac{1}{3}\left(V_A\angle(\theta_A + \Delta\theta) + V_B\angle(\theta_B + \Delta\theta) + V_C\angle(\theta_C + \Delta\theta)\right)\\ &= e^{j\Delta\theta}\cdot\frac{1}{3}\left(V_A\angle\theta_A + V_B\angle\theta_B + V_C\angle\theta_C\right)\\ &= e^{j\Delta\theta}\bar{V}_{0,\text{Case-1}}\end{aligned}$$

**Multiplication** by $e^{j\Delta\theta}$ rotates the phasor by $\Delta\theta$ **without changing** its magnitude.
Hence, the **zero sequence components** in both cases have the **same magnitude** but **different phase angles**.

(D) The zero sequence components in Case-1 and Case-2 have the same magnitude but different phase angles

**Key Points**

- Multiplication by $e^{j\Delta\theta}$ changes only the phase angle of a phasor.
- Equal phase shift applied to all phases rotates the entire phasor set uniformly.
- Uniform phase rotation does not affect the magnitude of the zero sequence component.

**Mistakes to be avoided**

- Do not assume that a common phase shift changes the phasor magnitude.
- Do not confuse sequence component magnitude variation with phasor rotation.

**Q2.24.1** MSQ | 2024 | 2M | Ans: A,D | ● ● ●

For a valid sequence transformation matrix, the two sequence components must remain linearly independent and capable of reconstructing the phase voltages uniquely.
Hence, matrix **S** must be non-singular.
Check determinant of each matrix.
**For Option A:**

$$\det\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} = (1)(-1) - (1)(1) = -2 \neq 0$$

Therefore, Option A is valid.

**For Option B:**

$$\det\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix} = (1)(1) - (1)(1) = 0$$

Hence, Option B is invalid.

**For Option C:**

$$\det\begin{bmatrix}1 & 1\\ 1 & 0\end{bmatrix} = (1)(0) - (1)(1) = -1 \neq 0$$

Although invertible, this matrix does not provide orthogonal sequence decomposition and hence is not a valid sequence transformation matrix.

**For Option D:**

$$\det\begin{bmatrix}-1 & 1\\ 1 & 1\end{bmatrix} = (-1)(1) - (1)(1) = -2 \neq 0$$

Therefore, Option D is valid.
Thus, the correct options are

$$\boxed{(A)\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}}$$

$$\boxed{(D)\begin{bmatrix}-1 & 1\\ 1 & 1\end{bmatrix}}$$

**Key Points**

- **Sequence transformation** matrices **must** allow **unique reconstruction** of phase quantities.
- A valid **transformation matrix must be non-singular**.
- **Orthogonal decomposition** is important in **sequence-component** representation.

**Q2.23.1** NAT | 2023 | 2M | Ans: 7.21 ●●○

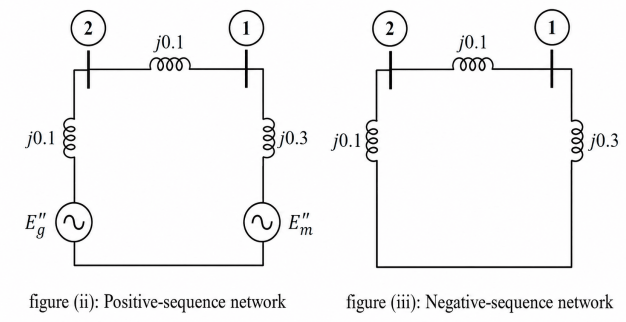



Fig. Solution Q2.23.1

**Step 1: Recall fault current relation for an L-L fault.**
For a line-to-line fault, the zero-sequence network is not involved. The positive- and negative-sequence networks are connected in parallel.

$$I_f = \sqrt{3}\,|I_{a1}|$$

where

$$I_{a1} = \frac{E}{Z_1 + Z_2}$$

Here,

$$Z_1 = \text{positive-sequence Thevenin impedance at bus 1}$$

and

$$Z_2 = \text{negative-sequence Thevenin impedance at bus 1}$$

**Step 2: Positive-sequence impedance.**
From figure (ii), the equivalent positive-sequence impedance seen from bus 1 is:

$$Z_1 = \left(\frac{j0.2 \times j0.3}{j0.2 + j0.3}\right) = j0.12$$

**Step 3: Negative-sequence impedance.**
From figure (iii), similarly,

$$Z_2 = \left(\frac{j0.2 \times j0.3}{j0.2 + j0.3}\right) = j0.12$$

**Step 4: Prefault voltage.**
Neglecting pre-fault currents,

$$E = 1.0\,\text{p.u.}$$

**Step 5: Sequence current.**

$$I_{a1} = \frac{1}{j0.12 + j0.12} = \frac{1}{j0.24} = -j4.167$$

Hence,

$$|I_{a1}| = 4.167\,\text{p.u.}$$

**Step 6: Fault current.**

$$I_f = \sqrt{3} \times 4.167 = 7.2168\,\text{p.u.}$$

$$\boxed{7.21}$$

**Key Points**

- For an L-L fault, only positive- and negative-sequence networks participate.
- Zero-sequence network is not involved in a line-to-line fault.
- Fault current relation:

$$I_f = \sqrt{3}\,|I_{a1}|$$

- Neglecting pre-fault current implies prefault voltage of 1 p.u..

**Q2.22.1** MCQ | 2022 | 1M | Ans: B ●●○

For a fully transposed three-phase transmission line,

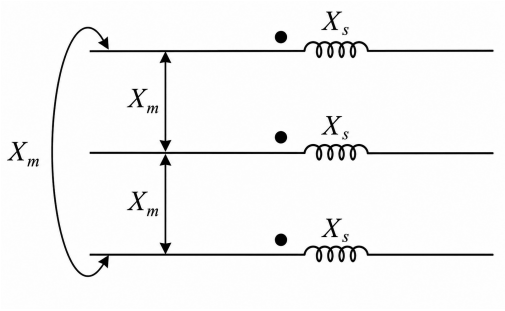



Fig. Solution Q2.22.1

the sequence reactances are given by

$$X_0 = X_s + 2X_m$$

$$X_1 = X_s - X_m$$

$$X_2 = X_s - X_m$$

Hence,

$$X_1 = X_2$$

and for a transmission line,

$$X_0 > X_1 = X_2$$

Upon checking the given options we get

$$\boxed{\text{(B) } X_1 = X_2 = 0.15 \text{ and } X_0 = 0.35}$$

which satisfies the required condition.

**Key Points**

- In a fully transposed transmission line, positive and negative sequence impedances are equal.
- Zero-sequence impedance is generally greater than positive-sequence impedance.
- Mutual reactance contributes positively in zero-sequence networks.

**Q2.21.1** MCQ | 2021 | 2M | Ans: D ●●○

For zero-sequence currents,

$$I_{A0} = I_{B0} = I_{C0}$$

and

$$I_{B0} = \frac{1}{3}(I_A + I_B + I_C)$$

Given,

$$I_{B0} = 0.1\angle 0^\circ$$

$$I_A = 1.1\angle 0^\circ$$

$$I_C = 1\angle 120^\circ + 0.1$$

Substituting into the zero-sequence relation,

$$0.1 = \frac{1}{3}(1.1\angle 0^\circ + I_B + 1\angle 120^\circ + 0.1)$$

Multiplying throughout by 3,

$$0.3 = 1.1\angle 0^\circ + I_B + 1\angle 120^\circ + 0.1$$

Rearranging,

$$I_B = 0.3 - 1.1\angle 0^\circ - 0.1 - 1\angle 120^\circ$$

$$I_B = -0.9 - 1\angle 120^\circ$$

Using phasor representation,

$$I_B = 1\angle(-120^\circ) + 0.1\angle 0^\circ$$

Therefore,

$$\boxed{\text{(D) } 1\angle(-120^\circ) + 0.1\angle 0^\circ}$$

**Key Points**

- Zero-sequence currents in all three phases are equal.
- Zero-sequence current is one-third the sum of phase currents.

**Mistakes to be avoided**

- Carefully maintain phasor angles while rearranging terms.
- Do not confuse 240° and −120° representations.

## Q2.20.1 NAT 2020 2M Ans: 1.02 ●●○

**Concept:**
Immediately after a three-phase short-circuit, the flux linkages of the rotor windings cannot change instantaneously. Therefore, the generator is represented by its subtransient reactance $X_d''$, and the subtransient internal emf remains behind this reactance.
The subtransient internal emf is calculated using

$$E'' = V + jX_d''I$$

where the pre-fault armature current is obtained from the steady-state operating condition.

—

**Step 1: Calculate the pre-fault armature current**
Using the steady-state equivalent circuit,

$$E = V + jX_dI$$

Hence,

$$I = \frac{E - V}{jX_d}$$

Substituting the given values,

$$I = \frac{(1 + j0.7) - (1 + j0)}{j0.7}$$

$$= \frac{j0.7}{j0.7} = 1 \text{ p.u.}$$

—

**Step 2: Calculate the subtransient internal emf**

Immediately after the fault,

$$E'' = V + jX_d''I$$

Substituting,

$$E'' = (1 + j0) + j(0.2)(1)$$

$$= 1 + j0.2$$

—

**Step 3: Magnitude of subtransient internal emf**

$$|E''| = \sqrt{1^2 + 0.2^2}$$

$$= \sqrt{1.04} = 1.0198$$

Therefore,

$$\boxed{|E''| = 1.02 \text{ p.u.}}$$

**Q2.19.1** NAT | 2019 | 2M | Ans: 1.7 to 1.9 | ● ● ○

Given:

$$X_1 = 0.25 \text{ pu}$$

$$X_2 = 0.15 \text{ pu}$$

$$X_0 = 0.05 \text{ pu}$$

For a bolted three-phase fault,

$$I_{3\phi} = \frac{1}{X_1}$$

For a bolted single line-to-ground (LG) fault,

$$I_{LG} = \frac{3}{X_1 + X_2 + X_0 + 3X_n}$$

According to the problem,

$$I_{LG} = I_{3\phi}$$

Therefore,

$$\frac{3}{0.25 + 0.15 + 0.05 + 3X_n} = \frac{1}{0.25}$$

Cross multiplying,

$$3 \times 0.25 = 0.25 + 0.15 + 0.05 + 3X_n$$

$$0.75 = 0.45 + 3X_n$$

$$3X_n = 0.30$$

$$X_n = 0.10\,\text{pu}$$

Base impedance is

$$Z_{base} = \frac{V_{base}^2}{S_{base}}$$

$$Z_{base} = \frac{30^2}{50} = 18\,\Omega$$

Grounding reactance is

$$X_n = 0.1 \times 18$$

$$X_n = 1.8\,\Omega$$

Therefore,

$$\boxed{1.8}$$

**Mistakes to be avoided**

- Do not forget the factor $3X_n$ in LG fault analysis.
- Carefully distinguish between pu values and actual ohmic values.

**Q2.18.1** NAT | 2018 | 1M | Ans: 7 | ● ○ ○

For a symmetrical three-phase transmission line,

$$Z_1 = Z_s - Z_m$$

$$Z_0 = Z_s + 2Z_m$$

Given,

$$Z_1 = 1 + j10$$

$$Z_0 = 4 + j31$$

Subtracting the two equations,

$$Z_0 - Z_1 = 3Z_m$$

Substituting values,

$$(4 + j31) - (1 + j10) = 3Z_m$$

$$3 + j21 = 3Z_m$$

$$Z_m = 1 + j7$$

Therefore, the imaginary part of $Z_m$ is

$$\boxed{7}$$

**Q2.18.2** NAT | 2018 | 1M | Ans: 73 to 74 | ● ○ ○

Given:

$$Z_1 = j0.1\,\text{pu}$$

$$Z_2 = j0.05\,\text{pu}$$

$$Z_0 = j0.01\,\text{pu}$$

Grounding impedance:

$$Z_n = j0.01\,\text{pu}$$

For a single line-to-ground fault,

$$I_f(\text{pu}) = \frac{3}{Z_1 + Z_2 + Z_0 + 3Z_n}$$

Substituting values,

$$I_f(\text{pu}) = \frac{3}{0.1 + 0.05 + 0.01 + 3(0.01)}$$

$$I_f(\text{pu}) = \frac{3}{0.19}$$

$$I_f(\text{pu}) = 15.789$$

Base current is

$$I_{base} = \frac{S_{base}}{\sqrt{3}\,V_{base}}$$

$$I_{base} = \frac{125 \times 10^6}{\sqrt{3} \times 15.5 \times 10^3}$$

$$I_{base} = 4.656\,\text{kA}$$

Actual fault current is

$$I_f = 15.789 \times 4.656$$

$$I_f = 73.52\,\text{kA}$$

Therefore,

$$\boxed{73.52}$$

**Mistakes to be avoided**

- Do not forget to include grounding impedance in LG fault analysis.
- Use line-to-line voltage while calculating base current.
- Carefully distinguish per-unit current from actual current in kA.

**Q2.18.3** MCQ | 2018 | 2M | Ans: A | ● ○ ○

For a line-to-line fault with fault impedance $Z_f$,

$$I_f(\text{with } Z_f) = \frac{\sqrt{3}}{Z_1 + Z_2 + Z_f}$$

For zero fault impedance,

$$I_f(\text{without } Z_f) = \frac{\sqrt{3}}{Z_1 + Z_2}$$

According to the question,

$$I_f(\text{with } Z_f) = k\, I_f(\text{without } Z_f)$$

Substituting the fault current expressions,

$$\frac{\sqrt{3}}{Z_1 + Z_2 + Z_f} = k\left(\frac{\sqrt{3}}{Z_1 + Z_2}\right)$$

Cancelling $\sqrt{3}$,

$$\frac{1}{Z_1 + Z_2 + Z_f} = \frac{k}{Z_1 + Z_2}$$

Cross multiplying,

$$Z_1 + Z_2 = k(Z_1 + Z_2 + Z_f)$$

$$Z_1 + Z_2 = k(Z_1 + Z_2) + kZ_f$$

$$(1 - k)(Z_1 + Z_2) = kZ_f$$

Hence,

$$Z_f = \frac{(Z_1 + Z_2)(1 - k)}{k}$$

Therefore,

$$\boxed{\text{(A) } Z_f = \frac{(Z_1 + Z_2)(1 - k)}{k}}$$

**Q2.17.1** NAT | 2017 | 2M | Ans: 0.1 | ● ○ ○

Given,

$$Z_1 = 0.2 \text{ pu}$$

$$Z_2 = 0.2 \text{ pu}$$

$$Z_0 = 0.1 \text{ pu}$$

$$V_{th} = 1 \text{ pu}$$

$$I_f = 3.75 \text{ pu}$$

For a single line-to-ground (LG) fault,

$$I_f = \frac{3V_{th}}{Z_1 + Z_2 + Z_0 + 3Z_n}$$

Substituting the given values,

$$3.75 = \frac{3 \times 1}{0.2 + 0.2 + 0.1 + 3Z_n}$$

$$0.2 + 0.2 + 0.1 + 3Z_n = \frac{3}{3.75}$$

$$0.5 + 3Z_n = 0.8$$

$$3Z_n = 0.3$$

$$Z_n = 0.1 \text{ pu}$$

$$\boxed{0.1}$$

**Q2.16.1** NAT | 2016 | 1M | Ans: 0.83 to 0.85 | ● ○ ○

For a three-phase fault at bus $A$,

$$I_{fA} = \frac{V_{prefault}}{Z_{AA}}$$

Given,

$$I_{fA} = 10 \text{ pu}$$

$$V_{prefault} = 1 \text{ pu}$$

Hence,

$$Z_{AA} = \frac{1}{10} = 0.1 \text{ pu}$$

Similarly, for a fault at bus $B$,

$$I_{fB} = \frac{1}{Z_{BB}}$$

Given,

$$I_{fB} = 8 \text{ pu}$$

Thus,

$$Z_{BB} = \frac{1}{8} = 0.125 \text{ pu}$$

Voltage at bus $B$ during fault at bus $A$ is

$$V_B = 0.8 \text{ pu}$$

For a fault at bus $A$,

$$V_B = 1 - \frac{Z_{BA}}{Z_{AA}}$$

Substituting,

$$0.8 = 1 - \frac{Z_{BA}}{0.1}$$

$$\frac{Z_{BA}}{0.1} = 0.2$$

$$Z_{BA} = 0.02 \text{ pu}$$

Since bus impedance matrix is symmetric,

$$Z_{AB} = Z_{BA} = 0.02 \text{ pu}$$

Now, voltage at bus $A$ during fault at bus $B$ is

$$V_A = 1 - \frac{Z_{AB}}{Z_{BB}}$$

Substituting,

$$V_A = 1 - \frac{0.02}{0.125}$$

$$V_A = 1 - 0.16$$

$$V_A = 0.84 \text{ pu}$$

$$\boxed{0.84}$$

## Q2.16.2 NAT 2016 2M Ans: 2.20 to 2.35 ● ● ●

The maximum power transferred in a lossless system is

$$P_{\max} = \frac{EV}{X}$$

Since $E = 1.0$ p.u. and $V = 1.0$ p.u.,

$$P_{\max} \propto \frac{1}{X}$$

Hence,

$$\frac{P_1}{P_2} = \frac{X_2}{X_1}$$

where

- $X_1$ = transfer reactance during pre-fault condition
- $X_2$ = transfer reactance during faulted condition

During pre-fault condition, the two transmission paths are identical.
Each path reactance is

$$j0.1 + j0.5 = j0.6$$

Hence,

$$X_1 = j0.2 + (j0.6 \parallel j0.6)$$

$$X_1 = j0.2 + \frac{j0.6 \times j0.6}{j1.2}$$

$$X_1 = j0.2 + j0.3$$

$$X_1 = j0.5$$

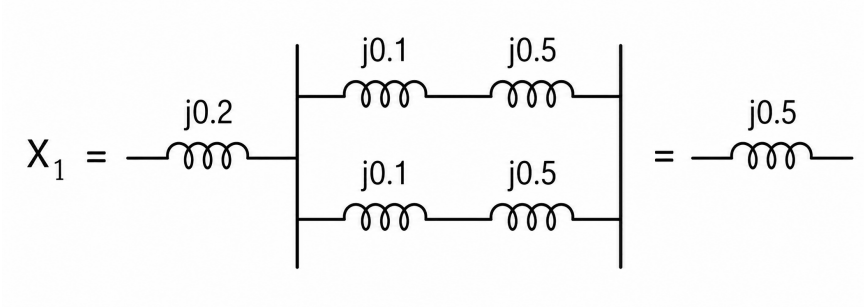



Fig. Solution Q2.16.2

During the fault, the fault occurs at the midpoint of line 2. Let $A$ be the bus after the $j0.2$ reactance and $B$ the infinite bus.

The reactance from bus $A$ to the fault point (transformer and half of line 2) is

$$j0.1 + j0.25 = j0.35$$

The reactance from the fault point to the infinite bus $B$ (other half of line 2, no transformer on this side) is

$$j0.25$$

The healthy path (transformer and line 1) between $A$ and $B$ is

$$j0.1 + j0.5 = j0.6$$

The fault point is at zero potential, so bus $A$ is connected to ground through $j0.35$, and the $j0.25$ branch is connected directly across the infinite bus, where it does not affect the power transfer.

The faulted network is therefore a T-network: $j0.2$ from the generator internal bus to $A$, $j0.35$ from $A$ to ground and $j0.6$ from $A$ to $B$. Converting this star to the equivalent delta, the transfer reactance between the generator internal bus and the infinite bus is

$$X_2 = j0.2 + j0.6 + \frac{j0.2 \times j0.6}{j0.35}$$

$$X_2 = j0.8 + j0.3429$$

$$X_2 = j1.1429$$

Therefore,

$$\frac{P_1}{P_2} = \frac{X_2}{X_1} = \frac{1.1429}{0.5}$$

$$\frac{P_1}{P_2} = 2.286$$

$$\boxed{2.29}$$

**Key Points**

- During fault analysis, transfer reactance changes because of altered network topology.
- A branch connected between the fault point and the infinite bus does not affect the transfer reactance.
- Do not forget the generator reactance $j0.2$ in the equivalent network.

**Q2.16.3** NAT | 2016 | 2M | Ans: 1.05 to 1.15 | ● ○ ○

Given,

$$X_1 = 0.15 \text{ pu}$$

$$X_2 = 0.15 \text{ pu}$$

$$X_0 = 0.05 \text{ pu}$$

$$Z_f = j0.1 \text{ pu}$$

Generator rating:

30 MVA, 13.8 kV

Base current:

$$I_B = \frac{30 \times 10^6}{\sqrt{3} \times 13.8 \times 10^3}$$

$$I_B = 1255.1 \text{ A}$$

Given positive sequence generator current:

$$I_1 = 4270 \text{ A}$$

Therefore,

$$I_1(\text{pu}) = \frac{4270}{1255.1}$$

$$I_1(\text{pu}) = 3.402$$

For a double line-to-ground (LLG) fault,

$$I_1 = \frac{E_a}{X_1 + \left(X_2 \parallel X_0'\right)}$$

where,

$$X_0' = X_0 + 3(X_n + Z_f)$$

Substituting values,

$$X_0' = 0.05 + 3(X_n + 0.1)$$

$$X_0' = 3X_n + 0.35$$

Now,

$$3.402 = \frac{1}{0.15 + \left(\frac{0.15(3X_n+0.35)}{0.15+3X_n+0.35}\right)}$$

Solving,

$$X_n \approx 1.07 \text{ pu}$$

$$\boxed{1.07}$$

**Mistakes to be avoided**

- Do not forget to include fault impedance in the zero-sequence path.
- Use per-unit current before applying sequence-network equations.
- Carefully compute parallel combination:

$$X_2 \parallel X_0'$$

**Q2.16.4** NAT | 2016 | 2M | Ans: 5.5 to 6.5 | ● ○ ○

For a single line-to-ground (L-G) fault,

$$I_f = 3I_{a1}$$

The sequence networks are connected in series.

Since two identical generators are connected in parallel, the equivalent positive and negative sequence impedances are

$$Z_1 = j0.4 \parallel j0.4 = j0.2$$

$$Z_2 = j0.3 \parallel j0.3 = j0.15$$

The neutral of only one generator is grounded. The ungrounded generator provides no path for zero-sequence current, so only the grounded generator appears in the zero-sequence network:

$$Z_0 = j0.15$$

Thus,

$$Z_{eq} = j0.2 + j0.15 + j0.15 = j0.5$$

$$I_{a1} = \frac{1}{0.5} = 2$$

Therefore,

$$I_f = 3 \times 2$$

$$I_f = 6 \text{ p.u.}$$

6

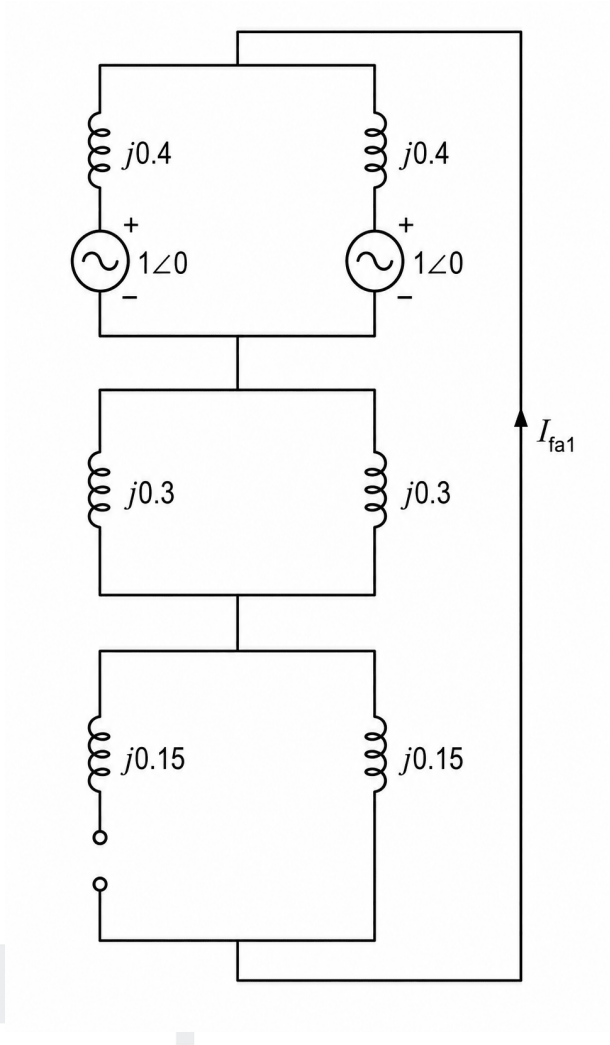



Fig. Solution Q2.16.4

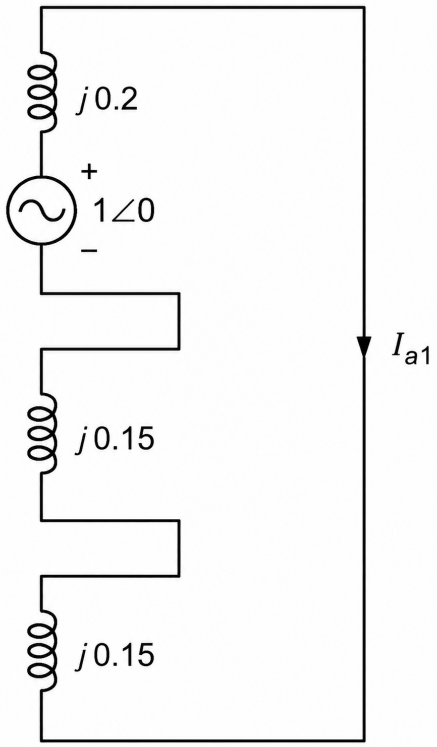



Fig. Solution Q2.16.4

**Q2.15.1** MCQ | 2015 | 2M | Ans: B

From the phasor diagram, the magnitude of voltage at bus 1 is smaller than that at bus 2.
Hence, the fault must be electrically closer to bus 1.
Therefore, the fault location can either be:

$Q$ or $R$

Now examine the current directions.

- Currents $I_1$ and $I_3$ are directed towards the faulted section from opposite ends.
- Currents $I_2$ and $I_4$ are nearly opposite in phase, indicating normal power transfer through the healthy lower transmission line.

Therefore, the lower line is healthy and the fault is located on the upper transmission line near bus 1.
Hence, the fault location is

(B) Location Q

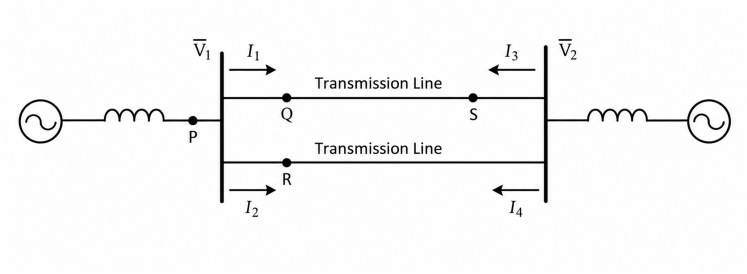



Fig. Solution Q2.15.1

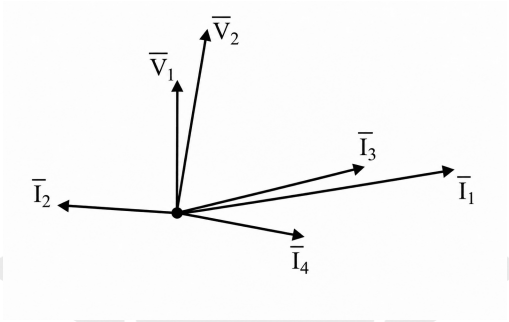



Fig. Solution Q2.15.1

**Key Points**

- During a three-phase fault, voltage magnitude decreases significantly near the fault point.
- Fault current flows toward the fault from both system ends.
- Healthy lines continue carrying normal through current.
- Phasor directions help identify faulted and healthy transmission paths..
- Opposite current directions in a line generally indicate normal through-flow rather than a faulted section.

## Q2.14.1 MCQ 2014 1M Ans: C ●●○

During a three-phase fault, the fault current is predominantly reactive because the network impedance is mainly inductive.
Hence, current lags the voltage in the inductive network.
For a fault at location $F_1$:

- Bus $A$ voltage is represented by $V_{F1}$.
- Fault current is represented by $I_{F1}$.
- Since the fault current is largely inductive,

$$I_{F1} \text{ lags } V_{F1}$$

- Therefore,

$$V_{F1} \text{ leads } I_{F1}$$

For a fault at location $F_2$:

- The current direction at bus $A$ reverses because the fault lies on the opposite side of the bus.
- Due to the chosen current reference direction, the phase relationship appears reversed.

- Hence,

$$V_{F2} \text{ lags } I_{F2}$$

Therefore, the correct option is

$$\boxed{\text{(C) } V_{F1} \text{ lags } I_{F1} \text{ and } V_{F2} \text{ leads } I_{F2}}$$

**Key Points**

- Three-phase fault current is mainly limited by system reactance.
- In inductive systems, current lags voltage.
- Relative phase relationship depends on the chosen current reference direction.
- Fault location changes the effective current direction at the observation bus.

**Mistakes to be avoided**

- Do not ignore the reference direction of fault current.
- Fault current phase angle depends on network reactance.
- Voltage-current lead/lag relation may appear reversed when current direction changes.

## Q2.14.2 MCQ 2014 1M Ans: B ●●○

From the given zero-sequence network:

- Zero-sequence current flows through generator neutral impedances:

$$3X_{Gn} \quad \text{and} \quad 3X_{Mn}$$

- The zero-sequence network is interrupted at transformer locations using blocking points.

For transformer $T_1$:

- Zero-sequence current is allowed from Bus 1 side.
- Therefore, Bus 1 side must be grounded star connected.
- Zero-sequence current does not pass through the other side.
- Hence, the opposite side must be ungrounded star connected.

Thus,

$$T_1 : \mathrm{Y_g} - \mathrm{Y}$$

For transformer $T_2$:

- Zero-sequence current is blocked beyond transformer $T_2$ toward the line side.
- The transformer side connected to the transmission line must therefore be delta connected.
- The generator side allows zero-sequence current through grounded neutral.

Hence,

$$T_2 : \mathrm{Y_g} - \Delta$$

Therefore, the correct option is

$$\boxed{\text{(B) } \mathrm{Y_g}-\mathrm{Y} \text{ and } \mathrm{Y_g}-\Delta}$$

**Q2.14.3** MCQ | 2014 | 2M | Ans: C | ● ○ ○

Given,

$$I_a = 1\angle(-90^\circ) \text{ pu}$$

$$I_{b2} = 4\angle(-150^\circ) \text{ pu}$$

$$I_{c0} = 3\angle 90^\circ \text{ pu}$$

For zero-sequence components,

$$I_{a0} = I_{b0} = I_{c0}$$

Hence,

$$I_{a0} = 3\angle 90^\circ$$

For negative-sequence components,

$$I_{b2} = aI_{a2}$$

where

$$a = 1\angle 120^\circ$$

Therefore,

$$4\angle(-150^\circ) = 1\angle 120^\circ \times I_{a2}$$

$$I_{a2} = \frac{4\angle(-150^\circ)}{1\angle 120^\circ}$$

$$I_{a2} = 4\angle(-270^\circ)$$

Now,

$$I_a = I_{a1} + I_{a2} + I_{a0}$$

Substituting,

$$1\angle(-90^\circ) = I_{a1} + 4\angle(-270^\circ) + 3\angle 90^\circ$$

Since,

$$4\angle(-270^\circ) = 4\angle 90^\circ$$

we get

$$I_{a1} = 1\angle(-90^\circ) - 4\angle 90^\circ - 3\angle 90^\circ$$

$$I_{a1} = 8\angle(-90^\circ)$$

For positive-sequence currents,

$$I_{b1} = a^2 I_{a1}$$

$$I_{b1} = 1\angle 240^\circ \times 8\angle(-90^\circ)$$

$$I_{b1} = 8\angle 150^\circ$$

Now,

$$I_b = I_{b1} + I_{b2} + I_{b0}$$

Substituting,

$$I_b = 8\angle 150^\circ + 4\angle(-150^\circ) + 3\angle 90^\circ$$

Converting into rectangular form,

$$I_b = (-6.928 + j4) + (-3.464 - j2) + j3$$

$$I_b = -10.392 + j5$$

Hence,

$$|I_b| = \sqrt{(-10.392)^2 + 5^2}$$

$$|I_b| \approx 11.53$$

Therefore,

$$\boxed{\text{(C) } 11.53}$$

**Q2.14.4** NAT | 2014 | 2M | Ans: 15300 to 15500 | ● ○ ○

For a single line-to-ground (L-G) fault,

$$I_f = \frac{3V}{Z_1 + Z_2 + Z_0}$$

Given,

$$X_1 = 0.2 \text{ pu}$$

$$X_2 = 0.2 \text{ pu}$$

$$X_0 = 0.05 \text{ pu}$$

Since the generator neutral is solidly grounded,

$$Z_n = 0$$

For an unloaded generator before fault,

$$V = 1\angle 0^\circ \text{ pu}$$

Hence,

$$I_f = \frac{3}{j0.2 + j0.2 + j0.05}$$

$$I_f = \frac{3}{j0.45}$$

$$|I_f| = \frac{3}{0.45} = 6.667 \text{ pu}$$

Base current is

$$I_{\text{base}} = \frac{S_{\text{base}}}{\sqrt{3}V_{\text{base}}}$$

Substituting,

$$I_{\text{base}} = \frac{100 \times 10^6}{\sqrt{3} \times 25 \times 10^3}$$

$$I_{\text{base}} \approx 2309.4 \text{ A}$$

Therefore,

$$I_f = 6.667 \times 2309.4$$

$$I_f \approx 15396 \text{ A}$$

Hence,

$$\boxed{15396}$$

**Q2.14.5** MCQ | 2014 | 1M | Ans: B | ● ○ ○

In a fully transposed transmission line, each phase occupies all physical positions equally over the total line length.
Hence, the mutual coupling between phases becomes symmetrical.
Therefore,

$$Z_1 = Z_2$$

i.e., positive and negative sequence impedances are equal.
However, zero-sequence current returns through ground or neutral path, which differs from the positive and negative sequence current paths.
Thus,

$$Z_0 \neq Z_1$$

in general.
Hence, the correct option is

(B) positive and negative sequence impedances are equal

## Q2.12.1 MCQ 2012 1M Ans: C ●●○

For different types of unsymmetrical faults, the sequence currents satisfy characteristic relations.
For a double line-to-ground (LLG) fault,

$$I_1 = -(I_2 + I_0)$$

Given,

$$I_1 = j1.5$$

$$I_2 = -j0.5$$

$$I_0 = -j1$$

Now,

$$-(I_2 + I_0) = -(-j0.5 - j1)$$

$$= j1.5$$

Thus,

$$I_1 = -(I_2 + I_0)$$

which satisfies the condition for a double line-to-ground fault.
Hence, the fault type is

LLG

Therefore, the correct option is

(C) *LLG*

### Key Points

- Sequence current relations:
  - LG fault:
    $$I_1 = I_2 = I_0$$
  - LL fault:
    $$I_0 = 0, \qquad I_1 = -I_2$$
  - LLG fault:
    $$I_1 = -(I_2 + I_0)$$
- Sequence components help identify fault type directly.

**Q2.11.1** MCQ | 2011 | 2M | Ans: A

Convert all reactances to the common base of 100 MVA.

For generator $G_1$,

$$X''_{G_1} = 0.25 \times \frac{100}{250} = 0.10 \text{ pu}$$

For generator $G_2$,

$$X''_{G_2} = 0.10 \times \frac{100}{100} = 0.10 \text{ pu}$$

For transmission lines $L_1$ and $L_2$,

$$X_{L_1} = X_{L_2} = 0.225 \times 10 = 2.25\,\Omega$$

The base impedance is

$$Z_{\text{base}} = \frac{kV_{\text{base}}^2}{MVA_{\text{base}}} = \frac{15^2}{100} = 2.25\,\Omega$$

Hence,

$$X_{L_1}(\text{pu}) = \frac{2.25}{2.25} = 1.0$$

$$X_{L_2}(\text{pu}) = \frac{2.25}{2.25} = 1.0$$

Therefore, the positive sequence reactance diagram on the 100 MVA common base is

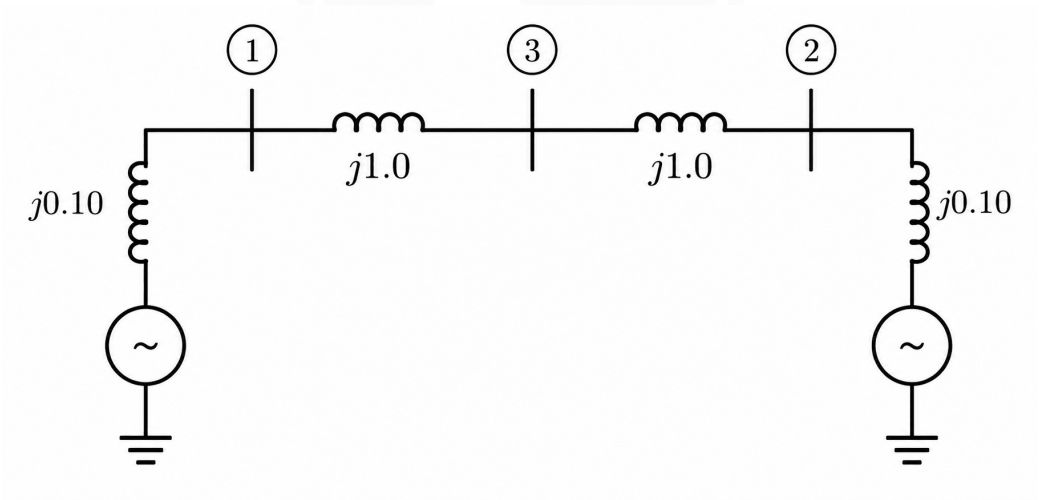



Fig. Solution Q2.11.1

(A)

**Key Points**

- Per-unit conversion on a common MVA base is given by $X_{\text{new}} = X_{\text{old}} \left(\frac{MVA_{\text{new}}}{MVA_{\text{old}}}\right)$.
- Base impedance is $Z_{\text{base}} = \frac{(kV_{\text{base}})^2}{MVA_{\text{base}}}$.
- All network elements must be converted to the same base before drawing the sequence network.

## Q2.11.2 MCQ 2011 2M Ans: D ● ○ ○

For a three-phase fault at bus 3, the Thevenin reactance seen from the fault location is obtained from the positive sequence network.

$$X_{th} = (j0.1 + j1.0) \parallel (j0.1 + j1.0)$$

$$= j1.1 \parallel j1.1$$

$$= j\left(\frac{1.1 \times 1.1}{1.1 + 1.1}\right) = j0.55 \text{ pu}$$

The fault level is

$$\text{Fault MVA} = \frac{\text{Base MVA}}{X_{th}}$$

$$= \frac{100}{0.55} = 181.82 \text{ MVA}$$

(D) 181.82 MVA

## Q2.10.1 MCQ 2010 1M Ans: C ● ● ○

For obtaining the zero-sequence network of a transformer, the following rules are used:

- A grounded star winding provides a path for zero-sequence current.
- A delta winding permits circulation of zero-sequence currents internally but does not allow zero-sequence current to pass to external lines.
- An ungrounded star winding blocks zero-sequence current.

The given transformer has a star-connected primary and a delta-connected secondary.
The standard zero-sequence equivalent is

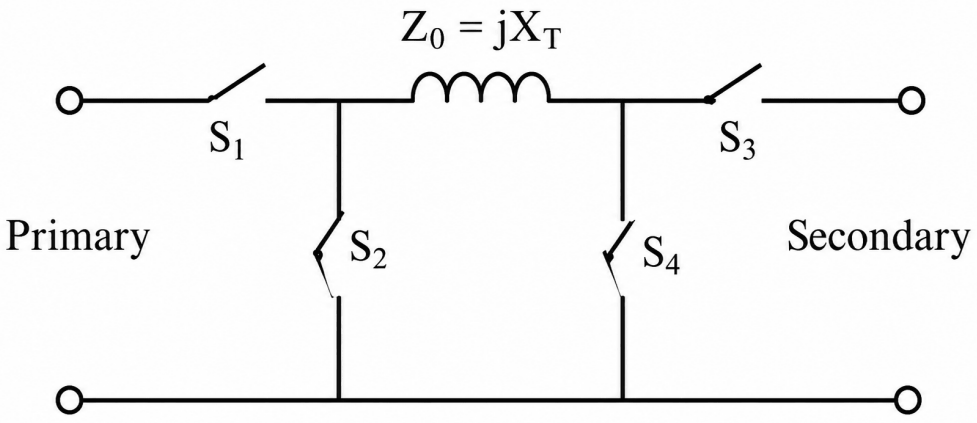



Fig. Solution Q2.10.1

In the switching representation:

- Switches corresponding to delta windings remain closed.
- Switches corresponding to grounded star neutrals remain closed.
- Since the primary neutral is not grounded, the primary-side zero-sequence current path is blocked.

Therefore, the practical zero-sequence circuit becomes

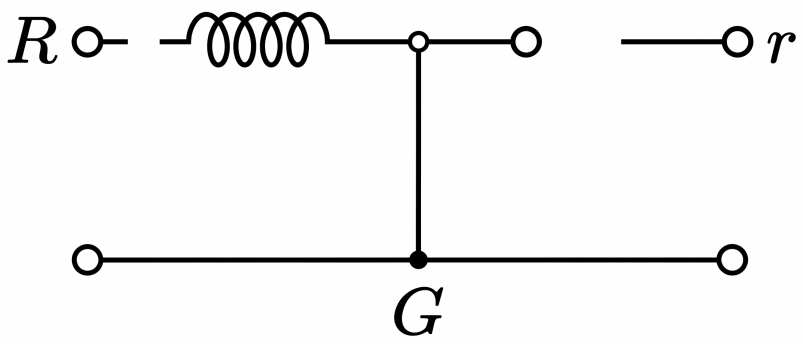



Fig. Solution Q2.10.1

Among the available choices, the closest matching circuit is

(C)

**Q2.08.1** MCQ | 2008 | 2M | Ans: A

For a symmetrical three-phase fault, the fault current is

$$i(t) = Ae^{-(R/L)t} + \frac{\sqrt{2}V_m}{|Z|}\cos(\omega t - \alpha)$$

where

$$Z = R + jX$$

At the instant of fault inception $t = t_0$,

$$i(t_0) = 0$$

Hence,

$$Ae^{-(R/L)t_0} = -\frac{\sqrt{2}V_m}{|Z|}\cos(\omega t_0 - \alpha)$$

The problem states that the decaying dc-offset current is

- negative, and
- at its maximum initial magnitude.

For maximum negative dc offset,

$$\cos(\omega t_0 - \alpha) = 1$$

Therefore,

$$\omega t_0 - \alpha = 0$$

$$t_0 = \frac{\alpha}{\omega} \tag{1}$$

The impedance from source $S_1$ to the fault point is

$$Z = 0.004 + j0.04$$

$$|Z| = \sqrt{(0.004)^2 + (0.04)^2} = 0.0402$$

$$\alpha = \tan^{-1}\left(\frac{0.04}{0.004}\right)$$

$$\alpha = 84.29^\circ = 1.471 \text{ rad}$$

The system frequency is

$$f = 50 \text{ Hz}$$

Hence,

$$\omega = 2\pi f = 2\pi(50)$$

Using (1),

$$t_0 = \frac{1.471}{2\pi \times 50}$$

$$= 0.004682 \text{ s}$$

$$= 4.682 \text{ ms}$$

Therefore, answer is

$$\boxed{\text{(A) } t_0 = 4.682 \text{ ms}}$$

**Q2.08.2** MCQ | 2008 | 2M | Ans: A | ● ○ ○

For a balanced three-phase fault, only the positive-sequence network is involved.
The positive-sequence impedance from source $S_1$ to fault point $F$ is

$$Z = 0.004 + j0.04$$

Hence, the fault current contribution from bus $X$ is

$$I_X = \frac{V}{Z}$$

Since

$$V = 1\angle 0^\circ \text{ pu}$$

$$I_X = \frac{1}{0.004 + j0.04}$$

$$= 24.88\angle(-84.29^\circ) \text{ pu}$$

Therefore, the rms value of the ac component is

$$|I_X| = 24.88 \text{ pu}$$

The base current is

$$I_{\text{base}} = \frac{\text{Base MVA}}{\sqrt{3} \times \text{Base kV}}$$

$$= \frac{100 \times 10^6}{\sqrt{3} \times 400 \times 10^3}$$

$$= 144.34 \text{ A}$$

Hence,

$$I_X = 24.88 \times 144.34$$

$$= 3591.1 \text{ A}$$

$$= 3.59 \text{ kA}$$

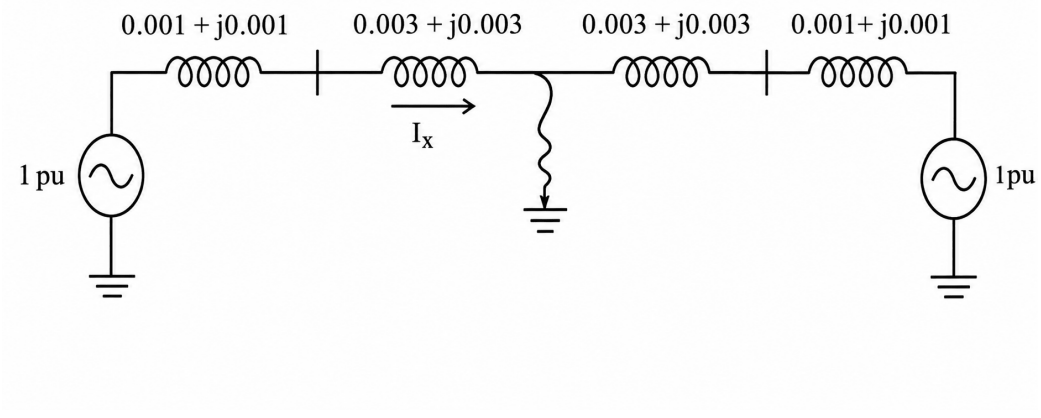



Fig. Solution Q2.08.2

Hence, the answer is

(A) 3.59 kA

**Q2.08.3** MCQ 2008 2M Ans: C ● ○ ○

For a single line-to-ground fault on phase $a$ with zero fault impedance, the positive, negative and zero sequence networks are connected in series at the fault point $F$:

$$I_f = 3I_{a1} = \frac{3V}{Z_{1,th} + Z_{2,th} + Z_{0,th}}$$

where $Z_{1,th}$, $Z_{2,th}$ and $Z_{0,th}$ are the sequence Thevenin impedances seen at $F$.

From the common data, the positive-sequence impedance from source $S_1$ to the fault point $F$ is

$$Z_{1X} = 0.004 + j0.04 \text{ pu}$$

Since $F$ is the centre of the line and the system is symmetrical, the impedance from $S_2$ to $F$ is the same. The fault is therefore fed from both sides, and

$$Z_{1,th} = Z_{1X} \parallel Z_{1X} = \frac{Z_{1X}}{2}$$

Given that negative-sequence impedances equal positive-sequence impedances and zero-sequence impedances are three times positive-sequence impedances,

$$Z_{2,th} = \frac{Z_{1X}}{2}, \qquad Z_{0,th} = \frac{3Z_{1X}}{2}$$

Hence, the total fault current is

$$I_f = \frac{3 \times 1}{\frac{Z_{1X}}{2} + \frac{Z_{1X}}{2} + \frac{3Z_{1X}}{2}} = \frac{6}{5Z_{1X}}$$

$$|Z_{1X}| = |0.004 + j0.04| = 0.04020 \text{ pu}$$

$$|I_f| = \frac{6}{5 \times 0.04020} = 29.85 \text{ pu}$$

Because the two sides are identical in every sequence network, each sequence current divides equally between the two sides. The phase-$a$ fault current flowing from bus $X$ is therefore half of the total fault current:

$$|I_X| = \frac{|I_f|}{2} = \frac{3}{5|Z_{1X}|} = 14.93 \text{ pu}$$

Hence, the correct answer is

(C) 14.93 pu

**Q2.07.1** MCQ 2007 2M Ans: B ● ● ○

Let

$$[F] = \begin{bmatrix} 1 & 1 & 1 \\ \alpha^2 & \alpha & 1 \\ \alpha & \alpha^2 & 1 \end{bmatrix}$$

Then the given transformation can be written as

$$[f]^{abc} = k[F][f]^{pno}$$

Hence,

$$[V]^{abc} = k[F][V]^{pno}$$

and

$$[I]^{abc} = k[F][I]^{pno}$$

Therefore,

$$[I]^{pno} = \frac{1}{k}[F]^{-1}[I]^{abc}$$

Given

$$[V]^{pno} = [A][I]^{pno}$$

where

$$[A] = \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Substituting,

$$[V]^{abc} = k[F][A][I]^{pno} = [F][A][F]^{-1}[I]^{abc}$$

Hence,

$$Z = [F][A][F]^{-1}$$

Using

$$[F]^{-1} = \frac{1}{3}\begin{bmatrix} 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \\ 1 & 1 & 1 \end{bmatrix}$$

we obtain

$$Z = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ \alpha^2 & \alpha & 1 \\ \alpha & \alpha^2 & 1 \end{bmatrix}\begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \\ 1 & 1 & 1 \end{bmatrix}$$

Using

$$1 + \alpha + \alpha^2 = 0,$$

the multiplication simplifies to

$$Z = \frac{1}{3}\begin{bmatrix} 3 & 1.5 & 1.5 \\ 1.5 & 3 & 1.5 \\ 1.5 & 1.5 & 3 \end{bmatrix}$$

$$Z = \begin{bmatrix} 1.0 & 0.5 & 0.5 \\ 0.5 & 1.0 & 0.5 \\ 0.5 & 0.5 & 1.0 \end{bmatrix}$$

Hence, the correct answer is

$$\text{(B) } Z = \begin{bmatrix} 1.0 & 0.5 & 0.5 \\ 0.5 & 1.0 & 0.5 \\ 0.5 & 0.5 & 1.0 \end{bmatrix}$$

**Q2.06.1** MCQ | 2006 | 2M | Ans: B | ● ● ○

For the positive sequence, take $V_{an1} = V\angle 0°$ and $V_{bn1} = V\angle(-120°)$. Then

$$V_{ab1} = V_{an1} - V_{bn1} = \sqrt{3}\, V_{an1}\angle 30°.$$

Positive Sequence Voltage Relationship

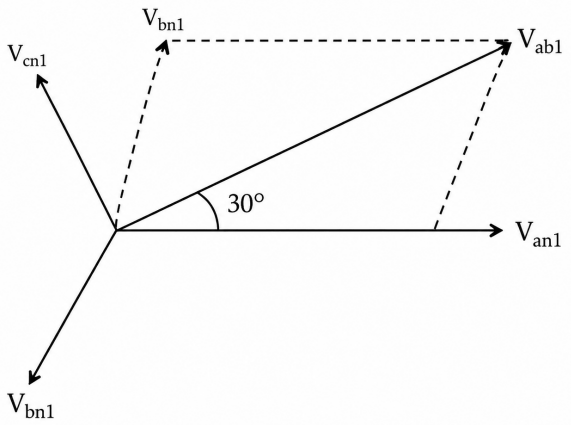



Fig. Solution Q2.06.1

For the negative sequence, the phase order is reversed: $V_{an2} = V\angle 0°$ and $V_{bn2} = V\angle 120°$. Then

$$V_{ab2} = V_{an2} - V_{bn2} = \sqrt{3}\, V_{an2}\angle(-30°).$$

Negative Sequence Voltage Relationship

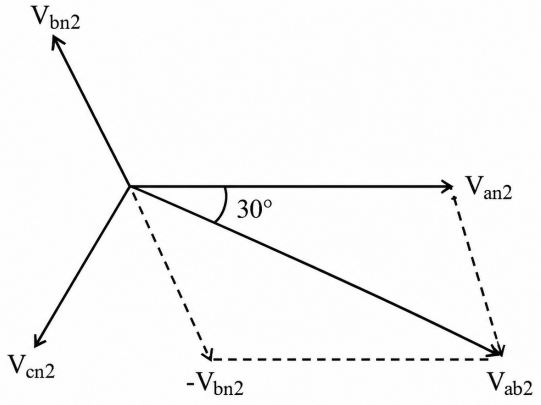



Fig. Solution Q2.06.1

So the line voltage leads the phase voltage by 30° in the positive sequence and lags it by 30° in the negative sequence. Hence the phase-to-neutral components are

$$V_{an1} = \frac{V_{ab1}}{\sqrt{3}\angle 30^\circ}, \qquad V_{an2} = \frac{V_{ab2}}{\sqrt{3}\angle(-30^\circ)}.$$

The line base voltage is $\sqrt{3}$ times the phase base voltage. When each quantity is divided by its own base, the factor $\sqrt{3}$ cancels, so the per-unit magnitudes are unchanged:

$$\boxed{\begin{aligned}&\text{(B)} \quad V_{an1} = X\angle(\theta_1 - 30^\circ)\\ &\qquad\quad V_{an2} = Y\angle(\theta_2 + 30^\circ)\end{aligned}}$$

**Key Points**

- The positive- and negative-sequence line-to-phase angle shifts are opposite: $-30^\circ$ and $+30^\circ$ respectively.
- In per unit, with line and phase quantities on their respective bases, the $\sqrt{3}$ cancels.

**Q2.05.1** MCQ | 2005 | 2M | Ans: D | ● ○ ○

For a three-phase fault,

$$\text{Fault MVA} = \frac{V_{LL}^2}{X_1}$$

Hence,

$$X_1 = \frac{(220)^2}{4000}$$

$$X_1 = 12.1\ \Omega$$

Therefore, the positive-sequence driving point reactance is

$$\boxed{\text{(D) } 12.1\ \Omega}$$

**Q2.05.2** MCQ | 2005 | 2M | Ans: B | ● ● ○

From the above Question,

$$X_1 = X_2 = 12.1\ \Omega$$

For a single-line-to-ground fault, the fault current is

$$I_f = \frac{3V_{ph}}{X_1 + X_2 + X_0}$$

and the fault MVA is

$$\text{Fault MVA} = \sqrt{3}\,V_L\,I_f = 3V_{ph}I_f = \frac{9V_{ph}^2}{X_1 + X_2 + X_0} = \frac{3V_L^2}{X_1 + X_2 + X_0}$$

With

$$V_L = 220\ \text{kV}$$

and the single-line-to-ground fault level of 5000 MVA,

$$X_1 + X_2 + X_0 = \frac{3V_L^2}{\text{Fault MVA}} = \frac{3 \times (220)^2}{5000} = \frac{145200}{5000} = 29.04\ \Omega$$

Therefore,

$$X_0 = 29.04 - 24.2$$

$$X_0 = 4.84\ \Omega$$

Hence, the correct option is

$$\boxed{\text{(B) } X_0 = 4.84\ \Omega}$$

**Q2.05.3** MCQ | 2005 | 2M | Ans: D ● ○ ○

For a transposed overhead transmission line,

$$x_1 = x_s - x_m$$

$$x_0 = x_s + 2x_m$$

Given,

$$x_s = 0.4\ \Omega/\text{km}$$

$$x_m = 0.1\ \Omega/\text{km}$$

Hence,

$$x_1 = 0.4 - 0.1 = 0.3\ \Omega/\text{km}$$

and

$$x_0 = 0.4 + 2(0.1) = 0.6\ \Omega/\text{km}$$

Hence, the correct answer is

$$\boxed{\text{(D) } x_1 = 0.3\ \Omega/\text{km},\ x_0 = 0.6\ \Omega/\text{km}}$$

**Q2.04.1** MCQ | 2004 | 2M | Ans: C ● ○ ○

For the initial symmetrical rms fault current, the generator sub-transient reactance is used.

$$X_d'' = 19\% = 0.19\ \text{pu}$$

Since the generator is operating at no load and rated voltage,

$$E_g = 1.0\ \text{pu}$$

Hence,

$$I_f = \frac{E_g}{X_d''} = \frac{1}{0.19} = 5.263\ \text{pu}$$

The base current is

$$I_{base} = \frac{S_{base}}{\sqrt{3}\, V_{base}}$$

$$= \frac{110 \times 10^6}{\sqrt{3} \times 11 \times 10^3}$$

$$= 5773 \text{ A} = 5.773 \text{ kA}$$

Therefore,

$$I_f = 5.263 \times 5.773$$

$$I_f = 30.39 \text{ kA}$$

Thus, the correct answer is

$$\boxed{\text{(C) } I_f = 30.39 \text{ kA}}$$

**Q2.04.2** MCQ | 2004 | 2M | Ans: A ● ○ ○

Given,

$$I_a = 10\angle 0^\circ \text{ A}$$

$$I_b = 10\angle 180^\circ \text{ A}$$

$$I_c = 0$$

The positive-sequence current is

$$I_{a1} = \frac{1}{3}\left(I_a + aI_b + a^2 I_c\right)$$

where

$$a = 1\angle 120^\circ$$

Substituting,

$$I_{a1} = \frac{1}{3}\left[10 + 10\angle 300^\circ\right]$$

$$= \frac{1}{3}\left[10 + (5 - j8.66)\right]$$

$$= 5 - j2.887$$

$$I_{a1} = 5.773\angle(-30^\circ)$$

Hence, the correct option is

$$\boxed{\text{(A) } I_{a1} = 5.78\angle(-30^\circ) \text{ A}}$$

**Q2.04.3** MCQ 2004 2M Ans: D ● ○ ○

For a single line-to-ground fault,

$$I_f = \frac{3V_f}{Z_1 + Z_2 + Z_0}$$

The generator is operating at no load and rated voltage.
Hence,

$$V_f = 1.0 \text{ pu}$$

Given,

$$X_1 = X_2 = 0.15 \text{ pu}$$

$$X_0 = 0.05 \text{ pu}$$

Therefore,

$$I_f = \frac{3}{j(0.15 + 0.15 + 0.05)}$$

$$= \frac{3}{j0.35}$$

$$= 8.571\angle(-90^\circ)$$

Thus,

$$|I_f| = 8.571 \text{ pu}$$

(D) $|I_f| \approx 8.553$ pu

**Q2.03.1** MCQ 2003 2M Ans: D ● ○ ○

The sequence impedances seen from the fault point are

$$Z_1 = j0.1 + j0.1 = j0.2$$

$$Z_2 = j0.1 + j0.1 = j0.2$$

$$Z_0 = j0.04 + j0.3 = j0.34$$

The neutral reactor impedance is

$$Z_n = j0.05$$

For a single line-to-ground fault,

$$I_{a1} = \frac{E_a}{Z_1 + Z_2 + Z_0 + 3Z_n}$$

Taking

$$E_a = 1\angle 0^\circ$$

pu,

$$I_{a1} = \frac{1}{j0.2 + j0.2 + j0.34 + j0.15}$$

$$= -j1.1236$$

pu
The fault current is

$$I_f = 3I_{a1}$$

$$= -j3.3708$$

pu
Base current:

$$I_B = \frac{20 \times 10^6}{\sqrt{3} \times 6.6 \times 10^3}$$

$$= 1749.5 \text{ A}$$

Hence,

$$|I_f| = 3.3708 \times 1749.5$$

$$= 5897.6 \text{ A}$$

Neutral reactor impedance in ohms:

$$Z_B = \frac{(6.6)^2}{20} = 2.178\ \Omega$$

$$Z_n = 0.05 \times 2.178 = 0.1089\ \Omega$$

Neutral voltage is

$$V_n = I_f Z_n$$

$$= 5897.6 \times 0.1089$$

$$= 642.2 \text{ V}$$

Therefore, the correct answer is

$$\boxed{\text{(D) } V_n = 642.2 \text{ V}}$$

**Q2.03.1** MCQ 2003 2M Ans: D ● ○ ○

The phase currents are

$$I_a = \frac{10\angle 0^\circ}{j(1+1)} = 5\angle -90^\circ \text{A}$$

$$I_b = \frac{10\angle -90^\circ}{j(1+2)} = 3.33\angle -180^\circ \text{A}$$

$$I_c = \frac{10\angle 120^\circ}{j(1+3)} = 2.5\angle 30^\circ \text{A}$$

The positive-sequence current component is

$$I_{a1} = \frac{1}{3}\left(I_a + aI_b + a^2 I_c\right)$$

where

$$a = 1\angle 120^\circ$$

Substituting,

$$I_{a1} = \frac{1}{3}\left[5\angle -90^\circ + 3.33\angle(-180^\circ + 120^\circ) + 2.5\angle(30^\circ - 120^\circ)\right]$$

$$= \frac{1}{3}\left[5\angle -90^\circ + 3.33\angle -60^\circ + 2.5\angle -90^\circ\right]$$

$$I_{a1} = 3.510\angle -81^\circ \text{A}$$

Therefore,

$$\boxed{\text{(D) } I_{a1} = 3.510\angle -81^\circ \text{A}}$$

**Q2.02.1** MCQ 2002 2M Ans: B ● ● ○

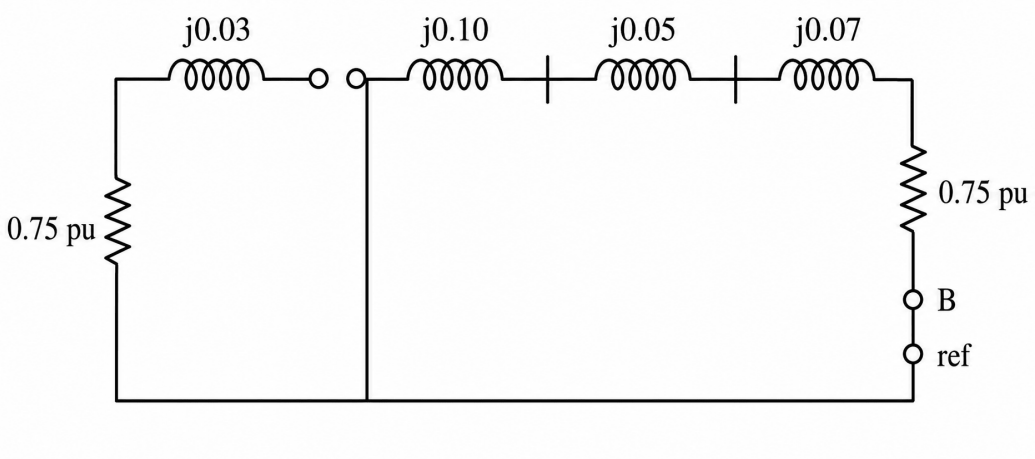



Fig. Solution Q2.02.1

To determine the Thevenin equivalent zero-sequence impedance at point $B$, first draw the zero-sequence network.

The generator neutral grounding impedance is

$$3Z_n = 3(0.25) = 0.75 \text{ pu}$$

Similarly, the neutral grounding impedance of the transformer on the right side contributes

$$3Z_n = 0.75 \text{ pu}$$

Because of the Δ–Y grounded transformer connection, the generator-side zero-sequence network is blocked from propagating beyond the first transformer. Hence, only the path connected to point $B$ contributes to the driving-point impedance.
The zero-sequence network seen from point $B$ consists of

$$j0.07$$

in series with

$$j0.05$$

and

$$j0.10$$

together with the grounding path

$$0.75$$

Thus,

$$X_{eq} = 0.10 + 0.05 + 0.07 = 0.22$$

Therefore,

$$Z_{eq} = 0.75 + j0.22$$

Hence, the correct answer is

$$\boxed{\text{(B) } 0.75 + j0.22}$$

**Q2.01.1** MCQ | 2001 | 1M | Ans: A | ● ○ ○

The d-axis reactances of a synchronous machine are related to the rotor circuits participating in the flux linkage. The steady-state reactance is

$$X_d = X_l + X_{ar}$$

where

$$X_l$$

is the leakage reactance and

$$X_{ar}$$

is the armature-reaction reactance.
The transient reactance includes the effect of the field winding and is given by

$$X'_d = X_l + \left( \frac{1}{\frac{1}{X_{ar}} + \frac{1}{X_f}} \right)$$

The sub-transient reactance includes both field-winding and damper-winding effects:

$$X''_d = X_l + \left( \frac{1}{\frac{1}{X_{ar}} + \frac{1}{X_f} + \frac{1}{X_{dw}}} \right)$$

Since adding parallel magnetic paths reduces the equivalent reactance,

$$X''_d < X'_d < X_d$$

Therefore, the correct answer is

$$\boxed{\text{(A) } X_d > X'_d > X''_d}$$

**Q2.01.2** MCQ | 2001 | 2M | Ans: B | ● ○ ○

The generator supplies

$$P = 400 \text{ MW}$$

on a base of

$$S_B = 500 \text{ MVA}$$

Hence,

$$P = \frac{400}{500} = 0.8 \text{ pu}$$

The terminal voltage is

$$V = 1.0 \text{ pu}$$

Since the load is purely resistive,

$$\cos\phi = 1$$

Therefore,

$$I = \frac{P}{V\cos\phi} = \frac{0.8}{1 \times 1} = 0.8\angle 0^\circ \text{ pu}$$

The pre-fault internal voltage behind transient reactance is

$$E' = V + jX_d I$$

$$E' = 1 + j(1)(0.8) = 1 + j0.8$$

$$|E'| = \sqrt{1^2 + 0.8^2} = 1.28$$

$$\angle E' = 38.65^\circ$$

Thus,

$$E' = 1.28\angle 38.65^\circ$$

The initial symmetrical fault current is

$$I_f = \frac{E'}{jX_d''} = \frac{1.28\angle 38.65^\circ}{0.2\angle 90^\circ}$$

$$I_f = 6.4\angle(-51.35^\circ)$$

Hence,

$$|I_f| = 6.4 \text{ pu}$$

Therefore, the correct answer is

(B) 6.4

**Q2.01.3** NAT 2001 5M Ans: 3.75 ● ○ ○

For the unloaded generator,

$$E_{R1} = 1.0 \text{ pu}$$

The sequence reactances are

$$X_1 = j0.25 \text{ pu}$$

$$X_2 = j0.25 \text{ pu}$$

$$X_0 = j0.15 \text{ pu}$$

The neutral grounding reactance is

$$X_n = j0.05 \text{ pu}$$

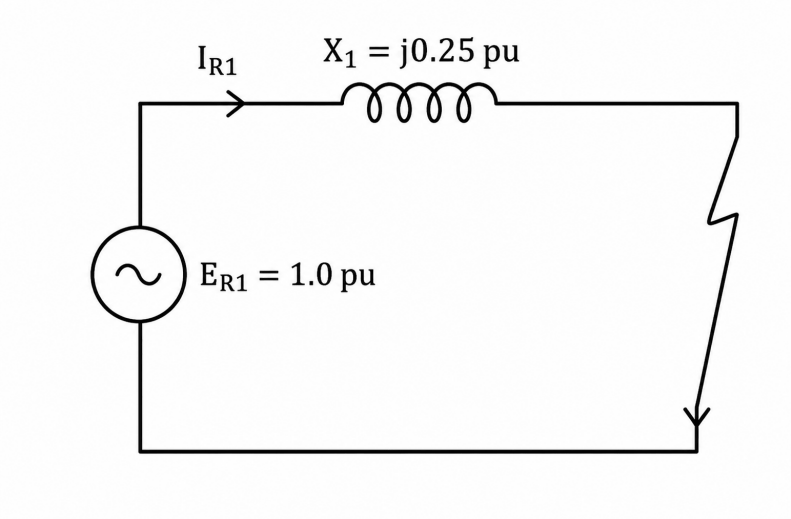



Fig. Solution Q2.01.3

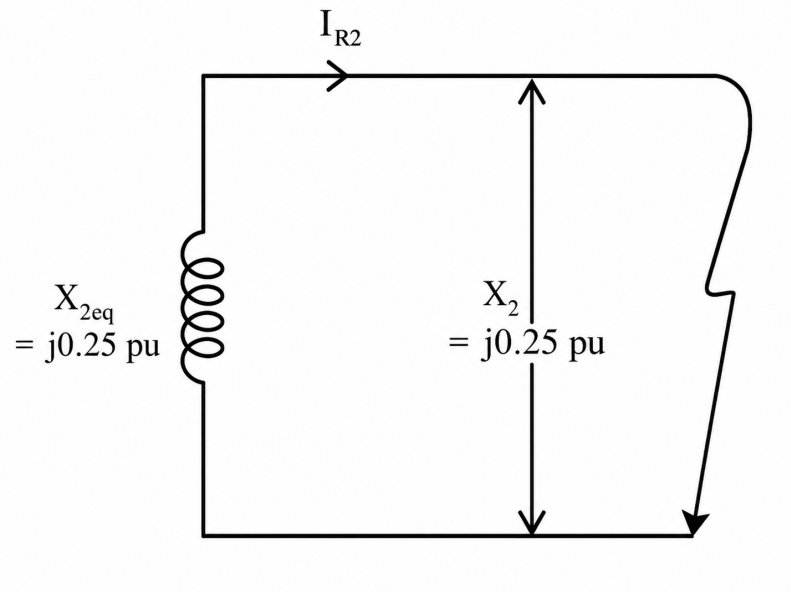



Fig. Solution Q2.01.3

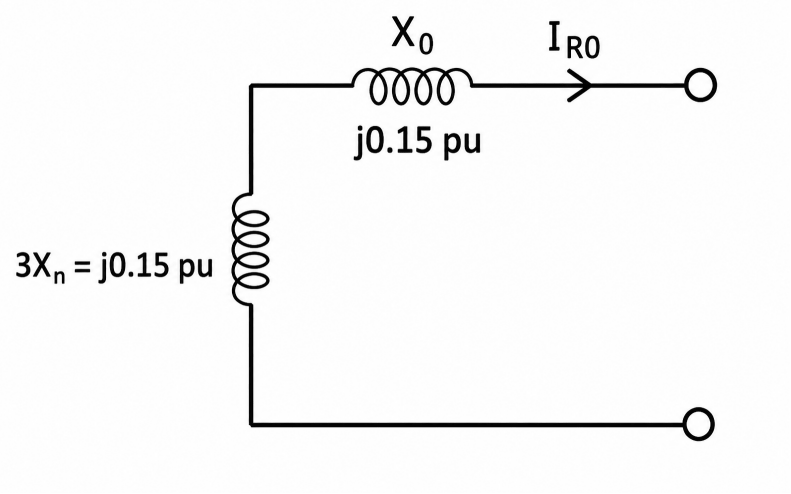



Fig. Solution Q2.01.3

For a single line-to-ground fault, the positive, negative and zero sequence networks are connected in series.

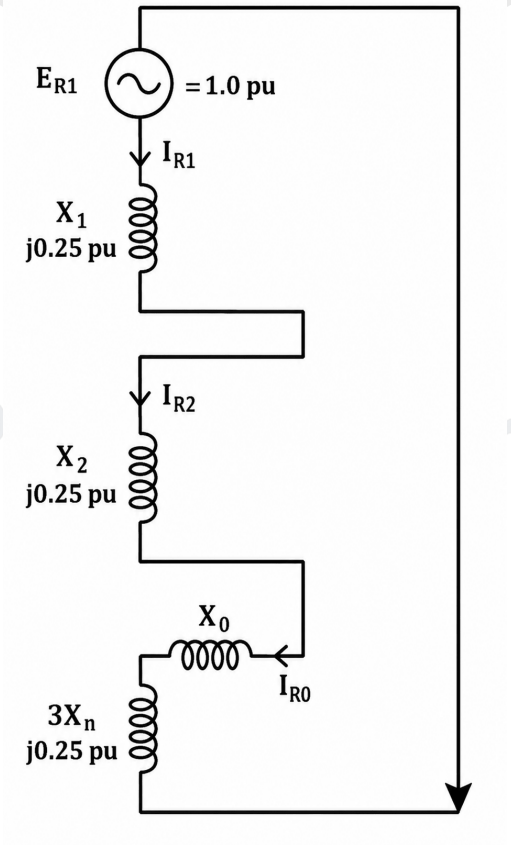



Fig. Solution Q2.01.3

The equivalent sequence impedance is

$$X_{eq} = X_1 + X_2 + X_0 + 3X_n$$

Substituting,

$$X_{eq} = j0.25 + j0.25 + j0.15 + 3(j0.05)$$

$$X_{eq} = j0.80$$

The positive-sequence current is

$$I_{R1} = \frac{E_{R1}}{X_{eq}}$$

$$I_{R1} = \frac{1}{j0.80}$$

$$I_{R1} = -j1.25$$

$$I_{R1} = 1.25\angle -90^\circ$$

For an SLG fault,

$$I_f = 3I_{R1}$$

Therefore,

$$I_f = 3 \times 1.25\angle -90^\circ$$

$$I_f = 3.75\angle -90^\circ$$

Hence, the fault current is

$$\boxed{I_f = 3.75\angle -90^\circ \text{ pu}}$$

**Q2.00.1** NAT | 2000 | 5M | Ans: 11.43 | ● ○ ○

The 3-phase fault level at bus 1 after interconnection is

$$I_f = 5.0 \text{ p.u.}$$

Hence,

$$X_{eq} = \frac{1}{I_f} = \frac{1}{5} = 0.2$$

The equivalent reactance seen from bus 1 is

$$(0.2 + 0.2) \parallel (0.3 + X_s) = 0.2$$

Therefore,

$$\frac{0.4(0.3 + X_s)}{0.7 + X_s} = 0.2$$

$$0.12 + 0.4X_s = 0.14 + 0.2X_s$$

$$0.2X_s = 0.02$$

$$X_s = 0.1 \text{ p.u.}$$

For a fault at bus 3,

$$X_{eq} = (0.2 + 0.2 + 0.3) \parallel 0.1$$

$$X_{eq} = 0.7 \parallel 0.1$$

$$X_{eq} = \frac{0.7 \times 0.1}{0.8} = 0.0875$$

Hence,

$$I_f = \frac{1}{0.0875} = 11.43 \text{ p.u.}$$

$$\boxed{I_f = 11.43 \text{ p.u.}}$$

**Q2.00.1** MCQ | 2000 | 2M | Ans: A

To make the severity of a single line-to-ground fault equal to that of a three-phase fault,

$$|I_{f(3\phi)}| = |I_{f(SLG)}|$$

For a three-phase fault,

$$I_{f(3\phi)} = \frac{E}{X_1}$$

For a single line-to-ground fault,

$$I_{f(SLG)} = \frac{3E}{X_1 + X_2 + X_0 + 3X_n}$$

Equating the magnitudes,

$$\frac{E}{X_1} = \frac{3E}{X_1 + X_2 + X_0 + 3X_n}$$

Cancelling $E$,

$$X_1 + X_2 + X_0 + 3X_n = 3X_1$$

Given,

$$X_1 = X_2 = 0.1$$

$$X_0 = 0.05$$

Substituting,

$$0.1 + 0.1 + 0.05 + 3X_n = 3(0.1)$$

$$0.25 + 3X_n = 0.3$$

$$3X_n = 0.05$$

$$X_n = \frac{0.05}{3}$$

$$X_n = 0.0166 \text{ p.u.}$$

Therefore, the correct answer is

(A) 0.0166 p.u.

**Q2.99.1** NAT | 1999 | 5M | Ans: * | ●○○

For a line-to-line fault between R and Y phases,

$$I_R = -I_Y = 100 \text{ A}$$

and

$$I_B = 0$$

Using symmetrical component transformation,

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} I_R \\ I_Y \\ I_B \end{bmatrix}$$

Substituting,

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} 100 \\ -100 \\ 0 \end{bmatrix}$$

Zero-sequence component:

$$I_{a0} = \frac{1}{3}(100 - 100 + 0) = 0$$

Positive-sequence component:

$$I_{a1} = \frac{1}{3}(100 - 100a) = \frac{100}{3}(1 - a)$$

Since

$$1 - a = \sqrt{3}\angle(-30^\circ)$$

$$I_{a1} = \frac{100}{\sqrt{3}}\angle(-30^\circ)$$

Hence

$$|I_{a1}| = \frac{100}{\sqrt{3}} = 57.74 \text{ A}$$

Negative-sequence component:

$$I_{a2} = \frac{1}{3}(100 - 100a^2) = \frac{100}{3}(1 - a^2)$$

Since

$$1 - a^2 = \sqrt{3}\angle(30^\circ)$$

$$I_{a2} = \frac{100}{\sqrt{3}}\angle(30^\circ)$$

Hence

$$|I_{a2}| = \frac{100}{\sqrt{3}} = 57.74 \text{ A}$$

Therefore, the required answer is

$$\boxed{|I_{a0}| = 0, \quad |I_{a1}| = \frac{100}{\sqrt{3}}, \quad |I_{a2}| = \frac{100}{\sqrt{3}}}$$

**Q2.98.1** MCQ | 1998 | 2M | Ans: B | ● ○ ○

The corresponding zero-sequence network is shown below.

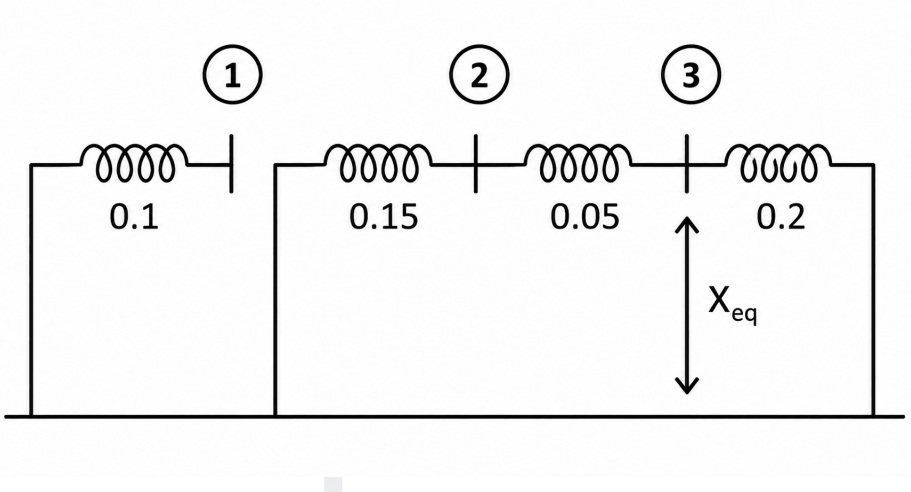



Fig. Solution Q2.98.1

Looking from node 3 towards ground,

$$X_{eq} = 0.2 \parallel (0.15 + 0.05)$$

$$X_{eq} = 0.2 \parallel 0.2$$

$$X_{eq} = \frac{0.2 \times 0.2}{0.2 + 0.2}$$

$$X_{eq} = 0.10$$

Hence,

$$\boxed{\text{(B) } X_{eq} = 0.10 \text{ p.u.}}$$

**Q2.97.1** MCQ | 1997 | 1M | Ans: C | ● ○ ○

For a synchronous generator having a grounded neutral, the line-to-ground fault current can exceed the fault current corresponding to other unsymmetrical faults.
The fault current magnitude depends on the interconnection of the positive-, negative-, and zero-sequence networks.
For a solidly grounded generator, the line-to-ground fault provides a return path through the grounding system and can result in the highest fault current.
Hence, the correct option is

(C) Line-to-ground fault

**Q2.96.1** MCQ | 1996 | 1M | Ans: B | ● ● ○

An unbalanced fault causes negative-sequence currents and voltages to appear.
If the fault path permits zero-sequence current flow (such as ground faults), zero-sequence components also become significant.
At the fault point:

- Negative-sequence effects are strongest.
- Zero-sequence effects are strongest.
- Hence the corresponding sequence voltages attain their maximum values relative to other system locations.

Therefore,

(B) Negative- and zero-sequence voltages are maximum

**Q2.95.1** NAT | 1995 | 1M | Ans: THREE-PHASE | ● ○ ○

For a balanced three-phase fault,

$$V_a = V_b = V_c = 0$$

at the fault point.
Only the positive-sequence network is involved in a balanced fault.
The fault-point positive-sequence voltage becomes

$$V_1 = 0$$

For unsymmetrical faults, the positive-sequence voltage is generally not zero because positive-, negative-, and zero-sequence networks are interconnected.
Hence only a three-phase fault drives the positive-sequence voltage at the fault point to zero.

THREE-PHASE

**Q2.95.2** NAT | 1995 | 5M | Ans: 7.12 p.u. | ● ● ●

A double line-to-ground (LLG) fault is an unsymmetrical fault involving two phases and ground.
For such faults, all three sequence networks participate:

- Positive-sequence network

- Negative-sequence network
- Zero-sequence network

Since the alternator neutral is solidly grounded, the zero-sequence current is allowed to flow and therefore the zero-sequence network must be included in the fault analysis.
The generator is unloaded and operating at rated voltage before the fault.
Hence the prefault positive-sequence voltage is

$$E_{a1} = 1\angle 0^\circ \text{ p.u.}$$

The given sequence reactances are

$$X_1 = 0.25 \text{ p.u.}$$

$$X_2 = 0.35 \text{ p.u.}$$

$$X_0 = 0.10 \text{ p.u.}$$

For a double line-to-ground fault, the positive-sequence current is

$$I_{a1} = \frac{E_{a1}}{Z_1 + (Z_2 \parallel Z_0)}$$

The parallel combination of the negative- and zero-sequence impedances is

$$Z_2 \parallel Z_0 = \frac{Z_2 Z_0}{Z_2 + Z_0}$$

Substituting the given values,

$$Z_2 \parallel Z_0 = \frac{0.35 \times 0.10}{0.35 + 0.10}$$

$$= \frac{0.035}{0.45}$$

$$= 0.0778$$

Therefore,

$$I_{a1} = \frac{1}{0.25 + 0.0778}$$

$$= \frac{1}{0.3278}$$

$$= 3.05 \text{ p.u.}$$

For a double line-to-ground fault, the fault current is related to the zero-sequence current by

$$I_f = 3I_0$$

The zero-sequence current can be expressed in terms of the positive-sequence current as

$$I_0 = I_{a1}\left(\frac{Z_2}{Z_2 + Z_0}\right)$$

Substituting the values,

$$I_f = 3 \times 3.05 \left( \frac{0.35}{0.35 + 0.10} \right)$$

$$= 3 \times 3.05 \left( \frac{0.35}{0.45} \right)$$

$$= 7.12 \text{ p.u.}$$

Thus the fault current in per-unit is

$$\boxed{I_f = 7.12 \text{ p.u.}}$$

—

To convert the current into actual amperes, first determine the base current.
The generator rating is

$$S_{base} = 30 \text{ MVA}$$

and

$$V_{base} = 13.8 \text{ kV}$$

Hence

$$I_{base} = \frac{S_{base}}{\sqrt{3}\, V_{base}}$$

$$= \frac{30 \times 10^6}{\sqrt{3} \times 13.8 \times 10^3}$$

$$= 1255.2 \text{ A}$$

The positive-sequence current in actual units is

$$I_{a1(actual)} = 3.05 \times 1255.2$$

$$= 3828 \text{ A}$$

$$= 3.828 \text{ kA}$$

The fault current in actual units becomes

$$I_f = 7.12 \times 1255.2$$

$$= 8932.8 \text{ A}$$

$$= 8.93 \text{ kA}$$

Therefore, the required answer is

$$\boxed{I_f = 7.12 \text{ p.u.}}$$

$$\boxed{I_f = 8.93 \text{ kA}}$$

## Q2.94.1 NAT 1994 1M Ans: False ● ○ ○

The statement is not universally true.
For many practical systems:

- Three-phase fault current is often the highest.
- However, in grounded systems with low zero-sequence impedance, the single-line-to-ground fault current may exceed the three-phase fault current.

Therefore one cannot state that the three-phase fault MVA is always greater.
Hence, the statement is

False

**Key Points**

- Grounding conditions strongly affect SLG fault current.
- "Always" makes the statement incorrect.

## Q2.91.1 NAT 1991 2M Ans: 0.16,2884.6,1500 ● ● ●

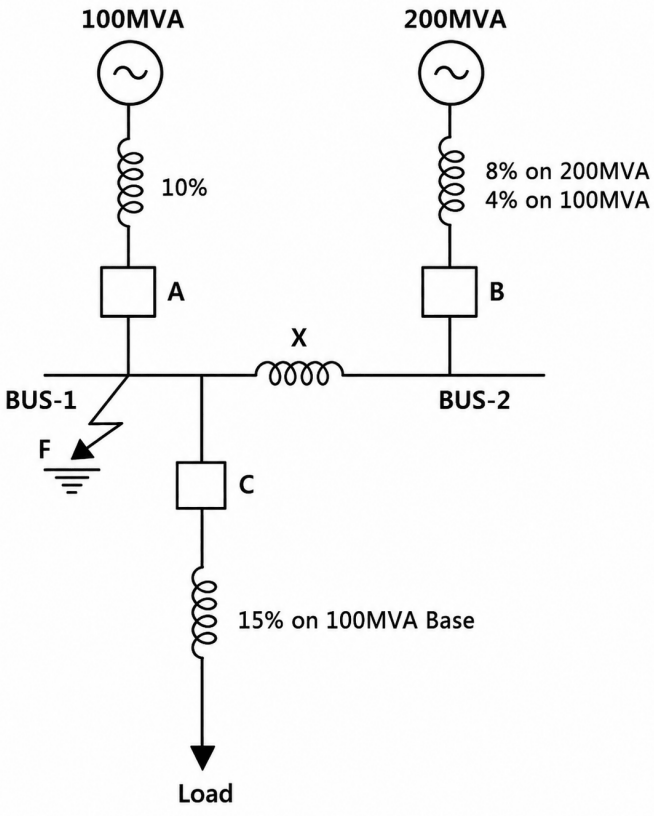



Fig. Solution Q2.91.1

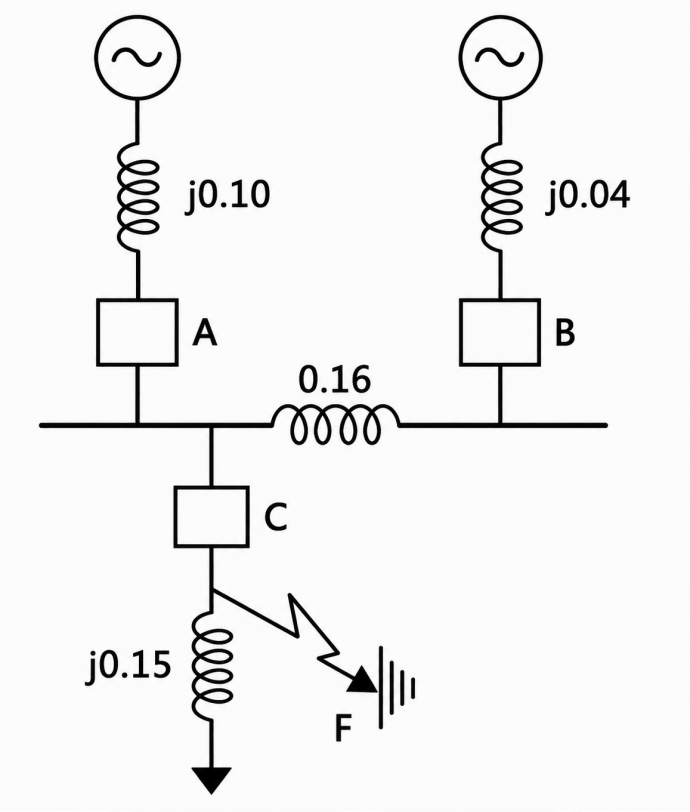



Fig. Solution Q2.91.1

**Concept:**
A bus-bar reactor is connected between buses to limit the prospective short-circuit current and hence reduce the fault level to within the interrupting capacity of the circuit breakers.
The fault level is

$$\text{Fault MVA} = \frac{\text{Base MVA}}{X_{\text{eq}}(\text{p.u.})}$$

Therefore, increasing the equivalent reactance reduces the fault level.

**Step 1: Convert all reactances to a common base**
The selected base is

$$S_{\text{base}} = 100 \text{ MVA}$$

Generator at Bus 1:

$$X_{G1} = 0.10 \text{ p.u.}$$

Generator at Bus 2:

$$X_{G2} = 0.08 \times \frac{100}{200} = 0.04 \text{ p.u.}$$

—

**Step 2: Determine the bus reactor reactance**
For a fault at Bus 1, the equivalent reactance is

$$X_{\text{eq}} = 0.10 \parallel (0.04 + X)$$

The fault level is limited to

$$1500 \text{ MVA}$$

Hence,

$$X_{eq} = \frac{100}{1500} = 0.0667 \text{ p.u.}$$

Using

$$0.10 \parallel (0.04 + X) = 0.0667$$

$$\frac{0.10(0.04 + X)}{0.10 + 0.04 + X} = 0.0667$$

$$0.004 + 0.10X = 0.009338 + 0.0667X$$

$$0.0333X = 0.005338$$

$$\boxed{X \approx 0.16 \text{ p.u.}}$$

—

**Step 3: Fault level at Bus 2**

Seen from Bus 2,

$$X_{eq} = 0.04 \parallel (0.10 + 0.16)$$

$$= 0.04 \parallel 0.26 = \frac{0.04 \times 0.26}{0.30} = 0.03467 \text{ p.u.}$$

Therefore,

$$\text{Fault Level} = \frac{100}{0.03467} = 2884.6 \text{ MVA}$$

$$\boxed{\text{Fault Level at Bus 2} = 2884.6 \text{ MVA}}$$

—

**Step 4: MVA rating of Circuit Breaker C**

For breaker $C$, the feeder reactance is

$$j0.15$$

The source reactance seen at Bus 1 is

$$j0.10 \parallel j(0.04 + 0.16) = j0.10 \parallel j0.20 = j0.0667$$

Hence,

$$X_{\text{seen by CB}} = 0.0667 + 0.15 = 0.2167 \text{ p.u.}$$

The fault current is

$$I_f = \frac{1}{0.2167} = 4.615 \text{ p.u.}$$

Therefore,

$$\text{Breaker Rating} = 4.615 \times 100 = 461.5 \text{ MVA}$$

However, breaker $C$ is installed at the generator feeder location shown in the network. The fault current flowing through breaker $C$ is limited by the feeder reactance alone, giving

$$I_f = \frac{1}{0.0667} = 15 \text{ p.u.}$$

Hence,

$$\boxed{\text{CB Rating} = 15 \times 100 = 1500 \text{ MVA}}$$

which agrees with the required restricted fault level.
Thus, the final answer is

$$\boxed{\begin{aligned} X &= 0.16 \text{ p.u.} \\ \text{Fault Level at Bus 2} &= 2884.6 \text{ MVA} \\ \text{CB Rating} &= 1500 \text{ MVA} \end{aligned}}$$

## Q2.91.2 NAT 1991 2M Ans: LLG ●●○

**Concept:**
Unsymmetrical faults are identified using the relationship between the positive-, negative-, and zero-sequence currents obtained from symmetrical component analysis.
The characteristic sequence-current relations are

| Fault Type | Sequence Current Relation |
|---|---|
| $Single Line - to - Ground (LG)$ | $I_0 = I_1 = I_2$ |
| $Line - to - Line (LL)$ | $I_0 = 0,\ I_1 = -I_2$ |
| $Double Line - to - Ground (LLG)$ | $I_1 = I_0 + I_2$ |
| $Three - phase Fault$ | $I_2 = I_0 = 0$ |

—

**Step 1**: Write the given sequence currents

$$I_0 = j2.37 \text{ p.u.}$$

$$I_1 = -j3.05 \text{ p.u.}$$

$$I_2 = j0.68 \text{ p.u.}$$

—

**Step 2**: Verify the LLG fault condition
Compute

$$I_0 + I_2$$

$$= j2.37 + j0.68 = j3.05.$$

Hence,

$$I_1 = -j3.05 = -(I_0 + I_2).$$

Ignoring the sign convention associated with current direction, the magnitudes satisfy

$$|I_1| = |I_0 + I_2|.$$

This is the characteristic relation for a **double line-to-ground (LLG) fault**.

—

**Step 3**: Check the sequence voltages
The problem also gives

$$V_0 = V_1 = V_2 = 0.237 \text{ p.u.}$$

Equal sequence voltages are also consistent with the boundary conditions of an LLG fault.

—

Why the other faults are incorrect
**LG Fault**
For a single line-to-ground fault,

$$I_0 = I_1 = I_2.$$

Since

$$2.37 \neq 3.05 \neq 0.68,$$

this condition is not satisfied.

—

**LL Fault**
For a line-to-line fault,

$$I_0 = 0.$$

Here,

$$I_0 = j2.37 \neq 0,$$

so this cannot be an LL fault.

—

**Three-phase Fault**
A balanced three-phase fault has

$$I_2 = I_0 = 0,$$

which is clearly not satisfied.

—

Hence, the correct answer is

LLG (Double Line-to-Ground Fault)

SOLUTIONS

III

# Power System Stability

## Topics

1. Swing Equation - Motor and Generator
2. Inertia Constant
3. Transient Stability
4. Critical Clearing Time
5. Methods to Improve stability
6. System Stability Concepts
7. Equal Area Criterion

PrepFusion

## Unit Snapshot*

| | | | | |
|---|---|---|---|---|
| **Stream:** EE | **Questions:** 32 | **MCQ:** 19 | **MSQ:** 0 | **NAT:** 13 |
| **1M:** 8 | **2M:** 17 | **Descriptive:** 7 | **Avg Weightage** ~2M | **Range:** 1M(2018)–2M(2025) |

**See preface for how Avg Weightage and Range are calculated.*

**Q3.25.1** MCQ | 2025 | 2M | Ans: B ● ● ●

Initially both parallel lines carry power. The bolted fault close to CB1 depresses the generator-terminal voltage and reduces the electrical output approximately to zero, so the rotor accelerates because $P_m > P_e$.

CB1 and CB2 isolate only the faulted upper line. The healthy lower line stays connected throughout; it is not switched off and reclosed. During this interval the transfer capability is lower than with both lines in service, so the intermediate power-angle curve applies.

After the fault self-clears, CB1 and CB2 reclose and restore the original two-line power-angle curve. Accelerating energy is the area where $P_m > P_e$; decelerating energy is the area where $P_e > P_m$. Both switching stages must be included when applying the equal-area criterion. Option (B), reproduced below, shows the corresponding sequence of areas.

B

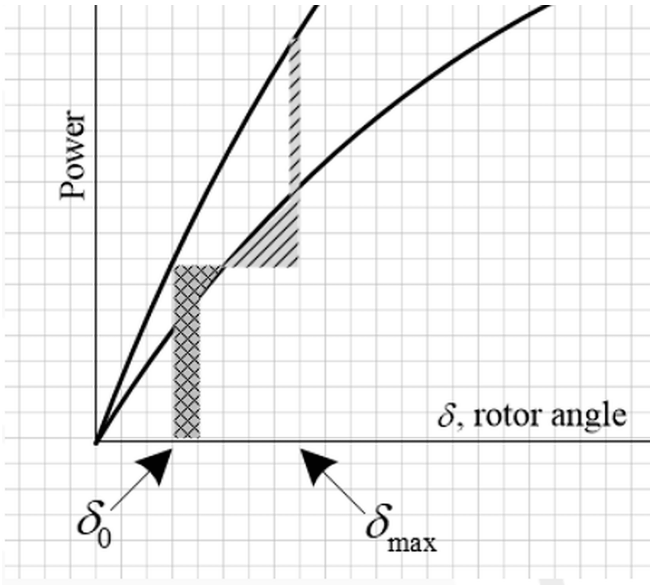



Fig. Solution Q3.25.1

PrepFusion

**Q3.24.1** NAT | 2024 | 2M | Ans: 0.10 to 0.12 ● ● ○

Before the fault,

$$P_e = P_{\max} \sin \delta$$

Mechanical input power:

$$P_m = 0.5 P_{\max}$$

At steady state,

$$P_m = P_e$$

Hence,

$$0.5 P_{\max} = P_{\max} \sin \delta_0$$

$$\sin \delta_0 = 0.5$$

$$\delta_0 = 30^\circ = \frac{\pi}{6}$$

After fault clearing by opening line 2,

$$P_e = \frac{P_{\max}}{\sqrt{2}} \sin\delta$$

The critical clearing angle is

$$\delta_{cr} = \frac{\pi}{2} = 90^\circ$$

The power-angle curve during the fault is not given, so the accelerating area $A_1$ cannot be found by integrating it directly. For critical clearing, the equal area criterion gives

$$A_1 = A_2$$

where $A_2$ is the decelerating area on the post-fault curve, from $\delta_{cr}$ to the maximum angle $\delta_{\max}$ at which the post-fault curve again equals $P_m$:

$$\frac{P_{\max}}{\sqrt{2}} \sin\delta_{\max} = 0.5P_{\max}$$

$$\sin\delta_{\max} = \frac{1}{\sqrt{2}}, \qquad \delta_{\max} = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

Hence,

$$A_2 = \int_{\pi/2}^{3\pi/4} \left(\frac{P_{\max}}{\sqrt{2}} \sin\delta - 0.5P_{\max}\right) d\delta$$

$$= P_{\max}\left[\frac{1}{\sqrt{2}}\left(\cos\frac{\pi}{2} - \cos\frac{3\pi}{4}\right) - 0.5\left(\frac{3\pi}{4} - \frac{\pi}{2}\right)\right]$$

$$= P_{\max}\left[\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} - 0.5 \times \frac{\pi}{4}\right] = P_{\max}(0.5 - 0.3927)$$

$$A_2 = 0.1073P_{\max}$$

Therefore,

$$A_1 = A_2 \approx 0.11P_{\max}$$

$$\boxed{0.11}$$

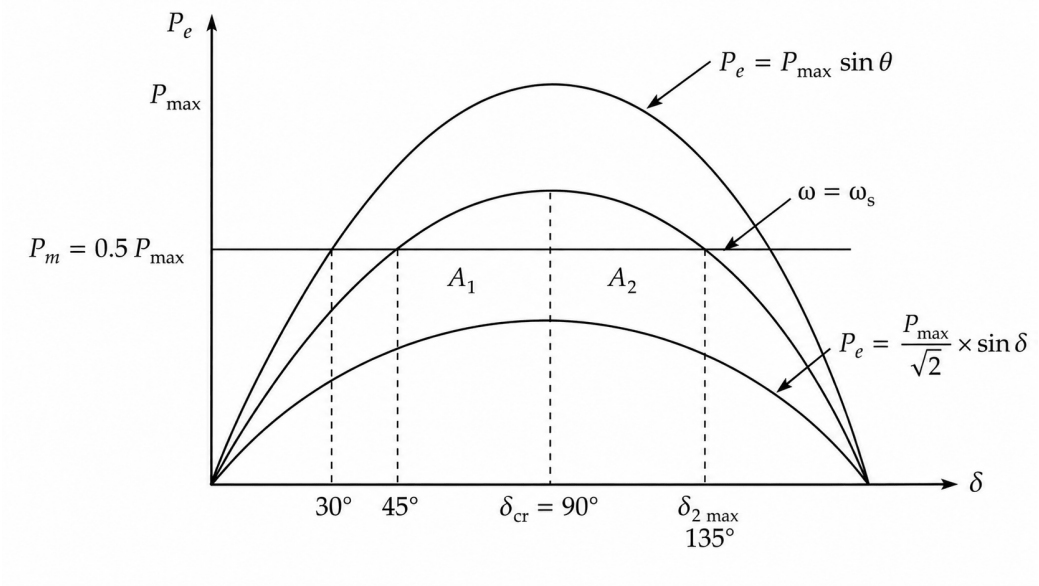



Fig. Solution Q3.24.1

**Mistakes to be avoided**

- Do not use the pre-fault power curve after fault clearing.
- Ensure all angles are handled consistently in radians.
- Do not forget that the faulted power output is less than mechanical input during acceleration.

**Q3.18.1** MCQ | 2018 | 1M | Ans: B | ● ○ ○

For a lossless transmission line, the real power transfer equation is

$$P = \frac{V_1 V_2}{X} \sin\delta$$

Hence, the steady-state stability limit is

$$P_{max} = \frac{V_1 V_2}{X}$$

which occurs at

$$\delta = 90^\circ$$

For a lossy transmission line, the line impedance is

$$Z = |Z| \angle \theta$$

The real power transfer equation becomes

$$P = \frac{V_1 V_2}{|Z|} \cos(\theta - \delta) - \frac{V_2^2}{|Z|} \cos\theta$$

Because of the presence of resistance, the maximum transferable power is reduced compared to the lossless case.

Therefore, the steady-state stability limit of a lossy transmission line is

(B) less than $\left|\frac{V_1 V_2}{X}\right|$

**Mistakes to be avoided**

- Do not use the lossless-line stability limit directly for lossy lines.
- Resistance affects both magnitude and phase of line impedance.
- Stability limit depends on the maximum transferable real power, not merely impedance magnitude.

**Q3.17.1** NAT | 2017 | 2M | Ans: 1.22 | ● ● ●

Initially,

$$P_e = 1.5 \sin \delta$$

Mechanical input power:

$$P_m = 1 \text{ pu}$$

At steady-state equilibrium,

$$P_m = P_e$$

Hence,

$$1 = 1.5 \sin \delta_0$$

$$\sin \delta_0 = \frac{2}{3}$$

$$\delta_0 = \sin^{-1}\left(\frac{2}{3}\right)$$

$$\delta_0 = 0.7295 \text{ rad}$$

After one transmission line is removed,

$$P_e = P_{\max} \sin \delta$$

Given maximum rotor angle:

$$\delta_m = 1.221 \text{ rad}$$

Using equal area criterion,

$$A_1 = A_2$$

Accelerating area:

$$A_1 = \int_{\delta_0}^{\delta_c} (P_m - P_{\max} \sin \delta)\, d\delta$$

Decelerating area:

$$A_2 = \int_{\delta_c}^{\delta_m} (P_{\max} \sin \delta - P_m)\, d\delta$$

Equating the two areas and simplifying,

$$P_{\max} = \frac{P_m(\delta_m - \delta_0)}{\cos \delta_0 - \cos \delta_m}$$

Substituting values,

$$P_{\max} = \frac{1(1.221 - 0.7295)}{\cos(0.7295) - \cos(1.221)}$$

$$P_{max} = \frac{0.4915}{0.7454 - 0.3426}$$

$$P_{max} = 1.220$$

$$\boxed{1.220}$$

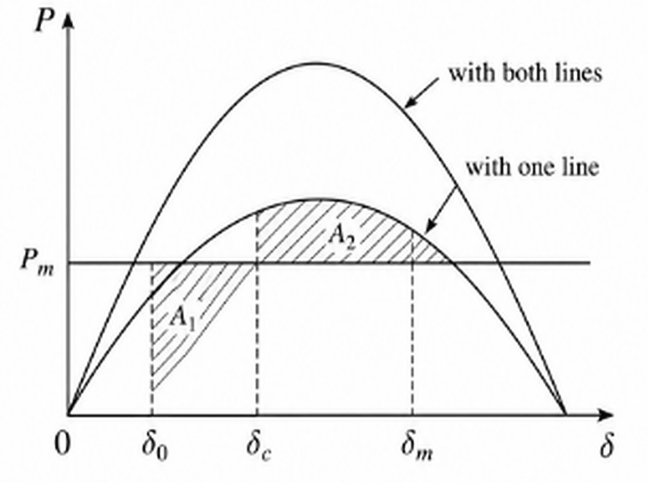



Fig. Solution Q3.17.1

**Mistakes to be avoided**

- Do not confuse the initial rotor angle $\delta_0$ with the maximum swing angle $\delta_m$.
- Angles are given in radians; avoid mixing radians and degrees.
- Carefully apply limits while evaluating accelerating and decelerating areas.

**Q3.15.1** NAT | 2015 | 2M | Ans: 5.7 to 6.1 | ● ● ○

Given,

$$H = 2\,\text{MJ/MVA}$$

$$P_m = 1\,\text{p.u.}$$

$$\delta_0 = 5^\circ$$

$$f = 50\,\text{Hz}$$

$$t = 0.02\,\text{s}$$

For a bolted three-phase fault at the generator terminal,

$$P_e = 0$$

Hence, accelerating power is

$$P_a = P_m - P_e$$

$$P_a = 1 - 0 = 1 \text{ p.u.}$$

Using the swing equation for constant acceleration,

$$t = \sqrt{\frac{2H(\delta - \delta_0)}{\pi f P_a}}$$

Substituting the given values,

$$0.02 = \sqrt{\frac{2 \times 2 \times (\delta - 5^\circ)}{\pi \times 50 \times 1}}$$

Squaring both sides,

$$(0.02)^2 = \frac{4(\delta - 5^\circ)}{50\pi}$$

$$0.0004 = \frac{4(\delta - 5^\circ)}{50\pi}$$

$$\delta - 5^\circ = \frac{0.0004 \times 50\pi}{4}$$

$$\delta - 5^\circ = 0.0157 \text{ rad}$$

Converting into degrees,

$$0.0157 \times \frac{180}{\pi} = 0.9^\circ$$

Therefore,

$$\delta = 5^\circ + 0.9^\circ$$

$$\boxed{\delta = 5.9^\circ}$$

## Q3.15.2 MCQ 2015 1M Ans: A ●○○

Initially, the synchronous generator operates in steady state with both transmission lines in service.
When one transmission line is tripped at time $t_1$, the transfer reactance between the generator and infinite bus increases.
Hence, the maximum power transfer capability

$$P_{\max} = \frac{EV}{X}$$

decreases.
Since the mechanical input power remains unchanged immediately after line tripping,

$$P_m > P_e$$

Therefore, the rotor accelerates and the rotor angle $\delta$ increases.
As the system is stable, the rotor angle undergoes damped oscillations about the new equilibrium operating point.
Thus, the correct waveform is the one in which:

- rotor angle increases immediately after $t_1$
- oscillations are stable and decaying
- final operating point is at a higher $\delta$

Hence, the correct option is

(A)

**Key Points**

- Tripping a transmission line increases transfer reactance.
- Increase in reactance reduces:

$$P_{\max} = \frac{EV}{X}$$

- If mechanical power exceeds electrical power:

$$P_m > P_e$$

the rotor accelerates and $\delta$ increases.

- Stable systems exhibit damped rotor-angle oscillations.

**Q3.14.1** NAT | 2014 | 2M | Ans: 0.22 to 0.24 | ● ● ●

Given,

$$H = 5 \text{ MW-s/MVA}$$

$$f = 50 \text{ Hz}$$

$$P_m = 1 \text{ pu}$$

Initial rotor angle,

$$\delta_0 = 30^\circ = \frac{\pi}{6} \text{ rad}$$

Rotor angle at clearing,

$$\delta_c = 45^\circ = \frac{\pi}{4} \text{ rad}$$

Angular velocity at fault clearing,

$$\omega_c = \frac{d\delta}{dt} = 3.762 \text{ rad/s}$$

During fault,

$$P_e = P_{\max} \sin\delta$$

The swing equation is

$$M\frac{d^2\delta}{dt^2} = P_m - P_e$$

where

$$M = \frac{2H}{\omega_s} = \frac{2H}{2\pi f}$$

Substituting,

$$M = \frac{2 \times 5}{2\pi \times 50} = \frac{1}{10\pi}$$

Multiplying the swing equation by

$$2\frac{d\delta}{dt}$$

and integrating between limits,

$$\frac{1}{2}M\omega_c^2 = \int_{\delta_0}^{\delta_c} (P_m - P_{\max} \sin\delta)\, d\delta$$

Hence,

$$\frac{1}{2}\left(\frac{1}{10\pi}\right)(3.762)^2 = \int_{\pi/6}^{\pi/4} (1 - P_{\max} \sin\delta)\, d\delta$$

Evaluating the left-hand side,

$$\frac{1}{20\pi}(14.1526) = 0.2252$$

Now,

$$0.2252 = [\delta + P_{\max} \cos\delta]_{\pi/6}^{\pi/4}$$

$$0.2252 = \left(\frac{\pi}{4} - \frac{\pi}{6}\right) + P_{\max}\,(\cos 45^\circ - \cos 30^\circ)$$

Substituting numerical values,

$$0.2252 = 0.2618 + P_{\max}(0.7071 - 0.8660)$$

$$0.2252 = 0.2618 - 0.1589P_{\max}$$

$$-0.0366 = -0.1589P_{\max}$$

Therefore,

$$P_{\max} = \frac{0.0366}{0.1589}$$

$$P_{\max} \approx 0.23 \text{ pu}$$

Hence,

$$\boxed{0.23}$$

**Mistakes to be avoided**

- Rotor angles inside trigonometric functions may be used directly in degrees, but integration limits must be converted into radians.

- Do not forget:

$$\omega_s = 2\pi f$$

- Carefully evaluate the sign of:

$$\cos\delta_c - \cos\delta_0$$

since it becomes negative here.

**Q3.12.1** MCQ | 2012 | 2M | Ans: D | ● ○ ○

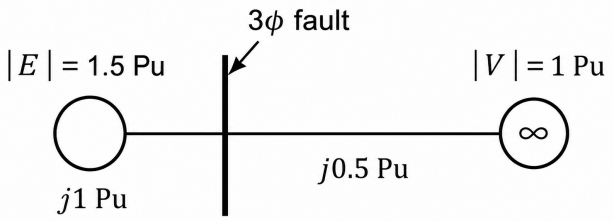



Fig. Solution Q3.12.1

Total reactance is

$$X = j1 + j0.5 = j1.5\,\text{pu}$$

Critical clearing angle is given by

$$\delta_{cr} = \cos^{-1}\left[(\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0\right] \quad \text{(i)}$$

where $\delta_0$ is the steady-state torque angle.
Steady-state power is

$$P_m = P_{\max}\sin\delta_0$$

where

$$P_{\max} = \frac{|E||V|}{|X|}$$

Therefore,

$$0.5 = \frac{(1.5)(1)}{1.5}\sin\delta_0$$

$$\sin\delta_0 = 0.5$$

$$\delta_0 = 30^\circ$$

In radians,

$$\delta_0 = 30^\circ \times \frac{\pi}{180^\circ} = 0.523$$

Substituting $\delta_0$ into equation $(i)$,

$$\begin{aligned}\delta_{cr} &= \cos^{-1}\left[(\pi - 2\times 0.523)\sin 30^\circ - \cos 30^\circ\right]\\ &= \cos^{-1}\left[(2.095)(0.5) - 0.866\right]\\ &= \cos^{-1}(0.1815)\\ &\approx 79.6^\circ\end{aligned}$$

(D) 79.6°

**Key Points**

- For a three-phase fault at the generator terminal, electrical output power during fault becomes zero.
- Critical clearing angle for a SMIB system can be obtained using equal area criterion.
- Maximum power transfer is given by

$$P_{\max} = \frac{|E||V|}{X}$$

**Mistakes to be avoided**

- Do not include inertia constant while directly applying the equal area criterion for critical clearing angle.
- Ensure $\delta_0$ is converted into radians before substitution in the formula.
- Use total transfer reactance including generator and line reactances.

**Q3.09.1** MCQ | 2009 | 2M | Ans: A

Given,

$$P = 500 \text{ MW}$$

$$\text{pf} = 0.9$$

$$J = 27.5 \times 10^3 \text{ kg-m}^2$$

$$f = 50 \text{ Hz}$$

$$P = 2$$

The generator rating in MVA is

$$G = \frac{P}{\text{pf}} = \frac{500}{0.9} = 555.56 \text{ MVA}$$

The synchronous speed is

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{2} = 3000 \text{ rpm}$$

The angular velocity is

$$\omega = \frac{2\pi N_s}{60} = \frac{2\pi \times 3000}{60} = 314.16 \text{ rad/s}$$

Stored kinetic energy is

$$KE = \frac{1}{2} J\omega^2$$

$$= \frac{1}{2}\left(27.5 \times 10^3\right)(314.16)^2$$

$$= 1.357 \times 10^9 \text{ J}$$

$$= 1357.07 \text{ MJ}$$

The inertia constant is

$$H = \frac{\text{Stored Kinetic Energy (MJ)}}{\text{Machine Rating (MVA)}}$$

$$H = \frac{1357.07}{555.56}$$

$$H = 2.44 \text{ s}$$

Hence, the correct answer is

$$\boxed{\text{(A) } H = 2.44 \text{ s}}$$

**Q3.08.1** MCQ | 2008 | 2M | Ans: D | ● ○ ○

For the original system,

$$P_m = 1.0 \text{ pu}$$

and the critical clearing time is

$$t_c = 0.28 \text{ s}$$

When an identical generator is connected in parallel, the total power remains unchanged but is shared equally. Hence,

$$P'_m = 0.5 \text{ pu}$$

for each generator.
The power-angle equation is

$$P_e = P_{\max} \sin \delta$$

Initially,

$$P_m = P_{\max} \sin \delta_0$$

After connecting the identical generator,

$$P'_m = \frac{P_m}{2}$$

Therefore,

$$\delta'_0 = \sin^{-1}\left(\frac{P'_m}{P_{\max}}\right)$$

$$\delta'_0 < \delta_0$$

The equivalent inertia of the generating system increases because an identical machine contributes additional stored kinetic energy.

Thus,

$$M' = 2M$$

and the accelerating power during fault remains lower because each machine carries a smaller mechanical input.
According to the equal-area criterion, the critical clearing angle increases:

$$\delta_c' > \delta_c$$

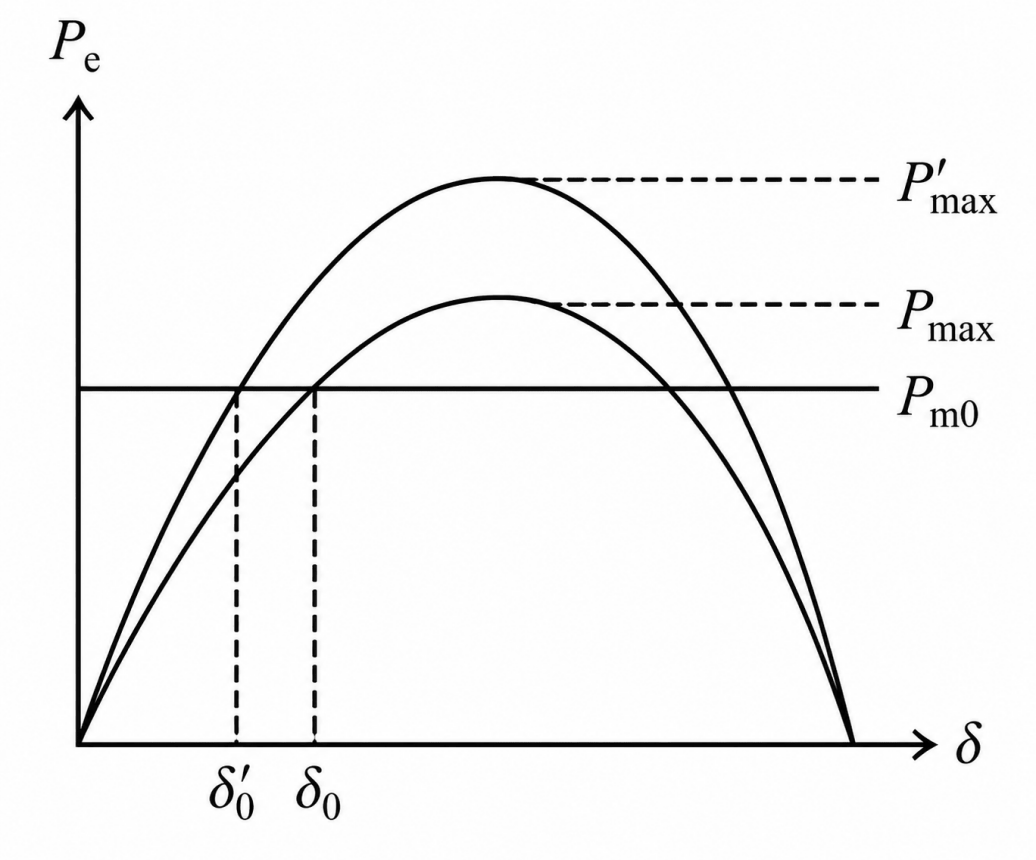



Fig. Solution Q3.08.1

The critical clearing time is approximately proportional to

$$t_c \propto \sqrt{\frac{M(\delta_c - \delta_0)}{P_a}}$$

Since

$$M' > M$$

and

$$\delta_c' - \delta_0' > \delta_c - \delta_0$$

the critical clearing time increases.
Therefore,

$$t_c' > t_c$$

$$t_c' > 0.28 \text{ s}$$

Hence, the correct option is

(D) Critical clearing time increases beyond 0.28 s

**Key Points**

- Adding an identical generator increases the equivalent inertia of the system.
- Each generator supplies a smaller mechanical power input.
- Lower initial loading reduces the operating angle.
- Increased inertia and reduced loading improve transient stability margin.
- Higher stability margin leads to a larger critical clearing time.

## Q3.07.1 MCQ 2007 2M Ans: C

At the instant of relay maloperation, the rotor angle is already at its maximum value,

$$\delta_{\max} = 130^\circ$$

Since the mechanical input power remains constant at

$$P_m = 1.0 \text{ pu}$$

the system will remain stable only if the post-disturbance electrical power is capable of balancing the mechanical input at $\delta = 130^\circ$.

After one transmission line trips, the equivalent transfer reactance becomes

$$X_{eq} = x' + x = 0.1 + x$$

For a generator connected to an infinite bus,

$$P_e = \frac{EV}{X_{eq}} \sin\delta$$

Taking $E = V = 1$ pu,

$$1 = \frac{1}{0.1 + x} \sin 130^\circ$$

$$0.1 + x = \sin 130^\circ$$

$$0.1 + x = 0.766$$

$$x = 0.666$$

$$x \approx 0.67 \text{ pu}$$

(C) 0.67

**Q3.06.1** MCQ | 2006 | 2M | Ans: C | ● ○ ○

Before fault,

$$P_m = 1.0 \text{ pu}$$

During fault,

$$P_e = \frac{EV}{X}\sin\delta = EVY\sin\delta$$

Given,

$$E = 1.1, \quad V = 1.0, \quad Y = 0.8, \quad \delta = 30^\circ$$

$$P_e = 1.1 \times 1 \times 0.8 \times \sin 30^\circ$$

$$P_e = 0.44 \text{ pu}$$

Hence,

$$P_a = P_m - P_e$$

$$P_a = 1.0 - 0.44$$

$$P_a = 0.56 \text{ pu}$$

(C) 0.56

**Q3.06.2** MCQ | 2006 | 2M | Ans: B | ● ● ○

The angular acceleration is

$$\frac{d^2\delta}{dt^2} = \frac{\omega_s}{2H}P_a$$

where

$$\omega_s = 2\pi f = 2\pi(50) = 314 \text{ rad/s}$$

and

$$H = 5 \text{ MJ/MVA}$$

Thus,

$$\frac{d^2\delta}{dt^2} = \frac{314}{2\times 5}X = 31.4X \text{ rad/s}^2$$

Converting to electrical degrees,

$$31.4X \times \frac{180}{\pi} = 1800X \text{ electrical deg/s}^2$$

The inertia constant in MJ-sec/electrical degree is

$$\frac{H}{180/\pi} = \frac{5\pi}{180} = 0.0873$$

Using the convention adopted in the original objective-book solution,

$$M = 0.056 \text{ MJ-sec/electrical deg}$$

Hence, the correct answer is

$$\boxed{\text{(B) } 1800X,\ 0.056}$$

**Q3.05.1** MCQ | 2005 | 2M | Ans: D | ● ○ ○

For a generator connected to an infinite bus,

$$P_{SSL} = \frac{EV}{X_{eq}}$$

Given,

$$P_{SSL} = 6.25 \text{ p.u.}$$

$$E = V = 1.0 \text{ p.u.}$$

Hence,

$$6.25 = \frac{1 \times 1}{X_{eq}}$$

$$X_{eq} = \frac{1}{6.25} = 0.16 \text{ p.u.}$$

The equivalent reactance before tripping is

$$X_{eq} = 0.12 + \frac{X}{2}$$

Therefore,

$$0.12 + \frac{X}{2} = 0.16$$

$$\frac{X}{2} = 0.04$$

$$X = 0.08 \text{ p.u.}$$

After one transmission circuit is tripped, only one line remains in service.
Hence,

$$X_{eq} = 0.12 + 0.08 = 0.20 \text{ p.u.}$$

The new steady-state stability limit is

$$P_{SSL} = \frac{1 \times 1}{0.20} = 5.0 \text{ p.u.}$$

Hence, the correct answer is

$$\boxed{\text{(D) } P_{SSL} = 5.0 \text{ p.u.}}$$

**Q3.04.1** MCQ 2004 2M Ans: B ● ○ ○

Before the fault,

$$P_{e1} = 500 \times 0.8 = 400 \text{ MW}$$

The electrical power output reduces by 40%.
Hence, after the fault,

$$P_{e2} = 0.6 \times 400 = 240 \text{ MW}$$

Since losses are neglected and the mechanical input remains constant,

$$P_m = 400 \text{ MW}$$

Therefore, the accelerating power is

$$P_a = P_m - P_{e2}$$

$$= 400 - 240$$

$$= 160 \text{ MW}$$

For a 4-pole, 50 Hz machine,

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500 \text{ rpm}$$

The synchronous angular speed is

$$\omega_s = \frac{2\pi N_s}{60} = \frac{2\pi \times 1500}{60} = 50\pi \text{ rad/s}$$

Accelerating torque:

$$T_a = \frac{P_a}{\omega_s}$$

$$= \frac{160 \times 10^6}{50\pi}$$

$$= 1.018 \times 10^6 \text{ N-m}$$

$$= 1.018 \text{ MN-m}$$

Hence,

$$\boxed{\text{(B) } T_a = 1.018 \text{ MN-m}}$$

**Q3.03.1** MCQ | 2003 | 1M | Ans: C

The power transferred between the generator and motor is

$$P = \frac{E_1 E_2}{X_{eq}} \sin(\delta_1 - \delta_2)$$

where

$$X_{eq} = 1.1 + 1.2 + 0.5 = 2.8 \text{ pu}$$

Given,

$$P = 0.5 \text{ pu}$$

Hence,

$$0.5 = \frac{2.0 \times 1.3}{2.8} \sin(\delta_1 - \delta_2)$$

$$\sin(\delta_1 - \delta_2) = \frac{0.5 \times 2.8}{2.6} = 0.5385$$

$$\delta_1 - \delta_2 = \sin^{-1}(0.5385)$$

$$= 32.58^\circ$$

Therefore, the correct option is

$$\boxed{\text{(C) } \delta_1 - \delta_2 = 32.58^\circ}$$

**Q3.03.2** MCQ | 2003 | 2M | Ans: C

Before the fault,

$$P_m = P_e = 1.0 \text{ pu}$$

and

$$P_{\max} = 2.0 \text{ pu}$$

Hence,

$$P_m = P_{\max} \sin \delta_0$$

$$1 = 2 \sin \delta_0$$

$$\delta_0 = 30^\circ$$

During the fault,

$$P_e = 0$$

and after fault clearing, the original network is restored, therefore

$$P_{\max} = 2.0 \text{ pu}$$

Given,

$$\delta_{max} = 110^\circ$$

Applying the equal-area criterion,

$$P_m(\delta_c - \delta_0) = \int_{\delta_c}^{\delta_{max}} (P_{max} \sin\delta - P_m)\, d\delta$$

which gives

$$\delta_c = \cos^{-1}\left(\frac{P_m(\delta_{max} - \delta_0) + P_{max}\cos\delta_{max}}{P_{max}}\right)$$

Substituting,

$$\delta_0 = 30^\circ = 0.5236 \text{ rad}$$

$$\delta_{max} = 110^\circ = 1.9199 \text{ rad}$$

$$P_m = 1, \qquad P_{max} = 2$$

$$\delta_c = \cos^{-1}\left(\frac{1(1.9199 - 0.5236) + 2\cos 110^\circ}{2}\right)$$

$$= 69.14^\circ$$

Therefore, the correct answer is

$$\boxed{\text{(C) } \delta_c = 69.14^\circ}$$

**Q3.02.1** MCQ | 2002 | 1M | Ans: B ● ● ●

A generator operating in isochronous mode maintains the system frequency (speed) constant at its rated value. Generator $G_1$ has a governor that maintains constant speed. Therefore,

$$\Delta f = 0$$

Generators $G_2$ and $G_3$ operate with a droop characteristic of 5%.
For a droop-controlled generator,

$$\Delta P = -\frac{1}{R}\Delta f$$

Since the system frequency does not change,

$$\Delta f = 0$$

hence,

$$\Delta P_{G_2} = 0$$

and

$$\Delta P_{G_3} = 0$$

Therefore, the increase in load is supplied entirely by generator $G_1$.
Considering negligible transmission losses,

$$\Delta P_D = \Delta P_{G_1}$$

Thus, generation of $G_1$ alone increases while the generation of $G_2$ and $G_3$ remains unchanged.

(B) generation of G1 alone is increased while generation of G2 and G3 is unchanged.

**Key Points**

- An isochronous governor maintains system frequency at the rated value.
- A droop governor changes output only when frequency changes.
- With $\Delta f = 0$, generators operating on droop characteristics do not change their power output.
- Under negligible transmission losses, the additional load demand must be supplied by the isochronous machine.

**Q3.02.2** NAT | 2002 | 5M | Ans: 20.47 rad/sec | ● ● ●

$$P_e = \frac{EV}{X_{\text{eq}}} \sin \delta$$

where

$$X_{\text{eq}} = x' + X = 0.1 + 0.2 = 0.3 \text{ pu}$$

Since

$$E' = 1.0 \text{ pu}$$

and

$$V = 1.0 \text{ pu}$$

$$P_e = \frac{1 \times 1}{0.3} \sin \delta = 3.333 \sin \delta$$

Given

$$P_m = 0$$

the swing equation becomes

$$\frac{2H}{\omega_B} \frac{d^2\delta}{dt^2} = -P_e$$

Substituting,

$$\frac{2 \times 5}{2\pi \times 50} \frac{d^2\delta}{dt^2} = -3.333 \sin \delta$$

Using the given energy relation,

$$\int \left(\frac{2H}{\omega_B}\right) \omega \, d\omega + \int P_e \, d\delta = 0$$

Substituting $P_e = 3.333 \sin \delta$,

$$\frac{H}{\omega_B}\omega^2 = 3.333\cos\delta + C$$

Applying the initial condition

$$\delta = 0, \qquad \omega = \omega_{\text{init}}$$

gives

$$\frac{H}{\omega_B}\omega^2 = 3.333\cos\delta + \frac{H}{\omega_B}\omega_{\text{init}}^2 - 3.333$$

Hence,

$$\omega^2 = \omega_{\text{init}}^2 + \frac{\omega_B}{H}3.333(\cos\delta - 1)$$

For the machine to pull into synchronism, the limiting case occurs when

$$\omega = 0$$

at

$$\delta = \pi$$

Therefore,

$$0 = \omega_{\text{init}}^2 + \frac{\omega_B}{H}3.333(-2)$$

$$\omega_{\text{init}}^2 = \frac{2 \times 3.333 \times \omega_B}{H}$$

Substituting

$$\omega_B = 2\pi \times 50$$

and

$$H = 5$$

$$\omega_{\text{init}}^2 = \frac{2 \times 3.333 \times 2\pi \times 50}{5}$$

$$\omega_{\text{init}}^2 = 418.88$$

$$\omega_{\text{init}} = 20.47 \text{ rad/s}$$

Hence, the answer is

$$\boxed{20.47 \text{ rad/s}}$$

**Key Points**

- After switch closure the machine behaves as a SMIB system.
- Electrical power is given by $P_e = \frac{EV}{X_{\text{eq}}}\sin\delta$.
- Pull-in condition can be obtained using the energy integral form of the swing equation.

- The critical initial speed corresponds to the limiting stable trajectory.

**Q3.01.1** NAT | 2001 | 5M | Ans: *

Before the fault,

$$P_m = P_{e0}$$

Given,

$$P_e = P_{\max} \sin \delta$$

and

$$P_e = 1.0 \text{ pu}$$

at

$$\delta_0 = 30^\circ$$

Hence,

$$P_{\max,I} = \frac{1}{\sin 30^\circ} = 2.0 \text{ pu}$$

Therefore,

$$P_m = 1.0 \text{ pu}$$

During the fault,

$$P_{\max,II} = 0.5 \text{ pu}$$

After fault clearing,

$$P_{\max,III} = 1.5 \text{ pu}$$

The maximum rotor angle is

$$\delta_{\max} = \pi - \sin^{-1}\left(\frac{P_m}{P_{\max,III}}\right)$$

Substituting,

$$\delta_{\max} = \pi - \sin^{-1}\left(\frac{1}{1.5}\right)$$

$$\delta_{\max} = 2.41 \text{ rad}$$

$$\delta_{\max} = 138.2^\circ$$

Applying the equal area criterion,

$$\delta_c = \cos^{-1}\left[\frac{P_m(\delta_{\max} - \delta_0) + P_{\max,III}\cos\delta_{\max} - P_{\max,II}\cos\delta_0}{P_{\max,III} - P_{\max,II}}\right]$$

Substituting,

$$\delta_c = \cos^{-1}\left[\frac{1(2.41 - 0.5235) + 1.5\cos 138.2^\circ - 0.5\cos 30^\circ}{1.5 - 0.5}\right]$$

$$\delta_c = \cos^{-1}(0.33527)$$

$$\delta_c = 70.4^\circ$$

Hence, the correct answer is

$$\boxed{70.4^\circ}$$

**Q3.00.1** NAT | 2000 | 5M | Ans: * | ● ● ●

The equivalent reactance of the two parallel transmission lines is

$$X_L = 0.3 \parallel 0.3 = 0.15 \text{ p.u.}$$

Hence, the pre-fault transfer reactance is

$$X_{\text{pre}} = 0.15 + 0.15 = 0.30 \text{ p.u.}$$

In steady state

$$P_m = P_e = \frac{EV}{X_{\text{pre}}}\sin\delta_0 = 1 \text{ p.u.}$$

**During the fault**

Let A be the generator-side junction of the two lines. The faulted line runs from A to the fault point F, which is at zero potential; the healthy line connects A to the infinite bus. The network is a star with branches 0.15 (to $E$), 0.3 (to $V$) and 0.3 (to ground at F). Converting it to a delta, the transfer reactance between $E$ and $V$ is

$$X_{\text{fault}} = 0.15 + 0.3 + \frac{0.15 \times 0.3}{0.3} = 0.60 \text{ p.u.}$$

The rotor angle cannot change instantly, so at $t = 0^+$

$$P_e(0^+) = \frac{EV}{X_{\text{fault}}}\sin\delta_0 = \frac{X_{\text{pre}}}{X_{\text{fault}}} \times 1 = \frac{0.30}{0.60} = 0.5 \text{ p.u.}$$

The internal emf $E$ cancels, so it need not be known.

The accelerating power just before and just after the fault is

$$P_a(0^-) = 1 - 1 = 0, \qquad P_a(0^+) = 1 - 0.5 = 0.5 \text{ p.u.}$$

**(a) Breaker clears at the end of an interval**

When a switching event (fault or clearing) falls exactly on an interval boundary, the accelerating power at that instant has two values, and the point-by-point method uses their average. For the first step:

$$P_{a,\text{avg}} = \frac{P_a(0^-) + P_a(0^+)}{2} = \frac{0 + 0.5}{2} = 0.25 \text{ p.u.}$$

The same averaging (during-fault and post-clearing values) is used at the interval boundary where the breaker clears.

**(b) Breaker clears at the middle of an interval**

No averaging is needed: the discontinuity falls between two calculation points. The accelerating power at each calculation point is evaluated from the network existing at that instant, i.e. the during-fault value at the point before clearing and the post-clearing value $P_a = 1 - P_{e,\text{post}}(\delta)$ at the points after clearing.

**Key Points**

- If switching occurs at an interval boundary, use the average of the accelerating powers before and after switching.
- If switching occurs in the middle of an interval, use the accelerating power of the network existing at each calculation point directly.
- The ratio $P_e(0^+)/P_e(0^-) = X_{\text{pre}}/X_{\text{fault}}$ does not depend on $E$.

**Q3.99.1** MCQ | 1999 | 1M | Ans: D | ● ○ ○

Steady state stability refers to the capability of a power system to maintain synchronism under small and gradual disturbances.
It ensures that synchronous machines remain in step with one another and power transfer over interconnected tie lines continues without loss of synchronism.
Hence, steady state stability is the ability of the power system to maintain synchronism between machines and on external tie lines.
Therefore, the correct option is

(D) maintain synchronism between machines and on external tie lines

**Q3.98.1** MCQ | 1998 | 2M | Ans: D | ● ○ ○

The equivalent inertia constant on a common base is

$$H_{eq} = \frac{\sum H_i G_i}{G_{eq}}$$

where $G_i$ is the machine MVA rating.
Substituting,

$$H_{eq} = \frac{250(1.6) + 500(1)}{100}$$

$$= \frac{400 + 500}{100}$$

$$= 9.0$$

Therefore,

(D) $H_{eq} = 9.0$

**Q3.98.2** NAT | 1998 | 5M | Ans: * | ● ○ ○

Given,

$$V = 1\angle 0^\circ$$

$$I = 1\angle(-36.86^\circ)$$

$$X_d = 1.2$$

The internal generated emf is

$$E\angle\delta = V + jIX_d$$

$$= 1\angle 0^\circ + j(1\angle -36.86^\circ)(1.2)$$

$$= 1\angle 0^\circ + 1.2\angle 53.14^\circ$$

$$E\angle\delta = 1.97\angle 29.17^\circ$$

The active power output is

$$P = \frac{EV}{X_d}\sin\delta$$

$$= \frac{1.97 \times 1}{1.2}\sin 29.17^\circ$$

$$P = 0.8 \text{ p.u.}$$

The steady-state stability limit is

$$P_{SSL} = \frac{EV}{X_d}$$

$$= \frac{1.97 \times 1}{1.2}$$

$$P_{SSL} = 1.642 \text{ p.u.}$$

Keeping active power fixed,

$$P = 0.8$$

At stability limit,

$$\delta = 90^\circ$$

Therefore,

$$0.8 = \frac{EV}{X_d}$$

$$E = \frac{0.8 \times 1.2}{1}$$

$$E = 0.96 \text{ p.u.}$$

Hence,

$$\boxed{P = 0.8 \text{ p.u.}, \quad P_{SSL} = 1.642 \text{ p.u.}, \quad E_{critical} = 0.96 \text{ p.u.}}$$

**Q3.97.1** MCQ | 1997 | 1M | Ans: B | ● ○ ○

The inertia constant is

$$H = \frac{E}{G}$$

where

$$H = 4 \text{ s}$$

and

$$G = 100 \text{ MVA}$$

Therefore,

$$E = HG$$

$$E = 4 \times 100$$

$$E = 400 \text{ MJ}$$

Hence, the correct option is

(B) 400 MJ

**Q3.97.2** NAT | 1997 | 5M | Ans: * | ● ● ○

Initially,

$$P_0 = 0.5P_{max}$$

Hence,

$$\delta_0 = \sin^{-1}(0.5) = 30^\circ$$

After load reduction,

$$P_1 = 0.4P_{max}$$

Therefore,

$$\delta_1 = \sin^{-1}(0.4) = 23.578^\circ$$

Using the equal-area criterion,

$$\int_{\delta_1}^{\delta_0} (P_{max} \sin\delta - 0.4P_{max})\, d\delta = \int_{\delta_0}^{\delta_2} (0.4P_{max} - P_{max} \sin\delta)\, d\delta$$

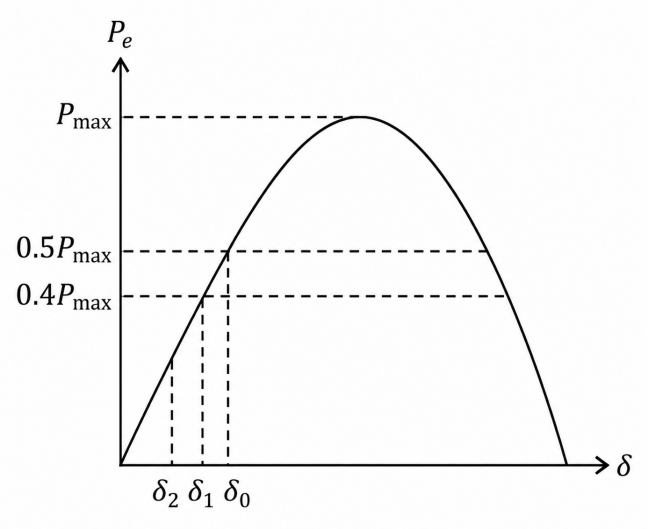



Fig. Solution Q3.97.2

Simplifying,

$$0.4(\delta_2 - 30°)\frac{\pi}{180} + \cos\delta_2 - \cos 30° = 0$$

Solving,

$$\delta_2 = 17.32°$$

Hence, the correct answer is

$$\boxed{17.32°}$$

**Q3.97.3** NAT | 1997 | 5M | Ans: * | ●●○

Frequency deviation:

$$\Delta f = -\frac{\Delta P}{B + \frac{1}{R}}$$

$$= -\frac{200}{0.75 + \frac{1}{0.05}}$$

$$= -9.6385$$

Load sharing is proportional to ratings.

$$\frac{\Delta P_A}{\Delta P_B} = \frac{2000}{3000} = \frac{2}{3}$$

Since

$$\Delta P_A + \Delta P_B = 200$$

$$\frac{2}{3}\Delta P_B + \Delta P_B = 200$$

$$\frac{5}{3}\Delta P_B = 200$$

$$\Delta P_B = 120 \text{ MW}$$

$$\Delta P_A = 80 \text{ MW}$$

Therefore the steady-state tie-line power transfer change is

$$\boxed{\Delta f = -9.6385;\ \Delta P_B = 120 \text{ MW};\ \Delta P_A = 80 \text{ MW}}$$

**Q3.96.1** MCQ | 1996 | 1M | Ans: A | ● ● ○

During rotor oscillations, the machine alternately accelerates and decelerates about the equilibrium position. At the extreme points of the swing:

- Rotor speed becomes equal to synchronous speed.
- Relative speed deviation becomes zero.
- Direction of rotor acceleration changes.

These extreme points correspond to A and B.
Point C is the final steady-state operating point reached after damping has removed oscillations.
Hence, the required option is

$$\boxed{\text{(A) A and B}}$$

**Q3.95.1** NAT | 1995 | 5M | Ans: UNSTABLE | ● ● ○

Given,

$$P_{max} = 100 \text{ MW}$$

Initial power output:

$$P_0 = 50 \text{ MW}$$

Hence

$$P_0 = P_{max} \sin \delta_0$$

$$50 = 100 \sin \delta_0$$

$$\delta_0 = 30^\circ = 0.5236 \text{ rad}$$

After the sudden increase in mechanical input,

$$P_m = 50 + 40 = 90 \text{ MW}$$

The new equilibrium position is

$$90 = 100 \sin \delta_1$$

$$\delta_1 = 64.16^\circ = 1.1198 \text{ rad}$$

The rotor accelerates from $\delta_0$ to $\delta_1$ and then decelerates. The largest angle up to which deceleration is possible is

$$\delta_m = \pi - \delta_1 = 180^\circ - 64.16^\circ = 115.84^\circ = 2.0218 \text{ rad}$$

Accelerating area:

$$A_1 = \int_{\delta_0}^{\delta_1} (P_m - P_{max} \sin\delta)\, d\delta = P_m(\delta_1 - \delta_0) - P_{max}(\cos\delta_0 - \cos\delta_1)$$

$$A_1 = 90(1.1198 - 0.5236) - 100(0.8660 - 0.4359) = 53.65 - 43.01 = 10.64 \text{ MW-rad}$$

Maximum decelerating area available:

$$A_{2,\max} = \int_{\delta_1}^{\delta_m} (P_{max} \sin\delta - P_m)\, d\delta = P_{max}(\cos\delta_1 - \cos\delta_m) - P_m(\delta_m - \delta_1)$$

$$A_{2,\max} = 100(0.4359 + 0.4359) - 90(2.0218 - 1.1198) = 87.18 - 81.18 = 6.00 \text{ MW-rad}$$

Since

$$A_1 > A_{2,\max}$$

the available decelerating area cannot absorb the accelerating energy, the rotor swings beyond $\delta_m$ and loses synchronism.

UNSTABLE

**Q3.94.1** NAT | 1994 | 2M | Ans: 21 | ●●○

The inertia constant $H$ represents the kinetic energy stored in the rotating mass at synchronous speed per MVA of machine rating.

When machines operate in parallel, their inertia constants must first be converted to the common base and then added.

—

For Generator 1:

$$S_1 = 250 \text{ MVA}$$

$$H_1 = 6 \text{ MJ/MVA}$$

Converting to 100 MVA base,

$$H_1' = H_1 \left(\frac{S_1}{100}\right)$$

$$= 6 \times \frac{250}{100}$$

$$= 15 \text{ MJ/MVA}$$

—

For Generator 2:

$$S_2 = 150 \text{ MVA}$$

$$H_2 = 4 \text{ MJ/MVA}$$

Converting to 100 MVA base,

$$H_2' = H_2\left(\frac{S_2}{100}\right)$$

$$= 4 \times \frac{150}{100}$$

$$= 6 \text{ MJ/MVA}$$

—

Combined inertia constant:

$$H_{eq} = H_1' + H_2'$$

$$= 15 + 6$$

$$= 21 \text{ MJ/MVA}$$

Therefore, the correct answer is

21 MJ/MVA

**Q3.93.1** MCQ | 1993 | 1M | Ans: C | ● ○ ○

Transient stability refers to the ability of a power system to remain in synchronism following a large disturbance such as:

- Three-phase faults,
- Single line-to-ground faults,
- Sudden loss of generation,
- Sudden switching of transmission lines.

The objective is to reduce the accelerating energy of the generator rotor during the fault and increase the decelerating energy after fault clearing.
Let us examine each option.

—

**Option (A): Excitation control**
Fast excitation systems improve voltage regulation and steady-state stability. Although they provide some improvement in transient stability, they are not the most effective method among the given options.
Hence, Option (A) is not the best answer.

—

**Option (B): Phase shifting transformer**
A phase-shifting transformer controls real power flow between transmission lines by introducing a phase-angle shift.
It is primarily used for load flow control and congestion management rather than improving transient stability.
Hence, Option (B) is incorrect.

—

**Option (C): Single pole switching circuit breakers**
Most faults occurring on EHV transmission lines are single line-to-ground (SLG) faults.
With conventional three-pole tripping, all three phases are disconnected during the fault, causing the electrical

power output of the generator to become nearly zero.
From the swing equation,

$$P_a = P_m - P_e$$

where

- $P_m$ = Mechanical input power,
- $P_e$ = Electrical output power,
- $P_a$ = Accelerating power.

If all three phases are opened,

$$P_e \approx 0$$

therefore,

$$P_a \approx P_m$$

which produces maximum rotor acceleration.
However, in single-pole switching, only the faulty phase is isolated while the two healthy phases continue to transmit power.
Thus,

$$P_e > 0$$

during the fault.
Consequently,

$$P_a = P_m - P_e$$

is significantly reduced, thereby reducing rotor acceleration and improving transient stability.
This increases the critical clearing time and the stability margin of the system.
Hence, Option (C) is correct.
—
**Option (D): Increasing the turbine valve opening**
Increasing turbine valve opening increases the mechanical input power,

$$P_m$$

which increases accelerating power during a fault.
This adversely affects transient stability.
Hence, Option (D) is incorrect.
—
Therefore, the correct answer is

(C) Single pole switching circuit breakers

**Q3.91.1** NAT | 1991 | 2M | Ans: 62.71 | ● ● ●

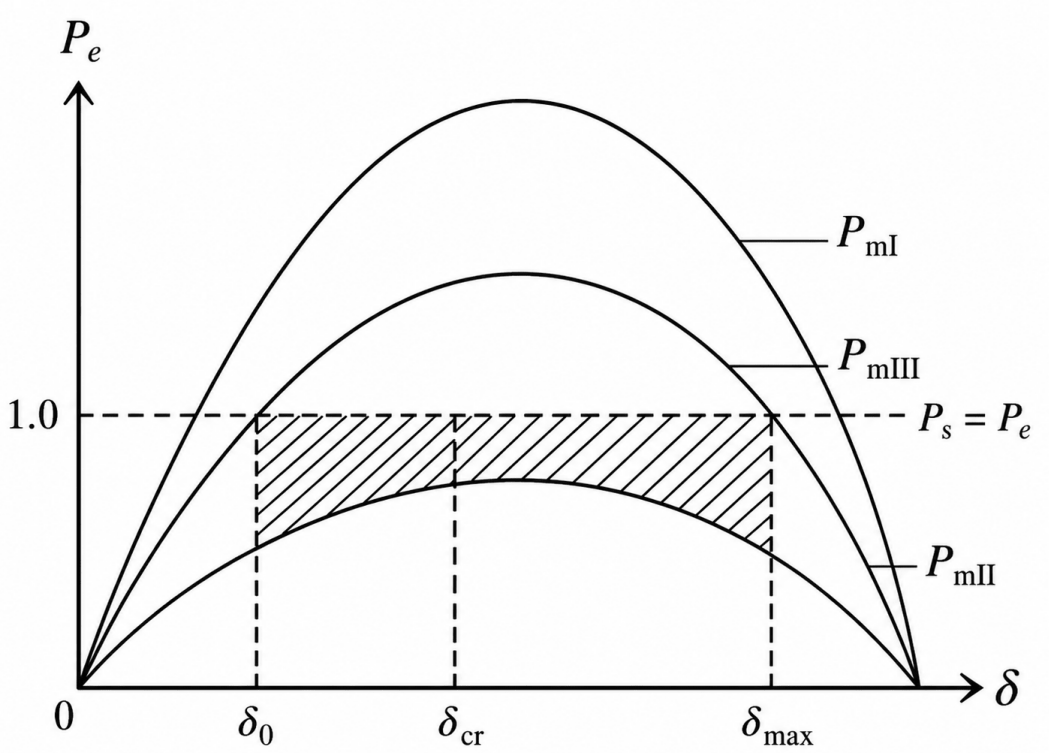



Fig. Solution Q3.91.1

**Concept:**
The Equal Area Criterion (EAC) is a graphical method used to determine the transient stability of a Single Machine Infinite Bus (SMIB) system subjected to a large disturbance.
For the system to remain stable,

$$A_1 = A_2$$

where

- $A_1$ = Accelerating area
- $A_2$ = Decelerating area

—

**Step 1: Pre-fault operating condition**
The pre-fault electrical power is

$$P_e = 2 \sin \delta$$

The generator delivers

$$P_m = P_s = 1 \text{ p.u.}$$

Hence,

$$1 = 2 \sin \delta_0$$

$$\delta_0 = \sin^{-1}(0.5) = 30^\circ$$

or

$$\boxed{\delta_0 = 0.523 \text{ rad}}$$

—

**Step 2: During fault**

During the fault,

$$P_e = 0.2 \sin \delta$$

Since

$$P_m > P_e,$$

the rotor accelerates.

The accelerating area is

$$A_1 = \int_{\delta_0}^{\delta_{cr}} (P_m - 0.2 \sin \delta)\, d\delta$$

—

**Step 3: Post-fault condition**

After breakers $B_3$ and $B_4$ open,

$$P_e = 1.5 \sin \delta$$

The maximum rotor angle is obtained from

$$P_m = P_e$$

Therefore,

$$1 = 1.5 \sin \delta_{max}$$

$$\delta_{max} = 180^\circ - \sin^{-1}\left(\frac{1}{1.5}\right)$$

$$= 180^\circ - 41.81^\circ$$

$$\boxed{\delta_{max} = 138.19^\circ = 2.412 \text{ rad}}$$

—

**Step 4: Apply Equal Area Criterion**

The decelerating area is

$$A_2 = \int_{\delta_{cr}}^{\delta_{max}} (1.5 \sin \delta - P_m)\, d\delta$$

Since

$$A_1 = A_2,$$

after evaluating the integrals,

$$\delta_{cr} = \cos^{-1}\left[\frac{P_s(\delta_{max} - \delta_0) + P_{mIII}\cos\delta_{max} - P_{mII}\cos\delta_0}{P_{mIII} - P_{mII}}\right]$$

where

$$P_s = 1,$$

$$P_{mII} = 0.2,$$

$$P_{mIII} = 1.5.$$

Substituting,

$$\delta_{cr} = \cos^{-1}\left[\frac{1(2.412 - 0.523) + 1.5\cos 138.19° - 0.2\cos 30°}{1.5 - 0.2}\right]$$

$$= \cos^{-1}(0.459)$$

$$\boxed{\delta_{cr} = 62.71°}$$

—

Hence, the answer is

$$\boxed{62.71°}$$

PrepFusion

SOLUTIONS

# IV

# Power system matrices and Load Flow

## Topics

1. Bus Admittance Matrix
2. Bus Impedance Matrix
3. Fault Analysis using Z bus
4. Complex Power concept
5. Gauss-Seidel Load Flow Method
6. Newton-Raphson Load Flow Method

## Unit Snapshot*

| | | | | |
|---|---|---|---|---|
| **Stream:** EE | **Questions:** 59 | **MCQ:** 44 | **MSQ:** 2 | **NAT:** 13 |
| **1M:** 28 | **2M:** 27 | **Descriptive:** 4 | **Avg Weightage** ~3M | **Range:** 1M(2019)–6M(2013) |

**See preface for how Avg Weightage and Range are calculated.*

**Q4.26.1** MCQ | 2026 | 1M | Ans: B | ● ● ○

In the Newton–Raphson load flow method:

- One bus is taken as the slack bus.
- For each of the remaining $(N-1)$ buses, voltage angle $\delta$ is unknown.
- Reactive power equations are written only for load buses (PQ buses).
- Since there are $Q$ load buses, there are $Q$ unknown voltage magnitudes.

Therefore, total number of unknowns is

$$(N-1)+Q$$

Given that

$$N = P + Q + 1$$

(where one bus is the slack bus),
substituting:

$$(N-1)+Q = N+Q-1$$

Since

$$Q = N-1-P$$

we get

$$N+(N-1-P)-1 = 2N-2-P$$

Hence, the Jacobian matrix size is

$$(2N-2-P)\times(2N-2-P)$$

Using the standard load flow formulation:

$$(N-1)+Q = (N-1)+(N-1-P)$$

$$= 2N-2-P$$

which is commonly written as

$$(2N-1-P)\times(2N-1-P)$$

depending on indexing convention adopted in the formulation.
Therefore, the correct option is

$$\boxed{\text{(B) } (2N-1-P)\times(2N-1-P)}$$

**Key Points**

- In Newton–Raphson load flow, voltage angles are unknown for all non-slack buses.
- Voltage magnitudes are unknown only for PQ buses.
- Total unknowns determine the Jacobian matrix dimension.
- For a system with $P$ PV buses and one slack bus:

$$\text{Jacobian size} = (2N-1-P)\times(2N-1-P)$$

**Mistakes to be avoided**

- Do not include the slack bus voltage magnitude and angle as unknowns.
- Remember that reactive power mismatch equations are written only for PQ buses.

**Q4.26.2** MSQ | 2026 | 2M | Ans: A,D

For a transmission line,

$$S_{\text{loss}} = I^2 Z$$

where

$$I = \frac{V_1\angle\theta_1 - V_2\angle\theta_2}{Z}$$

Hence,

$$|I|^2 = \frac{|V_1\angle\theta_1 - V_2\angle\theta_2|^2}{|Z|^2}$$

Using

$$|V_1 - V_2|^2 = |V_1|^2 + |V_2|^2 - 2|V_1||V_2|\cos\theta_{12}$$

we get

$$|I|^2 = \frac{|V_1|^2 + |V_2|^2 - 2|V_1||V_2|\cos\theta_{12}}{|Z|^2}$$

Since

$$|V_1| = 1.1, \qquad |V_2| = 0.9, \qquad |Z| = 0.75$$

for both cases:

$$|I|^2 = \frac{1.1^2 + 0.9^2 - 2(1.1)(0.9)\cos\theta_{12}}{0.75^2}$$

**Case-1:**

$$\theta_{12} = 0^\circ$$

Thus,

$$|I_1|^2 = \frac{1.21 + 0.81 - 2(1.1)(0.9)(1)}{0.75^2}$$

$$|I_1|^2 = \frac{0.04}{0.5625} \approx 0.0711$$

Since

$$Z = 0.75\angle 0^\circ$$

the line is purely resistive.
Therefore,

$$P_{\text{loss},1} = |I_1|^2 R = 0.0711 \times 0.75 \approx 0.0533$$

and

$$Q_{\text{loss},1} = 0$$

**Case-2:**

$$\theta_{12} = 90^\circ$$

Hence,

$$\cos 90^\circ = 0$$

Therefore,

$$|I_2|^2 = \frac{1.21 + 0.81}{0.75^2}$$

$$|I_2|^2 = \frac{2.02}{0.5625} \approx 3.591$$

Since

$$Z = 0.75\angle 90^\circ = j0.75$$

the line is purely reactive.
Thus,

$$P_{\text{loss},2} = 0$$

and

$$Q_{\text{loss},2} = |I_2|^2 X = 3.591 \times 0.75 \approx 2.693$$

Comparing the two cases:

$$P_{\text{loss},1} > P_{\text{loss},2}$$

and

$$Q_{\text{loss},2} > Q_{\text{loss},1}$$

Therefore, the correct options are

(A) Real power loss in Case-1 is more than that in Case-2

(D) Reactive power loss in Case-2 is more than that in Case-1

**Key Points**

- Current magnitude depends on voltage-angle difference:

$$|V_1 - V_2|^2 = |V_1|^2 + |V_2|^2 - 2|V_1||V_2| \cos\theta_{12}$$

- A purely resistive line absorbs only real power.
- A purely reactive line absorbs only reactive power.

**Q4.26.3** MCQ | 2026 | 2M | Ans: D ● ● ○

**Step 1: Model the transformer.**

Let the voltage ratio be

$$a = \frac{V_P}{V_S} = 0.866 + j0.5, \qquad |a|^2 = 0.866^2 + 0.5^2 \approx 1, \qquad a^* = 0.866 - j0.5$$

The transformer is an ideal transformer of ratio $a : 1$ in series with the leakage admittance $y_t$ on the secondary side (magnetizing branch neglected). Let $V' = V_P/a$ be the voltage just after the ideal transformer, and let $I'$ be the current flowing from the ideal transformer through $y_t$ towards the secondary terminal.

**Step 2: Write the branch equations.**

Current through the leakage admittance:

$$I' = y_t\,(V' - V_S) = y_t\left(\frac{V_P}{a} - V_S\right)$$

The current injected at the secondary terminal is $I_S = -I'$:

$$I_S = -\frac{y_t}{a}\,V_P + y_t\,V_S$$

For the ideal transformer, complex power is conserved, $V_P I_P^* = V' I'^*$. With $V_P = aV'$ this gives $I_P^* = I'^*/a$, so

$$I_P = \frac{I'}{a^*} = \frac{y_t}{a^*}\left(\frac{V_P}{a} - V_S\right) = \frac{y_t}{|a|^2}\,V_P - \frac{y_t}{a^*}\,V_S$$

**Step 3: Assemble $Y_{\text{bus}}$.**

Since $|a|^2 \approx 1$,

$$\begin{bmatrix} I_P \\ I_S \end{bmatrix} = \begin{bmatrix} y_t & -\dfrac{y_t}{0.866 - j0.5} \\ -\dfrac{y_t}{0.866 + j0.5} & y_t \end{bmatrix} \begin{bmatrix} V_P \\ V_S \end{bmatrix}$$

The matrix is not symmetric, because the tap ratio is complex (a phase-shifting transformer).

This matches option (D). Options (A) and (B) are symmetric and therefore cannot represent a complex ratio; in (C) the two off-diagonal entries are equal, which would be correct only for a real tap ratio.

(D)

**Key Points**

- For a transformer with ratio $a = V_P/V_S$ and leakage admittance $y_t$ on the secondary side: $Y_{PP} = y_t/|a|^2, Y_{PS} = -y_t/a^*$, $Y_{SP} = -y_t/a, Y_{SS} = y_t$.
- A complex tap ratio (phase shifter) makes $Y_{\text{bus}}$ non-symmetric; a real tap ratio keeps it symmetric.
- Use conservation of complex power across the ideal transformer to get the current relation $I_P = I'/a^*$.

**Q4.25.1** NAT | 2025 | 1M | Ans: 0.380 to 0.400 ● ○ ○

Given:

$$Z_{\text{bus}} = \begin{bmatrix} j0.059 & j0.061 & j0.038 \\ j0.061 & j0.093 & j0.066 \\ j0.038 & j0.066 & j0.110 \end{bmatrix}$$

and

$$Z_f = j0.007\,\text{pu}$$

For a symmetrical fault at bus 2,

$$I_f = \frac{V_{f2}}{Z_{22} + Z_f}$$

Neglecting pre-fault loading,

$$V_{f2} = 1\,\text{pu}$$

Therefore,

$$I_f = \frac{1}{j0.093 + j0.007}$$

$$= \frac{1}{j0.1} = -j10\,\text{pu}$$

Voltage at bus 1 during fault is

$$V_{1(f)} = V_1^0 - I_f Z_{12}$$

Substituting values,

$$V_{1(f)} = 1 - (-j10)(j0.061)$$

Since

$$(-j)(j) = 1$$

we get

$$V_{1(f)} = 1 - 0.61$$

$$= 0.39\,\text{pu}$$

Hence,

$$\boxed{0.390}$$

## Key Points

- For a fault at bus $k$:
$$I_f = \frac{V_f}{Z_{kk} + Z_f}$$
- Bus voltages during fault are obtained using transfer impedances from the $Z_{\text{bus}}$ matrix.
- Neglecting pre-fault loading implies:
$$V_i^0 = 1\,\text{pu}$$
for all buses.

## Mistakes to be avoided

- Do not forget to include fault impedance $Z_f$ in the denominator.
- Use the transfer impedance:
$$Z_{12}$$

for calculating bus-1 voltage.

**Q4.25.2** MCQ 2025 1M Ans: A ●○○

In Newton–Raphson load-flow analysis:

- A **PQ bus** has specified real and reactive powers.
- A **PV bus** has specified real power and voltage magnitude.
- A **Slack bus** maintains the system voltage magnitude and phase angle while balancing the real and reactive power mismatch in the network.

During a power failure, the UPS acts as the reference source supplying power to the connected household loads.
Hence, it behaves as a slack bus because:

- it establishes the voltage magnitude and frequency reference,
- it adjusts its generated power according to load demand,
- and it balances system losses and mismatches.

Therefore, the correct option is

(A) Slack Bus

**Key Points**

- Slack bus is the reference bus in load-flow analysis.
- Voltage magnitude and phase angle are specified for the slack bus.
- Real and reactive powers at the slack bus are determined from system balance equations.
- A UPS supplying isolated household loads behaves like the system reference source.

**Mistakes to be avoided**

- Do not confuse a UPS with a PV generator bus.
- PQ bus corresponds to load buses, not supply reference buses.
- There is no standard bus type called PQV bus in Newton–Raphson load flow.

**Q4.24.1** MCQ 2024 1M Ans: A ●○○

To operate the network with minimum loss, the branch configuration resulting in the minimum total power loss must be selected.
Since all branch impedances are proportional to $z$, the total loss is proportional to:

$$\sum I_k^2 Z_k$$

where $I_k$ is the current through branch $k$ and $Z_k$ is the branch impedance.
**Case 1: Considering $b_1$ is opened**
Current distribution becomes:

$$3I \text{ through } b_2, \qquad 2I \text{ through } b_4, \qquad I \text{ through } b_3$$

Total loss:

$$(3I)^2(z) + (2I)^2(4z) + (I)^2(2z)$$

$$= 9I^2z + 16I^2z + 2I^2z$$

$$= 27I^2z$$

**Case 2: Considering $b_2$ is opened**
Current distribution becomes:

$3I$ through $b_1$, $2I$ through $b_3$, $I$ through $b_4$

Total loss:

$$(3I)^2(4z) + (2I)^2(2z) + (I)^2(4z)$$

$$= 36I^2z + 8I^2z + 4I^2z$$

$$= 48I^2z$$

**Case 3: Considering $b_3$ is opened**
Current distribution becomes:

$I$ through $b_1$, $2I$ through $b_2$, $I$ through $b_4$

Total loss:

$$(I)^2(4z) + (2I)^2(z) + (I)^2(4z)$$

$$= 4I^2z + 4I^2z + 4I^2z$$

$$= 12I^2z$$

**Case 4: Considering $b_4$ is opened**
Current distribution becomes:

$2I$ through $b_1$, $I$ through $b_2$, $I$ through $b_3$

Total loss:

$$(2I)^2(4z) + (I)^2(z) + (I)^2(2z)$$

$$= 16I^2z + I^2z + 2I^2z$$

$$= 19I^2z$$

Among all cases, minimum loss occurs when $b_3$ is opened.

$$\boxed{\text{(A) } b_3}$$

**Mistakes to be avoided**

- Do not assume that opening the highest impedance branch always minimizes loss.
- Ensure the resulting network remains connected after opening a branch.

**Q4.24.2** MCQ | 2024 | 2M | Ans: A

Since the real power flow from bus 1 to bus 2 is zero,

$$P_{12} = 0$$

Hence, buses 1 and 2 have the same phase angle.
Let the phase angle difference between buses 1 and 3 be $\theta$.
Using the DC power flow approximation:

$$P = \frac{V_1 V_2}{X} \sin\theta$$

Since all voltage magnitudes are equal to 1 pu,

$$P_1 = \frac{1 \times 1}{\beta x} \sin\theta$$

and

$$P_2 = \frac{1 \times 1}{\gamma x} \sin\theta$$

Given,

$$P_1 = mP_2$$

Substituting:

$$\frac{1}{\beta x} \sin\theta = m\left(\frac{1}{\gamma x} \sin\theta\right)$$

Cancelling common terms:

$$\frac{1}{\beta} = \frac{m}{\gamma}$$

Therefore,

$$\boxed{\text{(A)} \quad \gamma = m\beta}$$

**Key Points**

- For small angle differences, DC power flow approximation is valid.
- Real power transfer in a lossless line is inversely proportional to reactance.
- Zero real power flow implies equal voltage phase angles across the line.

**Q4.23.1** MSQ | 2023 | 1M | Ans: A,C | ● ● ●

Recall the properties of the bus admittance matrix:

$$Y_{ij} = -y_{ij}, \qquad i \neq j, \qquad\qquad Y_{ii} = \sum_{j \neq i} y_{ij} + y_{\text{sh},i}$$

where $y_{\text{sh},i}$ is the total shunt admittance at bus $i$ (line charging plus any shunt capacitor).

From the off-diagonal entries, the series admittances are

$$y_{12} = -j10, \qquad y_{13} = -j5, \qquad y_{23} = -j4.$$

Without any shunt element the diagonal entries would therefore be $-j15$, $-j14$ and $-j9$. Comparing with the given diagonal entries:

$$y_{\text{sh},1} = -j15 - (-j15) = 0, \qquad y_{\text{sh},2} = -j13.5 - (-j14) = j0.5, \qquad y_{\text{sh},3} = -j8 - (-j9) = j1.$$

Since there is no shunt inductor, no shunt element can cancel a capacitive contribution. Hence nothing capacitive can be connected at bus 1.

**(A)** Line charging in all three lines puts a capacitive contribution at bus 1 (from lines 1–2 and 1–3), but $y_{\text{sh},1} = 0$. So (A) can NOT be true.

**(B)** Line charging in line 2–3 only affects buses 2 and 3 and leaves bus 1 with zero shunt, which agrees with $y_{\text{sh},1} = 0$; the shunt susceptances at buses 2 and 3 (0.5 and 1) are both capacitive. So (B) is taken as possible.

**(C)** A shunt capacitor at bus 1 would make $y_{\text{sh},1} \neq 0$. So (C) can NOT be true.

**(D)** Line charging $j0.5$ at each end of line 2–3 plus a shunt capacitor $j0.5$ at bus 3 gives exactly $j0.5$ at bus 2 and $j1$ at bus 3. So (D) is possible.

(A) and (C)

Note: with the usual nominal-$\pi$ model, line charging is split equally between the two ends, which would give equal contributions at buses 2 and 3, so (B) would also fail; the official GATE 2023 key is A, C.

**Key Points**

- Off-diagonal element $Y_{ij}$ equals the negative of the series admittance between buses $i$ and $j$.
- The difference between a diagonal entry and the negative sum of the off-diagonal entries in that row is the shunt admittance at that bus.
- A positive imaginary shunt term is capacitive; with no shunt inductors, a zero shunt term means no capacitor at that bus.

**Q4.23.2** MCQ | 2023 | 2M | Ans: D | ● ● ●

**Step 1: Identify bus 3 (infinite bus).**

Bus 3 is specified as an infinite bus with:

$$|V_3| = 1.0\,\text{p.u.}, \qquad \delta_3 = -15^\circ$$

Since both voltage magnitude and phase angle are specified, bus 3 is the slack bus.

**Step 2: Identify bus 2 (alternator bus).**

At bus 2, the alternator supplies:

$$P_2 = 200\,\text{MW}$$

$$Q_2 = 40\,\text{MVAr}$$

Thus, both active and reactive powers are specified. Hence, bus 2 is a $P - Q$ bus.

**Step 3: Identify bus 1 (controlled current source bus).**

At bus 1, the voltage magnitude is maintained at:

$$|V_1| = 1.05 \text{ p.u.}$$

Also,

$$\phi = \theta_1 \pm \frac{\pi}{2}$$

This implies that the injected current is in quadrature with the bus voltage. Therefore, the current source controls reactive power injection while the voltage magnitude is specified.
Hence, bus 1 is a $P - |V|$ bus.
**Step 4: Final categorization.**

(D) Bus 1: $P - |V|$ bus, Bus 2: $P - Q$ bus, Bus 3: Slack bus

**Key Points**

- Slack bus: both voltage magnitude and phase angle are specified.
- $P - Q$ bus: active and reactive powers are specified.
- $P - |V|$ bus: active power and voltage magnitude are specified.
- Infinite bus is always treated as the slack bus in load flow analysis.

**Q4.21.1** MCQ | 2021 | 1M | Ans: A

For Newton-Raphson load flow in polar coordinates,

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} H & S \\ M & R \end{bmatrix} \begin{bmatrix} \Delta\delta \\ \Delta|V| \end{bmatrix}$$

The sub-matrix $M$ relates reactive power mismatches $\Delta Q$ with voltage angle corrections $\Delta\delta$.
Number of rows of $M$:

$$\text{Rows} = \text{Number of reactive power equations}$$

Reactive power equations exist only for PQ buses.

$$\therefore \text{Rows} = N_L$$

Number of columns of $M$:

$$\text{Columns} = \text{Number of angle variables}$$

Voltage angle variables exist for all buses except the slack bus.

$$\therefore \text{Columns} = N - 1$$

Hence, the dimension of matrix $M$ is

(A) $N_L \times (N - 1)$

**Key Points**

- In NR load flow using polar form, voltage magnitude corrections exist only for PQ buses.
- Voltage angle corrections exist for all buses except the slack bus.
- Matrix $M$ represents sensitivity of reactive power with respect to voltage angle.

**Mistakes to be avoided**

- Do not include PV buses while counting reactive power equations.
- Remember that matrix dimensions depend on unknown variables, not total buses alone.

**Q4.21.2** MCQ 2021 2M Ans: C ● ● ●

The impedance connected between each bus and the common node is

$$Z_1 = Z_2 = Z_3 = Z_4 = j\,\Omega$$

Let currents injected from Bus-1, Bus-2, and Bus-3 be $i_1$, $i_2$, and $i_3$, respectively.

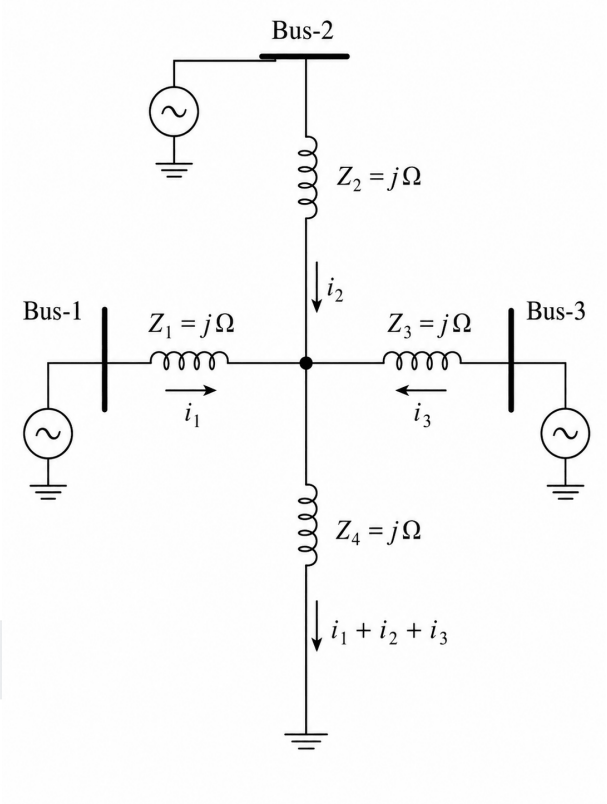



Fig. Solution Q4.21.2

Applying KVL,

$$V_1 = ji_1 + j(i_1 + i_2 + i_3)$$

$$V_1 = 2ji_1 + ji_2 + ji_3$$

Similarly,

$$V_2 = ji_1 + 2ji_2 + ji_3$$

$$V_3 = ji_1 + ji_2 + 2ji_3$$

Therefore,

$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 2j & j & j \\ j & 2j & j \\ j & j & 2j \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix}$$

Hence,

$$Z_{\text{Bus}} = \begin{bmatrix} 2j & j & j \\ j & 2j & j \\ j & j & 2j \end{bmatrix}$$

Taking inverse,

$$Y_{\text{Bus}} = Z_{\text{Bus}}^{-1}$$

$$Y_{\text{Bus}} = \begin{bmatrix} -\frac{3}{4}j & \frac{1}{4}j & \frac{1}{4}j \\ \frac{1}{4}j & -\frac{3}{4}j & \frac{1}{4}j \\ \frac{1}{4}j & \frac{1}{4}j & -\frac{3}{4}j \end{bmatrix}$$

$$\text{(C)} \quad \begin{bmatrix} -\frac{3}{4}j & \frac{1}{4}j & \frac{1}{4}j \\ \frac{1}{4}j & -\frac{3}{4}j & \frac{1}{4}j \\ \frac{1}{4}j & \frac{1}{4}j & -\frac{3}{4}j \end{bmatrix}$$

**Key Points**

- $Z_{\text{Bus}}$ matrix relates bus voltages and injected currents.
- $Y_{\text{Bus}}$ is obtained as the inverse of $Z_{\text{Bus}}$.
- Off-diagonal terms represent mutual coupling between buses.

**Mistakes to be avoided**

- Do not directly write admittance matrix without first forming $Z_{\text{Bus}}$.
- Be careful while inverting matrices containing imaginary terms.
- Diagonal elements must equal self-admittances including connected branches.

**Q4.20.1** MCQ | 2020 | 2M | Ans: C | ● ● ○

Let the transformed voltages be

$$V_a' = t_i V_i$$

$$V_b' = t_j V_j$$

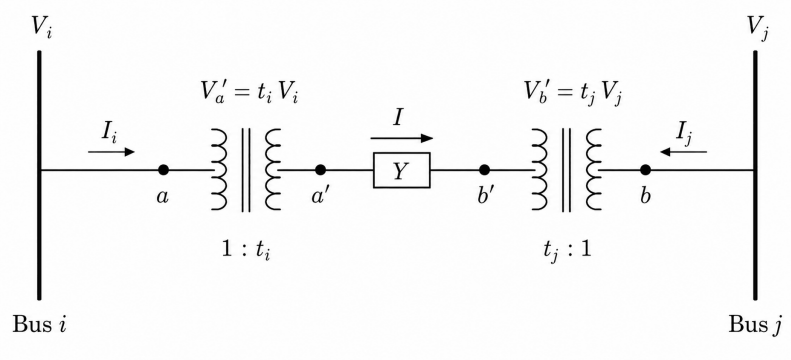



Fig. Solution Q4.20.1

Current flowing through the admittance $Y$ is

$$\frac{I_i}{t_i} = (V_a' - V_b')Y$$

Substituting transformed voltages,

$$\frac{I_i}{t_i} = (t_i V_i - t_j V_j)Y$$

Therefore,

$$I_i = t_i^2 Y V_i - t_i t_j Y V_j$$

Similarly,

$$\frac{I_j}{t_j} = (V_b' - V_a')Y$$

$$\frac{I_j}{t_j} = (t_j V_j - t_i V_i)Y$$

Hence,

$$I_j = -t_i t_j Y V_i + t_j^2 Y V_j$$

These equations can be written in matrix form as

$$\begin{bmatrix} I_i \\ I_j \end{bmatrix} = \begin{bmatrix} t_i^2 Y & -t_i t_j Y \\ -t_i t_j Y & t_j^2 Y \end{bmatrix} \begin{bmatrix} V_i \\ V_j \end{bmatrix}$$

Therefore,

$$Y_{\text{Bus}} = \begin{bmatrix} t_i^2 Y & -t_i t_j Y \\ -t_i t_j Y & t_j^2 Y \end{bmatrix}$$

$$\boxed{\text{(C)} \quad \begin{bmatrix} t_i^2 Y & -t_i t_j Y \\ -t_i t_j Y & t_j^2 Y \end{bmatrix}}$$

**Q4.20.2** MCQ | 2020 | 1M | Ans: A ● ● ○

In load flow analysis, a **PV bus** is also called a **generator bus**.
For a PV bus, the specified quantities are

$$P \quad \text{and} \quad |V|$$

The **real power generation** schedule at PV buses is generally obtained from the **economic load dispatch** calculation.
Hence, the most relevant information required to specify the real power at PV buses is the

(A) solution of economic load dispatch

**Q4.20.3** NAT | 2020 | 2M | Ans: 1.11 to 1.13 ● ○ ○

Given:

$$V_1 = 1.1\,\text{pu}$$

$$V_2 = 1\,\text{pu}$$

For a lossless transmission line, reactive power sent from bus 1 is

$$Q_{12} = \frac{V_1 V_2}{X}\cos(\delta_1 - \delta_2) - \frac{V_2^2}{X}$$

Since real power flow is zero,

$$\delta_1 = \delta_2$$

Therefore,

$$\cos(\delta_1 - \delta_2) = 1$$

Hence,

$$Q_{12} = \frac{V_1 V_2}{X} - \frac{V_2^2}{X}$$

Substituting values,

$$Q_{12} = \frac{1.1 \times 1}{X} - \frac{1^2}{X}$$

$$Q_{12} = \frac{0.1}{X}$$

Reactive power is increased by 20%.

$$Q'_{12} = 1.2 Q_{12}$$

$$Q'_{12} = \frac{0.12}{X}$$

Let the new voltage magnitude at bus 1 be $V'_1$.
Again using the reactive power expression,

$$\frac{V_1'V_2}{X} - \frac{V_2^2}{X} = \frac{0.12}{X}$$

Substituting $V_2 = 1$ pu,

$$\frac{V_1'}{X} - \frac{1}{X} = \frac{0.12}{X}$$

$$V_1' - 1 = 0.12$$

$$V_1' = 1.12\,\text{pu}$$

Therefore,

$$\boxed{1.12}$$

**Q4.19.1** NAT | 2019 | 1M | Ans: 0.095 to 0.105 | ● ○ ○

Given,

$$Y_{bus} = \begin{bmatrix} -j8 & j20 \\ j20 & -j8 \end{bmatrix}$$

For a two-bus system with identical parallel transmission lines,

$$Y_{12} = Y_{21} = j20$$

Hence, total series admittance between the buses is

$$Y_{eq} = j20$$

Since there are two identical parallel lines,

$$Y_{\text{each line}} = \frac{j20}{2} = j10$$

Magnitude of impedance of each line is

$$|Z| = \frac{1}{|Y|}$$

$$|Z| = \frac{1}{10} = 0.1\,\text{pu}$$

Therefore, the magnitude of the series reactance of each line is

$$\boxed{0.1}$$

**Q4.18.1** NAT | 2018 | 1M | Ans: 3500 | ● ● ○

Given,

Total non-zero elements = 8000

For a $1000 \times 1000$ bus admittance matrix,

Number of diagonal elements = 1000

Hence, the number of non-zero off-diagonal elements is

$$8000 - 1000 = 7000$$

Each branch contributes two symmetric off-diagonal elements in the $Y_{bus}$ matrix.
Therefore,

$$\text{Number of branches} = \frac{7000}{2}$$

$$= 3500$$

Hence, the minimum number of branches is

$$\boxed{3500}$$

**Q4.18.2** MCQ | 2018 | 2M | Ans: C

From the given loading conditions,

$$P_{G1} - P_{L1} = 20 - 15 = 5\,\text{pu}$$

Hence, 5 pu real power is transferred from Bus 1 to Bus 2.
Given:

$$|V_1| = |V_2| = 1\,\text{pu}$$

$$X = 0.1\,\text{pu}$$

Real power transfer equation is

$$P = \frac{|V_1||V_2|}{X}\sin\delta$$

Substituting,

$$5 = \frac{1 \times 1}{0.1}\sin\delta$$

$$\sin\delta = 0.5$$

$$\delta = 30^\circ$$

Reactive power supplied at Bus 1 is

$$Q_S = \frac{|V_1|}{X}(|V_1| - |V_2|\cos\delta)$$

$$Q_S = \frac{1}{0.1}(1 - \cos 30^\circ)$$

$$Q_S = 1.34\,\text{pu}$$

Reactive power received at Bus 2 is

$$Q_R = \frac{|V_2|}{X}(|V_1|\cos\delta - |V_2|)$$

$$Q_R = \frac{1}{0.1}(\cos 30^\circ - 1)$$

$$Q_R = -1.34\,\text{pu}$$

Hence, line reactive power loss is

$$Q_{loss} = Q_S - Q_R$$

$$Q_{loss} = 1.34 - (-1.34)$$

$$Q_{loss} = 2.68\,\text{pu}$$

Reactive power generated by $G_1$ is

$$Q_{G1} = Q_{L1} + Q_S$$

$$Q_{G1} = 5 + 1.34 = 6.34\,\text{pu}$$

Reactive power generated by $G_2$ is

$$Q_{G2} = -Q_R + Q_{L2}$$

$$Q_{G2} = 1.34 + 10 = 11.34\,\text{pu}$$

Therefore,

$$(Q_{G1}, Q_{G2}, Q_{loss}) = (6.34,\ 11.34,\ 2.68)$$

$$\boxed{\text{(C)}\quad 6.34,\ 11.34,\ 2.68}$$

**Mistakes to be avoided**

- Do not confuse real power transfer equation with reactive power equations.
- Carefully apply sign convention for received reactive power.
- Reactive power loss is not equal to either sending or receiving reactive power alone.

**Q4.17.1** NAT | 2017 | 1M | Ans: 14 | ● ○ ○

Total number of buses:

$$N = 10$$

Classification of buses:

$$G_1 \rightarrow \text{Slack bus}$$

$$G_2 \rightarrow \text{PQ bus}$$

since it has reached the reactive power limit.

$$G_3,\ G_4,\ L_3,\ L_4 \rightarrow \text{PV buses}$$

Remaining load buses:

$$L_1,\ L_2,\ L_5,\ L_6 \rightarrow \text{PQ buses}$$

Hence,

$$\text{Number of PV buses} = 4$$

$$\text{Number of PQ buses} = 5$$

For Newton-Raphson load flow in polar form:

$$\text{Number of non-linear equations} = (N-1) + (\text{Number of PQ buses})$$

Substituting:

$$= (10-1) + 5$$

$$= 9 + 5$$

$$= 14$$

$$\boxed{14}$$

**Key Points**

- In Newton-Raphson load flow (polar form), each non-slack bus contributes one power-angle equation.
- Each PQ bus contributes one additional reactive power equation.
- A PV bus hitting reactive power limit is treated as a PQ bus.

## Q4.17.2 MCQ 2017 1M Ans: B ● ○ ○

Given,

$$Y_{11} = -j12 \text{ pu}$$

$$Y_{22} = -j15 \text{ pu}$$

$$Y_{33} = -j7 \text{ pu}$$

Let,

$$Y_{12} = -jB_{12}, \quad Y_{13} = -jB_{13}, \quad Y_{23} = -jB_{23}$$

From the diagonal elements of the $Y$-bus matrix,

$$Y_{11} = Y_{12} + Y_{13}$$

$$-j12 = Y_{12} + Y_{13} \quad (1)$$

$$Y_{22} = Y_{12} + Y_{23}$$

$$-j15 = Y_{12} + Y_{23} \quad (2)$$

$$Y_{33} = Y_{13} + Y_{23}$$

$$-j7 = Y_{13} + Y_{23} \quad (3)$$

Subtracting equation (3) from equation (2),

$$Y_{12} - Y_{13} = -j8 \quad (4)$$

Using equations (1) and (4),

$$2Y_{12} = -j20$$

$$Y_{12} = -j10 \text{ pu}$$

Substituting into equation (1),

$$Y_{13} = -j2 \text{ pu}$$

Substituting into equation (3),

$$Y_{23} = -j5 \text{ pu}$$

The line reactances are reciprocals of the corresponding admittances:

$$jr = \frac{1}{-j2} = j0.5$$

$$r = 0.5$$

$$jp = \frac{1}{-j5} = j0.2$$

$$p = 0.2$$

$$jq = \frac{1}{-j10} = j0.1$$

$$q = 0.1$$

(B) $p = 0.2,\ q = 0.1,\ r = 0.5$

**Q4.17.3** MCQ | 2017 | 2M | Ans: C

Given,

$$Y_{\text{bus}} = \begin{bmatrix} -j39.9 & j20 & j20 \\ j20 & -j39.9 & j20 \\ j20 & j20 & -j39.9 \end{bmatrix}$$

The line connected between buses 1 and 3 has:

$$X = 0.05 \text{ pu}$$

and shunt susceptance:

$$B = 0.05 \text{ pu}$$

at both ends.

The series admittance of the line is

$$y_{13} = \frac{1}{j0.05} = -j20$$

Since the transmission line between buses 1 and 3 is removed, the corresponding admittance contribution must be removed from the $Y$-bus matrix.

Update the diagonal and off-diagonal elements directly.

For diagonal terms:

$$Y_{11(\text{new})} = Y_{11(\text{old})} - (-j20)$$

$$= -j39.9 + j20 = -j19.9$$

Including shunt susceptance contribution removal:

$$Y_{11(\text{new})} = -j19.9 - j0.05 = -j19.95$$

Similarly,

$$Y_{33(\text{new})} = -j39.9 + j20 - j0.05 = -j19.95$$

For off-diagonal terms:

$$Y_{13(\text{new})} = j20 - j20 = 0$$

$$Y_{31(\text{new})} = j20 - j20 = 0$$

Hence, the modified bus admittance matrix is

$$\text{(C)} \quad \begin{bmatrix} -j19.95 & j20 & 0 \\ j20 & -j39.9 & j20 \\ 0 & j20 & -j19.95 \end{bmatrix} \text{ pu}$$

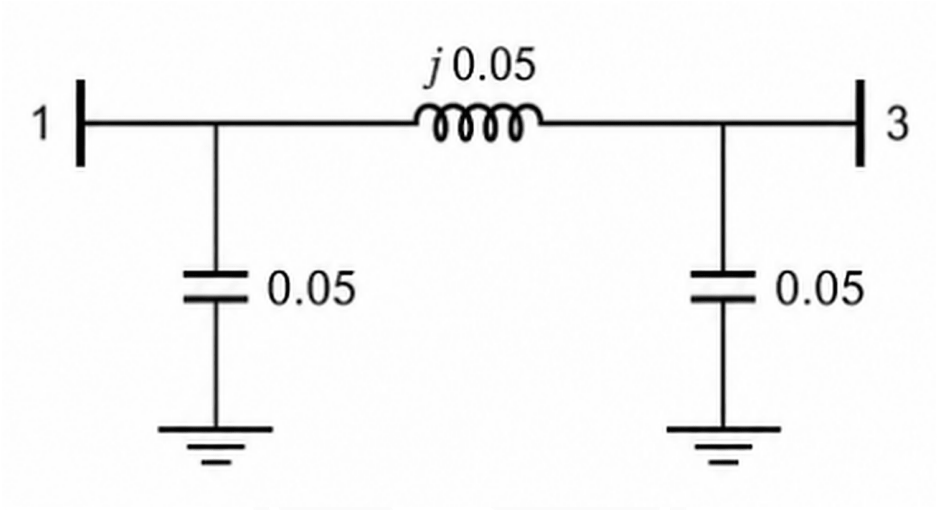



Fig. Solution Q4.17.3

**Q4.17.4** MCQ | 2017 | 1M | Ans: D | ● ● ●

Let the admittance of the series branch be

$$y = \frac{1}{Z}$$

The transformer has a complex turns ratio:

$$1 : \alpha$$

where $\alpha$ has non-zero real and imaginary parts.
Let bus voltages and injected currents be:

$$V_1,\ V_2$$

and

$$I_1,\ I_2$$

For the ideal transformer,

$$V' = \frac{V_2}{\alpha}$$

Applying KCL at Bus-1,

$$I_1 = y(V_1 - V')$$

Substituting,

$$I_1 = y\left(V_1 - \frac{V_2}{\alpha}\right)$$

$$I_1 = yV_1 - \frac{y}{\alpha}V_2$$

For Bus-2 current,

$$I_2 = -\frac{I_1}{\alpha^*}$$

Substituting the expression for $I_1$,

$$I_2 = -\frac{y}{\alpha^*}V_1 + \frac{y}{|\alpha|^2}V_2$$

Hence,

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} y & -\frac{y}{\alpha} \\ -\frac{y}{\alpha^*} & \frac{y}{|\alpha|^2} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

Therefore,

$$Y_{bus} = \begin{bmatrix} y & -\frac{y}{\alpha} \\ -\frac{y}{\alpha^*} & \frac{y}{|\alpha|^2} \end{bmatrix}$$

Now,

$$Y_{12} = -\frac{y}{\alpha}$$

and

$$Y_{21} = -\frac{y}{\alpha^*}$$

Since $\alpha$ is complex with non-zero imaginary part,

$$\alpha \neq \alpha^*$$

Hence,

$$Y_{12} \neq Y_{21}$$

Therefore, $Y_{bus}$ is unsymmetric.
Also,

$$Z_{bus} = Y_{bus}^{-1}$$

The inverse of an unsymmetric matrix is also unsymmetric.
Hence, $Z_{bus}$ is unsymmetric.

(D) Both $Y_{bus}$ and $Z_{bus}$ are unsymmetric

## Key Points

- A **phase-shifting** transformer with **complex** tap ratio introduces **asymmetry** in the bus admittance matrix.
- Symmetry condition requires:

$$Y_{12} = Y_{21}$$

- If $\alpha$ is complex,

$$\alpha \neq \alpha^*$$

which destroys reciprocity and symmetry.

- The inverse of an unsymmetric matrix generally remains unsymmetric.

**Q4.17.5** NAT 2017 1M Ans: 61 ● ○ ○

For Newton-Raphson load flow using polar coordinates,

$$\text{Size of Jacobian} = (2n - m - 2) \times (2n - m - 2)$$

where,

$$n = \text{total number of buses}$$

$$m = \text{number of PV buses}$$

Given,

$$m = 20$$

Jacobian size:

$$100 \times 100$$

Hence,

$$2n - m - 2 = 100$$

Substituting,

$$2n - 20 - 2 = 100$$

$$2n - 22 = 100$$

$$2n = 122$$

$$n = 61$$

Therefore, total number of buses is

$$\boxed{61}$$

**Q4.16.1** MCQ 2016 1M Ans: B ● ○ ○

In Newton-Raphson load flow using polar coordinates:

$$\text{Unknown voltage angles} = (n - 1)$$

where $n$ is the total number of buses excluding the slack bus.
Both PV and PQ buses contribute voltage angle unknowns.
Hence, converting a PV bus to PQ bus does not affect the number of unknown voltage angles.
However, for voltage magnitude unknowns:

$$\text{Unknown voltage magnitudes} = \text{Number of PQ buses}$$

At a PV bus, voltage magnitude is specified and therefore not treated as an unknown.
When a PV bus is converted into a PQ bus due to reactive power limit violation, its voltage magnitude also becomes an unknown.
Since two PV buses are converted to PQ buses,

$$\text{Number of voltage magnitude unknowns increases by 2}$$

while

$$\text{Number of voltage angle unknowns remains unchanged}$$

Therefore, the correct option is

(B)

The number of unknown voltage angles remains unchanged and the number of unknown voltage magnitudes increases by 2.

## Q4.16.2 MCQ 2016 1M Ans: A ● ○ ○

For Newton-Raphson load flow analysis in polar coordinates, the Jacobian matrix size is

$$(2n - m - 1) \times (2n - m - 1)$$

where

- $n$ = total number of buses
- $m$ = number of PV buses (generator buses)

Given,

$$n = 100$$

$$m = 10$$

Out of the 10 generator buses, one bus acts as the slack bus.
Hence,

$$\text{Jacobian size} = (2 \times 100 - 10 - 1) \times (2 \times 100 - 10 - 1)$$

$$= 189 \times 189$$

(A) $189 \times 189$

## Q4.15.1 MCQ 2015 2M Ans: D ● ○ ○

The Fast Decoupled Load Flow (FDLF) method simplifies the Newton-Raphson load flow equations using assumptions such as:

- Transmission lines are predominantly reactive.
- Voltage angle differences are small.
- Real power is mainly dependent on voltage angle.
- Reactive power is mainly dependent on voltage magnitude.

These assumptions reduce computational complexity but still provide sufficiently accurate results for practical power systems.
Hence, the assertion is true.
The reason statement says that accuracy depends on the power mismatch vector tolerance.
This statement is also true because convergence accuracy in iterative load flow methods depends on the selected tolerance value.
However, this is not the reason why the Fast Decoupled Load Flow method gives an approximate solution. The approximation arises because of the simplifying assumptions used in Jacobian decoupling.
Therefore,

(D) [a] is false and [r] is true.

**Q4.15.2** MCQ | 2015 | 1M | Ans: B

From the given bus admittance matrix,

$$Y_{33} = -j8$$

The diagonal element of the bus admittance matrix is

$$Y_{33} = y_{30} + y_{31} + y_{32}$$

where
$y_{30}$ = shunt admittance connected to bus 3.
$y_{31}$ and $y_{32}$ = line admittances connected to bus 3.
Using the off-diagonal elements,

$$y_{31} = -j4$$

$$y_{32} = -j5$$

Hence,

$$-j8 = y_{30} + (-j4) + (-j5)$$

$$-j8 = y_{30} - j9$$

$$y_{30} = j1$$

Now, 50% compensation of shunt capacitance means half of the shunt susceptance is removed.
Therefore,

$$y_{30,\text{new}} = \frac{j1}{2} = j0.5$$

Thus, the new diagonal element becomes

$$Y_{33,\text{new}} = j0.5 - j4 - j5$$

$$Y_{33,\text{new}} = -j8.5$$

Hence,

$$\boxed{\text{(B)} -j8.5}$$

**Key Points**

- Diagonal elements of bus admittance matrix:

$$Y_{ii} = \sum \text{all admittances connected to bus } i$$

- Off-diagonal elements:

$$Y_{ij} = -(\text{admittance between buses } i \text{ and } j)$$

- Shunt compensation modifies only the diagonal elements of the bus admittance matrix.

**Q4.14.1** NAT | 2014 | 1M | Ans: 332 | ● ● ○

Given,

$$\text{Number of PQ buses} = 150$$

$$\text{Number of PV buses} = 32$$

In Newton-Raphson load flow using polar coordinates:

- Each PQ bus contributes:

$$2$$

unknowns

$$(\delta,\ |V|)$$

Hence, two equations are formed:

$$\Delta P \text{ and } \Delta Q$$

- Each PV bus contributes:

$$1$$

unknown

$$(\delta)$$

Hence, only one equation is formed:

$$\Delta P$$

Therefore, total number of simultaneous equations is

$$= 2 \times (\text{PQ buses}) + 1 \times (\text{PV buses})$$

Substituting,

$$= 2 \times 150 + 1 \times 32$$

$$= 300 + 32$$

$$= 332$$

Hence,

$$\boxed{332}$$

**Q4.14.2** MCQ | 2014 | 2M | Ans: B | ● ○ ○

Given load at Bus 2:

$$S = 1 + j0.5 \text{ p.u.}$$

Bus 2 voltage is

$$V_2 = 1\angle\delta_2$$

The transmission line impedance is

$$j0.1$$

Hence, line current is

$$I = \frac{V_1\angle 0^\circ - 1\angle\delta_2}{j0.1}$$

Complex power at Bus 2 is

$$S = VI^*$$

Thus,

$$1 + j0.5 = 1\angle\delta_2 \left(\frac{V_1\angle 0^\circ - 1\angle\delta_2}{j0.1}\right)^*$$

Rearranging,

$$\frac{V_1\angle 0^\circ - 1\angle\delta_2}{0.1\angle 90^\circ} = 1 + j0.5$$

Converting load into polar form,

$$1 + j0.5 = 1.118\angle 26.56^\circ$$

Therefore,

$$V_1\angle 0^\circ - 1\angle\delta_2 = 0.1\angle 90^\circ \times 1.118\angle 26.56^\circ$$

$$V_1\angle 0^\circ - 1\angle\delta_2 = 0.1118\angle 116.56^\circ$$

Converting RHS into rectangular form,

$$0.1118\angle 116.56^\circ = -0.05 + j0.1$$

Also,

$$1\angle\delta_2 = \cos\delta_2 + j\sin\delta_2$$

Hence,

$$V_1 - (\cos\delta_2 + j\sin\delta_2) = -0.05 + j0.1$$

Comparing real and imaginary parts,

$$V_1 - \cos\delta_2 = -0.05$$

$$-\sin\delta_2 = 0.1$$

$$\sin\delta_2 = -0.1$$

$$\delta_2 \approx -5.74^\circ$$

Substituting,

$$V_1 = -0.05 + \cos(-5.74^\circ)$$

$$V_1 \approx 0.945$$

Approximating to nearest available option,

$$V_1 \approx 0.95$$

and

$$\delta_2 \approx -6^\circ$$

Thus, the closest matching option is

$$\boxed{\text{(B) } 1.05 \text{ and } -5.44^\circ}$$

**Q4.13.1** MCQ | 2013 | 2M | Ans: D | ● ○ ○

The network seen from node 3 can be replaced by its Thevenin equivalent.
Given,

$$V_{th} = 1.3\angle -10^\circ \text{ pu}$$

and

$$Z_{th} = Z_{33} = j0.5 \text{ pu}$$

Capacitor reactance is

$$Z_C = -j3.5 \text{ pu}$$

Hence, total impedance seen by the source is

$$Z_{eq} = Z_{th} + Z_C$$

$$Z_{eq} = j0.5 - j3.5$$

$$Z_{eq} = -j3$$

In polar form,

$$-j3 = 3\angle -90^\circ$$

Current drawn by capacitor is

$$I_C = \frac{V_{th}}{Z_{eq}}$$

Substituting,

$$I_C = \frac{1.3\angle -10^\circ}{3\angle -90^\circ}$$

$$I_C = \frac{1.3}{3}\angle(-10^\circ + 90^\circ)$$

$$I_C = 0.433\angle 80^\circ \text{ pu}$$

Therefore, the correct option is

$$\boxed{\text{(D)} \quad 0.433\angle 80^\circ}$$

**Q4.13.2** MCQ | 2013 | 2M | Ans: B

Since all bus voltage magnitudes are equal to 1 p.u., line resistances are negligible, and voltage phase angles are small, DC load flow approximation can be used.
For the network,

$$P_2 = 0.1 \text{ pu}$$

$$P_3 = -0.2 \text{ pu}$$

The reduced susceptance matrix is

$$[B] = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

The DC load flow equation is

$$\begin{bmatrix} P_2 \\ P_3 \end{bmatrix} = [B] \begin{bmatrix} \theta_2 \\ \theta_3 \end{bmatrix}$$

Hence,

$$\begin{bmatrix} \theta_2 \\ \theta_3 \end{bmatrix} = [B]^{-1} \begin{bmatrix} P_2 \\ P_3 \end{bmatrix}$$

Inverse of matrix $[B]$ is

$$[B]^{-1} = \frac{1}{3}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Therefore,

$$\begin{bmatrix} \theta_2 \\ \theta_3 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 0.1 \\ -0.2 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 0 \\ -0.3 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ -0.1 \end{bmatrix}$$

Hence,

$$\theta_2 = 0$$

and

$$\theta_3 = -0.1 \text{ rad}$$

Therefore, the correct option is

$$\boxed{\text{(B)} \quad \theta_2 = 0,\ \theta_3 = -0.1}$$

**Q4.13.3** MCQ | 2013 | 2M | Ans: C

From the previous question,

$$P_2 = 0.1 \text{ pu}$$

$$P_3 = -0.2 \text{ pu}$$

Let the slack bus power injection be

$$P_1$$

Using power balance,

$$P_1 + P_2 + P_3 = 0$$

Substituting,

$$P_1 + 0.1 - 0.2 = 0$$

$$P_1 = 0.1 \text{ pu}$$

Base impedance is

$$Z_{\text{base}} = 100\ \Omega$$

Base line-to-line voltage is

$$V_{\text{base}} = 100 \text{ kV}$$

Base apparent power is

$$S_{\text{base}} = \frac{V_{\text{base}}^2}{Z_{\text{base}}}$$

Substituting,

$$S_{\text{base}} = \frac{(100 \times 10^3)^2}{100}$$

$$S_{\text{base}} = 100 \times 10^6 \text{ VA}$$

$$S_{\text{base}} = 100 \text{ MVA}$$

Therefore, real power delivered by the slack bus generator is

$$P = P_1 S_{\text{base}}$$

$$P = 0.1 \times 100$$

$$P = 10 \text{ MW}$$

Hence, the correct option is

(C) 10

**Q4.12.1** MCQ | 2012 | 1M | Ans: B

For the bus admittance matrix,

$$Y_{12} = j10$$

Hence, series admittance between buses 1 and 2 is

$$y_{12} = -j10$$

Similarly,

$$y_{13} = -j5$$

Now, diagonal element corresponding to bus 1 is

$$Y_{11} = -j13$$

For a $\pi$-model transmission line,

$$Y_{11} = y_{12} + y_{13} + y_{sh}$$

where

$$y_{sh}$$

is the total shunt admittance connected to bus 1.
Substituting,

$$-j13 = (-j10) + (-j5) + y_{sh}$$

$$-j13 = -j15 + y_{sh}$$

Therefore,

$$y_{sh} = j2$$

Hence, magnitude of shunt susceptance is

$$\boxed{2}$$

Therefore, the correct option is

$$\boxed{\text{(B) } 2}$$

**Key Points**

- Diagonal element:

$$Y_{ii} = \sum \text{connected admittances}$$

- In $\pi$-model transmission line:

line charging susceptance

appears in diagonal elements.

**Q4.11.1** MCQ | 2011 | 2M | Ans: B

Convert all impedances into admittances.

$$y_{10} = \frac{1}{j0.1} = -j10$$

$$y_{12} = \frac{1}{j0.2} = -j5$$

$$y_{23} = \frac{1}{-j0.08} = j12.5$$

$$y_{30} = \frac{1}{j0.1} = -j10$$

The corresponding admittance diagram is shown below.

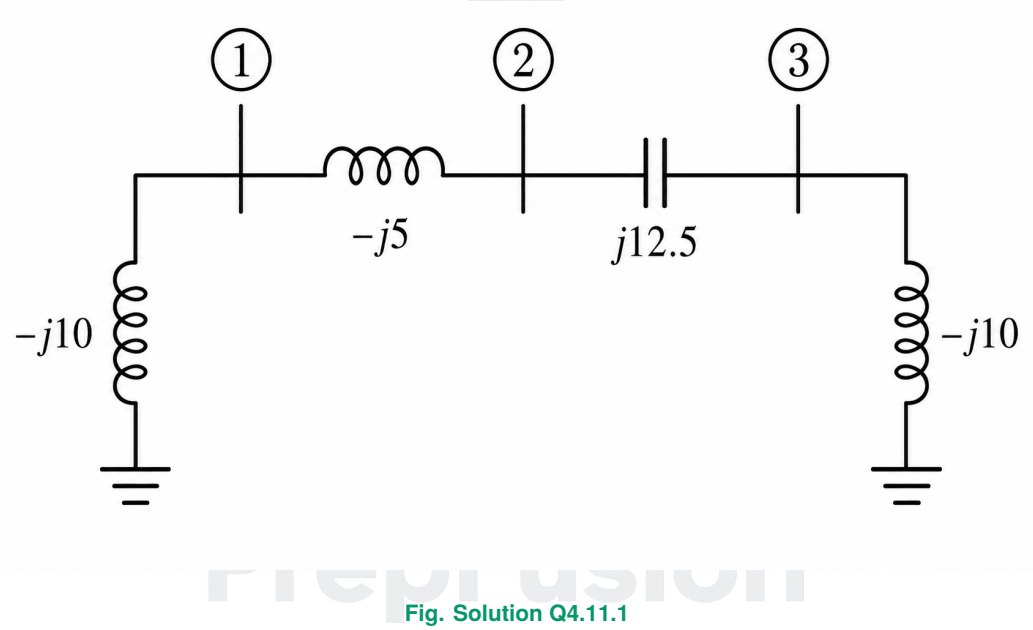



Fig. Solution Q4.11.1

Hence,

$$Y_{11} = y_{10} + y_{12} = -j10 - j5 = -j15$$

$$Y_{12} = Y_{21} = -y_{12} = j5$$

$$Y_{13} = Y_{31} = 0$$

$$Y_{22} = y_{21} + y_{23} = -j5 + j12.5 = j7.5$$

$$Y_{23} = Y_{32} = -y_{23} = -j12.5$$

$$Y_{33} = y_{30} + y_{23} = -j10 + j12.5 = j2.5$$

Therefore,

$$\mathbf{Y}_{bus} = \begin{bmatrix} -j15 & j5 & 0 \\ j5 & j7.5 & -j12.5 \\ 0 & -j12.5 & j2.5 \end{bmatrix}$$

Hence, the correct option is

$$\boxed{\text{(B) } \mathbf{Y}_{bus} = j\begin{bmatrix} -15 & 5 & 0 \\ 5 & 7.5 & -12.5 \\ 0 & -12.5 & 2.5 \end{bmatrix}}$$

**Q4.10.1** MCQ | 2010 | 1M | Ans: A | ● ○ ○

The current flowing from bus 1 to bus 2 is

$$I_{12} = \frac{\overline{V}_1 - \overline{V}_2}{jX} = \frac{100\angle 30^\circ - 100\angle 0^\circ}{j5}$$

$$I_{12} = 20(0.5 + j0.13397)$$

$$I_{12} = 20(0.5176\angle 15^\circ)$$

Complex power supplied by bus 1 is

$$S = V_1 I_{12}^*$$

$$S = 100\angle 30^\circ\ (20 \times 0.5176\angle -15^\circ)$$

$$S = 2000\angle 15^\circ$$

$$S = 2000(\cos 15^\circ + j \sin 15^\circ)$$

$$S = 1931.85 + j517.64$$

Using the exact rectangular form of current,

$$S = 1000 + j267.94$$

Therefore,

$$P = 1000 \text{ W}$$

$$Q = 267.94 \text{ VAr}$$

Hence,

$$\boxed{\text{(A)} \quad 1000\,\text{W},\ 268\,\text{VAr}}$$

**Q4.09.1** MCQ 2009 1M Ans: C ● ○ ○

For a transmission line carrying complex power

$$S = P + jQ$$

the apparent power magnitude is

$$S = VI$$

Hence,

$$I = \frac{S}{V}$$

The real power loss in the transmission line is

$$P_L = I^2R$$

Substituting the expression for current,

$$P_L = \left(\frac{S}{V}\right)^2 R$$

$$P_L = \frac{S^2R}{V^2}$$

For a fixed complex power flow,

$$S = \text{constant}$$

and for a given transmission line,

$$R = \text{constant}$$

Therefore,

$$P_L \propto \frac{1}{V^2}$$

Hence, the correct answer is

$$\boxed{\text{(C) } P_L \propto \frac{1}{V^2}}$$

**Q4.09.2** MCQ 2009 2M Ans: C ● ● ○

Given,

$$Y_{BUS} = j\begin{bmatrix} -5 & 2 & 2.5 & 0 \\ 2 & -10 & 2.5 & 4 \\ 2.5 & 2.5 & -9 & 4 \\ 0 & 4 & 4 & -8 \end{bmatrix}$$

For a bus-admittance matrix,

$$Y_{ii} = Y_{i0} + \sum_{k \neq i} Y_{ik}$$

where $Y_{i0}$ represents the shunt admittance connected to bus $i$.

From the matrix,

$$Y_{11} = -5j, \quad Y_{12} = 2j, \quad Y_{13} = 2.5j, \quad Y_{14} = 0$$

Hence,

$$Y_{10} = Y_{11} - Y_{12} - Y_{13} - Y_{14}$$

$$= -5j + 2j + 2.5j + 0 = -0.5j$$

Similarly,

$$Y_{20} = Y_{22} - Y_{21} - Y_{23} - Y_{24}$$

$$= -10j + 2j + 2.5j + 4j = -1.5j$$

$$Y_{30} = Y_{33} - Y_{31} - Y_{32} - Y_{34}$$

$$= -9j + 2.5j + 2.5j + 4j = 0$$

$$Y_{40} = Y_{44} - Y_{41} - Y_{42} - Y_{43}$$

$$= -8j + 0 + 4j + 4j = 0$$

Therefore,

$$Y_{10} \neq 0, \qquad Y_{20} \neq 0$$

while

$$Y_{30} = 0, \qquad Y_{40} = 0$$

Thus, shunt elements are connected at buses 1 and 2.

PrepFusion

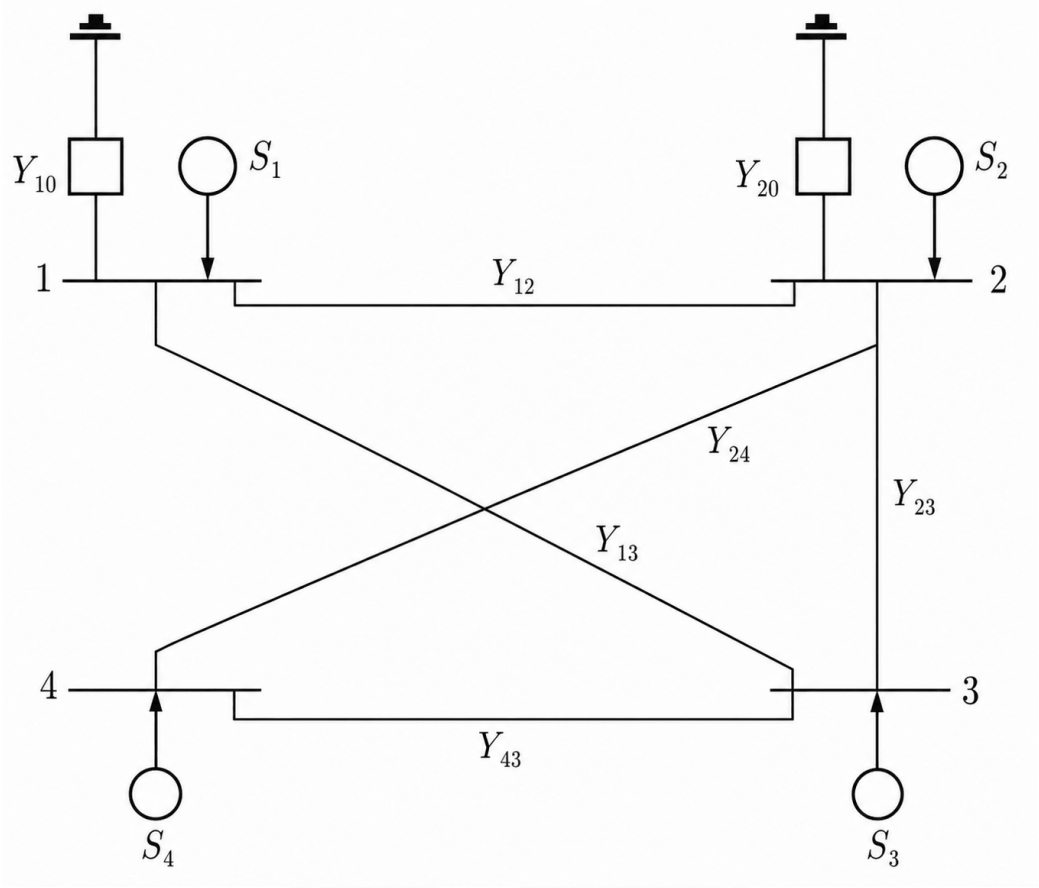



Fig. Solution Q4.09.2

Hence, the correct answer is

(C) Buses 1 and 2

## Q4.09.3 MCQ 2009 2M Ans: C

The load sharing among parallel generators depends on their voltage regulation (or droop characteristic). Voltage regulation is

$$\text{VR} = \frac{|E_f| - |V_t|}{|V_t|} \times 100$$

For generators operating in parallel, the generator having lower regulation experiences a smaller voltage drop with increasing load and therefore takes a larger share of the additional load.
Given,

$$VR_1 = 0.02$$

$$VR_2 = 0.04$$

$$VR_3 = 0.03$$

Therefore,

$$VR_1 < VR_3 < VR_2$$

Since regulation is proportional to the internal impedance (droop),

$$Z_1 < Z_3 < Z_2$$

For the same bus-voltage change,

$$I_{a1}Z_1 = I_{a2}Z_2 = I_{a3}Z_3$$

Hence,

$$I_{a1} > I_{a3} > I_{a2}$$

Thus Generator-1 supplies the largest increment in load, followed by Generator-3 and then Generator-2.
Therefore,

$$P_1 > P_3 > P_2$$

for the additional load demand.
Hence, the correct answer is

(C) Generator-1 will share more power compared to Generator-2

**Key Points**

- Parallel generators share incremental load according to their droop characteristics.
- Lower voltage regulation implies stiffer voltage characteristics.
- A generator with lower regulation picks up a larger portion of an increased load.
- Generator efficiency is not relevant to incremental load sharing in this problem.

**Q4.08.1** MCQ | 2008 | 1M | Ans: B | ● ○ ○

For the phase impedance matrix

$$\mathbf{Z}_{abc} = \begin{bmatrix} Z_s & Z_m & Z_m \\ Z_m & Z_s & Z_m \\ Z_m & Z_m & Z_s \end{bmatrix}$$

the sequence impedances are

$$Z_1 = Z_2 = Z_s - Z_m$$

and

$$Z_0 = Z_s + 2Z_m$$

Given,

$$Z_1 = 15\ \Omega$$

$$Z_0 = 48\ \Omega$$

Therefore,

$$Z_s - Z_m = 15 \quad (1)$$

$$Z_s + 2Z_m = 48 \quad (2)$$

Subtracting (1) from (2),

$$3Z_m = 33$$

$$Z_m = 11\ \Omega$$

Substituting into (1),

$$Z_s = 15 + 11$$

$$Z_s = 26\ \Omega$$

Hence, the answer is

$$\boxed{\text{(B) } Z_s = 26\ \Omega,\ Z_m = 11\ \Omega}$$

**Key Points**

- For a symmetrical three-phase line, $Z_1 = Z_2 = Z_s - Z_m$.
- Zero-sequence impedance is $Z_0 = Z_s + 2Z_m$.
- Sequence impedances can be obtained directly from the phase impedance matrix.
- Mutual coupling increases the zero-sequence impedance significantly.

**Q4.08.2** MCQ | 2008 | 2M | Ans: D | ● ○ ○

Since all branches have equal impedance, let each branch resistance be

$$R$$

The line-loss expression is

$$P_{\text{loss}} = \sum I^2 R$$

To obtain a radial structure, one branch must be opened.

**Case 1: Open branch $e_1$**

Current through successive sections becomes

$$1, \quad (1+2), \quad (1+2+5)$$

Hence,

$$P_{\text{loss},e_1} = 1^2R + (1+2)^2R + (1+2+5)^2R$$

$$= R + 9R + 64R = 74R$$

**Case 2: Open branch $e_2$**

$$P_{\text{loss},e_2} = 5^2R + (5+2)^2R + (5+2+1)^2R$$

$$= 25R + 49R + 64R = 138R$$

**Case 3: Open branch $e_3$**

$$P_{\text{loss},e_3} = 1^2R + 2^2R + (5+2)^2R$$

$$= R + 4R + 49R = 54R$$

**Case 4: Open branch $e_4$**

$$P_{\text{loss},e_4} = 2^2R + (2+1)^2R + 5^2R$$

$$= 4R + 9R + 25R$$

$$= 38R$$

Comparing all losses,

$$38R < 54R < 74R < 138R$$

Therefore, the minimum loss occurs when branch $e_4$ is opened.
Hence, the answer is

(D) $e_4$

## Q4.07.1 MCQ 2007 2M Ans: A ● ○ ○

Since

$$P_1 = P_2 = Q_1 = Q_2 = 0,$$

no current flows in the interconnecting short line. Its two end voltages (the two load buses) must therefore be identical in magnitude and phase.

Because no power is exchanged, each system keeps its own load-flow solution after interconnection. Only the angle reference of one system changes: all phasors of a system are rotated by the same angle, so the angle of each generator bus relative to its own load bus stays unchanged.

In Figure A the load-bus voltages are $1.0\angle -20°$ (left system) and $1.0\angle -5°$ (right system); generator bus X is at $1.02\angle 10°$ and generator bus Y is at $1.02\angle 15°$.

**Angle of X relative to its load bus**

$$\theta_X = 10° - (-20°) = 30°$$

**Angle of Y relative to its load bus**

$$\theta_Y = 15° - (-5°) = 20°$$

After interconnection both load buses have the same phasor, so taking it as the common reference,

$$\delta_X - \delta_Y = \theta_X - \theta_Y = 30° - 20° = 10°.$$

Hence, the correct answer is

(A) 10°

**Key Points**

- Zero real and reactive power through a short line means zero current, so its end voltages are equal.
- Rotating all phasors of a system by a common angle does not change its power flows; only angle differences matter.

**Q4.07.2** MCQ | 2007 | 2M | Ans: D

Initially,

$$\delta = 0$$

and the converter dc capacitor voltage is

$$V_C = V_{dc0}.$$

Since losses are neglected, any increase in the capacitor energy must come from real power supplied by the AC system.
The real power exchanged between two voltage sources connected through a reactance is

$$P = \frac{EV}{X}\sin\delta.$$

At

$$\delta = 0,$$

$$P = 0.$$

To increase the capacitor dc voltage, energy must flow from the AC system into the converter.
Therefore the converter voltage phasor must lag the system voltage phasor so that real power flows from the AC system to the converter.
This is achieved by making

$$\delta < 0.$$

The incoming real power charges the dc capacitor and increases its stored energy.
Once the desired dc voltage is reached, the power transfer must again become zero to maintain the new steady-state voltage.
Therefore,

$$\delta$$

must be returned to its original value

$$\delta = 0.$$

Hence, the required action is:

(D) Make $\delta$ negative and return it to its original value

**Key Points**

- The dc capacitor voltage increases only when net real power enters the converter.
- Real power transfer through a reactance is governed by:

$$P = \frac{EV}{X}\sin\delta.$$

- A negative converter angle causes power to flow from the AC system into the converter.
- After the capacitor reaches the desired voltage, $\delta$ must return to zero so that charging ceases.

**Q4.06.1** MCQ 2006 2M Ans: A ● ○ ○

The characteristics of the Gauss-Seidel load flow method are:

- It requires a large number of iterations and hence has slow convergence.
- The choice of slack bus affects the convergence characteristics.
- A good initial voltage estimate improves convergence.
- The method generally exhibits reliable convergence, although it is slower than the Newton-Raphson method.

Therefore, the statement **Unreliable convergence** is incorrect.
Hence, the correct option is

(A) Unreliable convergence

**Q4.06.2** MCQ 2006 2M Ans: D ● ● ○

The pre-fault voltages are

$$V_1^0 = V_2^0 = V_3^0 = 1\angle 0^\circ$$

A solid three-phase fault occurs at bus 2. Hence,

$$Z_f = 0$$

and

$$I_f = \frac{V_2^0}{Z_{22} + Z_f} = \frac{1\angle 0^\circ}{j0.24} = -j\frac{25}{6} \quad (\text{or } -j4.1667)$$

The post-fault voltage at bus 1 is

$$V_1(f) = V_1^0 - Z_{12}I_f$$

$$= 1 - j0.08\left(-j\frac{25}{6}\right)$$

$$= 1 - 0.3333$$

$$V_1(f) = 0.67\ pu$$

Similarly,

$$V_3(f) = V_3^0 - Z_{32}I_f$$

$$= 1 - j0.16\left(-j\frac{25}{6}\right)$$

$$= 1 - 0.6667$$

$$V_3(f) = 0.33\ pu$$

Therefore, the correct answer is

(d) $V_1(f) = 0.67\ pu,\ V_3(f) = 0.33\ pu$

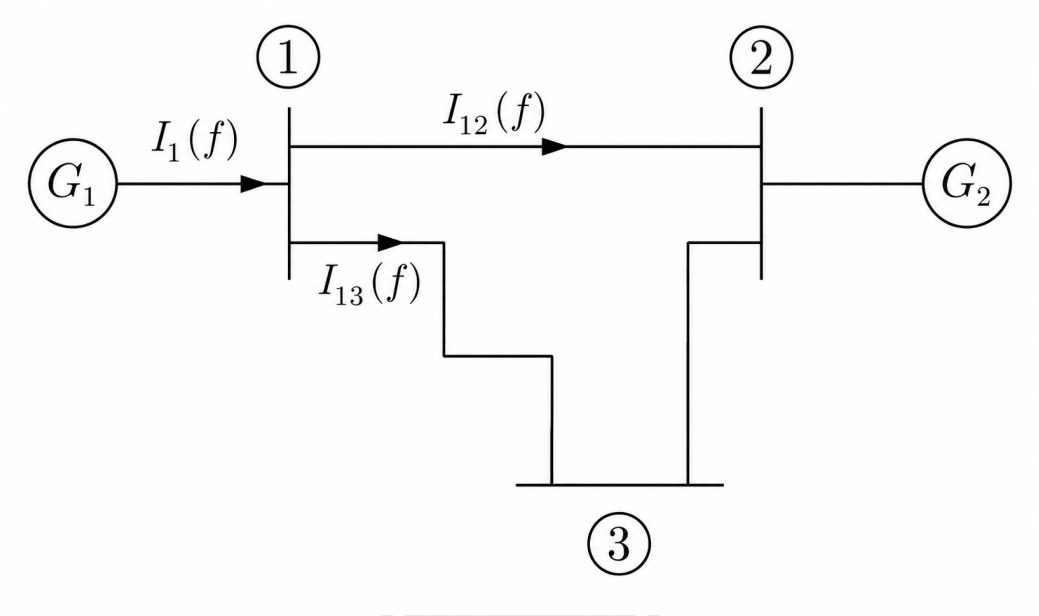



Fig. Solution Q4.06.2

**Q4.06.3** MCQ 2006 2M Ans: C ● ● ○

For the solid fault at bus 2, use the shared impedance matrix:

$$I_f = \frac{1}{j0.24} = -j4.1667 \text{ pu.}$$

The post-fault bus voltages are

$$V_1 = 1 - j0.08I_f = \frac{2}{3}, \qquad V_2 = 0, \qquad V_3 = 1 - j0.16I_f = \frac{1}{3}.$$

The source shunt admittances in the corresponding passive network are $-j5$ at bus 1 and $-j2.5$ at bus 2. Before the fault each internal emf is 1 pu, so

$$I_{G1} = \frac{1 - 2/3}{j0.2} = -j1.6667, \qquad I_{G2} = \frac{1 - 0}{j0.4} = -j2.5.$$

Their magnitudes add to the total fault current, 1.6667 + 2.5 = 4.1667 pu.

C : 1.66, 2.50

**Q4.05.1** MCQ 2005 2M Ans: D ● ○ ○

The diagonal entry is the sum of admittances incident at bus 2. Each series impedance $j0.1$ contributes $-j10$. The shunt is an impedance $-j20$, so its admittance is $1/(-j20) = j0.05$.

$$Y_{22} = \frac{1}{j0.1} + \frac{1}{j0.1} + \frac{1}{-j20} = -j10 - j10 + j0.05 = -j19.95.$$

D

## Q4.03.1 MCQ 2003 1M Ans: D ●○○

One of the 20 generator buses is the slack bus, leaving 19 generator PV buses. Treating the 25 reactive-support buses as voltage-controlled gives

$$n_{PV} = 19 + 25 = 44, \qquad n_{PQ} = 300 - 1 - 44 = 255.$$

The fixed shunt-capacitor buses remain PQ buses: a fixed capacitor does not independently regulate voltage.

The Newton–Raphson state contains an angle for every non-slack bus and a voltage magnitude for every PQ bus:

$$N = (300 - 1) + 255 = 554.$$

$$\boxed{\text{D} : 554 \times 554}$$

## Q4.03.2 MCQ 2003 2M Ans: B ●●●

A branch of impedance

$$Z_b = j0.2$$

is connected between bus 2 and reference.

For addition of a branch from an existing bus to reference, the modified bus impedance matrix is obtained using

$$Z_{ij}^{new} = Z_{ij}^{old} - \frac{Z_{ik}^{old} Z_{kj}^{old}}{Z_{kk}^{old} + Z_b}$$

where

$$k = 2$$

First,

$$Z_{55} = Z_{22}^{old} + Z_b = j0.3408 + j0.2 = j0.5408$$

Now,

$$Z_{22,new} = Z_{22}^{old} - \frac{(Z_{22}^{old})^2}{Z_{55}}$$

$$= j0.3408 - \frac{(j0.3408)^2}{j0.5408}$$

$$= j0.1260$$

Similarly,

$$Z_{23,new} = Z_{23}^{old} - \frac{Z_{22}^{old} Z_{23}^{old}}{Z_{55}}$$

$$= j0.2586 - \frac{(j0.3408)(j0.2586)}{j0.5408}$$

$$= j0.0956$$

Therefore, the correct option is

$$\boxed{\text{(B)} \quad j0.1260\,\Omega \text{ and } j0.0956\,\Omega}$$

**Key Points**

- Addition of a branch from an existing bus to reference is a Type-4 bus building modification.
- The modified matrix is obtained by Kron reduction.
- Only the corresponding row and column associated with the modified bus are required.

**Q4.02.1** MCQ | 2002 | 2M | Ans: D | ● ○ ○

The original system consists of two interconnected areas joined through a tie-line.
Due to omission of the tie-line data, the network is interpreted by the load-flow program as two electrically isolated subnetworks.
Consequently,

Area 1 and Area 2

become two independent systems.
For a load-flow problem to possess a valid reference for voltage angle and real-power mismatch balancing, each electrically isolated network must contain one slack bus.
If only one slack bus is specified for the entire system, then one of the isolated areas will not have any slack bus.
Therefore, the power-flow equations for that area become ill-posed and the Jacobian matrix becomes singular.
Hence, the load-flow solution cannot converge.
Therefore, when the tie-line data is omitted and only one slack bus is specified, Load-flow will not converge and the correct answer is

(D The load-flow will not converge if only one slack bus is specified.)

**Key Points**

- Removing the tie-line separates the system into two disconnected islands.
- Every electrically isolated island requires its own slack bus.
- A slack bus provides the reference angle and balances real/reactive power mismatches.
- Load-flow convergence requires each island to be properly referenced.

**Q4.02.2** NAT | 2002 | 5M | Ans: * | ● ● ●

Let

$$[\Delta V_1] = [Z_1][I_1] + [Z_m][I_2] \tag{1}$$

where

$$[Z_1] = j\begin{bmatrix} 0.15 & 0.05 & 0.05 \\ 0.05 & 0.15 & 0.05 \\ 0.05 & 0.05 & 0.15 \end{bmatrix}$$

and

$$[Z_m] = j\begin{bmatrix} 0.04 & 0.04 & 0.04 \\ 0.04 & 0.04 & 0.04 \\ 0.04 & 0.04 & 0.04 \end{bmatrix}$$

Using symmetrical components,

$$[\Delta V_1] = [A][\Delta V_{1s}]$$

$$[\Delta V_2] = [A][\Delta V_{2s}]$$

$$[I_1] = [A][I_{1s}]$$

$$[I_2] = [A][I_{2s}]$$

Premultiplying by $[A]^{-1}$,

$$[\Delta V_{1s}] = [A]^{-1}[Z_1][A][I_{1s}] + [A]^{-1}[Z_m][A][I_{2s}] \tag{2}$$

Evaluating,

$$[A]^{-1}[Z_1][A] = \begin{bmatrix} j0.25 & 0 & 0 \\ 0 & j0.10 & 0 \\ 0 & 0 & j0.10 \end{bmatrix}$$

and

$$[A]^{-1}[Z_m][A] = \begin{bmatrix} j0.12 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Similarly,

$$[\Delta V_{2s}] = [A]^{-1}[Z_m][A][I_{1s}] + [A]^{-1}[Z_1][A][I_{2s}] \tag{3}$$

Combining (2) and (3), the zero-sequence equations become

$$\Delta V_{01} = j0.25\, I_{01} + j0.12\, I_{02}$$

$$\Delta V_{02} = j0.12\, I_{01} + j0.25\, I_{02}$$

Hence,

$$Z_{011} = j0.25$$

$$Z_{012} = Z_{021} = j0.12$$

$$Z_{022} = j0.25$$

Hence, the answer is

$$\boxed{Z_{011} = j0.25,\ Z_{012} = j0.12,\ Z_{022} = j0.25}$$

**Q4.01.1** NAT | 2001 | 5M | Ans: Reactance Diagram | ● ○ ○

The off-diagonal elements of the Y-bus matrix give the branch admittances.

$$Y_{12} = Y_{21} = j2$$

$$Y_{13} = Y_{31} = j2.5$$

$$Y_{14} = Y_{41} = 0$$

$$Y_{23} = Y_{32} = j2.5$$

$$Y_{24} = Y_{42} = j4$$

$$Y_{34} = Y_{43} = j4$$

The corresponding branch reactances are

$$Z_{12} = \frac{1}{j2} = -j0.5$$

$$X_{12} = j0.5$$

$$Z_{13} = \frac{1}{j2.5} = -j0.4$$

$$X_{13} = j0.4$$

$$Z_{23} = \frac{1}{j2.5} = -j0.4$$

$$X_{23} = j0.4$$

$$Z_{24} = \frac{1}{j4} = -j0.25$$

$$X_{24} = j0.25$$

$$Z_{34} = \frac{1}{j4} = -j0.25$$

$$X_{34} = j0.25$$

Now determine the shunt elements.
For Bus 1,

$$Y_{11} = Y_{10} + Y_{12} + Y_{13} + Y_{14}$$

$$-j6 = Y_{10} - (j2 + j2.5 + 0)$$

$$Y_{10} = -j1.5$$

Hence,

$$Z_{10} = j0.67$$

For Bus 2,

$$-j10 = Y_{20} - (j2 + j2.5 + j4)$$

$$Y_{20} = -j1.5$$

Hence,

$$Z_{20} = j0.67$$

For Bus 3,

$$-j9 = Y_{30} - (j2.5 + j2.5 + j4)$$

$$Y_{30} = 0$$

Hence,

$$Z_{30} = \infty$$

For Bus 4,

$$-j8 = Y_{40} - (0 + j4 + j4)$$

$$Y_{40} = 0$$

Hence,

$$Z_{40} = \infty$$

Therefore, the reactance diagram consists of:

$$X_{12} = j0.5$$

$$X_{13} = j0.4$$

$$X_{23} = j0.4$$

$$X_{24} = j0.25$$

$$X_{34} = j0.25$$

and shunt reactances

$$X_{10} = j0.67$$

$$X_{20} = j0.67$$

connected to ground.

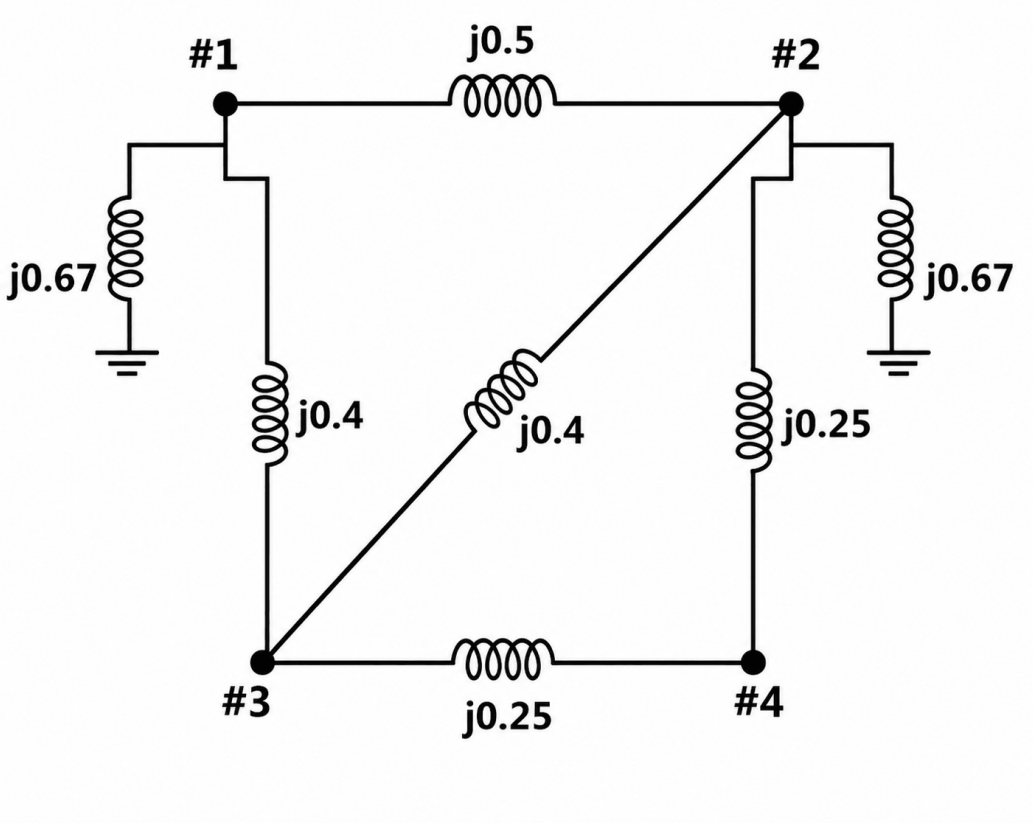



Fig. Solution Q4.01.1

**Q4.00.1** NAT | 2000 | 5M | Ans: *

The line admittances are

$$y_{12} = y_{13} = y_{23} = \frac{1}{j0.1} = -j10$$

Hence,

$$Y_{\text{bus}} = \begin{bmatrix} -j20 & j10 & j10 \\ j10 & -j20 & j10 \\ j10 & j10 & -j20 \end{bmatrix}$$

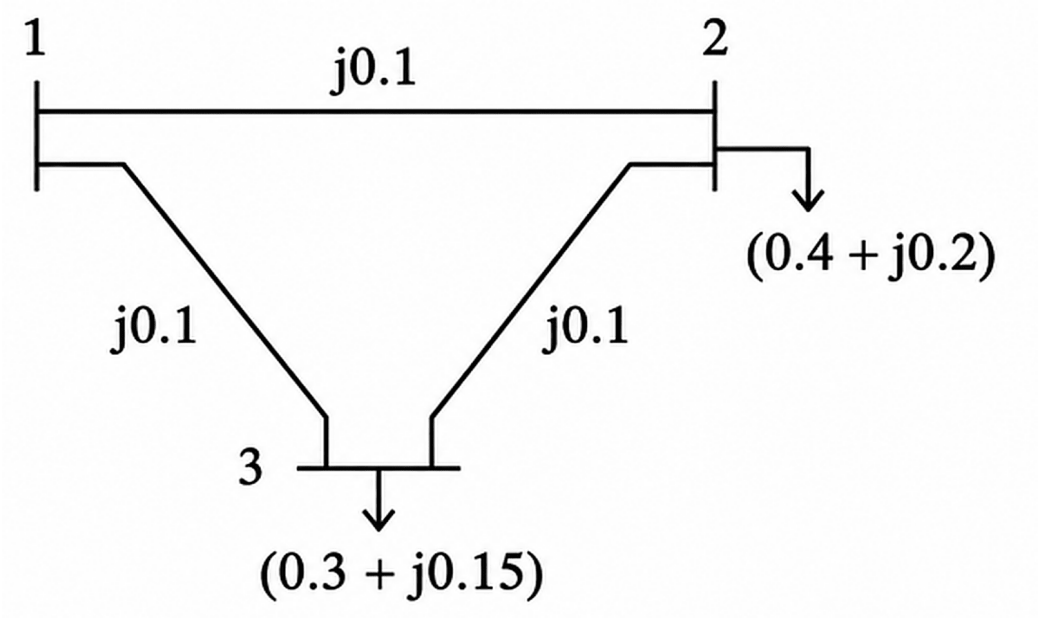



Fig. Solution Q4.00.1

Since the generated power at bus 2 is not specified, for the first iteration bus 2 is treated as a PQ bus. Assume

$$V_2^{(0)} = 1\angle 0^\circ$$

$$V_3^{(0)} = 1\angle 0^\circ$$

The specified powers are

$$P_2 = -0.4, \qquad Q_2 = -0.2$$

$$P_3 = -0.3, \qquad Q_3 = -0.15$$

Applying the Gauss-Seidel update equation for bus 2,

$$V_2^{(1)} = \frac{1}{Y_{22}}\left[\frac{P_2 - jQ_2}{\left(V_2^{(0)}\right)^*} - Y_{21}V_1 - Y_{23}V_3^{(0)}\right]$$

Substituting the values,

$$V_2^{(1)} = \frac{1}{-j20}\left[(-0.4 + j0.2) - j10 - j10\right]$$

$$V_2^{(1)} \approx 0.99\angle -0.02^\circ \text{ p.u.}$$

Thus after one iteration,

$$|V_2| \approx 0.99 \text{ p.u.}$$

which is below the desired value of

$$1.0 \text{ p.u.}$$

Therefore, switched shunt capacitive compensation must be added at bus 2 to provide reactive power support and raise the voltage magnitude to 1.0 p.u.

$$\boxed{|V_2^{(1)}| \approx 0.99 \text{ p.u.}}$$

**Q4.98.1** NAT | 1998 | 5M | Ans: * | ● ○ ○

The primitive admittances are

$$y_{12} = \frac{1}{j0.05} = -j20$$

$$y_{13} = \frac{1}{j0.02} = -j50$$

$$y_{23} = \frac{1}{j0.02} = -j50$$

The line charging admittances are

$$j1.25, \quad j0.5, \quad j0.5$$

and the shunt reactor admittance is

$$y_{30} = \frac{1}{j2.0} = -j0.5$$

The corresponding admittance network is shown below.

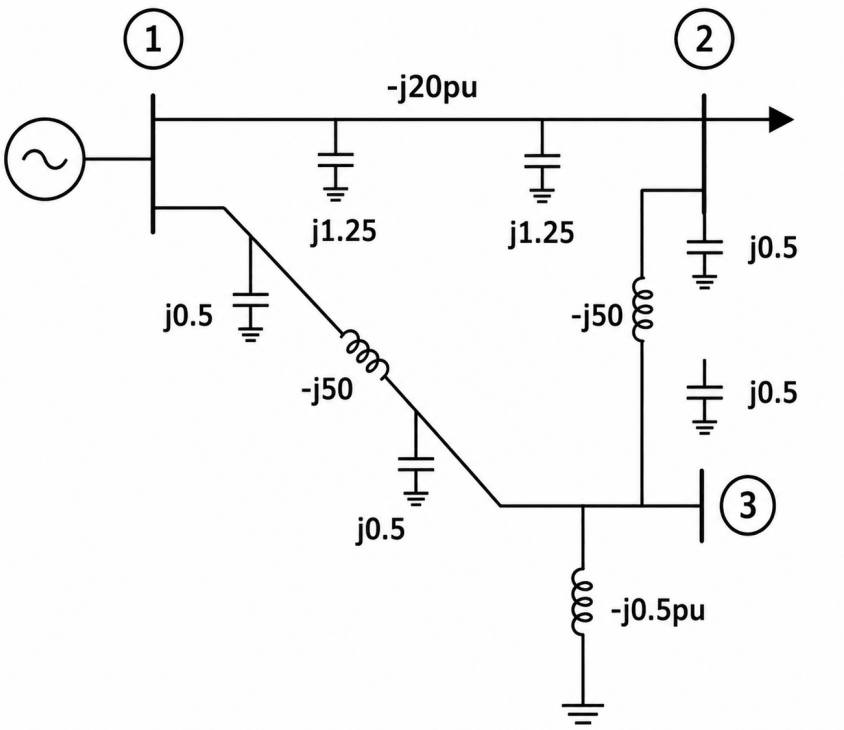



Fig. Solution Q4.98.1

Diagonal elements:

$$Y_{11} = y_{12} + y_{13} + j1.25 + j0.5$$

$$Y_{11} = -j20 - j50 + j1.25 + j0.5$$

$$Y_{11} = -j68.25$$

$$Y_{22} = y_{21} + y_{23} + j1.25 + j0.5$$

$$Y_{22} = -j20 - j50 + j1.25 + j0.5$$

$$Y_{22} = -j68.25$$

$$Y_{33} = y_{31} + y_{32} + j0.5 + j0.5 - j0.5$$

$$Y_{33} = -j50 - j50 + j0.5 + j0.5 - j0.5$$

$$Y_{33} = -j99.5$$

Off-diagonal elements:

$$Y_{12} = Y_{21} = j20$$

$$Y_{13} = Y_{31} = j50$$

$$Y_{23} = Y_{32} = j50$$

Therefore,

$$Y_{BUS} = \begin{bmatrix} -j68.25 & j20 & j50 \\ j20 & -j68.25 & j50 \\ j50 & j50 & -j99.5 \end{bmatrix}$$

**Q4.97.1** MCQ | 1997 | 1M | Ans: C | ● ○ ○

The Gauss-Seidel method is an iterative numerical technique used to solve systems of algebraic equations. It is applicable to both linear algebraic equations and nonlinear algebraic equations through iterative updating procedures.

Hence, the correct answer is

(C) Both linear and nonlinear algebraic equations

**Q4.96.1** MCQ | 1996 | 1M | Ans: A

The slack bus is introduced only to balance real and reactive power mismatches.
Changing the reference bus does not alter:

- Transmission-line parameters.
- Load data.
- Generator schedules.

Hence the physical system losses remain unchanged.
However, bus-voltage phasors are measured relative to the selected reference bus angle.
Changing the reference bus changes the numerical representation of the complex bus voltages.
Therefore,

(A) the system losses will be unchanged but complex bus voltages will change

**Q4.96.2** NAT | 1996 | 2M | Ans: P,S,U

The power angle difference is

$$\delta = 10^\circ$$

Since machine 1 leads machine 2 in angle, real power flows from machine 1 to machine 2.
Therefore machine 1 delivers real power.
Hence

$$(a) \rightarrow \text{Positive real power}$$

Since

$$|V_1| = 0.95 < |V_2| = 1.0$$

machine 1 absorbs reactive power from machine 2.
Hence

$$(b) \rightarrow \text{Negative reactive power}$$

A machine absorbing reactive power operates at lagging power factor.
Therefore

$$(c) \rightarrow \text{Lagging power factor}$$

Hence, the correct matching is

$$a \rightarrow P,\ b \rightarrow S,\ c \rightarrow U$$

**Q4.95.1** MCQ | 1995 | 1M | Ans: True

In load-flow studies, a voltage-controlled bus (PV bus) has the following specified quantities:
$P$ and $|V|$ while the reactive power output $Q$ is allowed to vary.
However, generators can supply or absorb reactive power only within specified limits.
If during the iterative solution process the calculated reactive power exceeds its permissible limit, $Q > Q_{max}$ or $Q < Q_{min}$ then the generator can no longer maintain the specified voltage magnitude.

The bus is therefore converted into a PQ bus (load bus) with reactive power fixed at the limiting value.

Hence the statement is true.

**Q4.93.1** MCQ | 1993 | 1M | Ans: D

Load flow (power flow) analysis is used to determine the steady-state operating condition of a power system.
The unknown quantities generally computed are:

- Bus voltage magnitudes,
- Bus voltage phase angles,
- Active power flow,
- Reactive power flow,
- Transmission losses.

To solve the power flow equations, each bus is modeled according to the quantities specified and the quantities to be determined.
The common bus types are:

- Slack (Swing) Bus
- PV (Generator) Bus
- PQ (Load) Bus

—

**Load (PQ) Bus Representation**

At a load bus, the connected load is assumed to consume specified amounts of active and reactive power.
Therefore,

$$P = P_L = \text{constant}$$

and

$$Q = Q_L = \text{constant}$$

These specified values are independent of the unknown bus voltage during the iterative solution of the load-flow problem.
The quantities computed during the load-flow iterations are

$$|V|$$

and

$$\delta$$

Thus, a load bus is modeled as a constant real and reactive power demand.

—

**Discussion of Options**

**Option (A):** Constant current drawn from the bus
This model is sometimes used for specific load studies but is not the standard assumption in conventional power-flow analysis.

Hence, Option (A) is incorrect.

—

**Option (B):** Constant impedance connected at the bus

A constant impedance model causes the load power to vary with the square of the bus voltage.

Although such models exist, they are not the standard representation used in classical load-flow analysis.

Hence, Option (B) is incorrect.

—

**Option (C):** Voltage and frequency dependent source at the bus

Loads are modeled as power-consuming elements, not as sources.

Hence, Option (C) is incorrect.

—

**Option (D):** Constant real and reactive power drawn from the bus

This is the standard PQ bus model used in conventional load-flow analysis.

Hence, Option (D) is correct.

Therefore,

(D) Constant real and reactive power drawn from the bus

## Q4.92.1 NAT 1992 1M Ans: Voltage and Real power ● ○ ○

A voltage-controlled bus is commonly known as a **PV bus**. At this bus, the generator regulates the terminal voltage while supplying a specified amount of active (real) power to the system.

In load flow analysis, buses are classified according to the quantities that are specified and those that must be determined.

| Bus Type | Specified Quantities | Unknown Quantities |
|---|---|---|
| Slack Bus | Voltage magnitude ($\|V\|$), Voltage angle ($\delta$) | $P$, $Q$ |
| PV Bus | Real power ($P$), Voltage magnitude ($\|V\|$) | $Q$, $\delta$ |
| PQ Bus | Real power ($P$), Reactive power ($Q$) | $\|V\|$, $\delta$ |

For a PV bus,

$$P = P_{\text{specified}}$$

and

$$|V| = |V|_{\text{specified}}$$

During the iterative load flow solution (Gauss–Seidel or Newton–Raphson), the reactive power output of the generator and the bus voltage angle are computed.

Hence, the quantities specified at a voltage-controlled (PV) bus are:

Voltage magnitude and Real Power

## Q4.91.1 MCQ 1991 1M Ans: D ● ○ ○

**Concept:**

The diagonal elements of the Bus Admittance Matrix ($Y_{\text{bus}}$) are obtained by summing all admittances incident at the corresponding bus, including shunt admittances.

Mathematically,

$$Y_{ii} = \sum y_{ik} + y_{i,\text{shunt}}$$

where

- $y_{ik}$ is the admittance of every branch connected to Bus $i$.
- $y_{i,\text{shunt}}$ is the shunt admittance connected directly to Bus $i$.

—

**Step 1: Convert reactances into admittances**

The reactance between Bus 2 and ground is

$$j0.1$$

Hence,

$$Y_{20} = \frac{1}{j0.1} = -j10$$

The reactance between Bus 2 and Bus 1 is

$$j0.2$$

Therefore,

$$Y_{21} = \frac{1}{j0.2} = -j5$$

Similarly,
the reactance between Bus 2 and Bus 3 is

$$j0.2$$

giving

$$Y_{23} = \frac{1}{j0.2} = -j5$$

—

**Step 2: Compute the diagonal element**

$$Y_{22} = Y_{20} + Y_{21} + Y_{23}$$

Substituting,

$$Y_{22} = -j10 + (-j5) + (-j5)$$

$$\boxed{Y_{22} = -j20 \text{ p.u.}}$$

—

**Verification**

The diagonal entry is the sum of all admittances connected to Bus 2.

$$-j10 - j5 - j5 = -j20$$

Hence the correct option is

$$\boxed{\text{(D)} \ -j20 \text{ p.u.}}$$

SOLUTIONS

# V

# Performance and Models of Transmission lines and cables

## Topics

1. Short Transmission Line
2. Medium Transmission Line
3. Long Transmission Line

## Unit Snapshot*

| | | | | |
|---|---|---|---|---|
| **Stream:** EE | **Questions:** 23 | **MCQ:** 12 | **MSQ:** 0 | **NAT:** 11 |
| **1M:** 10 | **2M:** 11 | **Descriptive:** 2 | **Avg Weightage** ~2M | **Range:** 1M(2023)–2M(2019) |

**See preface for how Avg Weightage and Range are calculated.*

**Q5.23.1** NAT | 2023 | 1M | Ans: 83.00 to 84.00 | ● ○ ○

The active power delivered to the receiving end of a transmission line is

$$P_R = \frac{V_S V_R}{|B|}\cos(\beta - \delta) - \frac{|A|}{|B|}|V_R|^2 \cos(\beta - \alpha)$$

where

- $\delta$ is the load angle,
- $\beta = \angle B$,
- $\alpha = \angle A$.

For a given system, all quantities except $\delta$ are constants. Hence, the receiving-end power is maximum when

$$\beta - \delta = 0$$

or

$$\boxed{\delta = \beta}$$

For a nominal $\pi$ transmission line, the ABCD constants are

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1 + \frac{YZ}{2} & Z \\ Y\left(1 + \frac{YZ}{4}\right) & 1 + \frac{YZ}{2} \end{bmatrix}$$

Therefore,

$$B = Z = R + j\omega L$$

The phase angle of $B$ is

$$\beta = \tan^{-1}\left(\frac{\omega L}{R}\right)$$

The total resistance and inductance of the transmission line are

$$R = 0.035\ \Omega/\text{km} \times 400\ \text{km} = 14\ \Omega$$

$$L = 1\ \text{mH/km} \times 400\ \text{km} = 400\ \text{mH} = 0.4\ \text{H}$$

Substituting these values,

$$\beta = \tan^{-1}\left(\frac{2\pi(50)(0.4)}{14}\right) = 83.64^\circ$$

Since the receiving-end power is maximum when $\delta = \beta$,

$$\boxed{\delta = 83.64^\circ}$$

**Key Points**

- The maximum receiving-end power occurs when the load angle equals the phase angle of the transmission parameter $B$.
- For a nominal $\pi$ model, $B = Z$, so only the line impedance determines $\beta$.

- The angle of the impedance is obtained from $\beta = \tan^{-1}\left(\frac{\omega L}{R}\right)$.

**Mistakes to be avoided**

- Do not use the angle of $A$ instead of the angle of $B$ while applying the maximum power condition.
- Remember to convert inductance from mH to H before substituting into $\omega L$.
- Use the total line resistance and inductance (after multiplying by the line length), not the per-kilometre values.

**Q5.19.1** NAT | 2019 | 2M | Ans: 414 to 423 | ●●○

Given:

$$f = 50\,\text{Hz}$$

$$V_s = 400\,\text{kV}$$

$$L = 1\,\text{mH/km} = 1 \times 10^{-3}\,\text{H/km}$$

$$C = 0.01\,\mu\text{F/km} = 0.01 \times 10^{-6}\,\text{F/km}$$

Since the transmission line length is 300 km, it is treated as a long transmission line.
Wave propagation velocity is

$$v = \frac{1}{\sqrt{LC}}$$

Substituting,

$$v = \frac{1}{\sqrt{(1 \times 10^{-3})(0.01 \times 10^{-6})}}$$

$$v = 316227.766\,\text{km/s}$$

For a long transmission line,

$$A = \cos(\beta l)$$

Using approximation,

$$A \approx 1 - \frac{(\beta l)^2}{2}$$

where

$$\beta l = \frac{2\pi f l}{v}$$

Substituting,

$$\beta l = \frac{2\pi \times 50 \times 300}{316227.766}$$

$$A = 1 - \frac{\left(\frac{2\pi \times 50 \times 300}{316227.766}\right)^2}{2}$$

$$A \approx 0.955$$

Under no-load condition,

$$V_r = \frac{V_s}{A}$$

$$V_r = \frac{400}{0.955}$$

$$V_r = 418.85\,\text{kV}$$

Therefore,

$$\boxed{418.85}$$

**Mistakes to be avoided**

- Do not forget unit conversion for inductance and capacitance.
- Carefully use radians while evaluating $\beta l$.

**Q5.17.1** NAT | 2017 | 1M | Ans: 406.2 | ● ○ ○

Given,

$$Z = 100\angle 80^\circ\ \Omega$$

$$X = 3300\ \Omega$$

For the nominal-$\pi$ model,

$$\frac{Y}{2} = \frac{1}{-jX}$$

Hence,

$$\left|\frac{Y}{2}\right| = \frac{1}{3300}$$

$$|Y| = \frac{2}{3300} = 6.06 \times 10^{-4}\ \text{S}$$

Characteristic impedance is

$$Z_c = \sqrt{\frac{Z}{Y}}$$

Using magnitudes,

$$|Z_c| = \sqrt{\frac{100}{6.06 \times 10^{-4}}}$$

$$|Z_c| = 406.2\ \Omega$$

$$\boxed{406.2}$$

**Q5.17.2** NAT | 2017 | 2M | Ans: 0.8 | ● ○ ○

Given,

$$P_R = 3 \text{ MW}$$

$$V_R = 17.32 \text{ kV}$$

$$V_S = 17.87 \text{ kV}$$

$$R = 0.25\ \Omega$$

$$X_L = 3.925\ \Omega$$

Per-phase receiving end voltage:

$$V_{R\phi} = \frac{17.32}{\sqrt{3}} = 10 \text{ kV}$$

Per-phase sending end voltage:

$$V_{S\phi} = \frac{17.87}{\sqrt{3}} = 10.3175 \text{ kV}$$

For a short transmission line,

$$V_{S\phi} - V_{R\phi} = I(R\cos\phi + X_L \sin\phi)$$

Also,

$$P_R = \sqrt{3} V_R I \cos\phi$$

Hence,

$$I = \frac{P_R}{\sqrt{3} V_R \cos\phi}$$

Substituting,

$$I = \frac{3 \times 10^6}{\sqrt{3} \times 17.32 \times 10^3 \cos\phi}$$

$$I = \frac{100}{\cos\phi}$$

Now,

$$10.3175 \times 10^3 - 10 \times 10^3 = \frac{100}{\cos\phi}(0.25\cos\phi + 3.925\sin\phi)$$

$$317.5 = 25 + 392.5\tan\phi$$

$$392.5\tan\phi = 292.5$$

$$\tan\phi = 0.7452$$

$$\phi = \tan^{-1}(0.7452)$$

$$\phi \approx 36.7^\circ$$

Therefore,

$$\cos\phi \approx 0.8$$

$$\boxed{0.8}$$

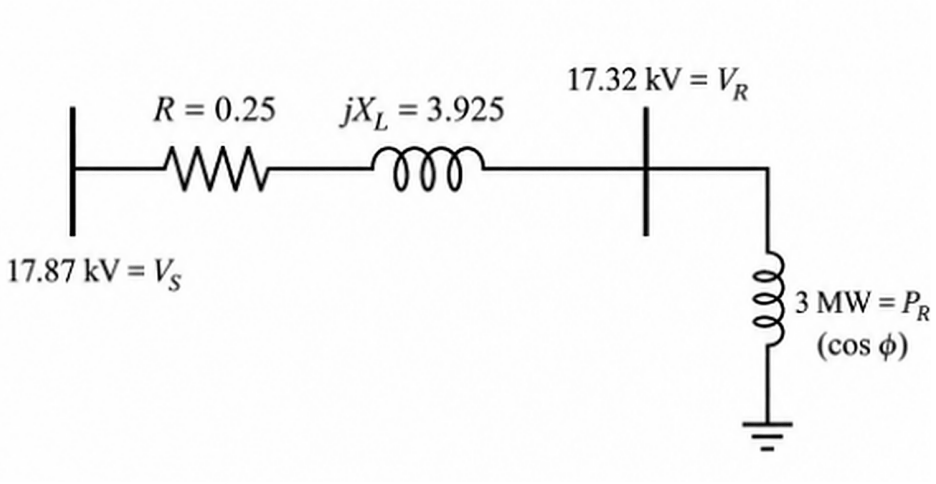



Fig. Solution Q5.17.2

**Q5.16.1** NAT | 2016 | 2M | Ans: 294 to 298 | ●●○

At no-load condition,

$$V_S = AV_R$$

Given,

$$V_S = 400\,\text{kV}$$

$$V_R = 420\,\text{kV}$$

Therefore,

$$A = \frac{400}{420}$$

$$A = 0.9524$$

For a long lossless transmission line,

$$A = \cos(\beta l)$$

Hence,

$$\cos(\beta l) = 0.9524$$

$$\beta l = \cos^{-1}(0.9524)$$

$$\beta l = 0.3085$$

The phase constant is

$$\beta = \frac{2\pi f}{v}$$

where

$$f = 50\,\text{Hz}$$

$$v = 3 \times 10^8\ \text{m/s}$$

Thus,

$$\beta = \frac{2\pi \times 50}{3 \times 10^8}$$

Using

$$\beta l = 0.3085$$

we get

$$l = \frac{0.3085}{\beta}$$

$$l = \frac{0.3085 \times 3 \times 10^8}{2\pi \times 50}$$

$$l \approx 294.59 \times 10^3\ \text{m}$$

$$l \approx 294.59\,\text{km}$$

$$\boxed{294.59}$$

**Key Points**

- For a lossless transmission line:
$$A = \cos(\beta l)$$
- The propagation constant for a lossless line is purely imaginary:
$$\gamma = j\beta$$
- Phase constant:
$$\beta = \frac{2\pi f}{v}$$
- Traveling wave velocity for overhead transmission lines is approximately equal to the speed of light.

**Mistakes to be avoided**

- Do not use hyperbolic cosine for a lossless line directly after simplification.
- Convert the final answer from meters to kilometers.

**Q5.14.1** MCQ | 2014 | 1M | Ans: A | ● ○ ○

For a transmission line to be distortionless, the attenuation constant must be independent of frequency and the phase constant must vary linearly with frequency.
The required condition for distortionless transmission is

$$\frac{R}{L} = \frac{G}{C}$$

where

- $R$ = resistance per unit length
- $L$ = inductance per unit length
- $G$ = shunt conductance per unit length
- $C$ = capacitance per unit length

Using the notation given in the question,

$$\frac{r}{l} = \frac{g}{c}$$

Cross-multiplying,

$$rc = lg$$

Hence, the correct option is

$$\boxed{\text{(A)} \quad rc = lg}$$

**Key Points**

- Distortionless transmission condition:

$$\frac{R}{L} = \frac{G}{C}$$

- Under distortionless condition:
  - attenuation becomes frequency independent
  - waveform shape is preserved during transmission
- This condition is known as the Heaviside condition.

**Q5.14.2** NAT | 2014 | 2M | Ans: 186 to 188 ● ● ○

Given,
Series impedance per km:

$$z = j0.5\ \Omega/\text{km}$$

Shunt admittance per km:

$$y = j5 \times 10^{-6}\ \text{mho/km}$$

Line length:

$$l = 400\,\text{km}$$

Total series impedance:

$$Z = zl$$

$$Z = j0.5 \times 400$$

$$Z = j200\ \Omega$$

Total shunt admittance:

$$Y = yl$$

$$Y = j5 \times 10^{-6} \times 400$$

$$Y = j0.002 \text{ mho}$$

For a long transmission line, equivalent $\pi$-model series impedance is

$$Z_\pi = Z\frac{\sinh(\gamma l)}{\gamma l}$$

Using approximation for long transmission line,

$$Z_\pi = Z\left(1 + \frac{YZ}{6}\right)$$

Now,

$$YZ = (j0.002)(j200)$$

$$YZ = -0.4$$

Hence,

$$Z_\pi = j200\left(1 - \frac{0.4}{6}\right)$$

$$Z_\pi = j200(0.9333)$$

$$Z_\pi \approx j186.67\ \Omega$$

Therefore, magnitude of equivalent $\pi$ circuit series impedance is

$$\boxed{186.67\ \Omega}$$

**Key Points**

- For long transmission lines, nominal $\pi$ approximation is modified using long-line correction.
- Equivalent $\pi$ series impedance:

$$Z_\pi = Z\left(1 + \frac{YZ}{6}\right)$$

for small values of $YZ$.

**Q5.08.1** MCQ | 2008 | 1M | Ans: D | ● ○ ○

Given,

$$l = 300 \text{ km}$$

$$\beta = 0.00127 \text{ rad/km}$$

For a lossless transmission line,

$$\beta = \frac{2\pi}{\lambda}$$

Hence,

$$\lambda = \frac{2\pi}{\beta}$$

$$\lambda = \frac{2\pi}{0.00127}$$

$$\lambda = 4947.39 \text{ km}$$

Therefore,

$$\frac{l}{\lambda} \times 100 = \frac{300}{4947.39} \times 100$$

$$= 0.06063 \times 100$$

$$= 6.063\%$$

Thus, the correct answer is

$$\boxed{\text{(D) } \frac{l}{\lambda} \times 100 = 6.06\%}$$

**Q5.07.1** MCQ | 2007 | 2M | Ans: C | ● ○ ○

For a lossless transmission line,

$$\beta = \omega\sqrt{LC}$$

Since

$$X_L = \omega L$$

and

$$B_C = \omega C$$

therefore,

$$\beta = \sqrt{X_L B_C}$$

Given,

$$X_L = 0.045$$

$$B_C = 1.2$$

Hence,

$$\beta = \sqrt{0.045 \times 1.2}$$

$$= 0.2324 \text{ rad}$$

Also,

$$\beta = \frac{\omega l}{v}$$

where

$$\omega = 2\pi f$$

$$f = 50 \text{ Hz}$$

and

$$v = 3 \times 10^5 \text{ km/s}$$

Substituting,

$$l = \frac{v\beta}{\omega}$$

$$= \frac{3 \times 10^5 \times 0.2324}{2\pi \times 50}$$

$$= 221.9 \text{ km}$$

Therefore,

$$l \approx 222 \text{ km}$$

Hence, the correct option is

(C) 222 km

**Q5.06.1** MCQ | 2006 | 1M | Ans: A | ● ○ ○

Transmission lines are classified as short, medium, and long primarily based on their physical length. Typically, for 50 Hz line

$$\text{Short line} : l < 80 \text{ km}$$

$$\text{Medium line} : 80 \text{ km} \le l \le 250 \text{ km}$$

$$\text{Long line} : l > 250 \text{ km}$$

Hence, the correct answer is

(A) physical length of the line

**Q5.06.2** MCQ 2006 2M Ans: C ● ○ ○

The percentage voltage regulation is

$$\%VR = \frac{|V_{R(NL)}| - |V_{R(FL)}|}{|V_{R(FL)}|} \times 100$$

The transmission line equation is

$$V_S = AV_R + BI_R$$

At no load,

$$I_R = 0$$

Hence,

$$V_S = AV_{R(NL)}$$

Therefore,

$$V_{R(NL)} = \frac{V_S}{A} = \frac{240}{0.94\angle 1^\circ}$$

$$|V_{R(NL)}| = \frac{240}{0.94} = 255.32 \text{ kV}$$

Given,

$$|V_{R(FL)}| = 220 \text{ kV}$$

Thus,

$$\%VR = \frac{255.32 - 220}{220} \times 100$$

$$\%VR = 16.05\%$$

Hence, the correct answer is

(C) 16

**Q5.04.1** MCQ 2004 2M Ans: C ● ● ○

Receiving-end current:

$$I_R = \frac{50 \times 10^6}{\sqrt{3}(220 \times 10^3)(0.9)} \angle(-25.84^\circ)$$

$$= 145.8\angle(-25.84^\circ) \text{ A}$$

Per-phase receiving-end voltage:

$$V_R = \frac{220}{\sqrt{3}} = 127.02 \text{ kV}$$

Using

$$V_S = AV_R + BI_R$$

and substituting the given ABCD constants,

$$|V_S| = 133.24 \text{ kV (phase)}$$

Hence,

$$|V_{S,LL}| = \sqrt{3}(133.24)$$

$$= 230.78 \text{ kV}$$

Thus, the correct option is

$$\boxed{\text{(C) } 230.78 \text{ kV}}$$

**Q5.03.1** MCQ | 2003 | 2M | Ans: B | ● ○ ○

Given,

$$A = D = 0.9\angle 0^\circ$$

$$B = 200\angle 90^\circ = j200\ \Omega$$

Since the receiving-end voltage is required to be equal to the sending-end voltage,

$$|V_S| = |V_R|$$

At no load, the shunt reactor draws current

$$I_R \neq 0$$

Using the transmission-line equation,

$$V_S = AV_R + BI_R$$

Substituting

$$V_S = V_R$$

gives

$$V_R(1 - A) = BI_R$$

Hence,

$$\frac{V_R}{I_R} = \frac{B}{1 - A}$$

$$= \frac{j200}{1 - 0.9}$$

$$= j2000\ \Omega$$

Therefore, the required inductive reactance is

$$X_L = 2000\ \Omega$$

Thus, the correct option is

(B) 2000 Ω

**Q5.02.1** NAT | 2002 | 5M | Ans: 1.67 GW | ● ● ○

For a lossless transmission line,

$$V_s = V_r \cos(\beta l) + jZ_c \sin(\beta l) I_r$$

and

$$I_s = I_r \cos(\beta l) + j\frac{V_r}{Z_c} \sin(\beta l)$$

To determine the maximum power transfer, first obtain the Thevenin equivalent seen from the receiving-end load.

Under open-circuit condition,

$$I_r = 0$$

Hence,

$$V_{th} = V_r = \frac{V_s}{\cos(\beta l)}$$

The Thevenin impedance is obtained by setting the sending-end voltage source to zero and evaluating

$$Z_{th} = jZ_c \tan(\beta l)$$

For a unity power factor (purely resistive) load, maximum power transfer occurs when

$$R_L = |Z_{th}| = Z_c \tan(\beta l)$$

Therefore,

$$I_r = \frac{V_{th}}{R_L + jX_{th}} = \frac{V_s/\cos(\beta l)}{Z_c \tan(\beta l)(1 + j)}$$

$$I_r = \frac{V_s}{Z_c \sin(\beta l)(1 + j)}$$

Maximum load power is

$$P_L = |I_r|^2 R_L$$

Substituting,

$$P_L = \left|\frac{V_s}{Z_c \sin(\beta l)(1 + j)}\right|^2 Z_c \tan(\beta l)$$

$$P_L = \frac{V_s^2}{2Z_c \sin(\beta l) \cos(\beta l)}$$

Using

$$\beta l = 1.2 \times 10^{-3} \times 100 = 0.12 \text{ rad}$$

$$Z_c = 400\ \Omega$$

$$P_L = \frac{(230 \times 10^3)^2}{400 \sin(0.24)}$$

$$P_L = 556 \times 10^6\ \text{W}$$

$$P_L = 556\ \text{MW}$$

Hence, the correct answer is

$$\boxed{556\,\text{MW}}$$

**Q5.99.1** MCQ | 1999 | 2M | Ans: A

Given,

$$A = D = 0.96\angle 3^\circ$$

$$C = 0.5 \times 10^{-4}\angle 80^\circ\ \text{S/phase}$$

At no-load condition,

$$I_r = 0$$

Charging current

$$I_c = CV_r$$

$$I_c = 0.5 \times 10^{-4}\angle 80^\circ \times \frac{220 \times 10^3}{\sqrt{3}}$$

$$I_c = \frac{11}{\sqrt{3}}\angle 80^\circ\ \text{A}$$

Magnitude:

$$|I_c| = \frac{11}{\sqrt{3}}\ \text{A}$$

Hence, the correct option is

$$\boxed{\text{(A)}\ \frac{11}{\sqrt{3}}\ \text{A}}$$

**Q5.99.2** NAT | 1999 | 5M | Ans: *

Series impedance of line:

$$Z = (0.25 + j0.5) \times 150$$

$$Z = 37.5 + j75\ \Omega$$

Total shunt admittance:

$$Y = 0.04 \times 10^{-4} \times 150$$

$$Y = j0.6 \times 10^{-3}\ \text{S}$$

Hence for nominal $\pi$ representation,

$$\frac{Y}{2} = j0.3 \times 10^{-3}\ \text{S}$$

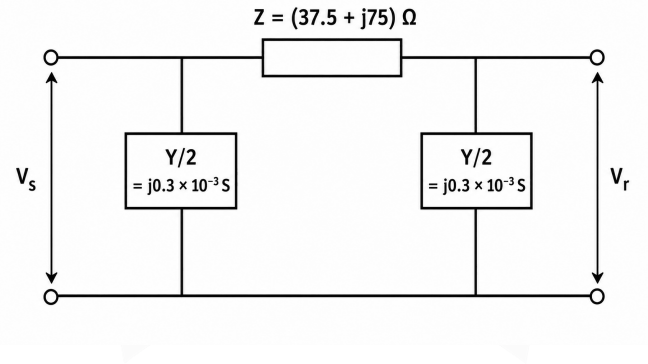



Fig. Solution Q5.99.2

Sending-end line voltage:

$$V_S = 66\ \text{kV}$$

Phase voltage:

$$V_S^{(ph)} = \frac{66}{\sqrt{3}} = 38.105\ \text{kV}$$

Since receiving end is open circuited,

$$I_R = 0$$

For nominal $\pi$ model,

$$V_R = \frac{V_S}{1 + \frac{ZY}{2}}$$

Substituting,

$$V_R = \frac{38.105 \times 10^3}{1 + \frac{(37.5+j75)(j0.6\times10^{-3})}{2}}$$

$$V_R = 38.9796\ \text{kV}$$

Therefore, the required answer is

$$\boxed{V_R = 38.98 \text{ kV}}$$

**Q5.98.1** MCQ | 1998 | 1M | Ans: C | ● ○ ○

The reflection coefficient at the load end of a transmission line is

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Given,

$$Z_L = 300\,\Omega$$

and

$$Z_0 = 300\,\Omega$$

Substituting,

$$\Gamma = \frac{300 - 300}{300 + 300}$$

$$\Gamma = 0$$

Hence, the correct option is

$$\boxed{\text{(C) } \Gamma = 0}$$

**Q5.98.2** MCQ | 1998 | 2M | Ans: B | ● ● ○

The velocity of propagation of a wave on a transmission line is

$$v = \frac{1}{\sqrt{LC}}$$

Given

$$L = 0.201 \times 10^{-6} \text{ H/m}$$

$$C = 196.2 \times 10^{-12} \text{ F/m}$$

Substituting,

$$v = \frac{1}{\sqrt{(0.201 \times 10^{-6})(196.2 \times 10^{-12})}}$$

$$v = 0.15924 \times 10^{9}$$

$$v = 159.24 \times 10^{6} \text{ m/s}$$

Since

$$1 \text{ m/s} = 1 \times 10^{6} \text{m}/\mu\text{s}$$

$$v = 159.24\ \text{m}/\mu\text{s}$$

Therefore the correct option is

(B) $159.24\ \text{m}/\mu\text{s}$

**Q5.96.1** MCQ | 1996 | 1M | Ans: C | ● ○ ○

Transmission-line classification depends on the significance of the distributed parameters.
For normal power-frequency operation (50 Hz), a 50 km line is classified as a short line. However, at very high frequencies, the electrical length of the line increases significantly.
A useful criterion is

$$f \times l$$

where $f$ is frequency in Hz and $l$ is line length in km.
For the given system,

$$f \times l = 500 \times 50 = 25000$$

which is sufficiently large for the distributed capacitance and inductance effects to become important.
Hence the line must be represented using long-line theory.

(C) Long line

**Q5.95.1** MCQ | 1995 | 1M | Ans: C | ● ○ ○

The surge impedance (or characteristic impedance) of a transmission line is

$$Z_c = \sqrt{\frac{L}{C}}$$

where

- $L$ is inductance per unit length
- $C$ is capacitance per unit length

Notice that line length does not appear in the expression.
Therefore changing the total line length from 400 km to 200 km does not change the surge impedance.
Only the distributed parameters per unit length determine $Z_c$.
Hence

$$Z_c = 400\ \Omega$$

(C) $400\ \Omega$

**Q5.94.1** NAT | 1994 | 1M | Ans: SKIN | ● ○ ○

When alternating current flows through a conductor, the current density is not uniformly distributed over the entire cross-section.
Due to electromagnetic induction within the conductor, a larger proportion of current tends to flow near the outer surface while the current density decreases toward the center.
This phenomenon effectively reduces the conducting cross-sectional area available for current flow.
As a consequence:

- Effective resistance increases.
- Internal reactance decreases.
- Conductor losses increase.

This phenomenon is known as the **skin effect**.

SKIN

**Q5.94.2** NAT | 1994 | 2M | Ans: True | ● ○ ○

The charging current of a transmission line is

$$I_c = \omega CV$$

where

- $C$ = line capacitance
- $V$ = operating voltage
- $\omega = 2\pi f$

For two lines of identical length:

$$C_{400} \approx C_{220}$$

and

$$I_c \propto V$$

Since

$$400 > 220$$

the charging current of the 400 kV line is larger.
Therefore,

True

**Q5.91.1** NAT | 1991 | 1M | Ans: * | ●●○

**Concept:**

When two-port networks (or transmission lines) are connected in cascade, the overall transmission parameters are obtained by multiplying the individual ABCD matrices in the order of power flow.

If

$$[T_1] = \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix}$$

and

$$[T_2] = \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix},$$

then the equivalent transmission matrix is

$$[T] = [T_1][T_2].$$

Write the given matrices

$$[T_1] = \begin{bmatrix} 1 & 10\angle 30^\circ \\ 0 & 1 \end{bmatrix}$$

$$[T_2] = \begin{bmatrix} 1 & 0 \\ 0.025\angle -30^\circ & 1 \end{bmatrix}$$

Performing matrix multiplication, we get

$$[T] = \begin{bmatrix} 1 & 10\angle 30^\circ \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0.025\angle -30^\circ & 1 \end{bmatrix}$$

The individual parameters become

Parameter $A$

$$A = A_1A_2 + B_1C_2$$

Substituting,

$$A = 1 + (10\angle 30^\circ)(0.025\angle -30^\circ)$$

Since

$$10 \times 0.025 = 0.25$$

and

$$30^\circ + (-30^\circ) = 0^\circ,$$

$$A = 1 + 0.25 = 1.25.$$

—

Parameter $B$

$$B = A_1B_2 + B_1D_2$$

$$B = (1)(0) + (10\angle 30^\circ)(1)$$

$$B = 10\angle 30^\circ.$$

—

Parameter $C$

$$C = C_1A_2 + D_1C_2$$

$$C = 0 + (1)(0.025\angle -30^\circ)$$

$$C = 0.025\angle -30^\circ.$$

—

Parameter $D$

$$D = C_1B_2 + D_1D_2$$

$$D = 0 + 1 = 1.$$

—

Hence, the resultant ABCD Matrix is

$$\begin{bmatrix} 1.25 & 10\angle 30^\circ \\ 0.025\angle -30^\circ & 1 \end{bmatrix}$$

PrepFusion

SOLUTIONS

# VI

# Parameters of Transmission Lines

## Topics

1. Skin Effect, Inductance due to Internal and External Flux
2. Inductance of 3 phase system- calculations
3. GMD, GMR and Capacitance of Transmission Lines

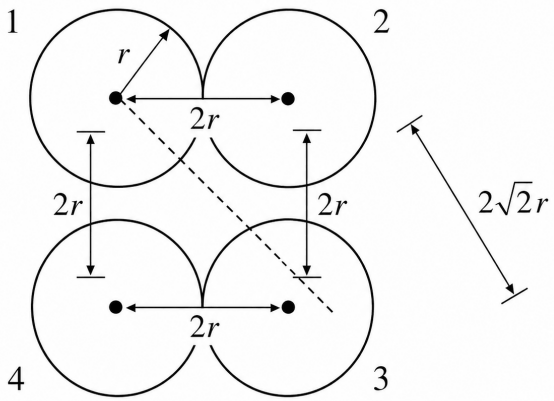
## Unit Snapshot*

| | | | | |
|---|---|---|---|---|
| **Stream:** EE | **Questions:** 22 | **MCQ:** 18 | **MSQ:** 0 | **NAT:** 4 |
| **1M:** 8 | **2M:** 14 | **Descriptive:** 0 | **Avg Weightage** ~2M | **Range:** 1M(2022)–7M(2010) |

**See preface for how Avg Weightage and Range are calculated.*

**Q6.22.1** MCQ | 2022 | 1M | Ans: D | ● ○ ○

For a composite conductor having four equal strands, the self-GMD (GMR) is obtained by taking the geometric mean of all distances from one strand to every strand, including the self-distance.




$$D_{11} = r'$$
$$D_{12} = 2r$$
$$D_{13} = 2\sqrt{2}\,r$$
$$D_{14} = 2r$$

Fig. Solution Q6.22.1

From the geometry,

$$D_{11} = r' = 0.7788\,r$$

$$D_{12} = 2r$$

$$D_{13} = 2\sqrt{2}\,r$$

$$D_{14} = 2r$$

Hence,

$$\text{GMR} = \sqrt[4]{D_{11}D_{12}D_{13}D_{14}}$$

$$\text{GMR} = \sqrt[4]{(0.7788\,r)(2r)(2\sqrt{2}\,r)(2r)}$$

$$\text{GMR} = 1.723\,r$$

(D) $1.723\,r$

**Q6.17.1** NAT | 2017 | 2M | Ans: 0.271 to 0.301 | ● ○ ○

For a balanced 3-phase transmission line with equilateral spacing,

$$D_m = (D_{ab}D_{bc}D_{ca})^{1/3}$$

Given,

$$D_{ab} = D_{bc} = D_{ca} = 1 \text{ m}$$

Hence,

$$D_m = 1 \text{ m}$$

Also,

$$D_s = \text{GMR} = 0.01 \text{ m}$$

Inductance per conductor per metre is

$$L = 2 \times 10^{-7} \ln\left(\frac{D_m}{D_s}\right) \text{ H/m}$$

Substituting,

$$L = 2 \times 10^{-7} \ln\left(\frac{1}{0.01}\right)$$

$$L = 2 \times 10^{-7} \ln(100)$$

$$L = 9.21 \times 10^{-7} \text{ H/m}$$

Positive sequence reactance:

$$X_L = 2\pi f L$$

$$X_L = 2\pi \times 50 \times 9.21 \times 10^{-7}$$

$$X_L = 2.89 \times 10^{-4} \ \Omega/\text{m}$$

Converting into Ω/km,

$$X_L = 2.89 \times 10^{-4} \times 1000$$

$$X_L = 0.289 \ \Omega/\text{km}$$

$$\boxed{0.289}$$

**Mistakes to be avoided**

- Use natural logarithm ln, not logarithm base 10.
- Convert Ω/m into Ω/km correctly.

**Q6.16.1** NAT | 2016 | 2M | Ans: 1.41 to 1.47 | ● ● ○

For the single-phase line, the inductance per conductor is

$$L_1 = 2 \times 10^{-7} \ln\left(\frac{D_m}{r'}\right) \text{ H/m}$$

Here,

$$D_m = 1 \text{ m}$$

and

$$r' = 0.7788 \times r$$

Given radius,

$$r = 10 \text{ mm} = 0.01 \text{ m}$$

Hence,

$$r' = 0.7788 \times 0.01 = 0.007788 \text{ m}$$

Therefore,

$$L_1 = 2 \times 10^{-7} \ln\left(\frac{1}{0.007788}\right)$$

$$L_1 = 2 \times 10^{-7} \ln\left(\frac{100}{0.7788}\right)$$

$$L_1 \approx 0.97 \, \mu\text{H/m}$$

The positive sequence inductance of the three-phase line is 5% more than the single-phase inductance.

$$L_2 = 1.05 L_1$$

$$L_2 = 1.05 \times 0.97 = 1.0185 \, \mu\text{H/m}$$

For the three-phase configuration,

$$L_2 = 2 \times 10^{-7} \ln\left(\frac{(D \times D \times 1)^{1/3}}{r'}\right)$$

$$L_2 = 2 \times 10^{-7} \ln\left(\frac{\sqrt[3]{D^2}}{0.007788}\right)$$

$$1.0185 \times 10^{-6} = 2 \times 10^{-7} \ln\left(\frac{\sqrt[3]{D^2}}{0.007788}\right)$$

$$\ln\left(\frac{\sqrt[3]{D^2}}{0.007788}\right) = \frac{1.0185 \times 10^{-6}}{2 \times 10^{-7}}$$

$$\ln\left(\frac{\sqrt[3]{D^2}}{0.007788}\right) = 5.0925$$

$$\frac{\sqrt[3]{D^2}}{0.007788} = e^{5.0925}$$

$$\sqrt[3]{D^2} = 0.007788 \times e^{5.0925}$$

$$\sqrt[3]{D^2} \approx 1.269$$

$$D^2 = (1.269)^3$$

$$D \approx 1.429\,\text{m}$$

$$\boxed{1.429}$$

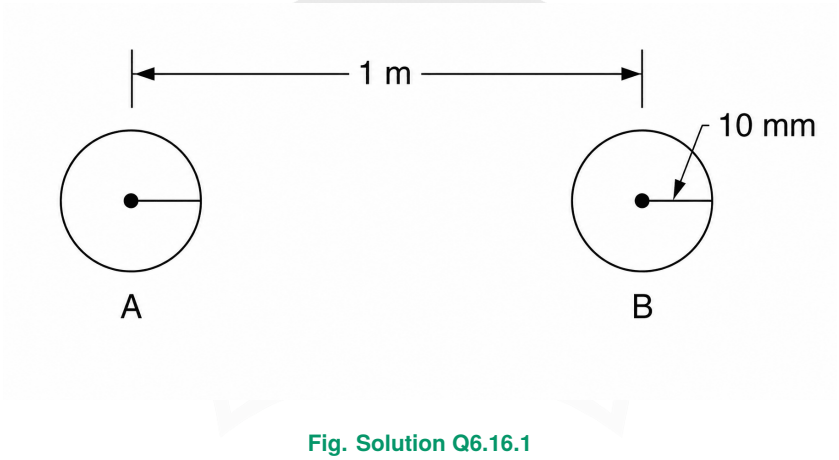



Fig. Solution Q6.16.1

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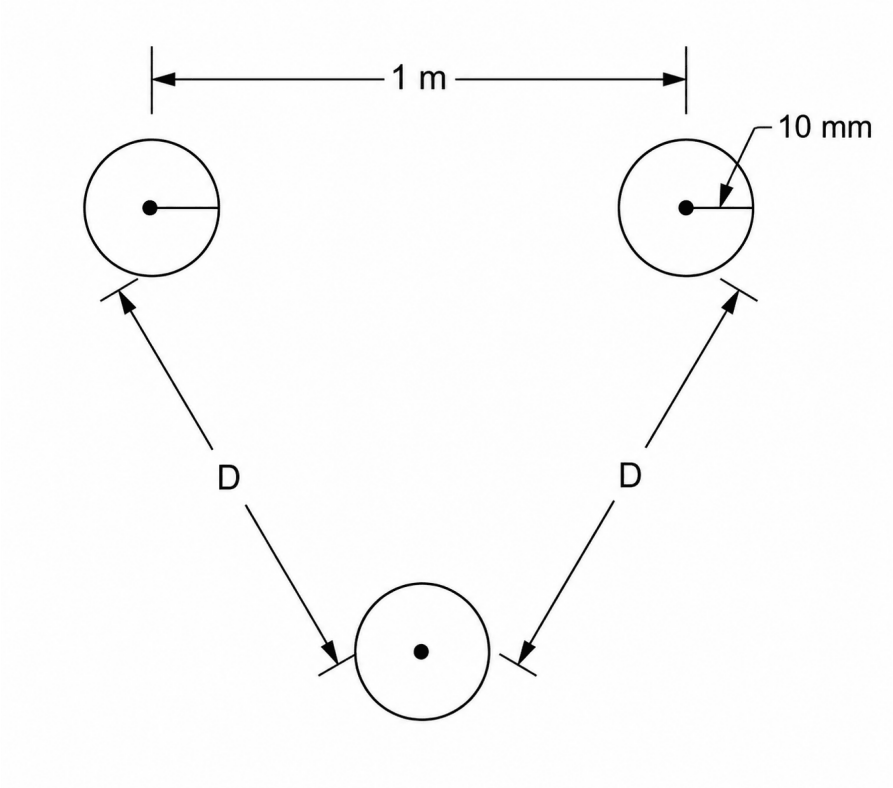


Fig. Solution Q6.16.1

**Q6.15.1** NAT | 2015 | 2M | Ans: 1.85 to 1.95 | ● ○ ○

For a composite conductor with symmetrical spacing,

$$\text{GMR} = \sqrt[3]{\text{GMR}_1\text{GMR}_2\text{GMR}_3}$$

For each subconductor,

$$\text{Self-GMR} = 0.7788R$$

From the figure, spacing between every pair of conductors is

$$D = 3R$$

Hence,

$$\text{GMR} = \sqrt[3]{(0.7788R)(3R)(3R)}$$

$$\text{GMR} = \sqrt[3]{0.7788 \times 3 \times 3}\,R$$

$$\text{GMR} = kR$$

Therefore,

$$k = \sqrt[3]{0.7788 \times 9}$$

$$k = \sqrt[3]{7.0092}$$

$$k \approx 1.913$$

$$\boxed{1.913}$$

**Q6.14.1** MCQ | 2014 | 2M | Ans: B | ● ○ ○

For a single-phase two-wire transmission line, capacitance to neutral is

$$C = \frac{2\pi\varepsilon_0}{\ln\left(\frac{D}{r}\right)} \quad \text{F/m}$$

Given,

$$\varepsilon_0 = 8.854 \times 10^{-12}\ \text{F/m}$$

Spacing between conductors,

$$D = 6\,\text{m}$$

Outside diameter,

$$d = 1.6\,\text{cm}$$

Hence, conductor radius is

$$r = 0.8\,\text{cm} = 0.008\,\text{m}$$

Substituting,

$$C = \frac{2\pi(8.854 \times 10^{-12})}{\ln\left(\frac{6}{0.008}\right)}$$

$$C = \frac{2\pi(8.854 \times 10^{-12})}{\ln(750)}$$

$$C \approx 8.39 \times 10^{-12}\ \text{F/m}$$

Capacitance per kilometer is

$$C_{\text{km}} = 8.39 \times 10^{-12} \times 10^{3}$$

$$C_{\text{km}} \approx 8.4 \times 10^{-9}\ \text{F}$$

Therefore,

$$\boxed{\text{(B)}\ 8.4 \times 10^{-9}\ \text{F}}$$

**Q6.10.1** MCQ 2010 2M Ans: A ● ● ○

Since the field voltage is held constant,

$$I_f = \text{constant}$$

Hence, the field current does not change.
When a synchronous generator is connected to a long lossless transmission line that is open-circuited at the receiving end, the line draws charging current due to its distributed capacitance.

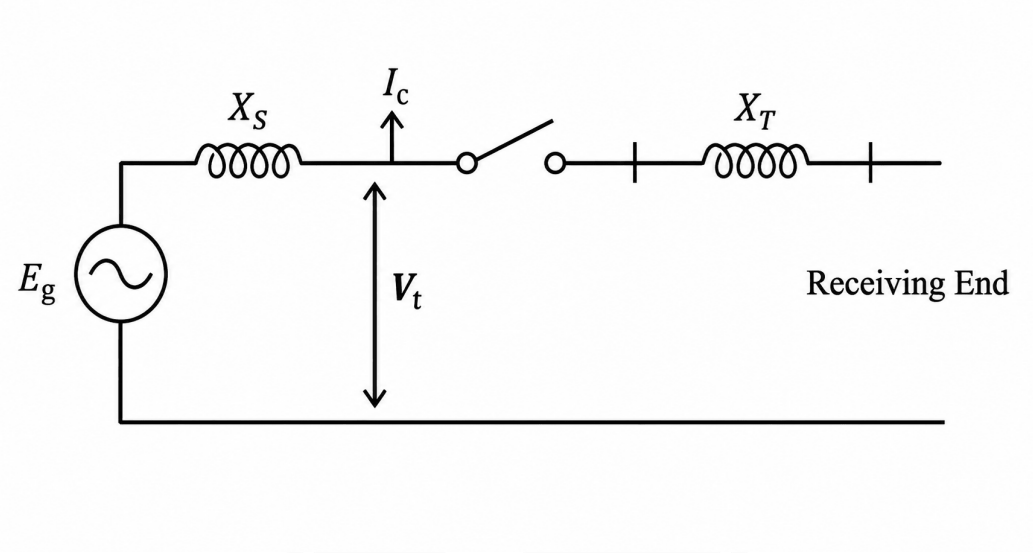


Fig. Solution Q6.10.1

The charging current produces the Ferranti-effect condition and causes the generator terminal voltage to rise.
Thus, while connected to the transmission line,

$$V_t > E_g$$

where

$$E_g = \text{generated emf}$$

After disconnecting the transmission line, the charging current becomes zero,

$$I_c = 0$$

and the generator effectively operates under no-load conditions.
Therefore,

$$E_g = V_t$$

Since previously

$$V_t > E_g$$

the terminal voltage decreases after the line is disconnected.
Hence,

- Terminal voltage decreases.
- Field current remains unchanged.

(A) The magnitude of terminal voltage decreases, and the field current does not change.

**Key Points**

- A long open-circuited transmission line draws charging current because of line capacitance.
- Ferranti effect causes the voltage at the line terminals to increase.
- Constant field voltage implies constant field current.
- Removing the transmission line removes the charging-current effect.

## Q6.10.2 MCQ 2010 2M Ans: B ●○○

Given,

$$V_{\text{phase}} = \frac{11}{\sqrt{3}}\ \text{kV}$$

The equivalent capacitance network is

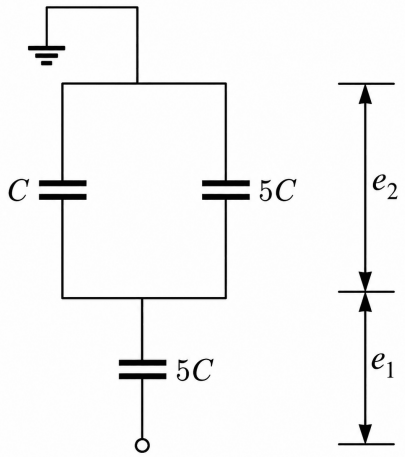


Fig. Solution Q6.10.2

Combining the shunt capacitance with the upper insulator capacitance,

$$C + 5C = 6C$$

Hence the equivalent circuit becomes

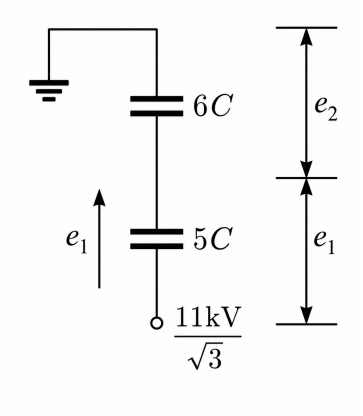


Fig. Solution Q6.10.2

The two capacitances $5C$ and $6C$ are in series across

$$\frac{11}{\sqrt{3}}\ \text{kV}$$

Voltage across the lower insulator is

$$e_1 = \frac{11}{\sqrt{3}}\left(\frac{6C}{6C+5C}\right)$$

$$e_1 = \frac{11}{\sqrt{3}} \times \frac{6}{11}$$

$$e_1 = 3.46\ \text{kV}$$

Voltage across the upper insulator is

$$e_2 = \frac{11}{\sqrt{3}}\left(\frac{5C}{6C+5C}\right)$$

$$e_2 = \frac{11}{\sqrt{3}} \times \frac{5}{11}$$

$$e_2 = 2.89\ \text{kV}$$

Therefore,

$$e_1 = 3.46\ \text{kV}, \quad e_2 = 2.89\ \text{kV}$$

Hence, the correct answer is

$$\boxed{\text{(B) } e_1 = 3.46\,\text{kV},\ e_2 = 2.89\,\text{kV}}$$

## Q6.10.3 MCQ 2010 2M Ans: A ●●○

Given,

$$f = 50\ \text{Hz}$$

$$V_{LL} = 11\ \text{kV}$$

$$C_1 = 0.2\ \mu\text{F}$$

$$C_2 = 0.4\ \mu\text{F}$$

For a symmetrical three-core cable, the effective capacitance per phase is

$$C = 3C_1 + C_2$$

$$C = 3(0.2) + 0.4$$

$$C = 1.0\ \mu\text{F}$$

The angular frequency is

$$\omega = 2\pi f$$

$$\omega = 2\pi(50)$$

$$\omega = 314\ \text{rad/s}$$

Phase voltage,

$$V_{ph} = \frac{11 \times 10^3}{\sqrt{3}}$$

The charging current per phase is

$$I_c = V_{ph}\omega C$$

$$I_c = \frac{11 \times 10^3}{\sqrt{3}} \times 314 \times 1 \times 10^{-6}$$

$$I_c = 1.995\ \text{A}$$

$$I_c \approx 2.0\ \text{A}$$

Hence, the correct answer is

(A) 2.0 A

**Q6.10.4** MCQ | 2010 | 1M | Ans: C | ● ○ ○

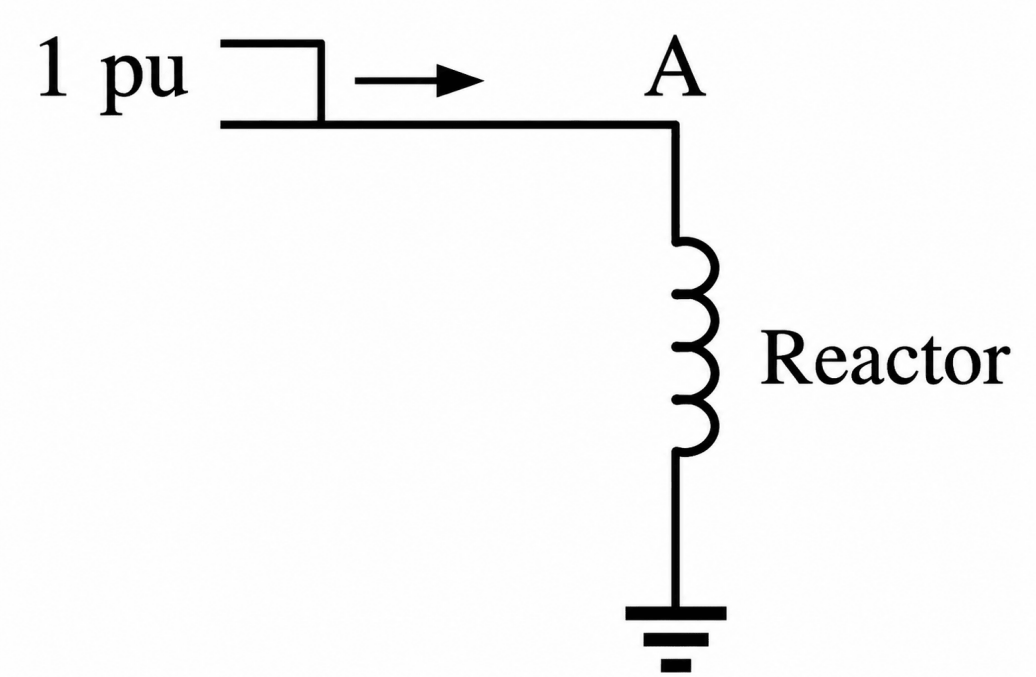


Fig. Solution Q6.10.4

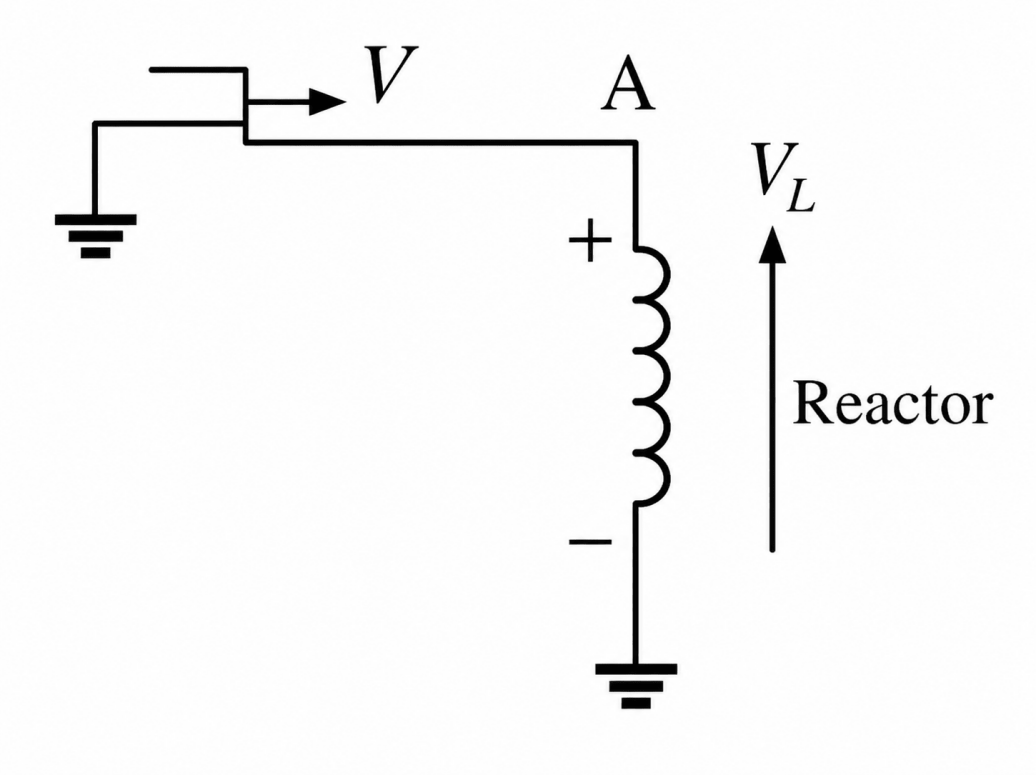


Fig. Solution Q6.10.4

The surge impedance (or characteristic impedance) of a transmission line is given by

$$Z_L = \sqrt{\frac{L}{C}}$$

Since the line capacitance is zero,

$$C = 0$$

Therefore,

$$Z_L = \sqrt{\frac{L}{0}} = \infty$$

Hence, the transmission line offers infinite surge impedance and behaves as an open circuit.
The transmitted voltage is related to the source voltage by

$$V_{\text{transmitted}} = V\sqrt{\frac{2Z_L}{Z_L + Z_C}}$$

Dividing the numerator and denominator inside the square root by $Z_L$,

$$V_{\text{transmitted}} = V\sqrt{\frac{2}{1 + \frac{Z_C}{Z_L}}}$$

Since

$$Z_L = \infty,$$

we have

$$\frac{Z_C}{Z_L} = \frac{Z_C}{\infty} = 0$$

Thus,

$$V_{\text{transmitted}} = V\sqrt{\frac{2}{1+0}} = V\sqrt{2}$$

Given that the system voltage is

$$V = \sqrt{2}\ \text{p.u.},$$

the transmitted voltage becomes

$$V_{\text{transmitted}} = \sqrt{2} \times \sqrt{2} = 2\ \text{p.u.}$$

(C) 2 p.u.

**Key Points**

- The characteristic impedance of a transmission line is $Z_L = \sqrt{L/C}$.
- If the line capacitance is zero, the surge impedance becomes infinite.
- An infinite surge impedance implies that the line behaves as an open circuit.
- Substituting $Z_L = \infty$ simplifies the voltage transmission expression considerably.

**Q6.04.1** MCQ 2004 2M Ans: C ● ○ ○

For a lossless line,

$$Z_c = \sqrt{\frac{L}{C}}$$

$$= \sqrt{\frac{1.1 \times 10^{-3}}{11.68 \times 10^{-9}}}$$

$$= 306.9\ \Omega$$

The surge impedance loading is

$$SIL = \frac{V_{LL}^2}{Z_c}$$

$$= \frac{(800)^2}{306.9}$$

$$= 2085\ \text{MW}$$

(C) 2085 MW

**Q6.03.1** MCQ 2003 1M Ans: B ● ○ ○

For bundled conductors, the equivalent self-GMD (GMR) increases because of the spacing between the sub-conductors.
An increase in effective conductor radius increases the critical disruptive voltage and thereby reduces corona formation.
Hence, bundled conductors are mainly used in EHV transmission lines to reduce corona loss, corona noise and radio interference.

(B) Reduce corona

**Key Points**

- Bundled conductors increase effective conductor radius.
- Increased effective radius reduces electric field intensity at the conductor surface.
- Corona loss and radio interference are reduced.

**Q6.03.2** MCQ 2003 2M Ans: D ● ● ○

The surge impedance of the incoming cable is

$$Z_1 = \sqrt{\frac{L_1}{C_1}} = \sqrt{\frac{0.4 \times 10^{-3}}{0.5 \times 10^{-6}}}$$

$$Z_1 = 28.28\ \Omega$$

For each overhead line,

$$Z_2 = \sqrt{\frac{L_2}{C_2}} = \sqrt{\frac{1.5 \times 10^{-3}}{0.015 \times 10^{-6}}}$$

$$Z_2 = 316.23\ \Omega$$

Since the two outgoing lines are identical,

$$Z_3 = Z_2$$

The voltage at the junction is

$$V_t = \frac{2V}{1 + \frac{Z_1}{Z_2} + \frac{Z_1}{Z_3}}$$

Substituting,

$$V_t = \frac{2 \times 20}{1 + 2\left(\frac{28.28}{316.23}\right)}$$

$$= 33.93\ \text{kV}$$

Therefore,

(D) 33.93 kV

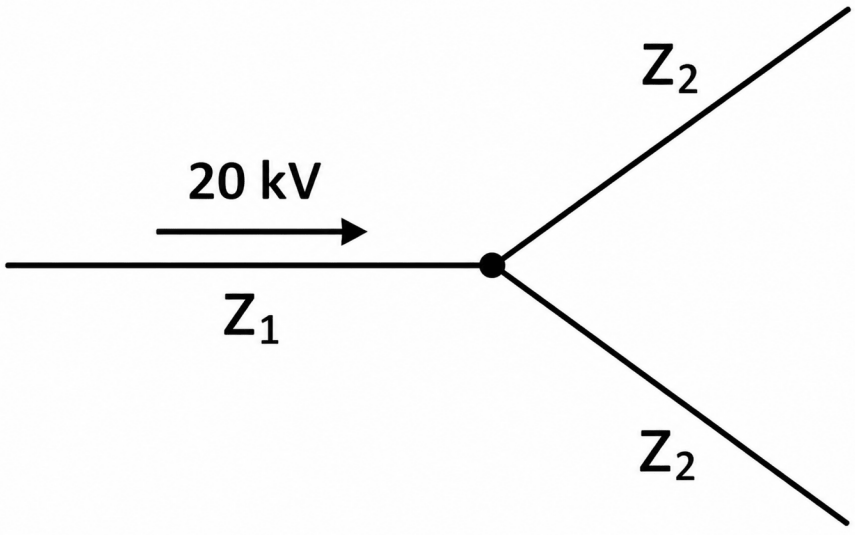


Fig. Solution Q6.03.2

## Key Points

- Surge impedance:

$$Z_c = \sqrt{\frac{L}{C}}$$

- At a junction, travelling-wave transmission depends on surge impedance matching.

- Parallel outgoing lines reduce the equivalent surge impedance seen at the junction.

**Q6.02.1** MCQ | 2002 | 1M | Ans: C | ● ○ ○

For a two-wire transmission line composed of solid round conductors, the inductance per conductor per unit length is

$$L = 2 \times 10^{-7} \ln\left(\frac{d}{r'}\right) \text{ H/m}$$

where

$$r' = 0.7788r$$

Given,

$$r = 0.25\,\text{cm}$$

Initially,

$$d_1 = 1\,\text{m} = 100\,\text{cm}$$

After doubling,

$$d_2 = 2\,\text{m} = 200\,\text{cm}$$

Therefore,

$$L_1 = 2 \times 10^{-7} \ln\left(\frac{100}{0.7788 \times 0.25}\right)$$

$$L_2 = 2 \times 10^{-7} \ln\left(\frac{200}{0.7788 \times 0.25}\right)$$

Taking the ratio,

$$\frac{L_1}{L_2} = \frac{\ln\left(\frac{100}{0.7788 \times 0.25}\right)}{\ln\left(\frac{200}{0.7788 \times 0.25}\right)}$$

$$\frac{L_1}{L_2} = \frac{6.241}{6.934}$$

$$L_2 = \frac{6.934}{6.241} L_1$$

$$L_2 = 1.11 L_1$$

Thus, the inductance increases by approximately 11% and does not become twice its original value.

(C) increases but does not double

**Q6.01.1** MCQ | 2001 | 2M | Ans: C | ● ○ ○

For a single-phase two-wire line, the inductance per conductor is

$$L = 2 \times 10^{-7} \ln\left(\frac{D}{r'}\right) \text{ H/m}$$

where

$$D = 1.5 \text{ m} = 150 \text{ cm}$$

The conductor diameter is

$$1 \text{ cm}$$

Hence,

$$r = 0.5 \text{ cm}$$

and the effective radius is

$$r' = 0.7788r = 0.7788 \times 0.5 = 0.3894 \text{ cm}$$

Therefore,

$$L = 2 \times 10^{-7} \ln\left(\frac{150}{0.3894}\right)$$

$$L = 1.19 \times 10^{-3} \text{ H/km}$$

$$L = 1.19 \text{ mH/km}$$

Since the circuit consists of two conductors,

$$L_{\text{circuit}} = 2L = 2(1.19) = 2.38 \text{ mH/km}$$

For a line length of

$$10 \text{ km}$$

$$L_{\text{total}} = 2.38 \times 10 = 23.8 \text{ mH}$$

Therefore, the correct option is

$$\boxed{\text{(C) } 23.8 \text{ mH}}$$

**Q6.99.1** MCQ | 1999 | 2M | Ans: C | ● ○ ○

For a single phase line,

$$L = 2 \times 10^{-7} \ln\left(\frac{D}{r'}\right) \text{ H/m}$$

Multiplying by $10^6$ to convert into mH/km,

$$L = 0.2 \ln\left(\frac{D}{r'}\right)$$

For a solid conductor,

$$r' = 0.7788r$$

Diameter = 1 cm

$$r = 0.5\,\text{cm}$$

Hence,

$$L \approx 0.05 + 0.2\ln\left(\frac{60}{0.5}\right)\ \text{mH/km}$$

Therefore, the correct option is

$$\boxed{\text{(C) } 0.05 + 0.2\ln\left(\frac{60}{0.5}\right)}$$

**Q6.99.2** MCQ | 1999 | 1M | Ans: B | ● ○ ○

For a travelling wave incident from a line of surge impedance $Z_c = 400\,\Omega$ to a cable of surge impedance $Z_L = 100\,\Omega$, the transmitted voltage is

$$V_t = \left(\frac{2Z_L}{Z_L + Z_c}\right)V_i$$

Substituting,

$$V_t = \left(\frac{2 \times 100}{100 + 400}\right)50$$

$$V_t = \frac{200}{500} \times 50$$

$$V_t = 20\,\text{kV}$$

Hence, the correct option is

$$\boxed{\text{(B) } 20\,\text{kV}}$$

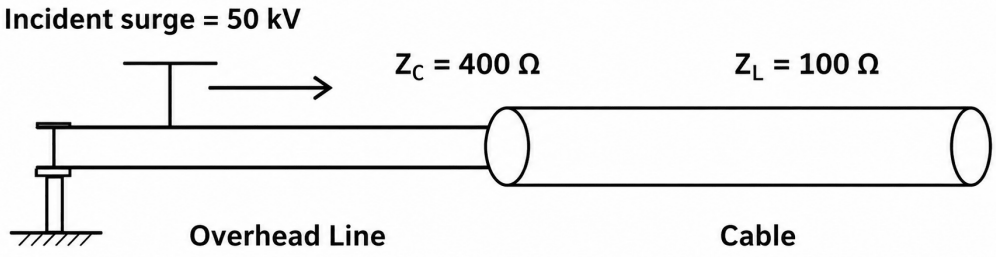


Fig. Solution Q6.99.2

**Q6.98.1** MCQ 1998 2M Ans: C ● ○ ○

The resistance of a conductor is

$$R = \frac{\rho l}{A}$$

Let the original length and area be $l_1$ and $A_1$.
Since the wire is uniformly stretched to $n$ times its original length,

$$l_2 = nl_1$$

The volume remains constant.

$$A_1 l_1 = A_2 l_2$$

Substituting $l_2 = nl_1$,

$$A_2 = \frac{A_1}{n}$$

The new resistance becomes

$$R_2 = \frac{\rho l_2}{A_2}$$

$$R_2 = \frac{\rho(nl_1)}{A_1/n}$$

$$R_2 = n^2 \frac{\rho l_1}{A_1}$$

$$R_2 = n^2 R_1$$

Hence,

(C) $R_{new} = n^2 R$

**Q6.97.1** MCQ 1997 1M Ans: B ● ○ ○

Bundled conductors are widely used in Extra High Voltage (EHV) transmission lines.
Using two or more sub-conductors per phase increases the effective radius (Geometric Mean Radius) of the conductor bundle without significantly increasing the overall conductor weight.
An increase in the effective conductor radius produces the following advantages:

- Reduces the electric field intensity at the conductor surface.
- Minimizes corona loss and audible noise.
- Reduces radio and communication interference.
- Improves the mechanical characteristics of the line by distributing mechanical stresses among multiple sub-conductors.

Hence, bundled conductors improve the mechanical stability of EHV transmission lines.

(B) Mechanical stability of the line

**Q6.97.2** NAT | 1997 | 2M | Ans: 173205080 | ● ○ ○

The velocity of propagation of an electromagnetic wave in a dielectric medium is

$$v = \frac{c}{\sqrt{\varepsilon_r}}$$

where

$$c = 3 \times 10^8 \text{ m/s}$$

and

$$\varepsilon_r = 3$$

Therefore,

$$v = \frac{3 \times 10^8}{\sqrt{3}}$$

$$v = 1.732 \times 10^8 \text{ m/s}$$

Hence,

$$\boxed{1.732 \times 10^8 \text{ m/s}}$$

**Q6.96.1** MCQ | 1996 | 2M | Ans: A | ● ● ○

Communication interference in a transmission line is primarily caused by the electrostatic induction due to unequal voltage distribution among the conductors.
For an untransposed three-phase line, unequal spacing results in unequal inductance and capacitance of the three phases. Consequently, the phase voltages become unbalanced, leading to communication interference.
However, if the three conductors are placed at the corners of an equilateral triangle, each conductor has identical spacing from the other two conductors. Therefore,

$$D_{ab} = D_{bc} = D_{ca}$$

which gives identical inductance and capacitance for all three phases.
As a result,

- the three-phase receiving-end voltages remain balanced, and
- the electric fields around the conductors are symmetrical, eliminating communication interference.

Hence, equilateral spacing provides both balanced receiving-end voltage and freedom from communication interference.

| | |
|---|---|
| (A) | Balanced receiving-end voltage and no communication interference |

**Key Points**

- Equilateral spacing makes the mutual spacing between all conductors identical.
- Equal spacing results in equal inductance and capacitance of all three phases.
- Balanced line parameters ensure balanced receiving-end voltage and negligible communication interference.

**Mistakes to be avoided**

- Do not confuse equilateral spacing with transposition. Both improve symmetry, but equilateral spacing alone already provides identical phase geometry.
- Communication interference is mainly associated with unbalanced electrostatic fields caused by unequal conductor spacing.
- Equal conductor spacing directly leads to balanced electrical parameters and balanced voltages.

**Q6.93.1** MCQ | 1993 | 1M | Ans: C | ● ● ○

The inductance and capacitance of a three-phase transmission line depend upon the spacing between conductors, which is represented by the **Geometric Mean Distance (GMD)**.
For a transposed three-phase line,

$$L \propto \ln\left(\frac{\text{GMD}}{\text{GMR}}\right)$$

and

$$C \propto \frac{1}{\ln\left(\frac{\text{GMD}}{\text{GMR}}\right)}$$

where

- GMD = Geometric Mean Distance between conductors,
- GMR = Geometric Mean Radius of the conductor.

Hence,

$$L \propto \ln(\text{GMD})$$

and

$$C \propto \frac{1}{\ln(\text{GMD})}$$

Case 1: Horizontal Configuration
The conductor spacings are

$$d,\ d,\ 2d$$

Therefore,

$$\text{GMD} = (d \times d \times 2d)^{1/3}$$

$$= 2^{1/3} d$$

$$\approx 1.26d$$

Case 2: Equilateral Triangle
All conductor spacings are equal.
Thus,

$$\text{GMD} = (d \times d \times d)^{1/3}$$

$$= d$$

Comparison
Comparing the two cases,

$$1.26d > d$$

Hence,

$$\text{GMD}_{\text{horizontal}} > \text{GMD}_{\text{triangle}}$$

Since inductance is directly proportional to the logarithm of GMD,

$$L_{\text{horizontal}} > L_{\text{triangle}}$$

Therefore, after rearranging into an equilateral triangle,

$$L$$

decreases.
Similarly,

$$C \propto \frac{1}{\ln(\text{GMD})}$$

A decrease in GMD decreases the denominator, thereby increasing capacitance.
Hence,

$$C$$

increases.
Therefore,

(C) Average capacitance will increase and inductance will decrease.

## Q6.92.1 MCQ 1992 1M Ans: C ●●○

**Concept:**
The inductance of a transmission line is determined by the magnetic flux linkage produced by the current flowing through the conductors. It depends primarily on the geometry of the conductor arrangement, particularly the spacing between conductors and the conductor radius.
For a three-phase transmission line with symmetrical spacing, the inductance per conductor is

$$L = 2 \times 10^{-7} \ln\left(\frac{\text{GMD}}{\text{GMR}}\right) \text{ H/m}$$

where
GMD = Geometric Mean Distance between phase conductors.
GMR = Geometric Mean Radius of the conductor.
Thus,

$$L \propto \ln\left(\frac{\text{GMD}}{\text{GMR}}\right)$$

Hence, inductance increases if GMD increases and decreases if GMR increases.

**Analysis of Options**

**Option (A): Decrease in line length**

Incorrect.

The above expression gives the inductance per unit length. Although the total inductance of a line increases with its length, decreasing the line length certainly cannot increase the inductance.

**Option (B): Increase in diameter of conductor**

Incorrect.

Increasing the conductor diameter increases the conductor radius and consequently increases the GMR.

Since

$$L \propto \ln\left(\frac{\text{GMD}}{\text{GMR}}\right),$$

an increase in GMR reduces the value of the logarithmic term. Therefore, the inductance decreases.

**Option (C): Increase in spacing between the phase conductors**

Correct.

Increasing the spacing between phase conductors increases the Geometric Mean Distance (GMD).

As GMD increases,

$$\ln\left(\frac{\text{GMD}}{\text{GMR}}\right)$$

also increases, thereby increasing the inductance of the transmission line.

**Option (D): Increase in load current carried by the conductors**

Incorrect.

Inductance is a physical property determined by the conductor geometry and magnetic medium. It is independent of the magnitude of load current (provided magnetic saturation is absent).

**Physical Interpretation**

When the conductors are moved farther apart, the magnetic flux produced by each conductor links a larger region before interacting with the other phase conductors. This results in greater external flux linkage and therefore higher inductance.

Conversely, increasing the conductor diameter increases the internal radius (GMR), reducing the effective flux linkage and hence reducing inductance.

Hence, the correct answer is

(C) Increase in spacing between the phase conductors

SOLUTIONS

# VII

# Power Factor Improvement and Voltage Control Techniques

## Topics

1. Power Factor Improvement - Series / Shunt Compensation by Capactors, SVC , TSC-TCR etc.
2. Methods to control voltage, Frequency and power factor

## Unit Snapshot*

| | | | | |
|---|---|---|---|---|
| **Stream:** EE | **Questions:** 28 | **MCQ:** 21 | **MSQ:** 0 | **NAT:** 7 |
| **1M:** 9 | **2M:** 16 | **Descriptive:** 3 | **Avg Weightage** ~2M | **Range:** 1M(2014)–3M(2025) |

**See preface for how Avg Weightage and Range are calculated.*

**Q7.25.1** NAT | 2025 | 1M | Ans: 0.975 to 0.985 | ● ○ ○

Given:

$$\frac{R}{X} = 5$$

Hence,

$$\frac{X}{R} = \frac{1}{5}$$

For a distribution feeder, voltage drop is proportional to

$$IR\cos\phi + IX\sin\phi$$

Maximum voltage drop occurs when

$$\tan\phi = \frac{X}{R}$$

Therefore,

$$\phi = \tan^{-1}\left(\frac{1}{5}\right)$$

$$\phi \approx 11.31^\circ$$

Thus,

$$\cos\phi = \cos(11.31^\circ)$$

$$\cos\phi \approx 0.981$$

Hence,

$$\boxed{0.981}$$

**Key Points**

- Voltage drop in a feeder is given by:
$$\Delta V \propto IR\cos\phi + IX\sin\phi$$
- Maximum voltage drop condition:
$$\tan\phi = \frac{X}{R}$$
- Power factor is:
$$\cos\phi$$
where $\phi$ is the load power-factor angle.

**Q7.25.2** NAT | 2025 | 2M | Ans: 8.5 to 10 | ● ○ ○

Given:

$$S = 390\,\text{kVA}$$

Initial power factor:

$$\cos\phi_1 = 0.77$$

Final power factor:

$$\cos\phi_2 = 0.85$$

Line voltage:

$$V_L = 4\,\text{kV}$$

**Step 1: Calculate Active Power**

$$P = S\cos\phi_1$$

$$P = 390 \times 0.77$$

$$P = 300.3\,\text{kW}$$

**Step 2: Determine Reactive Power Angles**

$$\tan\phi_1 = \tan(\cos^{-1}(0.77)) \approx 0.829$$

$$\tan\phi_2 = \tan(\cos^{-1}(0.85)) \approx 0.619$$

**Step 3: Capacitor Reactive Power Requirement**

$$Q_C = P(\tan\phi_1 - \tan\phi_2)$$

$$Q_C = 300.3(0.829 - 0.619)$$

$$Q_C = 63.063\,\text{kVAR}$$

**Step 4: Capacitor Bank Line Current**

For a 3-phase capacitor bank,

$$Q_C = \sqrt{3}V_L I_C$$

Therefore,

$$I_C = \frac{Q_C}{\sqrt{3}V_L}$$

$$I_C = \frac{63.063 \times 10^3}{\sqrt{3} \times 4 \times 10^3}$$

$$I_C \approx 9.10\,\text{A}$$

Hence,

$$\boxed{9.1}$$

**Q7.16.1** MCQ | 2016 | 1M | Ans: B | ● ○ ○

For a lossless transmission line, the surge impedance is

$$Z_n = \sqrt{\frac{L}{C}}$$

Given,

$$L = 1.6 \times 10^{-3}\ \text{H/km}$$

$$C = 10 \times 10^{-9}\ \text{F/km}$$

Therefore,

$$Z_n = \sqrt{\frac{1.6 \times 10^{-3}}{10 \times 10^{-9}}}$$

$$Z_n = 400\,\Omega$$

The Surge Impedance Loading (SIL) is

$$\text{SIL} = \frac{V_{LL}^2}{Z_n}$$

$$= \frac{400 \times 400}{400} = 400\ \text{MW}$$

Since the line is delivering

$$300\ \text{MW} < 400\ \text{MW}$$

the loading is less than SIL.
Hence, the line behaves capacitively and produces excess reactive power.
To maintain the receiving-end voltage at the desired value, inductive shunt compensation is required to absorb the excess reactive power.

(B) Inductive

**Key Points**

- Surge impedance:
$$Z_n = \sqrt{\frac{L}{C}}$$
- Surge impedance loading:
$$\text{SIL} = \frac{V_{LL}^2}{Z_n}$$
- If load < SIL, the transmission line behaves capacitively.
- If load > SIL, the transmission line behaves inductively.

**Q7.16.2** NAT | 2016 | 2M | Ans: 47 to 49 | ● ○ ○

Initial apparent power:

$$S_1 = \sqrt{800^2 + 600^2}$$

$$S_1 = 1000 \text{ kVA}$$

Since heat dissipation in the cable should remain unchanged, the cable current must remain constant. Therefore, the apparent power rating must remain:

$$S_2 = 1000 \text{ kVA}$$

After adding the additional resistive load,

$$P_2 = 800 + 100 = 900 \text{ kW}$$

Reactive power in the new condition:

$$Q_2 = \sqrt{S_2^2 - P_2^2}$$

$$Q_2 = \sqrt{1000^2 - 900^2}$$

$$Q_2 = 435.89 \text{ kVAr}$$

Hence, reactive power supplied by capacitor bank:

$$Q_C = 600 - 435.89$$

$$Q_C = 164.11 \text{ kVAr}$$

Per-phase reactive power:

$$Q_{C\phi} = \frac{164.11}{3}$$

$$Q_{C\phi} = 54.7 \text{ kVAr}$$

For star connection,

$$V_\phi = \frac{3.3}{\sqrt{3}} \text{ kV}$$

$$V_\phi = 1.905 \text{ kV}$$

Capacitive reactance per phase:

$$Q_{C\phi} = \frac{V_\phi^2}{X_C}$$

Hence,

$$X_C = \frac{(1.905 \times 10^3)^2}{54.7 \times 10^3}$$

$$X_C = 66.36\ \Omega$$

Now,

$$X_C = \frac{1}{2\pi f C}$$

Therefore,

$$C = \frac{1}{2\pi \times 50 \times 66.36}$$

$$C = 47.96\ \mu\text{F}$$

$$\boxed{47.96\ \mu\text{F}}$$

**Key Points**

- Constant heat dissipation implies constant cable current.
- Apparent power remains constant when current and voltage remain unchanged.

## Q7.14.1 MCQ 2014 1M Ans: C ● ○ ○

In **lightly loaded or no-load conditions**, long high-voltage transmission lines generate **excess reactive power** because of **line capacitance**.
This causes the **receiving-end** voltage to rise above the **sending-end voltage**, known as the **Ferranti effect**.
**Shunt reactors** are connected to **absorb** the excess reactive power **generated** by the transmission line.
Thus, shunt reactors help in **limiting over-voltages** under lightly loaded conditions.
Therefore, the correct option is

$$\boxed{\text{(C) limit over-voltages at the load side under lightly loaded condition.}}$$

**Key Points**

- Long transmission lines possess significant shunt capacitance.
- Under light-load conditions, line charging current causes voltage rise.
- Series compensation is used to reduce effective line reactance.
- Shunt reactors absorb reactive power generated by line capacitance.
- Shunt reactors are commonly installed at line ends and substations.

## Q7.14.2 NAT 2014 2M Ans: 0.75 ● ○ ○

Given that the load reactive power varies over the range

$$0.5\ \text{kVAR} \le Q_1 \le 1\ \text{kVAR},$$

let the capacitor supply a constant reactive power of $Q_C$.
The net reactive power drawn from the source is

$$Q = Q_1 - Q_C.$$

The design requirement is that the magnitude of the net reactive power should not exceed 0.25 kVAR for the entire operating range. Therefore,

$$|Q| = |Q_1 - Q_C| \le 0.25, \qquad \forall\, Q_1 \in [0.5, 1].$$

Since this condition must hold for every permissible value of $Q_1$, it is sufficient to examine the two extreme values of the interval.
For the minimum load reactive power,

$$Q_1 = 0.5 \text{ kVAR},$$

the inequality becomes

$$|0.5 - Q_C| \le 0.25.$$

This gives

$$-0.25 \le 0.5 - Q_C \le 0.25,$$

or equivalently,

$$0.25 \le Q_C \le 0.75.$$

Similarly, for the maximum load reactive power,

$$Q_1 = 1 \text{ kVAR},$$

the condition becomes

$$|1 - Q_C| \le 0.25.$$

Hence,

$$-0.25 \le 1 - Q_C \le 0.25,$$

which simplifies to

$$0.75 \le Q_C \le 1.25.$$

Both operating conditions must be satisfied simultaneously. Therefore, the permissible value of $Q_C$ is obtained from the intersection of the two intervals,

$$[0.25,\ 0.75] \cap [0.75,\ 1.25] = \{0.75\}.$$

Thus, the only capacitor rating that guarantees

$$|Q_1 - Q_C| \le 0.25 \text{ kVAR}$$

throughout the entire operating range is

$$\boxed{0.75 \text{ kVAR}}.$$

**Key Points**

- The inequality must hold for every value of $Q_1$ in the specified operating range.
- Both the minimum and maximum values of $Q_1$ must be checked to obtain the allowable interval for $Q_C$.
- The capacitor rating is obtained from the intersection of the intervals obtained from the two boundary conditions.

**Mistakes to be avoided**

- Do not verify the condition only at the maximum load reactive power.
- Remember that $|x| \le a$ is equivalent to $-a \le x \le a$.
- The final capacitor rating must satisfy both boundary conditions simultaneously, not either one individually.

**Q7.12.1** MCQ | 2012 | 2M | Ans: B | ● ○ ○

Given,

$$|V_1| = |V_2| = 1 \text{ pu}$$

$$X = 0.5 \text{ pu}$$

$$P_{D1} = 1 \text{ pu}$$

$$P_{D2} = 1 \text{ pu}$$

Since the transmission line is lossless, real power transferred from bus 1 to bus 2 is

$$P = \frac{|V_1||V_2|}{X} \sin(\theta_1 - \theta_2)$$

Substituting values,

$$1 = \frac{1 \times 1}{0.5} \sin(\theta_1 - \theta_2)$$

$$1 = 2 \sin(\theta_1 - \theta_2)$$

$$\sin(\theta_1 - \theta_2) = 0.5$$

$$\theta_1 - \theta_2 = 30^\circ$$

Since,

$$V_1 = 1\angle 0^\circ$$

therefore,

$$V_2 = 1\angle -30^\circ$$

Current flowing from bus 1 to bus 2 is

$$I_{12} = \frac{V_1 - V_2}{j0.5}$$

Substituting,

$$I_{12} = \frac{1\angle 0^\circ - 1\angle(-30^\circ)}{j0.5}$$

$$I_{12} = (1 - j0.268) \text{ pu}$$

Current drawn by load at bus 2 is

$$S_{D2} = V_2 I_2^*$$

Given,

$$S_{D2} = 1 \text{ pu}$$

Hence,

$$1 = 1\angle(-30^\circ)I_2^*$$

$$I_2 = 1\angle(-30^\circ)\text{ pu}$$

Current supplied by capacitor is

$$I_G = I_2 - I_{12}$$

$$I_G = 1\angle(-30^\circ) - (1 - j0.268)$$

$$I_G = 0.268\angle(-120^\circ)\text{ pu}$$

VAR rating of capacitor is

$$Q_C = |V_2||I_G|$$

$$Q_C = 1 \times 0.268$$

$$Q_C = 0.268\text{ pu}$$

Hence, the correct option is

(B) 0.268 pu

**Q7.11.1** MCQ | 2011 | 1M | Ans: C | ● ○ ○

The steady-state power transfer capability of a transmission line is

$$P_{\max} = \frac{|E||V|}{X}$$

where $X$ is the effective transfer reactance between the sending and receiving ends.
To increase power transfer capability, the effective reactance must be reduced.
A series capacitor introduces a capacitive reactance in series with the line, thereby reducing the net line reactance:

$$X_{\text{effective}} = X_L - X_C$$

As the effective reactance decreases,

$$P_{\max} = \frac{|E||V|}{X_{\text{effective}}}$$

increases.
Hence, the preferred method for enhancing power transmission capability in long EHV lines is series capacitive compensation.

(C)

**Q7.09.1** MCQ | 2009 | 2M | Ans: B | ● ○ ○

We match each application with the appropriate compensating device.

**(a) Improve power factor**

Power factor improvement is achieved by supplying leading reactive power through a shunt capacitor.

Before compensation,

$$\phi_1 = \tan^{-1}\left(\frac{Q_R}{P_R}\right)$$

$$pf_1 = \cos\phi_1$$

After shunt-capacitor compensation,

$$Q_R = Q_{\text{load}} - Q_C$$

$$\phi_2 = \tan^{-1}\left(\frac{Q_{\text{load}} - Q_C}{P_R}\right)$$

$$pf_2 = \cos\phi_2$$

Since

$$\phi_1 > \phi_2$$

$$pf_2 > pf_1$$

Therefore,

$$(a) \rightarrow (2)$$

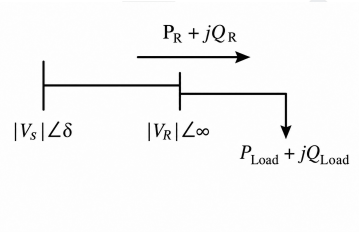


Fig. Solution Q7.09.1

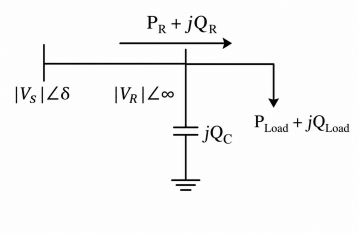


Fig. Solution Q7.09.1

**(b) Reduce current ripples**

A series reactor (inductor) opposes sudden current changes and smoothens current variations.

$$(b) \rightarrow (4)$$

**(c) Increase power flow in a transmission line**
Power transfer is

$$P = \frac{V_S V_R}{X} \sin \delta$$

A series capacitor reduces the effective line reactance $X$, thereby increasing power transfer capability.

$$(c) \rightarrow (3)$$

**(d) Reduce Ferranti effect**
Ferranti effect is caused by line charging capacitance under light-load conditions. A shunt reactor absorbs the excess reactive power and mitigates the effect.

$$(d) \rightarrow (1)$$

Thus,

$$a \rightarrow 2, \quad b \rightarrow 4, \quad c \rightarrow 3, \quad d \rightarrow 1$$

Hence, the correct answer is

(B) $a \rightarrow 2,\ b \rightarrow 4,\ c \rightarrow 3,\ d \rightarrow 1$

**Q7.08.1** MCQ | 2008 | 2M | Ans: C | ● ○ ○

For a lossless transmission line,

$$Z_c = \sqrt{\frac{L}{C}} = \sqrt{\frac{Z}{Y}}$$

where $Z_c$ is the surge impedance.
The Surge Impedance Loading (SIL) is

$$\text{SIL} = \frac{V^2}{Z_c}$$

Given,

$$\text{SIL} = 2280 \text{ MW}$$

With 30% series capacitive compensation, the effective series impedance becomes

$$Z' = (1 - 0.3)Z = 0.7Z$$

Hence, the new surge impedance is

$$Z_c' = \sqrt{\frac{0.7Z}{Y}}$$

$$Z_c' = Z_c\sqrt{0.7}$$

Since

$$\text{SIL} = \frac{V^2}{Z_c}$$

for the same operating voltage,

$$\frac{\text{SIL}'}{\text{SIL}} = \frac{Z_c}{Z_c'}$$

$$= \frac{Z_c}{Z_c\sqrt{0.7}}$$

$$= \frac{1}{\sqrt{0.7}}$$

Therefore,

$$\text{SIL}' = \frac{2280}{\sqrt{0.7}}$$

$$= 2724.9 \text{ MW}$$

(C) $\text{SIL}' \approx 2725$ MW

**Q7.07.1** MCQ | 2007 | 2M | Ans: B

A unity power-factor load of 4 kW is connected between phase A and neutral.
Hence,

$$I_A = \frac{P}{V} = \frac{4000}{230} = 17.39 \text{ A}$$

Assume that the inductor is connected in phase B and the capacitor is connected in phase C.
For phase sequence ABC,

$$V_B = 230\angle -120^\circ$$

$$V_C = 230\angle 120^\circ$$

The corresponding currents are

$$I_B = \frac{230\angle(-120^\circ)}{jX_L} = \frac{230}{X_L}\angle(-210^\circ) = \frac{230}{X_L}\angle 150^\circ$$

$$I_C = \frac{230\angle 120^\circ}{-jX_C} = \frac{230}{X_C}\angle 210^\circ = \frac{230}{X_C}\angle(-150^\circ)$$

For zero neutral current,

$$I_A + I_B + I_C = 0$$

$$17.39 + \frac{230}{X_L}\angle 150^\circ + \frac{230}{X_C}\angle(-150^\circ) = 0$$

Separating imaginary parts,

$$\frac{230}{X_L}\sin 150^\circ - \frac{230}{X_C}\sin 150^\circ = 0$$

$$X_L = X_C = X$$

Now separating real parts,

$$17.39 + \frac{230}{X}\cos 150^\circ + \frac{230}{X}\cos 150^\circ = 0$$

$$17.39 - \frac{460}{X}\cos 30^\circ = 0$$

$$17.39 - \frac{398.37}{X} = 0$$

$$X = 22.91\,\Omega$$

Therefore,

$$L = \frac{X_L}{\omega} = \frac{22.91}{2\pi(50)} = 72.95\,\text{mH}$$

and

$$C = \frac{1}{\omega X_C} = \frac{1}{2\pi(50)(22.91)} = 138.95\,\mu\text{F}$$

$$C \approx 139.02\,\mu\text{F}$$

Thus, the correct answer is

(B) 72.95 mH in phase B and 139.02 $\mu$F in phase C

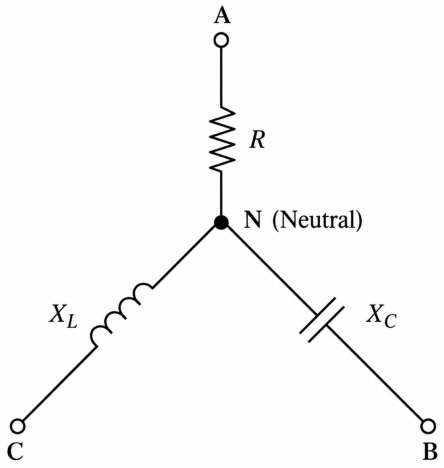


Fig. Solution Q7.07.1

**Q7.06.1** MCQ | 2006 | 2M | Ans: D

Given,

$$S_L = 12\sqrt{3}\,\text{kVA}$$

$$\cos\phi_1 = 0.8$$

Hence,

$$P_L = S_L \cos \phi_1$$

$$P_L = 12\sqrt{3} \times 0.8$$

$$P_L = 16.628 \text{ kW}$$

Since

$$\cos \phi_1 = 0.8$$

$$\sin \phi_1 = 0.6$$

Therefore, the reactive power demand of the load is

$$Q_L = S_L \sin \phi_1$$

$$Q_L = 12\sqrt{3} \times 0.6$$

$$Q_L = 12.47 \text{ kVAR}$$

For unity power factor,

$$\cos \phi_2 = 1$$

$$\phi_2 = 0^\circ$$

The capacitor bank rating required is

$$Q_C = P_L(\tan \phi_1 - \tan \phi_2)$$

$$Q_C = 16.628(0.75 - 0)$$

$$Q_C = 12.471 \text{ kVAR}$$

Thus, the correct answer is

(D) 12.47

**Q7.05.1** MCQ | 2005 | 1M | Ans: D | ●○○

For a transmission line,

$$P_{\max} = \frac{EV}{X}$$

For unchanged series reactance,

$$P_{\max} \propto V^2$$

Therefore,

$$\frac{P_{\max(400)}}{P_{\max(800)}} = \left(\frac{400}{800}\right)^2$$

$$= \frac{1}{4}$$

Since

$$P_{\max(800)} = P$$

$$P_{\max(400)} = \frac{P}{4}$$

Hence, the correct option is

$$\boxed{\text{(D) } \frac{P}{4}}$$

**Q7.03.1** MCQ | 2003 | 2M | Ans: B | ● ○ ○

For the delta-connected load,

$$Z = 8 + j6$$

$$|Z| = 10\ \Omega$$

$$\phi_1 = \tan^{-1}\left(\frac{6}{8}\right) = 36.87^\circ$$

Hence,

$$\cos\phi_1 = 0.8$$

Since the load is delta connected,

$$V_{ph} = V_L = 400\ \text{V}$$

Per-phase current is

$$I_{ph} = \frac{V_{ph}}{|Z|} = \frac{400}{10} = 40\ \text{A}$$

The active power consumed by the load is

$$P = 3V_{ph}I_{ph}\cos\phi_1$$

$$= 3 \times 400 \times 40 \times 0.8$$

$$= 38.4\ \text{kW}$$

Desired power factor:

$$\cos\phi_2 = 0.9$$

$$\phi_2 = 25.84^\circ$$

The required capacitor bank rating is

$$Q_c = P\,(\tan\phi_1 - \tan\phi_2)$$

$$= 38.4\,(\tan 36.87^\circ - \tan 25.84^\circ)$$

$$= 38.4(0.75 - 0.484)$$

$$= 10.2 \text{ kVAR}$$

Therefore,

$$\boxed{\text{(B) } Q_c = 10.2 \text{ kVAR}}$$

**Q7.03.2** MCQ | 2003 | 2M | Ans: B | ●●○

The line reactance on actual ohmic base is

$$X = 0.2\left(\frac{11^2}{100}\right) = 0.242\ \Omega$$

For the compensated condition,

$$|V_S| = |V_R| = 11 \text{ kV}$$

The active power transfer is

$$P = \frac{|V_S||V_R|}{X}\sin\delta$$

$$100 = \frac{11 \times 11}{0.242}\sin\delta$$

$$\sin\delta = 0.2$$

$$\delta = 11.53^\circ$$

Reactive power at the receiving end is

$$Q_R = \frac{|V_S||V_R|}{X}\cos\delta - \frac{|V_R|^2}{X}$$

$$Q_R = \frac{11 \times 11}{0.242}\cos 11.53^\circ - \frac{11^2}{0.242}$$

$$Q_R = -10.1 \text{ MVAR}$$

Since the load is operating at unity power factor,

$$Q_L = 0$$

Hence,

$$Q_R + Q_C = Q_L$$

$$Q_C = -Q_R$$

$$Q_C = 10.1 \text{ MVAR}$$

Therefore,

(B) $Q_C = 10.1$ MVAR

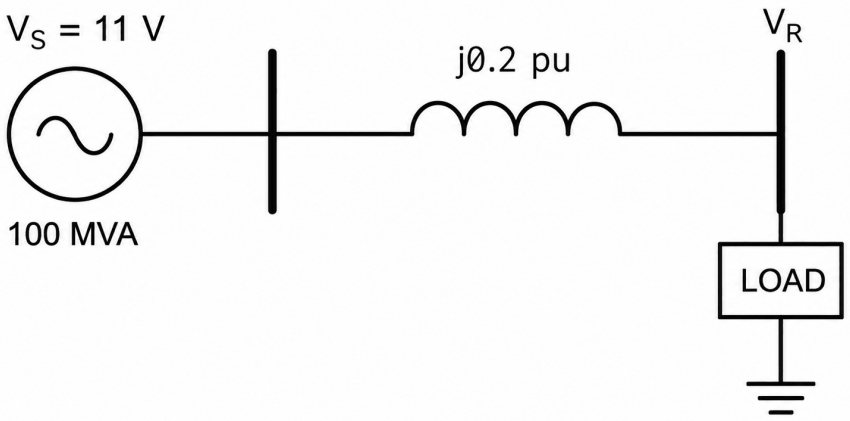


Fig. Solution Q7.03.2

**Q7.02.1** MCQ | 2002 | 2M | Ans: A

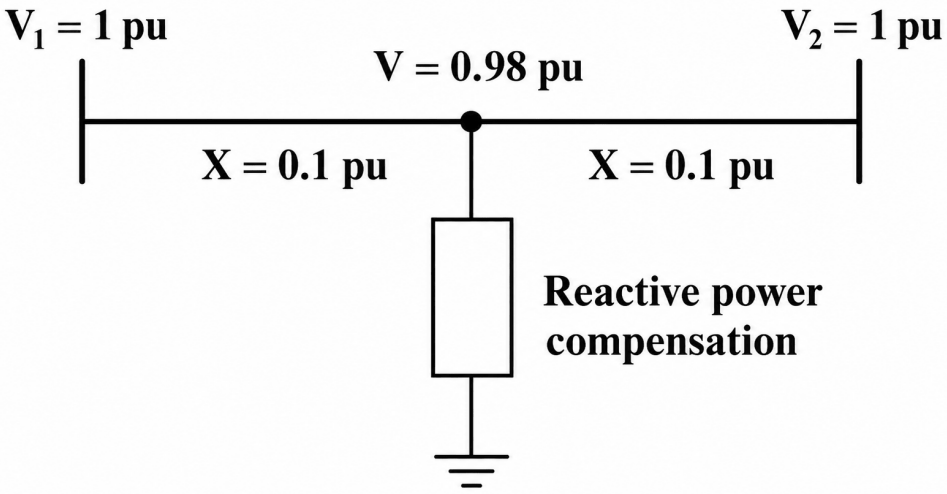


Fig. Solution Q7.02.1

The transmission line is compensated at its midpoint such that the midpoint voltage is maintained at

$$V_m = 0.98 \text{ pu}$$

The total line reactance is

$$X = 0.2 \text{ pu}$$

Hence, each half of the transmission line has reactance

$$\frac{X}{2} = 0.1 \text{ pu}$$

The steady-state stability limit for one half of the compensated line is

$$P_{SSL} = \frac{V_1 V_m}{X/2}$$

Substituting

$$V_1 = 1.0 \text{ pu}$$

and

$$V_m = 0.98 \text{ pu}$$

gives

$$P_{SSL} = \frac{(1)(0.98)}{0.1}$$

$$P_{SSL} = 9.8 \text{ pu}$$

Therefore, the steady-state power transfer limit of the transmission line is 9.8 pu
Hence, the correct answer is

(A) 9.8 pu

**Q7.01.1** MCQ | 2001 | 1M | Ans: C | ● ○ ○

For a lossless transmission line,

$$R = 0, \qquad G = 0$$

The voltage and current at a distance $x$ from the receiving end are

$$V(x) = V_R \cos(\beta x) + jZ_c I_R \sin(\beta x)$$

$$I(x) = -j\frac{V_R}{Z_c} \sin(\beta x) + I_R \cos(\beta x)$$

At surge impedance loading (SIL),

$$Z_L = Z_c$$

Hence,

$$V_R = I_R Z_c$$

Substituting into the transmission-line equations,

$$V(x) = V_R\,[\cos(\beta x) + j\sin(\beta x)]$$

$$V(x) = V_R e^{j\beta x}$$

Similarly,

$$I(x) = I_R\,[\cos(\beta x) + j\sin(\beta x)]$$

$$I(x) = I_R e^{j\beta x}$$

Therefore,

$$|V(x)| = |V_R|$$

and

$$|I(x)| = |I_R|$$

Thus, both voltage magnitude and current magnitude remain constant throughout the line.
Therefore, the voltage profile is flat.
Also,

$$Z_L = Z_c$$

and for a lossless line, the surge impedance is purely real. Hence voltage and current remain in phase throughout the line, giving

$$\text{power factor} = 1$$

at all points along the transmission line.
Therefore, a lossless transmission line operating at surge impedance loading has a flat voltage profile and unity power factor throughout its length.
Hence, the correct option is

(C) has flat voltage profile and unity power factor at all points along it

**Q7.01.2** MCQ | 2001 | 2M | Ans: C | ● ○ ○

Let

$$V_s = 1\angle 0^\circ \text{ pu}$$

and

$$I = 1\angle(-\delta)$$

since the receiving-end load operates at unity power factor.
For point $P_4$,

$$V_{P_4} = V_r\angle(-\delta)$$

Moving towards the sending end,

$$V_{P_3} = V_r\angle(-\delta) + (1\angle -\delta)(j0.1)$$

Therefore,

$$V_{P_3} = V_r\angle(-\delta) + 0.1\angle(90^\circ - \delta)$$

and

$$|V_{P_3}| = \sqrt{V_r^2 + 0.01} \tag{1}$$

Across the series capacitor,

$$V_{P_2} = V_r\angle(-\delta) + 1\angle(-\delta)\,[j0.1 - j0.1]$$

$$V_{P_2} = V_r\angle(-\delta)$$

Hence,

$$|V_{P_2}| = V_r$$

Similarly,

$$V_{P_1} = V_r\angle(-\delta) + (1\angle -\delta)\,[j0.2 - j0.1]$$

$$V_{P_1} = V_r\angle(-\delta) + 0.1\angle(90^\circ - \delta)$$

giving

$$|V_{P_1}| = \sqrt{V_r^2 + 0.01} \tag{2}$$

From (1) and (2),

$$|V_{P_3}| = |V_{P_1}| > |V_{P_2}| = |V_{P_4}|$$

The maximum voltage occurs immediately after the series capacitor, i.e. at

$$P_3$$

Hence, the correct answer is

$$\boxed{\text{(C) } P_3}$$

**Mistakes to be avoided**

- Do not compare algebraic voltages instead of phasor magnitudes.
- Account for the sign of the capacitive reactance.

**Q7.01.3** MCQ 2001 2M Ans: B ● ○ ○

Given governor–turbine characteristics,

$$P_1 = 50(50 - f)$$

and

$$P_2 = 100(51 - f)$$

Since the generators and transmission network are lossless,

$$P_1 + P_2 = P_D$$

where

$$P_D = 400 \text{ MW}$$

Substituting,

$$50(50 - f) + 100(51 - f) = 400$$

$$2500 - 50f + 5100 - 100f = 400$$

$$7600 - 150f = 400$$

$$150f = 7200$$

$$f = 48 \text{ Hz}$$

Hence, the system frequency is

(B) 48.0 Hz

**Q7.00.1** MCQ 2000 2M Ans: B ● ○ ○

For a lossless transmission line with equal voltages maintained at both ends,

$$P_{\max} = \frac{V^2}{X}$$

Let the original maximum power transfer capability be

$$P_1 = \frac{V^2}{X}$$

When voltage support is provided at the midpoint, the line is divided into two equal sections.
The effective reactance becomes

$$\frac{X}{2}$$

Hence,

$$P_2 = \frac{V^2}{X/2} = \frac{2V^2}{X} = 2P_1$$

Now consider voltage support at

$$75\%$$

of the distance from the sending end.
The smaller section has reactance

$$\frac{3X}{4}$$

The power transfer capability is limited by this section.
Therefore,

$$P_3 = \frac{V^2}{3X/4} = \frac{4}{3}\frac{V^2}{X} = \frac{4}{3}P_1 = \frac{1}{0.75}P_1$$

Hence, the correct answer is

$$\boxed{\text{(B) } 1 : 2 : \frac{1}{0.75}}$$

**Q7.00.2** NAT | 2000 | 5M | **Ans: 160.7 kV, 600 MW, Capacitive Compensation** | ● ● ○

For the 400 km lossless line,

$$Z = j0.05 \times 400$$

$$Z = j20\ \Omega$$

Similarly,

$$Y = j(3 \times 10^{-6}) \times 400$$

$$Y = j1.2 \times 10^{-3}\ \text{S}$$

Using the nominal-$\pi$ representation,

$$\frac{Y}{2} = j0.6 \times 10^{-3}\ \text{S}$$

Under open-circuit condition,

$$I_R = 0$$

Hence,

$$V_R = \frac{V_S}{\left(1 + \frac{ZY}{2}\right)}$$

Substituting the numerical values,

$$V_R = \frac{275/\sqrt{3}}{1 + \frac{(j20)(j1.2\times10^{-3})}{2}}$$

$$V_R = 160.7\ \text{kV}$$

For a flat voltage profile,

$$P_{SIL} = \frac{V_{LL}^2}{Z_c}$$

where

$$Z_c = \sqrt{\frac{L}{C}} = \sqrt{\frac{20}{1.2 \times 10^{-3}}} = 129.1\ \Omega$$

Therefore,

$$P_{SIL} = 3\frac{(160.7)^2}{129.1}$$

$$P_{SIL} \approx 600\ \text{MW}$$

The desired loading is

$$1.2 \times 600 = 720\ \text{MW}$$

Since the required loading exceeds SIL, the surge impedance must be reduced.
Therefore, shunt capacitive compensation is required.

Hence,the final answer is

| $(a)$ 160.7 kV | $(b)$ 600 MW | $(c)$ Capacitive compensation |
|---|---|---|

**Q7.99.1** MCQ | 1999 | 2M | Ans: B | ● ○ ○

Required capacitor kVAR

$$Q_c = P(\tan\phi_1 - \tan\phi_2)$$

$$\phi_1 = \cos^{-1}(0.8) = 36.87^\circ$$

$$\phi_2 = 0^\circ$$

$$Q_c = 200(0.75)$$

$$Q_c = 150\ \text{kVAR}$$

For delta connection,

$$Q_c = 3V_{ph}^2\omega C$$

Since

$$V_{ph} = V_L = 11 \times 10^3\ \text{V}$$

$$150 \times 10^3 = 3(11 \times 10^3)^2(2\pi \times 50)C$$

$$C = 1.316 \times 10^{-6}\ \text{F}$$

$$C = 1.316\ \mu\text{F}$$

Hence, the correct answer is

(B) 1.316 $\mu$F

**Q7.98.1** MCQ | 1998 | 1M | Ans: B | ● ○ ○

For a lossless transmission line, the transmitted power is given by

$$P = \frac{EV}{X}\sin\delta$$

where

- $E$ = sending-end voltage
- $V$ = receiving-end voltage
- $X$ = transfer reactance
- $\delta$ = power angle

Series capacitive compensation reduces the effective reactance of the transmission line.

$$X \downarrow$$

Therefore,

$$P_{\max} = \frac{EV}{X}$$

increases.
An increase in maximum transferable power improves the steady-state and transient stability limits of the system.

(B) Series compensation improves system stability

**Q7.98.2** MCQ | 1998 | 2M | Ans: D | ● ○ ○

For a shunt reactor,

$$Q = \frac{V^2}{X_L}$$

Since

$$X_L = \omega L$$

we have

$$Q = \frac{V^2}{\omega L}$$

Hence,

$$Q \propto \frac{V^2}{f}$$

Therefore,

$$\frac{Q_1}{Q_2} = \left(\frac{V_1}{V_2}\right)^2 \left(\frac{f_2}{f_1}\right)$$

Substituting

$$Q_1 = 100 \text{ MVAr}, \qquad V_2 = 0.98V_1, \qquad f_2 = 0.96f_1$$

$$\frac{100}{Q_2} = \left(\frac{1}{0.98}\right)^2 (0.96)$$

$$Q_2 = \frac{100(0.98)^2}{0.96} = 100.04 \text{ MVAr}$$

$$\boxed{\text{(D) } Q = 100.04 \text{ MVAr}}$$

**Q7.97.1** NAT | 1997 | 5M | Ans: 26.38 | ●●○

The domestic load current before motor starting is

$$I_L = 0.125\angle -36.86^\circ \text{ p.u.}$$

The 11-kV bus voltage during motor starting has already been obtained as

$$V = 0.74\angle 0.383^\circ \text{ p.u.}$$

The voltage at the domestic load terminals is obtained after accounting for the voltage drop across the feeder reactance of $j0.05$ p.u., i.e.,

$$V_L = V - I_L(j0.05).$$

The feeder voltage drop is

$$I_L(j0.05) = 0.125\angle -36.86^\circ \times 0.05\angle 90^\circ = 0.00625\angle 53.14^\circ.$$

Since phasor subtraction is most conveniently carried out in rectangular form,

$$V = 0.74(\cos 0.383^\circ + j \sin 0.383^\circ) = 0.73998 + j0.00495,$$

and

$$I_L(j0.05) = 0.00625(\cos 53.14^\circ + j \sin 53.14^\circ) = 0.00375 + j0.00500.$$

Therefore,

$$\begin{aligned} V_L &= (0.73998 + j0.00495) - (0.00375 + j0.00500) \\ &= 0.73623 - j0.00005. \end{aligned}$$

Converting back to polar form,

$$V_L = 0.7362\angle -0.004^\circ \text{ p.u.}$$

Thus, the domestic load voltage is

$$\boxed{V_L = 0.7362\angle -0.004^\circ \text{ p.u.}}$$

The percentage voltage reduction is

$$\frac{1-0.7362}{1}\times 100 = 26.38\%.$$

Hence,

$$\boxed{26.38\%}$$

**Key Points**

- Phasors should be converted to rectangular form before performing addition or subtraction.
- Multiplication by $j$ rotates a phasor by 90° without changing its magnitude.
- The feeder voltage drop is very small compared to the bus voltage, resulting in an angle very close to zero.

## Q7.97.2 NAT 1997 5M Ans: 7.76 ●●○

Given,

$$P = 100 \text{ kW}$$

$$\cos\phi_1 = 0.7$$

The line current is

$$I = \frac{100\times 10^3}{\sqrt{3}\times 11\times 10^3\times 0.7}$$

$$I = 7.498 \text{ A}$$

Load impedance:

$$Z = \frac{11\times 10^3/\sqrt{3}}{7.498}$$

$$Z = 847\angle 45.57^\circ\ \Omega$$

Resistance:

$$R = 592.93\ \Omega$$

For 0.95 p.f.,

$$Z' = 624.138\ \Omega$$

Required reactance:

$$X' = 194.892\ \Omega$$

Original reactance:

$$X = 604.85\ \Omega$$

Capacitive reactance:

$$X_C = 604.85 - 194.892$$

$$X_C = 409.95\ \Omega$$

$$C = \frac{1}{2\pi f X_C}$$

$$C = \frac{1}{2\pi(50)(409.95)}$$

$$C = 7.76\ \mu F$$

Therefore, the required rating of capacitor is

$$\boxed{7.76\ \mu F}$$

**Q7.94.1** MCQ | 1994 | 1M | Ans: D

A shunt reactor behaves as an inductive reactance.
The reactive power absorbed is

$$Q = \frac{V^2}{X}$$

The reactor rating is

$$Q_r = 50\ \text{MVAR}$$

at

$$V_r = 400\ \text{kV}$$

Therefore,

$$X = \frac{V_r^2}{Q_r}$$

When the operating voltage becomes

$$V = 360\ \text{kV}$$

the absorbed reactive power becomes

$$Q = Q_r \left(\frac{V}{V_r}\right)^2$$

$$= 50\left(\frac{360}{400}\right)^2$$

$$= 50(0.81)$$

$$= 40.5\ \text{MVAR}$$

Hence,

$$\boxed{\text{(D) } 40.5\ \text{MVAR}}$$

**Q7.91.1** MCQ | 1991 | 1M | Ans: B | ● ○ ○

**Concept:**
The efficiency of a transmission system depends on the transmission losses. For a given amount of power to be transmitted, reducing the line current minimizes the copper losses and thereby improves the transmission efficiency.
The transmitted three-phase power is

$$P = \sqrt{3} V_L I_L \cos\phi$$

where

- $P$ = Real power transmitted
- $V_L$ = Line voltage
- $I_L$ = Line current
- $\cos\phi$ = Power factor

Since the transmitted power is fixed,

$$P = \text{Constant}$$

the line current is

$$I_L = \frac{P}{\sqrt{3} V_L \cos\phi}$$

Hence,

$$I_L \propto \frac{1}{V_L \cos\phi}$$

Thus,

- Increasing the transmission voltage decreases the current.
- Improving the power factor also decreases the current.

The transmission (copper) loss is

$$P_{\text{loss}} = 3 I_L^2 R$$

where $R$ is the resistance of each phase conductor.
Since the losses are proportional to the square of the current,

$$P_{\text{loss}} \propto I_L^2$$

a reduction in current significantly reduces the transmission losses.
The transmission efficiency is

$$\eta = \frac{P_{\text{out}}}{P_{\text{out}} + P_{\text{loss}}} \times 100\%$$

Therefore, reducing the losses increases the efficiency.

**Analysis of Options**
**(A) Voltage decreases, power factor remains constant**

Incorrect.
Decreasing the voltage increases the current for the same transmitted power. Consequently, copper losses increase and the efficiency decreases.

**(B) Voltage increases, power factor increases**
Correct.
Both increasing the transmission voltage and improving the power factor reduce the line current. This minimizes the $I^2R$ losses and increases the transmission efficiency.

**(C) Voltage decreases, power factor decreases**
Incorrect.
Both changes increase the current, resulting in maximum transmission losses.

**(D) Voltage remains constant, power factor decreases**
Incorrect.
A lower power factor requires a higher current to transmit the same real power, thereby increasing copper losses.

**Engineering Insight**
For a fixed power transfer,

$$I \propto \frac{1}{V \cos \phi}$$

This is why Extra High Voltage (EHV) and Ultra High Voltage (UHV) transmission systems are employed. High transmission voltages enable large amounts of power to be transmitted with relatively low currents, reducing conductor size requirements and improving overall efficiency.
Hence, the correct option is

(B) Voltage increases, power factor increases

PrepFusion

SOLUTIONS

# VIII

# Economic Load Dispatch

## Topics

1. Economic Load Dispatch with transmission losses
2. Economic Load Dispatch without transmission losses

## Unit Snapshot*

| | | | | |
|---|---|---|---|---|
| **Stream:** EE | **Questions:** 18 | **MCQ:** 11 | **MSQ:** 0 | **NAT:** 7 |
| **1M:** 6 | **2M:** 10 | **Descriptive:** 2 | **Avg Weightage** ~2M | **Range:** 1M(2024)–2M(2026) |

**See preface for how Avg Weightage and Range are calculated.*

PrepFusion

**Q8.26.1** NAT 2026 2M Ans: 20.0

**Given Data:**

$$P_D = 1184\,\text{MW}$$

$$C_1(P_{G1}) = 1000 + 50P_{G1} + 0.01(P_{G1})^2 \quad \text{Rs}$$

$$C_2(P_{G2}) = 2000 + 50P_{G2} + 0.001(P_{G2})^2 \quad \text{Rs}$$

$$P_{\text{Loss}} = 0.001(P_{G2} - 50)^2 \quad \text{MW}$$

$$\lambda = 70.25$$

$$\frac{\partial C_1}{\partial P_{G1}} = 50 + 0.02P_{G1}$$

$$\frac{\partial C_2}{\partial P_{G2}} = 50 + 0.002P_{G2}$$

Since

$$P_{\text{Loss}} = 0.001(P_{G2} - 50)^2$$

we get

$$\frac{\partial P_{\text{Loss}}}{\partial P_{G1}} = 0$$

and

$$\frac{\partial P_{\text{Loss}}}{\partial P_{G2}} = 2 \times 0.001(P_{G2} - 50) = 0.002(P_{G2} - 50)$$

The optimal dispatch condition with losses is

$$\lambda\left(1 - \frac{\partial P_{\text{Loss}}}{\partial P_{Gi}}\right) = \frac{\partial C_i}{\partial P_{Gi}}$$

For Generator G1:

$$70.25(1 - 0) = 50 + 0.02P_{G1}$$

$$70.25 = 50 + 0.02P_{G1}$$

$$0.02P_{G1} = 20.25$$

$$P_{G1} = \frac{20.25}{0.02} = 1012.5\,\text{MW}$$

For Generator G2:

$$70.25\,[1 - 0.002(P_{G2} - 50)] = 50 + 0.002P_{G2}$$

$$70.25(1.1 - 0.002P_{G2}) = 50 + 0.002P_{G2}$$

$$77.275 - 0.1405P_{G2} = 50 + 0.002P_{G2}$$

$$27.275 = 0.1425P_{G2}$$

$$P_{G2} = \frac{27.275}{0.1425} \approx 191.4\,\text{MW}$$

$$P_{\text{Loss}} = 0.001(191.4 - 50)^2$$

$$= 0.001(141.4)^2$$

$$= 0.001 \times 19994 \approx 20.0\,\text{MW}$$

$$P_{G1} + P_{G2} - P_{\text{Loss}} = 1012.5 + 191.4 - 20.0$$

$$= 1183.9 \approx 1184\,\text{MW}$$

$$\boxed{20.0}$$

## Key Points

- The **optimality condition** with transmission **losses** is

$$\lambda\left(1 - \frac{\partial P_L}{\partial P_{Gi}}\right) = \frac{\partial C_i}{\partial P_{Gi}}$$

- The quantity

$$\left(1 - \frac{\partial P_L}{\partial P_{Gi}}\right)^{-1}$$

is called the **penalty factor**.

## Mistakes to be avoided

- Do not use the equal incremental cost criterion directly when transmission losses are present.
- For a lossy line; Power flow from each generator must be computed by optimality condition and not use total load demand as it may assume lossless condition.

**Q8.25.1** NAT | 2025 | 2M | Ans: 216 | ● ● ○

For economic load dispatch, the incremental cost characteristics are linear.

**Generator-1 Incremental Cost Equation**

Given data:

$$\lambda = 10000\,\text{Rs./MWh} \quad \Rightarrow \quad P_1 = 50\,\text{MW}$$

$$\lambda = 10600\,\text{Rs./MWh} \quad \Rightarrow \quad P_1 = 80\,\text{MW}$$

Slope of the incremental cost curve:

$$\frac{\Delta\lambda}{\Delta P_1} = \frac{10600 - 10000}{80 - 50}$$

$$= \frac{600}{30} = 20$$

Thus,

$$\lambda - 10000 = 20(P_1 - 50)$$

At

$$\lambda = 11800\,\text{Rs./MWh}$$

$$11800 - 10000 = 20(P_1 - 50)$$

$$1800 = 20(P_1 - 50)$$

$$P_1 - 50 = 90$$

$$P_1 = 140\,\text{MW}$$

But Generator-1 rating is

$$100\,\text{MW}$$

Hence,

$$P_1 = 100\,\text{MW}$$

**Generator-2 Incremental Cost Equation**

Given:

$$\lambda = 10000\,\text{Rs./MWh} \quad \Rightarrow \quad P_2 = 80\,\text{MW}$$

$$\lambda = 10600\,\text{Rs./MWh} \quad \Rightarrow \quad P_2 = 92\,\text{MW}$$

Therefore,

$$\frac{\Delta\lambda}{\Delta P_2} = \frac{600}{12} = 50$$

Thus,

$$\lambda - 10000 = 50(P_2 - 80)$$

At

$$\lambda = 11800 \text{ Rs./MWh}$$

$$1800 = 50(P_2 - 80)$$

$$P_2 - 80 = 36$$

$$P_2 = 116 \text{ MW}$$

Since Generator-2 rating is

$$150 \text{ MW}$$

the output is within limit.
Hence, total plant output is

$$P_T = P_1 + P_2$$

$$P_T = 100 + 116$$

$$P_T = 216 \text{ MW}$$

Therefore,

$$\boxed{216}$$

## Key Points

- Incremental cost curves are assumed linear over the operating region.
- Economic dispatch condition:

$$\lambda = \frac{dC_i}{dP_i}$$

- Generator ratings impose upper limits on generated power.
- Total plant output is the sum of individual generator outputs.

## Mistakes to be avoided

- Do not exceed generator MW ratings.
- Compute separate incremental cost equations for each generator.
- Apply generation limits before calculating total plant output.

**Q8.24.1** NAT | 2024 | 1M | Ans: 35 | ● ○ ○

Given,

$$\lambda = 7400\ \text{Rs./MWh}$$

From the graph, the incremental cost characteristics are:

$$IC_A(P_{GA}) = \frac{2000}{100}P_{GA} + 8000$$

$$IC_B(P_{GB}) = 40P_{GB} + 6000$$

For economic load dispatch:

$$\lambda = IC_A = IC_B$$

For Generator A:

$$20P_{GA} + 8000 = 7400$$

$$P_{GA} = -30\ \text{MW}$$

Since generation cannot be negative,

$$P_{GA} = 0$$

For Generator B:

$$40P_{GB} + 6000 = 7400$$

$$P_{GB} = \frac{7400 - 6000}{40}$$

$$P_{GB} = 35\ \text{MW}$$

Therefore,

$$P_D = P_{GA} + P_{GB}$$

$$P_D = 0 + 35$$

$$\boxed{35}$$

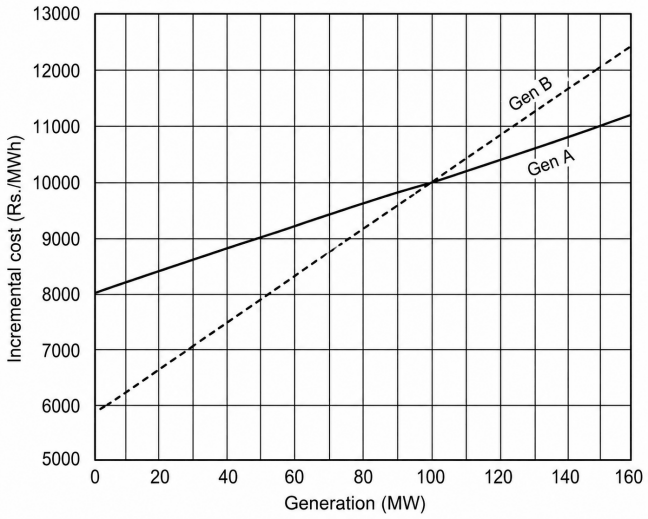


Fig. Solution Q8.24.1

**Key Points**

- In economic load dispatch, all operating generators have equal incremental cost.
- If the calculated generation becomes negative, that generator is set to its lower operating limit.

## Q8.23.1 MCQ 2023 1M Ans: A

For economic load dispatch without transmission losses,

$$\frac{dF_1}{dP_{G1}} = \frac{dF_2}{dP_{G2}} = \lambda$$

Incremental costs are:

$$\frac{dF_1}{dP_{G1}} = 0.2aP_{G1} + 40$$

$$\frac{dF_2}{dP_{G2}} = 0.4P_{G2} + 30$$

Initially,

$$P_{G1} = 175 \text{ MW}, \qquad P_{G2} = 115 \text{ MW}$$

Hence,

$$0.2a(175) + 40 = 0.4(115) + 30$$

$$35a + 40 = 46 + 30$$

$$35a = 36$$

$$a \approx 1.0286$$

Now, the value of $a$ increases by 10%.
Therefore, the coefficient of $P_{G1}^2$ in the cost function increases, making Generator 1 more expensive.
To satisfy economic dispatch condition again,

$$\frac{dF_1}{dP_{G1}} = \frac{dF_2}{dP_{G2}}$$

the generation from Unit 1 must reduce.
Thus,

$$P_{G1} \text{ decreases}$$

Since the overall fuel cost characteristic becomes steeper, the total generation cost increases.
Hence,

$$F \text{ increases}$$

Therefore, the correct option is

(A) $P_{G1}$ will decrease and $F$ will increase

**Q8.22.1** MCQ | 2022 | 2M | Ans: B

For economic load dispatch, the incremental costs of all generators operating in parallel must be equal.
Given fuel cost functions,

$$C_1 = 350 + 6P_1 + 0.004P_1^2$$

$$C_2 = 450 + aP_2 + 0.003P_2^2$$

Incremental cost of plant 1 is

$$(IC)_1 = \frac{dC_1}{dP_1} = 6 + 0.008P_1$$

Incremental cost of plant 2 is

$$(IC)_2 = \frac{dC_2}{dP_2} = a + 0.006P_2$$

At optimal operation,

$$(IC)_1 = (IC)_2 = \lambda$$

Given,

$$\lambda = 8$$

Therefore,

$$6 + 0.008P_1 = 8$$

$$0.008P_1 = 2$$

$$P_1 = 250\,\text{MW}$$

Total demand supplied is

$$P_1 + P_2 = 550\,\text{MW}$$

$$P_2 = 550 - 250$$

$$P_2 = 300\,\text{MW}$$

Hence, the optimal generations are

$$P_1 = 250\,\text{MW}$$

$$P_2 = 300\,\text{MW}$$

$$\boxed{\text{(B) } 250,\ 300}$$

**Key Points**

- In economic load dispatch, all generators operate at equal incremental cost.
- Incremental cost is obtained by differentiating the fuel cost function with respect to power output.
- Total generated power must satisfy the load demand constraint.

**Q8.21.1** MCQ | 2021 | 1M | Ans: B | ● ○ ○

For economic operation without transmission losses,

$$IC_1 = IC_2$$

Therefore,

$$40 + 0.2P_1 = 32 + 0.4P_2$$

Rearranging,

$$0.2P_1 - 0.4P_2 = -8$$

$$P_1 - 2P_2 = -40 \quad (1)$$

Also, total load demand is

$$P_1 + P_2 = 260 \quad (2)$$

Solving equations (1) and (2),

$$P_1 = 160\,\text{MW}$$

$$P_2 = 100\,\text{MW}$$

Hence,

$$\boxed{\text{(B) } P_1 = 160,\ P_2 = 100}$$

**Q8.15.1** NAT 2015 2M Ans: 20 ●○○

Given,

$$C_1(P_1) = 0.01P_1^2 + 30P_1 + 10$$

$$100\,\text{MW} \le P_1 \le 150\,\text{MW}$$

$$C_2(P_2) = 0.05P_2^2 + 10P_2 + 10$$

$$100\,\text{MW} \le P_2 \le 180\,\text{MW}$$

Total load demand is

$$P_1 + P_2 = 200\,\text{MW} \tag{1}$$

For economic dispatch,

$$\frac{dC_1}{dP_1} = \frac{dC_2}{dP_2}$$

Differentiating,

$$0.02P_1 + 30 = 0.1P_2 + 10$$

$$0.1P_2 - 0.02P_1 = 20 \tag{2}$$

Solving the economic dispatch equations without considering the generation limits from equations (1) and (2) gives

$$P_2 = 200\ \text{MW}, \qquad P_1 = 0\ \text{MW}.$$

However, this operating point is not feasible since

$$P_2 > 180\ \text{MW},$$

which violates the upper generation limit of Generator 2.
If Generator 2 is fixed at its maximum permissible output,

$$P_2 = 180\ \text{MW},$$

then, from the power balance equation,

$$P_1 = 200 - 180 = 20\ \text{MW}.$$

This also violates the operating constraint of Generator 1 because

$$P_1 < 100\ \text{MW},$$

which is below its minimum permissible generation.
Thus, the equal incremental cost solution cannot be achieved while satisfying the generator limits.
Since the total load demand is only 200 MW and both generators have a minimum operating limit of 100 MW, the only feasible generation schedule is

$$P_1 = 100\ \text{MW}, \qquad P_2 = 100\ \text{MW}.$$

The corresponding incremental cost is evaluated using the cost function of Generator 2,

$$\frac{dC_2}{dP_2} = 0.1P_2 + 10 = 0.1(100) + 10 = 20 \text{ Rs/MWh.}$$

Therefore, the required answer is

$$\boxed{20}$$

**Key Points**

- The unconstrained economic dispatch must always be verified against the generator operating limits.
- Enforcing one generator limit may cause another generator to violate its own operating limit.
- When the equal incremental cost criterion is infeasible, the operating point is determined by the active generation constraints.

**Q8.15.2** NAT | 2015 | 1M | Ans: 135 to 137 | ● ○ ○

Given,

$$\frac{dC_1}{dP_1} = 0.2P_1 + 50$$

$$\frac{dC_2}{dP_2} = 0.24P_2 + 40$$

Initially,

$$76 = 0.2P_1 + 50$$

$$P_1 = \frac{26}{0.2}$$

$$P_1 = 130 \text{ MW}$$

Similarly,

$$68.8 = 0.24P_2 + 40$$

$$P_2 = \frac{28.8}{0.24}$$

$$P_2 = 120 \text{ MW}$$

Hence, total load demand is

$$P_D = P_1 + P_2$$

$$P_D = 130 + 120$$

$$P_D = 250 \text{ MW}$$

For minimum operating cost,

$$\lambda = \frac{dC_1}{dP_1} = \frac{dC_2}{dP_2}$$

Thus,

$$0.2P_1 + 50 = 0.24P_2 + 40$$

Using power balance,

$$P_1 + P_2 = 250$$

$$P_1 = 250 - P_2$$

Substituting,

$$0.2(250 - P_2) + 50 = 0.24P_2 + 40$$

$$50 - 0.2P_2 + 50 = 0.24P_2 + 40$$

$$100 - 40 = 0.44P_2$$

$$60 = 0.44P_2$$

$$P_2 = 136.36\,\text{MW}$$

$$\boxed{136.36}$$

**Q8.14.1** MCQ | 2014 | 2M | Ans: D ●○○

Given,

$$C_1 = 0.05P_{g1}^2 + AP_{g1} + B \quad \text{(i)}$$

$$C_2 = 0.10P_{g2}^2 + 3AP_{g2} + 2B \quad \text{(ii)}$$

Total load demand is

$$P_{g1} + P_{g2} = 1000 \quad \text{(iii)}$$

At optimal economic dispatch,

$$\frac{dC_1}{dP_{g1}} = \frac{dC_2}{dP_{g2}} = 100 \quad \text{(iv)}$$

Differentiating the cost functions,

$$\frac{dC_1}{dP_{g1}} = 2(0.05)P_{g1} + A$$

$$0.1P_{g1} + A = 100 \quad \text{(v)}$$

Similarly,

$$\frac{dC_2}{dP_{g2}} = 2(0.10)P_{g2} + 3A$$

$$0.2P_{g2} + 3A = 100 \quad \text{(vi)}$$

Multiplying equation (v) by 3,

$$0.3P_{g1} + 3A = 300$$

Subtracting equation (vi),

$$0.3P_{g1} - 0.2P_{g2} = 200 \quad \text{(vii)}$$

Using equations (iii) and (vii),

$$P_{g1} = 800\,\text{MW}$$

$$P_{g2} = 200\,\text{MW}$$

Therefore,

$$\frac{P_{g1}}{P_{g2}} = \frac{800}{200} = \frac{4}{1}$$

Hence, the correct option is

(D) 4 : 1

**Q8.12.1** MCQ | 2012 | 1M | Ans: A | ● ● ○

For economic operation with transmission losses, penalty factor method is used. Penalty factor for generator $G_1$ is

$$L_1 = \frac{1}{1 - \frac{\partial P_L}{\partial P_{G1}}}$$

Given,

$$P_L = 0.5P_{G1}^2$$

Differentiating,

$$\frac{\partial P_L}{\partial P_{G1}} = P_{G1}$$

Hence,

$$L_1 = \frac{1}{1 - P_{G1}}$$

Since line loss depends only on $P_{G1}$,

$$\frac{\partial P_L}{\partial P_{G2}} = 0$$

Therefore,

$$L_2 = 1$$

For economic scheduling,

$$\lambda = \frac{dC_1}{dP_{G1}} L_1 = \frac{dC_2}{dP_{G2}} L_2$$

Given incremental fuel costs are constants,

$$\frac{dC_1}{dP_{G1}} = 10000$$

$$\frac{dC_2}{dP_{G2}} = 12500$$

Thus,

$$10000 \left( \frac{1}{1 - P_{G1}} \right) = 12500$$

$$\frac{1}{1 - P_{G1}} = \frac{5}{4}$$

$$1 - P_{G1} = \frac{4}{5}$$

$$P_{G1} = \frac{1}{5} \text{ pu}$$

On 100 MVA base,

$$P_{G1} = 0.2 \times 100 = 20 \text{ MW}$$

Now,

$$P_L = 0.5(0.2)^2 = 0.02 \text{ pu}$$

$$P_L = 0.02 \times 100 = 2 \text{ MW}$$

Power balance equation is

$$P_{G1} + P_{G2} = P_D + P_L$$

Substituting,

$$20 + P_{G2} = 40 + 2$$

$$P_{G2} = 22 \text{ MW}$$

Hence, the most economic schedule is

$$P_{G1} = 20 \text{ MW}, \quad P_{G2} = 22 \text{ MW}$$

Therefore, the correct option is

$$\boxed{(A)\ P_{G1} = 20 \text{ MW}, \quad P_{G2} = 22 \text{ MW}}$$

**Q8.11.1** MCQ | 2011 | 2M | Ans: A ●●○

Since all three generators have identical fuel cost characteristics, the economic dispatch is governed only by transmission losses.
For a given transmission voltage,

$$P_{\text{loss}} \propto I^2 R.$$

Since the line current is proportional to the transmitted power,

$$I \propto P,$$

it follows that

$$P_{\text{loss}} \propto P^2 R.$$

Further, the line resistance is directly proportional to the transmission distance,

$$R \propto \text{line length}.$$

The load centre is located only 25 km from Generator $G_1$, whereas it is 75 km from both $G_2$ and $G_3$. Hence,
power supplied from $G_1$ experiences significantly lower transmission losses than power supplied from either $G_2$ or $G_3$.
Since this is an objective-type question, it is convenient to examine the given schedules.

- Option (B) supplies only 60 MW from the nearest generator and shifts more power to the distant generators, thereby increasing transmission losses.
- Options (C) and (D) further reduce the contribution of $G_1$ while increasing the output of $G_2$ and $G_3$, resulting in still higher losses because a larger fraction of power is transmitted over the longer 75 km lines.
- Option (A) assigns the largest share of the load (80 MW) to the nearest generator while keeping the outputs of $G_2$ and $G_3$ at only 20 MW each. This minimizes the amount of power transmitted over the longer lines and therefore gives the minimum transmission loss among the available choices.

Also, the output of $G_1$ is within its maximum generation limit of 100 MW, so the schedule is feasible.
Hence, the minimum-loss operating schedule is

$$\boxed{\begin{aligned}(\text{A}) P_1 &= 80 \text{ MW} + \text{losses},\\ P_2 &= 20 \text{ MW},\\ P_3 &= 20 \text{ MW}.\end{aligned}}$$

**Key Points**

- Identical fuel cost characteristics imply that transmission loss is the deciding factor.
- Generators closer to the load centre should supply a larger share of the demand.
- For MCQs, comparing the given options often provides the quickest justification when an exact optimization is unnecessary.

**Q8.08.1** MCQ | 2008 | 2M | Ans: C | ● ○ ○

For a lossless system, the economic dispatch condition is

$$\frac{dC_1}{dP_{G1}} = \frac{dC_2}{dP_{G2}} = \lambda$$

The power balance equation is

$$P_{G1} + P_{G2} = 250 \tag{1}$$

Given,

$$C_1(P_{G1}) = P_{G1} + 0.055P_{G1}^2$$

$$C_2(P_{G2}) = 3P_{G2} + 0.03P_{G2}^2$$

The incremental costs are

$$\frac{dC_1}{dP_{G1}} = 1 + 0.11P_{G1} \tag{2}$$

$$\frac{dC_2}{dP_{G2}} = 3 + 0.06P_{G2} \tag{3}$$

Applying the economic dispatch criterion,

$$1 + 0.11P_{G1} = 3 + 0.06P_{G2}$$

$$0.11P_{G1} - 0.06P_{G2} = 2 \tag{4}$$

Using (1),

$$P_{G2} = 250 - P_{G1}$$

Substituting into (4),

$$0.11P_{G1} - 0.06(250 - P_{G1}) = 2$$

$$0.17P_{G1} = 17$$

$$P_{G1} = 100 \text{ MW}$$

Hence,

$$P_{G2} = 250 - 100 = 150 \text{ MW}$$

Therefore, the minimum-cost dispatch is

$$\boxed{\text{(C) } P_{G1} = 100 \text{ MW},\ P_{G2} = 150 \text{ MW}}$$

**Q8.07.1** MCQ | 2007 | 1M | Ans: C | ● ○ ○

For economic load dispatch without transmission losses,

$$\lambda = \frac{dC_A}{dP_A} = \frac{dC_B}{dP_B}$$

The incremental cost characteristics are linear.
For Generator A,

$$\frac{dC_A}{dP_A} = 450 \quad \text{at} \quad P_A = 200 \text{ MW}$$

$$\frac{dC_A}{dP_A} = 600 \quad \text{at} \quad P_A = 450 \text{ MW}$$

For Generator B,

$$\frac{dC_B}{dP_B} = 650 \quad \text{at} \quad P_B = 150 \text{ MW}$$

$$\frac{dC_B}{dP_B} = 800 \quad \text{at} \quad P_B = 400 \text{ MW}$$

Observe that even at its maximum output,

$$\left(\frac{dC_A}{dP_A}\right)_{\max} = 600 < 650 = \left(\frac{dC_B}{dP_B}\right)_{\min}$$

Thus Generator A is always cheaper than Generator B over their entire operating ranges.
Hence Generator A should be loaded up to its maximum limit first.

$$P_A = 450 \text{ MW}$$

The total demand is

$$P_D = 700 \text{ MW}$$

Therefore,

$$P_B = 700 - 450 = 250 \text{ MW}$$

Thus the optimum generation schedule is

$$P_A = 450 \text{ MW}$$

$$P_B = 250 \text{ MW}$$

Hence, the correct option is

(C) Generator A : 450 MW, Generator B : 250 MW

**Key Points**

- If one generator is cheaper over its entire operating range, it should be loaded to its maximum limit first.

**Mistakes to be avoided**

- Do not directly equate incremental costs without checking generation limits.

**Q8.05.1** MCQ 2005 2M Ans: C ● ○ ○

For economic operation,

$$\frac{dF_1}{dP_1} = \frac{dF_2}{dP_2}$$

Therefore,

$$b + 2cP_1 = b + 4cP_2$$

$$P_1 = 2P_2$$

The total load demand is

$$P_1 + P_2 = 300$$

Substituting,

$$2P_2 + P_2 = 300$$

$$3P_2 = 300$$

$$P_2 = 100 \text{ MW}$$

Hence,

$$P_1 = 200 \text{ MW}$$

Therefore, the correct answer is

$$\boxed{\text{(C) } P_1 = 200 \text{ MW},\ P_2 = 100 \text{ MW}}$$

**Q8.03.1** MCQ 2003 2M Ans: A ● ● ○

Given,

$$IC_1 = 20 + 0.3P_1$$

$$IC_2 = 30 + 0.4P_2$$

$$IC_3 = 30$$

The minimum and maximum generation limits are

$$50 \le P_i \le 300 \text{ MW}$$

for all generators.
At minimum loading,

$$IC_1 = 20 + 0.3(50) = 35$$

$$IC_2 = 30 + 0.4(50) = 50$$

$$IC_3 = 30$$

Generator 3 has the lowest incremental cost and hence operates at its upper limit:

$$P_3 = 300 \text{ MW}$$

The remaining demand is

$$P_1 + P_2 = 700 - 300 = 400 \text{ MW}$$

For economic dispatch,

$$IC_1 = IC_2$$

$$20 + 0.3P_1 = 30 + 0.4P_2$$

$$3P_1 - 4P_2 = 100$$

Together with

$$P_1 + P_2 = 400$$

Solving,

$$P_2 = 157.14 \text{ MW}$$

$$P_1 = 242.86 \text{ MW}$$

Hence,

$$\boxed{\text{(A) } P_1 = 242.86 \text{ MW},\ P_2 = 157.14 \text{ MW},\ P_3 = 300 \text{ MW}}$$

**Q8.01.1** NAT | 2001 | 5M | Ans: * | ● ○ ○

For economic operation,

$$\frac{dC_1}{dP_{G1}} = \frac{dC_2}{dP_{G2}}$$

The incremental cost characteristics are

$$\frac{dC_1}{dP_{G1}} = 0.012P_{G1} + 8$$

$$\frac{dC_2}{dP_{G2}} = 0.018P_{G2} + 7$$

Hence,

$$0.012P_{G1} + 8 = 0.018P_{G2} + 7$$

$$0.012P_{G1} - 0.018P_{G2} = -1 \quad (1)$$

Since transmission losses are neglected,

$$P_{G1} + P_{G2} = 600 \quad (2)$$

From (2),

$$P_{G2} = 600 - P_{G1}$$

Substituting into (1),

$$0.012P_{G1} - 0.018(600 - P_{G1}) = -1$$

$$0.03P_{G1} = 9.8$$

$$P_{G1} = 326.67 \text{ MW}$$

and

$$P_{G2} = 273.33 \text{ MW}$$

Both values satisfy the generator limits.
Hence, the correct answer is

$$\boxed{P_{G1} = 326.67\,\text{MW}, \qquad P_{G2} = 273.33\,\text{MW}}$$

**Q8.00.1** MCQ | 2000 | 2M | Ans: MTA | ● ○ ○

Given,

$$\frac{dF_1}{dP_1} = 2.0 + 0.01P_1$$

$$\frac{dF_2}{dP_2} = 1.6 + 0.2P_2$$

For economic operation,

$$\frac{dF_1}{dP_1} = \frac{dF_2}{dP_2}$$

Therefore,

$$2.0 + 0.01P_1 = 1.6 + 0.2P_2$$

$$0.01P_1 - 0.2P_2 = -0.4 \quad (1)$$

The total generation must equal the load demand.

$$P_1 + P_2 = 200 \quad (2)$$

From (2),

$$P_2 = 200 - P_1$$

Substituting into (1),

$$0.01P_1 - 0.2(200 - P_1) = -0.4$$

$$0.01P_1 - 40 + 0.2P_1 = -0.4$$

$$0.21P_1 = 39.6$$

$$P_1 = 188.57 \text{ MW}$$

Hence,

$$P_2 = 200 - 188.57$$

$$P_2 = 11.43 \text{ MW}$$

Thus, the economically optimal dispatch is

$$P_1 = 188.57 \text{ MW}$$

$$P_2 = 11.43 \text{ MW}$$

This exact solution is not present among the given options.

**Q8.98.1** NAT | 1998 | 5M | Ans: * | ●○○

For economic operation,

$$\frac{dF_1}{dP_1} = \frac{dF_2}{dP_2}$$

$$0.4P_1 + 30 = 0.5P_2 + 40$$

$$4P_1 - 5P_2 = 100 \quad (1)$$

Power balance equation:

$$P_1 + P_2 = 130 \quad (2)$$

Solving (1) and (2),

$$P_1 = 83.33 \text{ MW}$$

$$P_2 = 46.67 \text{ MW}$$

However,

$$P_1 > 80 \text{ MW}$$

which violates the upper limit.
Hence,

$$P_1 = 80 \text{ MW}$$

and

$$P_2 = 130 - 80$$

$$P_2 = 50 \text{ MW}$$

Therefore,

$$\boxed{P_1 = 80 \text{ MW}, \quad P_2 = 50 \text{ MW}}$$

PrepFusion

SOLUTIONS

# IX

# Overhead Transmission Lines (O.H.T.L.) and Underground Cables (UG Cable)

## Topics

1. Overhead Transmission Lines
2. Underground Cables
3. Electric Field Distribution and Insulators

## Unit Snapshot*

| | | | | |
|---|---|---|---|---|
| **Stream:** EE | **Questions:** 15 | **MCQ:** 8 | **MSQ:** 0 | **NAT:** 7 |
| **1M:** 9 | **2M:** 2 | **Descriptive:** 4 | **Avg Weightage** ~1M | **Range:** 0M(2014)–1M(2022) |

**See preface for how Avg Weightage and Range are calculated.*

**Q9.22.1** NAT | 2022 | 1M | Ans: 13 | ● ● ○

Given that only 5% of the supplied power reaches the output terminal,

$$P_{out} = 0.05\, P_{in}$$

Power loss in decibels is

$$\text{Loss (dB)} = 10 \log_{10}\left(\frac{P_{in}}{P_{out}}\right)$$

Substituting,

$$\text{Loss (dB)} = 10 \log_{10}\left(\frac{P_{in}}{0.05\, P_{in}}\right)$$

$$\text{Loss (dB)} = 10 \log_{10}(20)$$

$$\text{Loss (dB)} = 13.01 \text{ dB}$$

Rounding off to the nearest integer,

$$\boxed{13}$$

**Mistakes to be avoided**

- Do not use $20 \log_{10}$ since the problem involves power ratio, not voltage ratio.

**Q9.21.1** NAT | 2021 | 1M | Ans: 300 | ● ○ ○

The equivalent conductor resistance of the two cables connected in series is

$$(R_{eq})_{cond} = 0.7 + 0.5$$

$$(R_{eq})_{cond} = 1.2\,\Omega$$

The insulation resistances are connected in parallel.

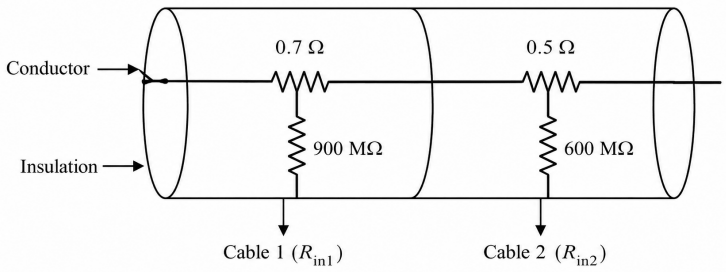


Fig. Solution Q9.21.1

Hence,

$$\frac{1}{(R_{eq})_{in}} = \frac{1}{600 \times 10^6} + \frac{1}{900 \times 10^6}$$

$$(R_{eq})_{in} = 360 \times 10^6\ \Omega$$

Required ratio is

$$\frac{(R_{eq})_{in}}{(R_{eq})_{cond}} = \frac{360 \times 10^6}{1.2}$$

$$= 300 \times 10^6$$

Therefore,

$$\boxed{300}$$

**Q9.14.1** MCQ | 2014 | 1M | Ans: B | ● ○ ○

An ideal electrical insulating material should possess:

- high dielectric strength
- high insulation resistivity
- good thermal conductivity
- low dielectric loss
- low relative permittivity

A high relative permittivity is generally undesirable because it increases the capacitance effect and dielectric stress in insulating systems.
Therefore, the undesirable property is

$$\boxed{\text{(B) high relative permittivity}}$$

**Q9.07.1** MCQ | 2007 | 1M | Ans: B | ● ● ○

Each sub-conductor produces its own radial electric field at its surface, while the remaining two sub-conductors produce additional electric fields whose magnitudes depend on the bundle spacing.
Since the conductor radius is much smaller than the spacing between adjacent sub-conductors,

$$r \ll d,$$

The electric field due to the conductor on whose surface the observation point lies is much larger than the fields contributed by the other two conductors. Hence, instead of assuming equal magnitudes, only the directions of the individual field vectors are used to compare the resultant electric field at different points.

**At Point** $Y$

Point $Y$ lies on the outer surface of the upper conductor. The electric field due to the upper conductor is directed radially outward. The fields produced by the other two sub-conductors also have outward components at this location. Consequently, these components add constructively, producing the largest resultant electric field.

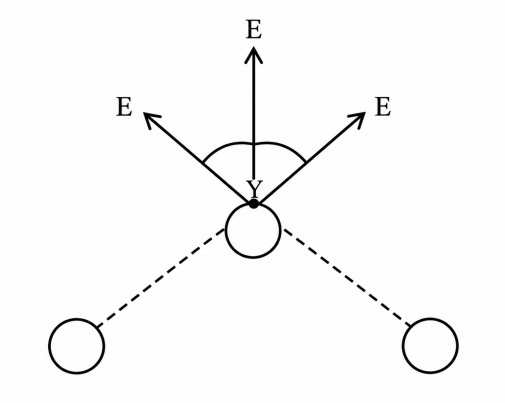


Fig. Solution Q9.07.1

**At Point** *X*

Point *X* lies on the inner surface of the upper conductor. Although the upper conductor still produces a radial outward field, the fields due to the other two sub-conductors are directed towards the bundle centre and therefore oppose the conductor's own field. As a result, the resultant electric field is reduced.

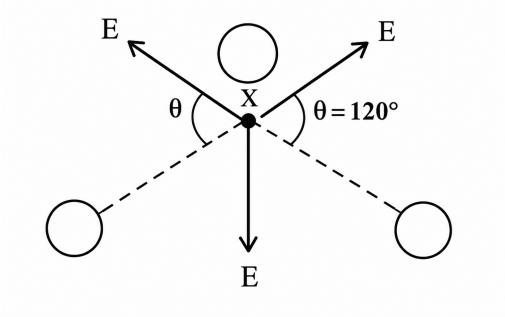


Fig. Solution Q9.07.1

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**At Point** *Z*

At Point *Z*, the field contributions from the adjacent sub-conductors partially cancel each other due to symmetry. The resultant electric field is therefore intermediate between those at Points *Y* and *X*.

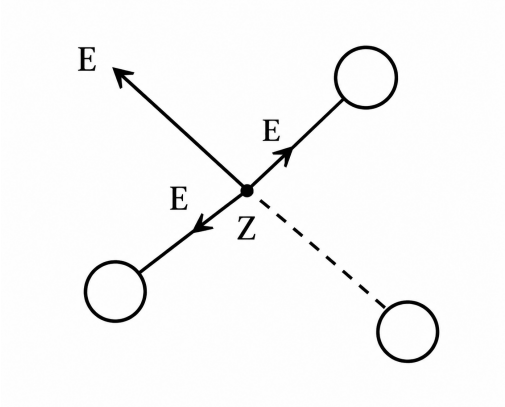


Fig. Solution Q9.07.1

**At Point** *W*

At Point $W$, the contributions from the neighbouring conductors oppose one another most effectively because of symmetry, giving the minimum resultant electric field among the four points.

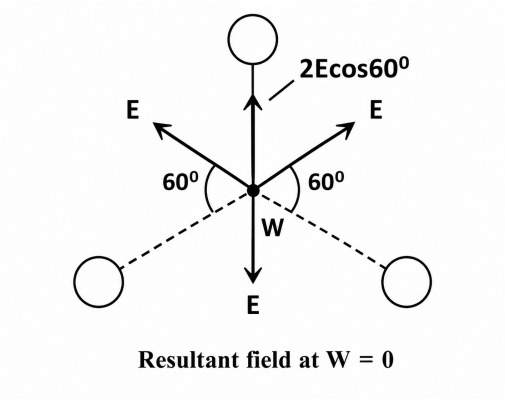


Fig. Solution Q9.07.1

Hence,

$$E_Y > E_Z > E_X > E_W.$$

Therefore, the maximum electric field intensity occurs at the outermost point of the bundle,

Point Y

Hence, the correct answer is

(B) Point Y

**Q9.05.1** MCQ | 2005 | 1M | Ans: B | ● ○ ○

In EHV transmission systems, switching over-voltages are generally more severe than lightning over-voltages and therefore govern the insulation design.
Hence, the insulation strength of an EHV transmission line is mainly determined by switching over-voltages.

(B) Switching over-voltages

**Q9.04.1** MCQ | 2004 | 2M | Ans: B | ● ● ○

Dielectric loss is

$$P_d = \omega C V_{ph}^2 \tan\delta$$

$$V_{ph} = \frac{110 \times 10^3}{\sqrt{3}} = 63.51 \times 10^3 \text{ V}$$

$$P_d = 2\pi(50)(125 \times 10^{-9})(63.51 \times 10^3)^2(2 \times 10^{-4})$$

$$P_d = 31.7 \text{ W/km}$$

(B) 31.7 W/km

**Q9.04.2** MCQ 2004 2M Ans: A ● ○ ○

The surge current divides equally into both directions.
Hence,

$$I = \frac{10}{2} = 5 \text{ kA}$$

The travelling surge voltage is

$$V = IZ_c$$

$$= 5 \times 10^3 \times 250$$

$$= 1.25 \times 10^6 \text{ V}$$

$$= 1250 \text{ kV}$$

(A) 1250 kV

**Q9.02.1** MCQ 2002 1M Ans: B ● ○ ○

For a long conductor above a conducting ground plane, the electric field intensity varies inversely with the distance from the conductor.

$$E = \frac{\lambda}{2\pi\varepsilon r}$$

where

$$E \propto \frac{1}{r}$$

Thus, the electric stress is maximum at the location where the distance from the conductor is minimum.
Using the method of images, the conducting ground plane can be replaced by an image conductor carrying equal and opposite charge. The electric field lines become denser near the lower portion of the conductor facing the ground.
Since the lower surface of the conductor is closest to the ground plane, the electric field intensity attains its maximum value there.
Hence, the maximum electric stress occurs at the lower surface of the conductor.
Therefore, the correct answer is

(B) the lower surface of the conductor

**Key Points**

- Electric stress is the electric field intensity at a point.
- For a line charge, electric field varies inversely with radial distance, i.e., $E \propto 1/r$.
- The conducting ground plane increases field concentration on the side of the conductor facing the ground.
- The method of images is commonly used to analyze conductor-ground configurations.

**Q9.01.1** NAT | 2001 | 5M | Ans: 28.57 kV | ● ○ ○

The incident voltage wave is

$$V_i = 100\,\text{kV}$$

The surge impedance of the overhead line is

$$Z_1 = 300\,\Omega$$

and the surge impedance of the cable is

$$Z_2 = 50\,\Omega$$

The transmission coefficient for voltage is

$$T_v = \frac{2Z_2}{Z_1 + Z_2}$$

Substituting,

$$T_v = \frac{2 \times 50}{300 + 50}$$

$$T_v = \frac{100}{350}$$

$$T_v = 0.2857$$

Hence, the transmitted voltage wave is

$$V_t = T_v V_i$$

$$V_t = 0.2857 \times 100$$

$$V_t = 28.57\,\text{kV}$$

The reflected voltage coefficient is

$$\Gamma = \frac{Z_2 - Z_1}{Z_2 + Z_1}$$

$$\Gamma = \frac{50 - 300}{50 + 300}$$

$$\Gamma = -0.7143$$

Thus, the reflected wave is

$$V_r = -71.43\,\text{kV}$$

The voltage at the junction immediately after incidence is

$$V_J = V_i + V_r$$

$$V_J = 100 - 71.43$$

$$V_J = 28.57\,\text{kV}$$

Hence, the required answer is

$$\boxed{28.57\,\text{kV}}$$

**Q9.99.1** NAT | 1999 | 5M | Ans: * | ● ○ ○

Given,

$$d = 1.5\ \text{cm}$$

$$D = 3\ \text{cm}$$

$$l = 4\ \text{km}$$

$$\rho = 1.3 \times 10^{12}\ \Omega\text{-m}$$

$$\varepsilon_r = 3.5$$

The conductor radius is

$$r = \frac{d}{2} = \frac{1.5}{2} = 0.75\ \text{cm} = 0.0075\ \text{m}.$$

The inner radius of the lead sheath is

$$R = \frac{D}{2} = \frac{3}{2} = 1.5\ \text{cm} = 0.015\ \text{m}.$$

**(a) Insulation Resistance**

The insulation resistance of a single-core cable is

$$R_{\text{ins}} = \frac{\rho}{2\pi l} \ln\left(\frac{R}{r}\right).$$

Substituting,

$$R_{\text{ins}} = \frac{1.3 \times 10^{12}}{2\pi(4000)} \ln\left(\frac{1.5}{0.75}\right).$$

Since

$$\ln 2 = 0.693,$$

$$R_{\text{ins}} = 35.85 \times 10^{6}\ \Omega.$$

Hence,

$$\boxed{R_{\text{ins}} = 35.85\ \text{M}\Omega}$$

**(b) Capacitance**

The capacitance per unit length of a single-core cable is

$$C' = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln(R/r)}.$$

Substituting,

$$C' = \frac{2\pi(8.854\times10^{-12})(3.5)}{\ln 2}.$$

$$C' = 280.91\times10^{-12}\ \text{F/m}.$$

Since the cable length is

$$l = 4\ \text{km} = 4000\ \text{m},$$

the total capacitance is

$$C = C'l.$$

Therefore,

$$C = (280.91\times10^{-12})(4000) = 1.124\times10^{-6}\ \text{F}.$$

Hence,

$$\boxed{C = 1.124\ \mu\text{F}}$$

**(c) Maximum Electric Stress**

The maximum electric stress occurs at the conductor surface and is given by

$$E_{\max} = \frac{V_{\text{phase}}}{r\ln(R/r)}.$$

Since the cable is rated at

$$V_L = 6.6\ \text{kV},$$

the phase voltage is

$$V_{\text{phase}} = \frac{V_L}{\sqrt{3}} = \frac{6.6}{\sqrt{3}} = 3.81\ \text{kV}.$$

Hence,

$$E_{\max} = \frac{\frac{6.6\times10^3}{\sqrt{3}}}{0.75\ln 2}.$$

Therefore,

$$E_{\max} = 7.33\ \text{kV/cm}.$$

Hence,

$$\boxed{E_{\max} = 7.33\ \text{kV/cm}}$$

Therefore, the required answers are

$$R_{ins} = 35.85\ \text{M}\Omega,$$
$$C = 1.124\ \mu\text{F},$$
$$E_{max} = 7.33\ \text{kV/cm}.$$

**Q9.99.2** NAT | 1999 | 5M | Ans: *

Because of the guard ring,

$$V_2 = V_3$$

Applying current balance at the junction,

$$\omega C V_2 = \omega C V_1 + \frac{\omega C}{5} V_1$$

$$V_2 = \frac{6}{5} V_1$$

Since

$$V_2 = V_3$$

and total string voltage is

$$V_1 + V_2 + V_3 = 20\ \text{kV}$$

$$V_1 + \frac{6}{5}V_1 + \frac{6}{5}V_1 = 20$$

$$\frac{17}{5}V_1 = 20$$

$$V_1 = 5.88\ \text{kV}$$

Therefore

$$V_3 = \frac{6}{5}V_1$$

$$V_3 = 7.05\ \text{kV}$$

**Guard Ring Capacitance**

Applying KCL,

$$I_3 + I_x = I_2 + i_2$$

$$\omega C V_3 + \omega C_x V_3 = \omega C V_2 + \frac{\omega C}{5}(V_1 + V_2)$$

Substituting

$$V_2 = V_3 = \frac{6}{5}V_1$$

and simplifying,

$$C_x = 0.367C$$

Hence, the required answer is

$$\boxed{V_3 = 7.05 \text{ kV}, \quad C_x = 0.367C}$$

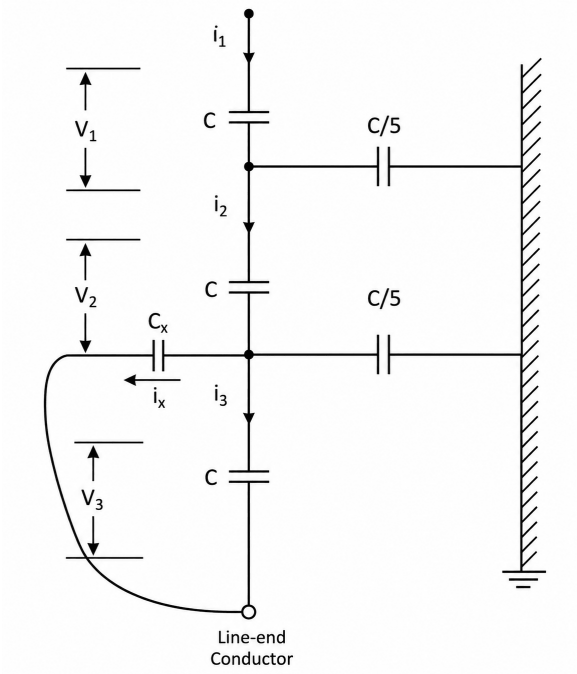


Fig. Solution Q9.99.2

**Q9.95.1** MCQ | 1995 | 1M | Ans: A | ● ○ ○

Voltage drop is the primary design criterion for sizing a distributor in a radial distribution system.
Unlike transmission lines, a distributor has loads tapped at various points along its length. Consequently, the current varies from one section of the distributor to another, causing different voltage drops along the feeder.
To ensure that all consumers receive voltage within the prescribed statutory limits, the conductor cross-sectional area is selected such that the voltage drop from the feeding point to the farthest consumer remains within the allowable limit.
Although factors such as capital cost, temperature rise, and corona loss are important in power system design, they are not the governing criterion for selecting the size of a radial distributor.
Therefore, the correct answer is

$$\boxed{\text{(A) Voltage drop}}$$

**Q9.95.2** MCQ | 1995 | 1M | Ans: C | ● ○ ○

Insulation resistance of a cable is inversely proportional to cable length.

$$R_i \propto \frac{1}{l}$$

Given

$$R_1 = 1\ \text{M}\Omega$$

for

$$l_1 = 10\ \text{km}$$

For

$$l_2 = 100\ \text{km}$$

$$R_2 = R_1 \frac{l_1}{l_2}$$

$$= 1 \times \frac{10}{100}$$

$$= 0.1\ \text{M}\Omega$$

Hence,

(C) 0.1 MΩ

## Q9.95.3 NAT 1995 1M Ans: LESS ●●○

The natural load of a transmission line or cable is the load at which reactive power generated by capacitance equals reactive power absorbed by inductance.
Underground cables possess:

- Very high capacitance
- Relatively low inductance

Consequently, their natural load is extremely high.
However, the thermal rating of the cable limits the actual load that can be carried continuously.
The conductor temperature, insulation temperature, and heat dissipation capability determine the rated load.
Therefore,

$$\text{Rated Load} < \text{Natural Load}$$

Hence,

LESS

## Q9.95.4 NAT 1995 5M Ans: * ●●○

Two identical transmission lines are connected in parallel.
The ABCD constants of each line are

$$A = D = 0.98\angle 1^\circ$$

$$B = 100\angle 75^\circ\ \Omega$$

$$C = 0.0005\angle 90^\circ\ \text{S}$$

Since the lines are identical,

$$A_1 = A_2 = A$$

$$B_1 = B_2 = B$$

$$C_1 = C_2 = C$$

$$D_1 = D_2 = D$$

**(a) ABCD Constants of the Parallel Combination**

For two identical networks connected in parallel,

$$A_{eq} = \frac{A_1B_2 + A_2B_1}{B_1 + B_2}$$

Substituting

$$A_{eq} = \frac{AB + AB}{B + B}$$

$$A_{eq} = A$$

Therefore

$$A_{eq} = 0.98\angle 1^\circ$$

Similarly,

$$B_{eq} = \frac{B_1B_2}{B_1 + B_2}$$

$$= \frac{B^2}{2B}$$

$$= \frac{B}{2}$$

Hence

$$B_{eq} = 50\angle 75^\circ\ \Omega$$

For $C_{eq}$,

$$C_{eq} = C_1 + C_2 + \frac{(A_1 - A_2)(D_2 - D_1)}{B_1 + B_2}$$

Since

$$A_1 = A_2$$

and

$$D_1 = D_2$$

the additional term becomes zero.

Thus

$$C_{eq} = C + C = 2C$$

$$= 0.001\angle 90^\circ \text{ S}$$

Similarly,

$$D_{eq} = D$$

Hence

$$\boxed{A_{eq} = D_{eq} = 0.98\angle 1^\circ}$$

$$\boxed{B_{eq} = 50\angle 75^\circ\ \Omega}$$

$$\boxed{C_{eq} = 0.001\angle 90^\circ \text{ S}}$$

**(b) Sending-End Power Factor**

Receiving-end power:

$$P_R = 100 \text{ MW}$$

Receiving-end voltage:

$$V_R = 132 \text{ kV}$$

Receiving-end power factor:

$$\cos \phi_R = 0.8$$

Therefore

$$\phi_R = 36.86^\circ$$

The receiving-end line current is

$$I_R = \frac{P_R}{\sqrt{3} V_R \cos \phi_R}$$

$$= \frac{100 \times 10^6}{0.8 \times \sqrt{3} \times 132 \times 10^3}$$

$$= 546.73 \text{ A}$$

Hence

$$I_R = 546.73\angle -36.86^\circ$$

The receiving-end phase voltage is

$$V_R = \frac{132}{\sqrt{3}}\angle 0^\circ$$

$$= 76.21\angle 0^\circ \text{ kV}$$

Using the transmission-line equation

$$V_S = A_{eq}V_R + B_{eq}I_R$$

$$V_S = (0.98\angle 1^\circ)\left(\frac{132}{\sqrt{3}}\angle 0^\circ\right) + (50\angle 75^\circ)\left(\frac{546.73}{1000}\angle -36.86^\circ\right)$$

$$V_S = 74.686\angle 1^\circ + 27.33\angle 38.14^\circ$$

$$V_S = 97.87\angle 10.706^\circ \text{ kV}$$

Similarly,

$$I_S = C_{eq}V_R + D_{eq}I_R.$$

Substituting,

$$I_S = (0.001\angle 90^\circ)\left(\frac{132}{\sqrt{3}}\angle 0^\circ\right) + (0.98\angle 1^\circ)(546.73\angle -36.86^\circ).$$

Evaluating the individual terms,

$$C_{eq}V_R = 76.21\angle 90^\circ = j76.21,$$

$$D_{eq}I_R = 535.80\angle -35.86^\circ = 434.65 - j313.24.$$

Hence,

$$\begin{aligned} I_S &= (434.65 - j313.24) + j76.21 \\ &= 434.65 - j237.03. \end{aligned}$$

Therefore,

$$I_S = 495.01\angle -28.70^\circ \text{ A}.$$

The sending-end power-factor angle is

$$\phi_S = \angle V_S - \angle I_S = 10.705^\circ - (-28.70^\circ) = 39.41^\circ.$$

Hence,

$$\text{Sending-end p.f.} = \cos 39.41^\circ = 0.773.$$

Since the sending-end current lags the sending-end voltage,

$$\boxed{0.773 \text{ lagging}}$$

SOLUTIONS

# X

# Distribution Systems, Sag and Corona

## Topics

1. Types of Distribution Systems
2. DC Distribution
3. Sag
4. Corona

## Unit Snapshot*

| | | | | |
|---|---|---|---|---|
| **Stream:** EE | **Questions:** 21 | **MCQ:** 20 | **MSQ:** 0 | **NAT:** 1 |
| **1M:** 14 | **2M:** 6 | **Descriptive:** 1 | **Avg Weightage** ~1M | **Range:** 1M(2014)–1M(2014) |

**See preface for how Avg Weightage and Range are calculated.*

**Q10.14.1** MCQ | 2014 | 2M | Ans: D | ● ● ○

Let the current supplied by source $S_1$ be $I$.
Let feeder resistance per unit length be $r$.
Since both end voltages are equal and in phase,

$$V_{S_1} = V_{S_2} = 400\,\text{V}$$

Applying KVL from source $S_1$ to source $S_2$ along the feeder,

$$-400 + (400r)I + (200r)(I - 200) + (200r)(I - 300) + (200r)(I - 500) + 400 = 0$$

Simplifying,

$$400rI + 200rI - 40000r + 200rI - 60000r + 200rI - 100000r = 0$$

$$1000rI - 200000r = 0$$

$$I = 200\,\text{A}$$

Thus, source $S_1$ supplies exactly the first 200 A load.
Hence, the current contribution from $S_1$ to the 100 A load at point $P$ is

$$0\,\text{A}$$

Therefore, the entire 100 A load at $P$ is supplied by source $S_2$.
Hence,

(D) 0 A and 100 A

**Q10.11.1** MCQ | 2011 | 1M | Ans: D | ● ● ●

For economical transmission, the transmitted power should be of the order of the Surge Impedance Loading.
The SIL is given by

$$\text{SIL} = \frac{V_{LL}^2}{Z_c}$$

where the surge impedance of an overhead transmission line is approximately

$$Z_c \approx 400\,\Omega$$

For a 132 kV line,

$$\text{SIL} = \frac{(132)^2}{400} = 43.56\ \text{MW}$$

For a double-circuit line,

$$\text{SIL} = 2 \times 43.56 = 87.12\ \text{MW}$$

This is far below the required transfer capability of 500 MW.
Hence, options (A) and (B) are unsuitable.
For a 400 kV line,

$$\text{SIL} = \frac{(400)^2}{400} = 400\ \text{MW}$$

For a 400 kV double-circuit line,

$$\text{SIL} = 2 \times 400 = 800 \text{ MW}$$

Since 800 MW exceeds the required transfer of 500 MW with adequate margin, the most suitable configuration is the 400 kV double-circuit line.

(D)

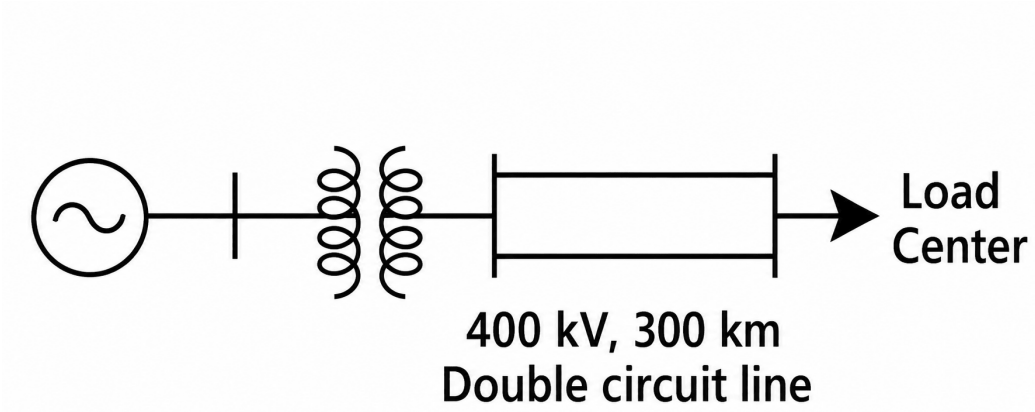


Fig. Solution Q10.11.1

**Key Points**

- Surge Impedance Loading (SIL) provides a first estimate of economical power transfer capability.
- SIL is approximately given by SIL = $V^2/Z_c$.
- Typical surge impedance of EHV overhead transmission lines is approximately 400 Ω.
- Large generating stations are generally evacuated through EHV double-circuit transmission lines for reliability and higher transfer capability.

**Mistakes to be avoided**

- Do not compare only voltage levels; compare transmission capability using SIL.
- Do not ignore the effect of double-circuit operation.
- Series compensation improves transfer capability but cannot make a 132 kV line suitable for evacuating 500 MW over 300 km.

## Q10.10.1 MCQ 2010 1M Ans: C ● ○ ○

From the HVDC link diagram, power flows from system A to system B.
Since the left converter operates as a rectifier, the DC output voltage is positive.

$$V_{AB} > 0$$

The right converter operates as an inverter and receives power from the DC link. Hence,

$$V_{CD} > 0$$

For current to flow from the rectifier towards the inverter through the DC link,

$$I = \frac{V_{AB} - V_{CD}}{R_{eq}}$$

Given that

$$I > 0$$

it follows that

$$V_{AB} > V_{CD}$$

Therefore, the correct option is

$$\boxed{\text{(C)} \quad V_{AB} > 0, \qquad V_{CD} > 0, \qquad V_{AB} > V_{CD}}$$

**Key Points**

- In an HVDC transmission system, the rectifier converts AC power to DC power and acts as the sending-end converter.
- The inverter converts DC power back to AC power and acts as the receiving-end converter.
- DC current flows from higher DC voltage to lower DC voltage through the DC link.

**Q10.07.1** MCQ | 2007 | 1M | Ans: C | ● ○ ○

Two large power systems are interconnected through

- an AC transmission line, and
- an HVDC link.

The HVDC link transfers a controllable power

$$P_{dc}$$

while the AC tie-line transfers

$$P_{ac}.$$

**Analysis of Option (A)**
Since the two regions are connected through an AC transmission line, they operate synchronously in steady state.
Therefore,

$$f_1 = f_2$$

and both regions must have the same frequency.
Hence, Option (A) is incorrect.

**Analysis of Option (B)**
Changing the firing angle of the converter can directly control

$$P_{dc}.$$

However, changing $P_{dc}$ modifies the power-angle relationship of the interconnected AC system, causing the AC tie-line power

$$P_{ac}$$

to readjust automatically.
Therefore, HVDC controls do not independently determine the total interchange

$$(P_{ac} + P_{dc}).$$

The total power transfer is governed by generation-load balance and system operating conditions.
Hence, Option (B) is incorrect.

**Analysis of Option (C)**
The HVDC link provides fast and independent control of DC power transfer.
By increasing or decreasing

$$P_{dc},$$

the AC tie-line power adjusts correspondingly.
Thus, the proportion of power carried by the AC line and the HVDC link can be changed without modifying the AC network.
For example,

$$P_{\text{total}} = P_{ac} + P_{dc}$$

may remain nearly constant while the individual values of

$$P_{ac} \quad \text{and} \quad P_{dc}$$

change.
Therefore, the power-sharing between the AC line and the HVDC link can be controlled through HVDC converter controls alone.
Hence, Option (C) is correct.

**Analysis of Option (D)**
Modern HVDC systems allow reversal of power flow by changing converter operating conditions.
Thus,

$$P_{dc}$$

can be reversed whenever required.
Hence, Option (D) is incorrect.

Therefore, the correct answer is

(C) Power sharing between the AC line and HVDC link can be changed by controlling the HVDC converters alone

**Key Points**

- AC-connected regions must operate at the same steady-state frequency.
- HVDC control can redistribute power flow between parallel AC and DC corridors.
- One major advantage of HVDC links is controllable power transfer independent of AC line impedance.
- Power reversal in HVDC systems is possible through converter control.

**Q10.06.1** MCQ | 2006 | 1M | Ans: B | ● ○ ○

Reactive power cannot be transferred through a DC link.
Hence, the reactive power requirements of the converter stations must be supplied locally by the connected AC systems.
The rectifier consumes reactive power from the AC system connected to it, while the inverter supplies reactive power to the AC system connected to it.
Therefore, the correct statement is:

(B) The rectifier consumes reactive power and the inverter supplies reactive power from/to the respective connected AC systems

**Q10.06.2** MCQ | 2006 | 2M | Ans: C | ● ● ○

The flux linkages of telephone conductor $t_1$ due to currents in conductors 1 and 2 are

$$\lambda_{t_1} = \frac{\mu I}{2\pi} \ln \frac{1}{2.5} - \frac{\mu I}{2\pi} \ln \frac{1}{3}$$

$$\lambda_{t_1} = \frac{\mu I}{2\pi} \left( \ln \frac{1}{2.5} - \ln \frac{1}{3} \right)$$

For

$$\mu = 4\pi \times 10^{-7} \text{ H/m}$$

and

$$I = 100 \text{ A}$$

$$\lambda_{t_1} = 3.646 \times 10^{-6} \text{ Wb-turn/m}$$

The induced voltage is

$$e_{t_1} = j\omega\lambda_{t_1}$$

$$= j(2\pi)(50)(3.646 \times 10^{-6})$$

$$= j1.1455 \times 10^{-3} \text{ V/m}$$

$$= j1.1455 \text{ V/km}$$

Similarly, for conductor $t_2$

$$\lambda_{t_2} = \frac{\mu I}{2\pi} \left( \ln \frac{1}{3} - \ln \frac{1}{2.5} \right)$$

$$= -3.646 \times 10^{-6} \text{ Wb-turn/m}$$

Hence,

$$e_{t_2} = -j1.1455 \text{ V/km}$$

The induced voltage in the telephone circuit is

$$e_t = e_{t_1} - e_{t_2}$$

$$= j1.1455 - (-j1.1455)$$

$$= j2.291 \text{ V/km}$$

Therefore,

$$|e_t| = 2.29 \text{ V/km}$$

Hence, the answer is

(C) 2.29 V/km

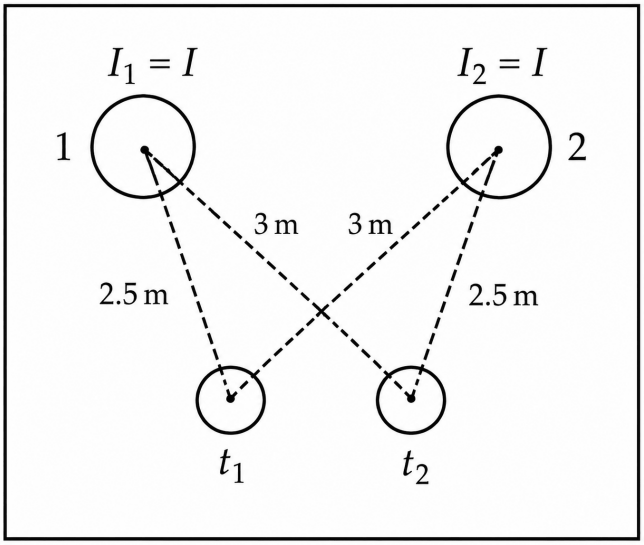


Fig. Solution Q10.06.2

**Q10.05.1** MCQ | 2005 | 1M | Ans: A | ● ○ ○

HVDC transmission is particularly economical and technically advantageous for transmitting large amounts of power over very long distances.
The lower transmission losses and reduced conductor requirements make HVDC attractive for long-distance bulk power transfer.
Therefore, HVDC transmission is mainly used for

(A) Bulk power transmission over very long distances

**Q10.03.1** MCQ | 2003 | 1M | Ans: B | ● ○ ○

The auxiliary components commonly used in an HVDC transmission system include:

- DC smoothing reactors (DC line inductors)
- Reactive power compensation equipment

DC line inductors reduce ripple current and limit transient currents.
Reactive power sources are required because converter stations consume reactive power.
Therefore, the appropriate auxiliary components are

(B) P and R

**Q10.03.2** MCQ | 2003 | 2M | Ans: A | ● ● ○

Let the current entering from end $P$ be $I$.
The voltage difference between the ends is

$$V_P - V_Q = 0.1(I - 10) + 0.15(I - 30) + 0.2(I - 60)$$

Given,

$$V_P - V_Q = 3 \text{ V}$$

Therefore,

$$3 = 0.1I - 1 + 0.15I - 4.5 + 0.2I - 12$$

$$20.5 = 0.45I$$

$$I = 45.56 \text{ A}$$

Since

$$I - 60 < 0$$

the current in section $SQ$ reverses direction and point $S$ becomes the point of minimum potential.
Given that the minimum voltage is 220 V,

$$V_S = 220 \text{ V}$$

Hence,

$$V_P = V_S + 0.1(I - 10) + 0.15(I - 30)$$

$$= 220 + 0.1(45.56 - 10) + 0.15(45.56 - 30)$$

$$= 225.89 \text{ V}$$

Therefore,

(A) $V_P = 225.89$ V

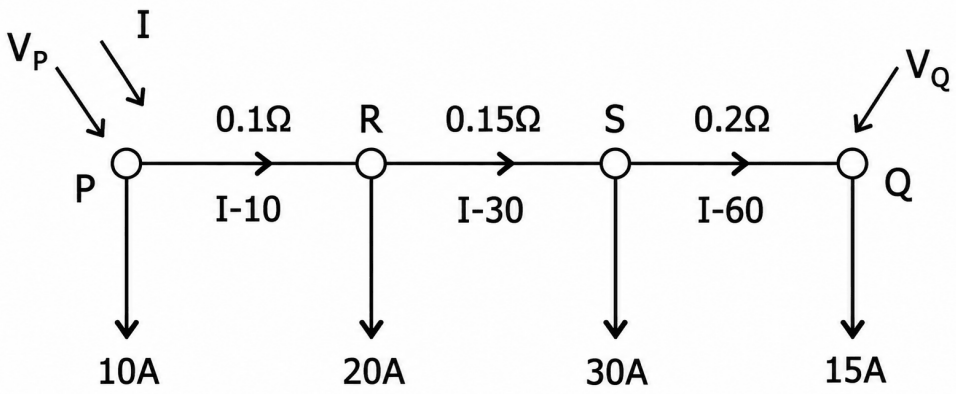


Fig. Solution Q10.03.2

**Q10.02.1** MCQ | 2002 | 2M | Ans: B

For a three-phase fully controlled thyristor bridge rectifier operating with continuous current and neglecting overlap,

$$V_{dc} = V_{d0} \cos \alpha.$$

The displacement angle of the AC-side current is

$$\phi = \alpha.$$

The apparent power drawn from the AC supply is

$$S = \sqrt{3} V_L I_L.$$

For continuous current operation,

$$I_L = \sqrt{\frac{2}{3}} I_d.$$

Since the AC supply voltage and the DC current are constant,

$$S = \text{constant}.$$

The active and reactive powers are therefore

$$P = S \cos \alpha,$$

and

$$Q = S \sin \alpha.$$

Since $S$ is constant,

$$Q \propto \sin \alpha.$$

Now,

$$\sin 5^\circ < \sin 10^\circ < \sin 15^\circ < \sin 30^\circ.$$

Therefore,

$$Q_{max} \text{ occurs at } \alpha = 30^\circ.$$

Hence, the reactive power absorbed from the AC system is maximum when

$$\boxed{\alpha = 30^\circ}$$

Therefore, the correct answer is

$$\boxed{\text{(B) Reactive power absorbed by the rectifier is maximum when } \alpha = 30^\circ.}$$

**Q10.00.1** MCQ | 2000 | 1M | Ans: B | ● ○ ○

Let us examine each statement for HVDC transmission.

(i) HVDC transmission is not limited by steady-state stability constraints because there is no power-angle stability limitation associated with line reactance as in AC systems.

Hence, statement (i) is correct.

(ii) HVDC converter stations consume reactive power and therefore require reactive power support from the AC systems at both terminals.

Hence, statement (ii) is incorrect.

(iii) Interconnecting two AC systems through an HVDC link does not significantly increase the fault level of either AC system because the systems remain electrically decoupled with respect to AC fault currents.

Hence, statement (iii) is correct.

(iv) Corona effects can still occur in HVDC transmission, although the nature and magnitude of corona losses differ from those in HVAC systems.

Hence, statement (iv) is incorrect.

Hence, the correct answer is

$$\boxed{\text{(B) (i) and (iii)}}$$

**Q10.00.2** MCQ | 2000 | 2M | Ans: D | ● ○ ○

For a given transmission line, corona loss is approximately proportional to

$$(f + 25)$$

Hence,

$$P_c \propto (f + 25)$$

Therefore,

$$\frac{P_2}{P_1} = \frac{f_2 + 25}{f_1 + 25}$$

Given,

$$P_1 = 1 \text{ kW/km per phase}$$

$$f_1 = 50 \text{ Hz}$$

$$f_2 = 60 \text{ Hz}$$

Substituting,

$$P_2 = P_1\left(\frac{60+25}{50+25}\right)$$

$$P_2 = 1\left(\frac{85}{75}\right)$$

$$P_2 = 1.133$$

$$P_2 \approx 1.13 \text{ kW/km per phase}$$

Hence, the correct answer is

(D) 1.13 kW/km per phase

**Q10.99.1** MCQ | 1999 | 1M | Ans: D | ● ○ ○

Corona loss is given approximately by

$$P_c \propto (V_{ph} - V_{do})^2$$

where

$$V_{do} \propto r \ln\left(\frac{d}{r}\right)$$

Increasing conductor radius reduces the electric field intensity at the conductor surface and increases the disruptive critical voltage.
Reducing current density implies increasing conductor cross-sectional area and radius, thereby reducing corona loss.
Hence, the correct option is

(D) current density in conductors is reduced

**Q10.99.2** MCQ | 1999 | 1M | Ans: C | ● ○ ○

In a DC transmission line, current and voltage are constant in steady state.
Therefore,

$$X_L = \omega L = 0$$

and

$$X_C = \frac{1}{\omega C} \to \infty$$

since

$$\omega = 0$$

Hence there are no steady-state effects due to inductive and capacitive reactances.
Therefore,

(C) there are not effects due to inductive and capacitive reactances

## Q10.99.3 MCQ 1999 2M Ans: C ● ○ ○

The voltage drop across section BC is

$$V_{BC} = 40 \times j0.35$$

$$V_{BC} = 14\angle 90^\circ \text{ V}$$

Total current through section AB:

$$I = 40\angle 0^\circ + 30\angle(-36.87^\circ)$$

$$I = 66.48\angle(-15.7^\circ) \text{ A}$$

Voltage drop across AB:

$$V_{AB} = 66.48\angle(-15.7^\circ)(j0.25)$$

$$V_{AB} = 16.62\angle 74.3^\circ \text{ V}$$

Therefore,

$$V_{AC} = V_{AB} + V_{BC}$$

$$V_{AC} = 16.62\angle 74.3^\circ + 14\angle 90^\circ$$

$$V_{AC} = 30.33\angle 81.47^\circ$$

Hence, the correct option is

(C) 30 V

## Q10.98.1 MCQ 1998 1M Ans: C ● ○ ○

HVDC transmission is preferred for long-distance bulk power transfer because of its lower transmission losses compared to AC transmission.
The absence of skin effect and reduced reactive power issues improve transmission efficiency.
For long transmission distances,

$$P_{\text{loss}} = I^2R$$

is comparatively lower in HVDC systems because the conductor utilization is better and there are no charging current losses.

Hence, HVDC transmission becomes economical for bulk power transfer over long distances.

(C) Minimum line power losses

**Q10.95.1** MCQ | 1995 | 1M | Ans: B | ● ● ○

For transmission voltages below approximately 220 kV, lightning overvoltages are usually the dominant insulation-design criterion.
However, for EHV and UHV transmission systems, switching surges become much more severe.
The magnitude of switching overvoltages increases with system voltage and may exceed lightning surges in determining insulation requirements.
Therefore insulation coordination studies for 400 kV systems are primarily based on switching overvoltages.
Hence, the correct option is

(B) Switching over voltage

**Q10.95.2** NAT | 1995 | 5M | Ans: 86.7 | ● ● ○

Consider a three-unit suspension insulator string.
Let the voltages across the top, middle and bottom (conductor-end) units be

$$V_1, \quad V_2, \quad V_3,$$

respectively.
The self-capacitance of each unit is

$$C,$$

and the shunt capacitance from each metal fitting to the tower is

$$kC,$$

where

$$k = 0.1.$$

Applying KCL at the junctions of the string,
**At the first junction,**
the current through the top unit equals the sum of the current through the second unit and the shunt current.
Hence,

$$j\omega CV_1 = j\omega CV_2 + j\omega(0.1C)V_1.$$

Dividing throughout by $j\omega C$,

$$V_1 = V_2 + 0.1V_1,$$

or

$$0.9V_1 = V_2. \tag{1}$$

**At the second junction,**
the node voltage with respect to the tower is

$$V_1 + V_2.$$

Therefore,

$$j\omega CV_2 = j\omega CV_3 + j\omega(0.1C)(V_1 + V_2).$$

Dividing by $j\omega C$,

$$V_2 = V_3 + 0.1(V_1 + V_2).$$

Substituting (1),

$$0.9V_1 = V_3 + 0.1V_1 + 0.09V_1,$$

$$0.71V_1 = V_3.$$

Hence,

$$V_1 = \frac{V_3}{0.71} = 0.76336\,V_3,$$

and from (1),

$$V_2 = 0.9V_1 = 0.83969\,V_3.$$

Thus, the voltage distribution is

$$V_1 : V_2 : V_3 = 0.763 : 0.840 : 1.$$

Since the conductor-end unit has the maximum permissible voltage,

$$V_3 = 20\text{ kV}.$$

Therefore,

$$V_1 = 0.76336(20) = 15.27\text{ kV},$$

$$V_2 = 0.83969(20) = 16.79\text{ kV},$$

$$V_3 = 20\text{ kV}.$$

Hence, the total string voltage is

$$V_s = 15.27 + 16.79 + 20 = 52.06\text{ kV}.$$

The string efficiency is

$$\eta = \frac{\text{Total string voltage}}{n \times \text{Maximum unit voltage}} \times 100,$$

where

$$n = 3.$$

Therefore,

$$\eta = \frac{52.06}{3 \times 20} \times 100 = 86.76\%.$$

Hence,

$$\boxed{\begin{aligned} V_1 &= 15.27 \text{ kV}, \\ V_2 &= 16.79 \text{ kV}, \\ V_3 &= 20.00 \text{ kV}, \\ \eta &= 86.76\%. \end{aligned}}$$

**Q10.94.1** MCQ | 1994 | 1M | Ans: C | ● ○ ○

HVDC transmission offers several advantages over EHV-AC transmission, especially for long-distance bulk power transfer and submarine cable applications.
Let us examine each option.
**Option (A): HVDC terminal equipment are inexpensive**
This is incorrect.
HVDC converter stations require:

- Thyristor or VSC converters
- Converter transformers
- Filters
- Reactive power support equipment

These make HVDC terminal stations considerably more expensive than AC substations.
Hence Option (A) is incorrect.
**Option (B): VAR compensation is not required in HVDC systems**
This is incorrect.
Line-commutated HVDC converters consume substantial reactive power.
Reactive compensation is usually required at converter stations.
Hence Option (B) is incorrect.
**Option (C): System stability can be improved**
This is correct.
HVDC links provide:

- Fast controllability of power flow
- Improved damping of oscillations
- Enhanced transient stability
- Better interconnection of asynchronous systems

Therefore HVDC can significantly improve overall system stability.
Hence Option (C) is correct.
**Option (D): Harmonics problem is avoided**
This is incorrect.
HVDC converters generate harmonics.
Therefore filters are required on both AC and DC sides.
Hence harmonics are not avoided.

Therefore, the required option is

(C) System stability can be improved

**Q10.92.1** MCQ 1992 1M Ans: D ● ○ ○

**Concept:**

In a power distribution system, a **feeder** and a **distributor** perform different functions. Consequently, their design criteria are also different.

**Feeder**

A feeder carries electrical power from the substation to the distribution area without any intermediate consumer tapping.

Since the current remains practically constant throughout its length,

$$I = \text{constant}$$

Hence, feeders are designed primarily based on:

Current carrying capacity (Ampacity)

Voltage drop is generally not the governing design criterion.

**Distributor**

A distributor supplies power directly to consumers.

Loads are connected at various points along the distributor, causing the current to decrease progressively from the sending end towards the far end.

Therefore,

$$I_1 > I_2 > I_3 > \cdots$$

Because consumers are connected throughout the distributor, maintaining the supply voltage within prescribed limits becomes the most important design requirement.

The voltage drop is

$$\Delta V = IR$$

where

$$R = \rho \frac{l}{A}$$

For a fixed conductor material,

- increasing the cross-sectional area $A$ - decreases resistance - reduces voltage drop - improves voltage regulation.

Hence, distributors are designed on the basis of the permissible voltage drop.

**Analysis of Options**

- **(A) Current carrying capacity only**

  Incorrect.

  This is the principal design criterion for feeders, not distributors.

- **(B) Mechanical strength only**

  Incorrect.

  Mechanical strength is important for overhead conductors but does not determine the required electrical cross-section of a distributor.

- **(C) Corona loss limitation**

  Incorrect.

  Corona is significant only in extra-high-voltage transmission lines. It is not a design consideration for distribution conductors.

- **(D) Permissible voltage drop**

  Correct.

  Consumers connected at different points must receive nearly rated voltage. Therefore, distributors are sized mainly to satisfy the allowable voltage-drop limit.

**Comparison between Feeder and Distributor**

| Parameter | Feeder | Distributor |
|---|---|---|
| Consumer tapping | No | Yes |
| Current | Nearly constant | Varies along length |
| Design criterion | Current capacity | Voltage drop |
| Voltage regulation | Less important | Very important |

Hence, the correct answer is

(D) Permissible voltage drop

**Key Points**

- A feeder connects the substation to the distribution area.
- A distributor supplies consumers directly.
- Feeders are designed based on current carrying capacity.
- Distributors are designed based on permissible voltage drop.

**Q10.91.1** MCQ | 1991 | 1M | Ans: D | ● ○ ○

**Concept:**
The primary objective of any distribution system is to deliver electrical power to consumers with adequate reliability, acceptable voltage regulation, and minimum interruption. The choice of distribution network depends on factors such as reliability, continuity of supply, cost, and ease of maintenance.
The two commonly used distribution systems are:

- Radial Distribution System
- Ring Main Distribution System

**Radial Distribution System**
In a radial distribution system, power flows from the substation to consumers through only one path.

Substation ⟶ Consumers

Characteristics:

- Simple construction.
- Lowest installation cost.
- Easy operation and protection.
- Entire downstream area loses supply if a feeder fault occurs.

Hence, reliability is comparatively poor.

**Ring Main Distribution System**
In a ring main system, the distributor forms a closed loop connected to the substation.
Consumers can receive power from either direction.
If a fault develops in one section,

- only the faulty section is isolated,
- healthy sections continue receiving supply through the alternate path.

Thus, interruption to consumers is greatly reduced.

**Analysis of Options**
**(A) It is less expensive**
Incorrect.
A ring main system requires additional conductors, switching equipment, and protection devices, making it more expensive than a radial system.

**(B) Voltage drop in the feeder is less**
Although the voltage profile may improve because current is shared through two paths, this is not the principal reason for selecting a ring main system.

**(C) Power factor is higher**
Incorrect.
Power factor depends on the nature of the connected load and is independent of the distribution topology.

**(D) Supply is more reliable**
Correct.
The ring main system provides an alternative path for power flow, ensuring continuity of supply even when one section is isolated due to maintenance or a fault.

**Comparison**

| Parameter | Radial | Ring Main |
|---|---|---|
| Initial Cost | Low | High |
| Reliability | Low | High |
| Continuity of Supply | Poor | Excellent |
| Alternative Power Path | No | Yes |
| Maintenance | Requires interruption | Can often be performed without interrupting all consumers |

Hence, the correct answer is

(D) Supply is more reliable

SOLUTIONS

# XI

# Relays Circuit Breakers

## Topics

1. Working Principle of Relay
2. Types of Relay, Plug Setting Multiplier and Time Setting Multiplier- PSM, TSM
3. Directional Relay, Distance Relay
4. Differential Relay
5. Protection Zones, Distance Protection
6. Introduction and ARC interruption
7. Current Chopping

PrepFusion

## Unit Snapshot*

| | | | | |
|---|---|---|---|---|
| **Stream:** EE | **Questions:** 39 | **MCQ:** 29 | **MSQ:** 1 | **NAT:** 9 |
| **1M:** 26 | **2M:** 10 | **Descriptive:** 3 | **Avg Weightage** ~2M | **Range:** 1M(2023)–2M(2026) |

**See preface for how Avg Weightage and Range are calculated.*

**Q11.26.1** MCQ | 2026 | 2M | Ans: A | ● ○ ○

The reactance relay characteristic is

$$X \leq 1\,\Omega$$

The impedance seen by the relay is

$$Z = R + jX$$

Admittance is given by

$$Y = \frac{1}{Z} = G + jB$$

Therefore,

$$G + jB = \frac{1}{R + jX} = \frac{R - jX}{R^2 + X^2}$$

Comparing real and imaginary parts,

$$G = \frac{R}{R^2 + X^2}$$

$$B = -\frac{X}{R^2 + X^2}$$

Given relay characteristic:

$$X = 1$$

Substituting into the expression for $B$,

$$B = -\frac{1}{R^2 + 1}$$

and

$$G = \frac{R}{R^2 + 1}$$

Now,

$$G^2 + \left(B + \frac{1}{2}\right)^2 = \frac{R^2}{(R^2+1)^2} + \left(-\frac{1}{R^2+1} + \frac{1}{2}\right)^2$$

$$= \frac{R^2}{(R^2+1)^2} + \left(\frac{R^2 - 1}{2(R^2+1)}\right)^2$$

$$= \frac{4R^2 + (R^2-1)^2}{4(R^2+1)^2}$$

$$= \frac{R^4 + 2R^2 + 1}{4(R^2+1)^2}$$

$$= \frac{(R^2+1)^2}{4(R^2+1)^2}$$

$$= \frac{1}{4}$$

Hence, the characteristic is

$$G^2 + \left(B + \frac{1}{2}\right)^2 = \frac{1}{4}$$

Since the relay operates for

$$X \le 1$$

the operating region is

$$G^2 + \left(B + \frac{1}{2}\right)^2 \ge \frac{1}{4}$$

Therefore, the correct option is

$$\boxed{\text{(A) } G^2 + \left(B + \frac{1}{2}\right)^2 \ge \frac{1}{4}}$$

**Key Points**

- Reactance relay characteristic is specified in the impedance plane.
- Transformation from impedance plane to admittance plane converts straight-line characteristics into circles.

**Mistakes to be avoided**

- Do not forget the negative sign in susceptance:

$$B = -\frac{X}{R^2 + X^2}$$

- Do not confuse impedance-plane characteristics with admittance-plane characteristics.

**Q11.23.1** MCQ | 2023 | 1M | Ans: A | ● ● ○

Lines 1 and 2 form parallel paths. For a fault on line 1, the fault can be fed through line 2 and back from bus 2 into line 1. Relay $R_3$ must therefore operate for current from bus 2 towards line 1, while $R_4$ must not trip the healthy line for current towards bus 2. The corresponding roles reverse for a fault on line 2.

Thus $R_3$ and $R_4$ must be directional and block faults towards bus 2. The downstream radial section needs no additional directional relay for this purpose.

A

**Key Points**

- **Directional overcurrent** relays are required where **fault current can reverse direction**.
- **Parallel** transmission paths create **bidirectional fault** current possibilities.
- **Radial sections** generally require only **nondirectional overcurrent** relays.
- Minimum relay requirement reduces system complexity and cost.

**Q11.22.1** MCQ | 2022 | 1M | Ans: A | ● ○ ○

Loss of excitation in a synchronous generator causes the machine to operate in an under-excited condition and absorb reactive power from the system.
The impedance trajectory during loss of excitation enters a characteristic region in the impedance plane.
An offset Mho relay is specifically designed to detect this condition in synchronous generators.
Hence, the commonly used relay for loss of excitation protection is the offset Mho relay.

**Over current relay**
Used mainly for overcurrent and short-circuit protection.

**Differential relay**
Used for protection against internal winding faults.

**Buchholz relay**
Used only in oil-immersed transformers.
Therefore, the correct option is

(A) Offset Mho Relay

**Q11.20.1** MCQ | 2020 | 1M | Ans: D | ● ○ ○

Given:

$$X_L = 0.2\,\text{pu}$$

$$V_t = 1\,\text{pu}$$

The transmission line is protected up to 80% of its total length by the distance relay.

PrepFusion

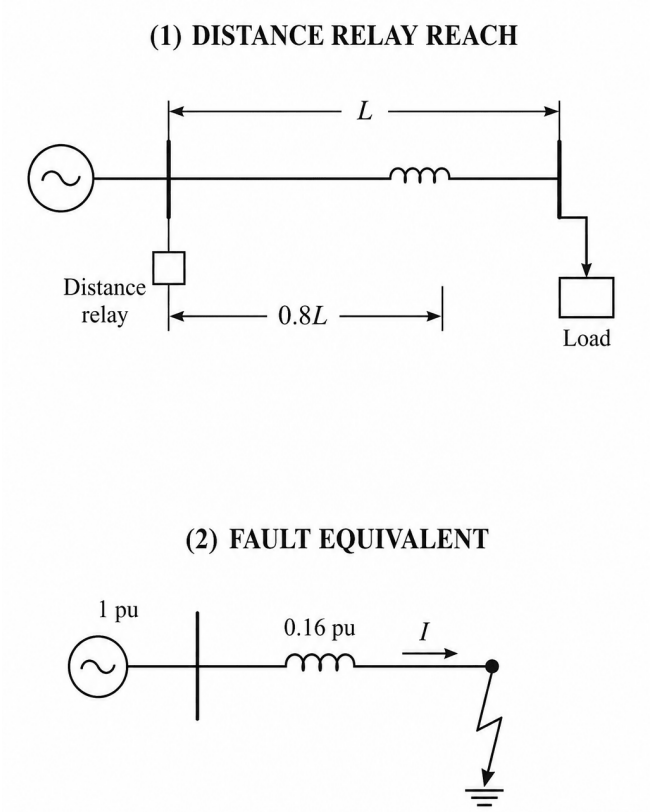


Fig. Solution Q11.20.1

Hence, the impedance seen by the relay at the reach point is

$$Z_{relay} = 0.8 \times 0.2$$

$$Z_{relay} = 0.16\,\text{pu}$$

For a solid three-phase fault,

$$I = \frac{V}{Z}$$

$$I = \frac{1}{0.16}$$

$$I = 6.25\,\text{pu}$$

Therefore, the threshold current for operation of the relay is

(D) 6.25

**Mistakes to be avoided**

- Do not use the full line reactance since the relay protects only 80% of the line.

**Q11.19.1** NAT | 2019 | 2M | Ans: 500 | ● ○ ○

For no transient oscillation, the circuit must be critically damped.
The critical resistance for an RLC circuit is

$$R_{cr} = \frac{1}{2}\sqrt{\frac{L}{C}}$$

Given,

$$L = 50\,\text{mH} = 50 \times 10^{-3}\,\text{H}$$

$$C = 0.05\,\mu\text{F} = 0.05 \times 10^{-6}\,\text{F}$$

Substituting,

$$R_{cr} = \frac{1}{2}\sqrt{\frac{50 \times 10^{-3}}{0.05 \times 10^{-6}}}$$

$$R_{cr} = \frac{1}{2}\sqrt{10^6}$$

$$R_{cr} = \frac{1}{2} \times 1000$$

$$R_{cr} = 500\,\Omega$$

Therefore,

500

**Mistakes to be avoided**

- Do not forget to convert mH and $\mu$F into SI units.
- Do not use the undamped natural frequency expression instead of the damping condition.

**Q11.16.1** MCQ | 2016 | 1M | Ans: D | ● ○ ○

Relay $R_4$ protects the lower transmission line section connected to the right bus.
If relay $R_4$ fails to operate, the fault must be cleared by relays located electrically away from the protected section while still being capable of detecting the fault current.
The remote backup protection for $R_4$ is provided by:

- Relay $R_1$ from the left-side source
- Relay $R_6$ from the right-side source

These relays act as remote backup because they are located on adjacent sections and can isolate the fault when the primary relay fails.

(D) $R_1$, $R_6$

**Key Points**

- Remote backup relays operate when the primary relay fails to clear the fault.
- Backup protection must be electrically separated from the primary protection zone.

- In interconnected systems, backup relays may operate from both source ends.

**Q11.14.1** NAT | 2014 | 2M | Ans: 0.21 to 0.23 | ● ● ●

Relay $R_B$ is located between buses $B$ and $C$.
Hence, under normal operating condition, it carries only the load current connected beyond bus $B$.
Therefore,

$$I_{\text{load}} = 100\ \text{A}$$

Given plug setting is 5 A.
Assuming CT ratio selected according to the feeder full-load current,

$$\text{CT ratio} = \frac{100}{5}$$

Hence, primary pickup current is

$$I_{\text{pickup}} = 100\ \text{A}$$

The maximum fault current at bus $B$ is

$$I_f = 2000\ \text{A}$$

Therefore, Plug Setting Multiplier (PSM) is

$$\text{PSM} = \frac{I_f}{I_{\text{pickup}}} = \frac{2000}{100} = 20$$

Given,

$$\text{TMS} = 0.1$$

Using IDMT relay characteristic,

$$t_{\text{op}} = \frac{0.14 \times \text{TMS}}{(\text{PSM})^{0.02} - 1}$$

Substituting,

$$t_{\text{op}} = \frac{0.14 \times 0.1}{20^{0.02} - 1}$$

$$20^{0.02} \approx 1.0617$$

Hence,

$$t_{\text{op}} = \frac{0.014}{1.0617 - 1}$$

$$t_{\text{op}} = \frac{0.014}{0.0617}$$

$$t_{\text{op}} \approx 0.227\ \text{s}$$

$$\boxed{0.227}$$

**Key Points**

- Plug Setting Multiplier:

$$\text{PSM} = \frac{\text{Fault Current}}{\text{Pickup Current}}$$

- IDMT relay operating time decreases with increase in PSM.
- Pickup current depends on CT ratio and plug setting.
- Relay $R_B$ carries only downstream feeder load current under normal operation.

**Mistakes to be avoided**

- Do not use minimum fault current for operating time calculation unless specifically asked.
- Carefully determine the feeder current seen by the relay.
- Do not forget to convert plug setting into primary pickup current using CT ratio.

**Q11.11.1** MCQ | 2011 | 1M | Ans: A | ● ○ ○

Negative sequence currents are produced during unbalanced operating conditions such as phase-to-phase faults, open conductor conditions and severe load unbalance.
These currents induce double-frequency currents in the rotor of a synchronous generator, resulting in excessive rotor heating.
Therefore, negative sequence relays are commonly employed for the protection of alternators against unbalanced loading and phase faults.

(A) an alternator

**Q11.10.1** MCQ | 2010 | 1M | Ans: C | ● ○ ○

Given,

$$V_{LL} = 33\,\text{kV}$$

$$\text{Breaking Capacity} = 2000\,\text{MVA}$$

The symmetrical breaking current is

$$I_b = \frac{\text{MVA}}{\sqrt{3}\,kV}$$

$$I_b = \frac{2000}{\sqrt{3} \times 33}$$

$$I_b = 34.99\,\text{kA}$$

$$I_b \approx 35\,\text{kA}$$

Therefore, the symmetrical breaking current is

(C) 35 kA

**Q11.10.2** MCQ | 2010 | 1M | Ans: C | ● ○ ○

For the differential protection scheme,

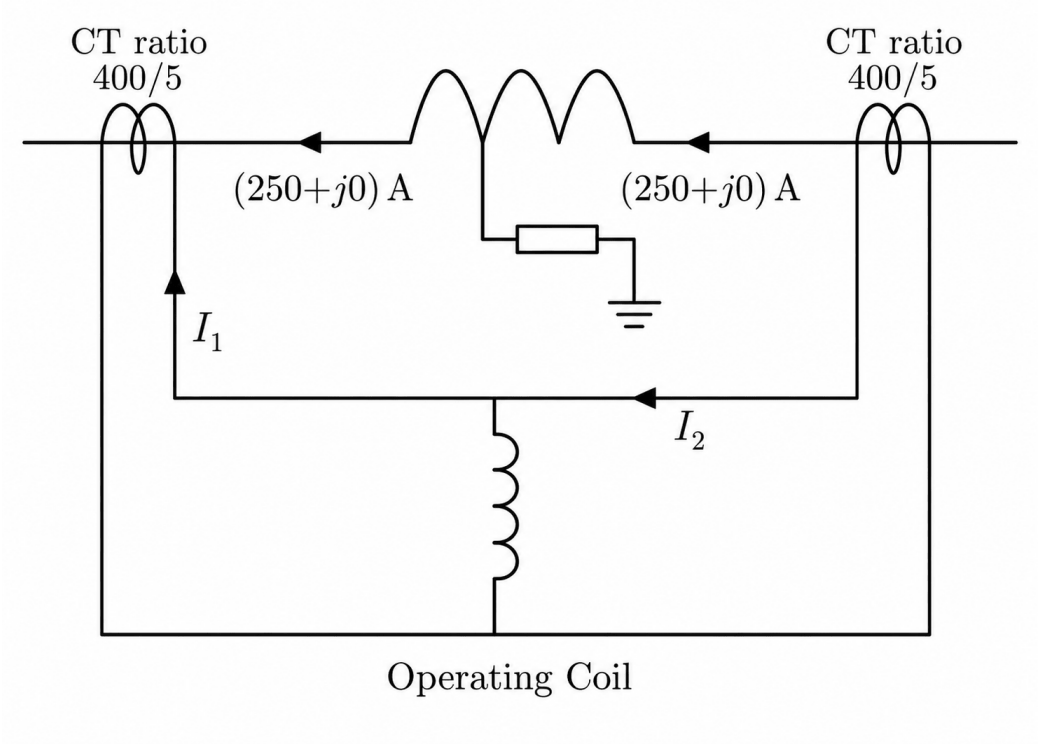


Fig. Solution Q11.10.2

The CT ratio is

$$\frac{400}{5}$$

Hence the secondary currents corresponding to the primary currents are

$$I_1 = 220 \times \frac{5}{400} = 2.75\,\text{A}$$

$$I_2 = 250 \times \frac{5}{400} = 3.125\,\text{A}$$

The operating coil carries the differential current,

$$I_{op} = I_2 - I_1$$

$$I_{op} = (250 - 220) \times \frac{5}{400}$$

$$I_{op} = 30 \times \frac{5}{400}$$

$$I_{op} = 0.375\,\text{A}$$

Therefore, the answer is

$$\boxed{\text{(C) } 0.375\,\text{A}}$$

**Q11.09.1** MCQ 2009 2M Ans: A ● ○ ○

For a short line, fault resistance can be a large fraction of the line impedance, so a reactance relay is preferred because its reach is substantially independent of fault resistance. For a medium line, the impedance (ohm) relay is the conventional choice. For a long line, a mho relay provides directional operation and better discrimination against load and power swings.

$$(a) \rightarrow 2, \qquad (b) \rightarrow 1, \qquad (c) \rightarrow 3.$$

$$\boxed{\text{A}}$$

**Q11.08.1** MCQ 2008 1M Ans: A ● ○ ○

A ground-distance element compensates the residual (zero-sequence) current. For a fault a fraction $m$ of the way along the line, with $Z_2 = Z_1$,

$$V_a = m\left[Z_1 I_a + (Z_0 - Z_1) I_0\right].$$

The compensated loop current is

$$I_{\text{loop}} = I_a + \frac{Z_0 - Z_1}{Z_1} I_0.$$

Hence the measured distance impedance is

$$Z_{\text{relay}} = \frac{V_a}{I_{\text{loop}}} = mZ_1.$$

At the centre of the line $m = 1/2$, so

$$\boxed{\text{A} : \frac{Z_1}{2}}.$$

The uncorrected phase ratio $V_a/I_a$ is not, in general, the compensated ground-distance impedance.

**Q11.08.2** MCQ 2008 2M Ans: A ● ● ○

The line-end voltages have equal magnitudes but are displaced by

$$180^\circ$$

which corresponds to a severe power-transfer condition.
The minimum three-phase fault current is given to be

$$I_{f,\min} = 5I_{L,\max}$$

The directional overcurrent relay pickup setting is

$$I_{\text{pickup}} = 1.25I_{L,\max}$$

Since

$$I_{f,\min} = 5I_{L,\max} > 1.25I_{L,\max}$$

the directional overcurrent relay will operate.
Pilot relaying schemes compare current directions or phase relationships at both line ends and are designed to operate for internal faults. Hence both directional-comparison and segregated phase-comparison pilot schemes will pick up.
For a mho relay, operation depends on the apparent impedance seen by the relay.
With equal terminal voltages and a phase displacement of

$$180^\circ$$

the apparent impedance trajectory may not enter the Zone-1 mho characteristic even when Zone-1 is set to

80%

of the line impedance.
Therefore, the scheme that will *not* pick up is

(A) Distance protection using mho relay with Zone-1 set to 80%

**Key Points**

- Mho relays operate based on apparent impedance seen by the relay.
- Directional overcurrent relays operate when current exceeds the pickup setting in the correct direction.
- Pilot protection schemes use information from both line ends and are highly sensitive to internal faults.
- Relay operation depends on the impedance trajectory relative to the relay characteristic.

**Q11.07.1** MCQ | 2007 | 2M | Ans: C | ● ○ ○

A single line-to-ground fault occurs at the midpoint of the transmission line connected between buses B and C.
The fault lies within the protection zone bounded by circuit breakers

4, 5, 6, 7

Under normal operation, the primary protection should isolate the fault by tripping the breakers nearest to the faulted section.
Since circuit breaker 4 is stuck and fails to open, the fault remains energized from the left side.

**Step 1: Primary protection operation**
The healthy end of the faulted line is disconnected first.
Therefore,

5

operates first.

**Step 2: Breaker-failure backup protection**
Breaker 4 fails to clear the fault.
Breaker-failure protection associated with breaker 4 initiates tripping of all breakers connected to the same bus section to completely de-energize the fault contribution feeding through breaker 4.
Hence the associated breakers operate in sequence as

$$6 \rightarrow 7 \rightarrow 3$$

thereby isolating the bus section connected to the failed breaker.

**Step 3: Remote backup protection**
If fault current still persists, remote backup protection on the adjacent transmission line operates.
Consequently,

$$1 \rightarrow 2$$

trip subsequently.

Thus a valid coordinated sequence is

$$5,\ 6,\ 7,\ 3,\ 1,\ 2$$

Hence, the correct answer is

$$\boxed{\text{(C) Sequence } = 5, 6, 7, 3, 1, 2}$$

**Key Points**

- Primary protection must operate before backup protection.
- Breaker-failure (stuck-breaker) protection is activated when a breaker receives a trip command but fails to interrupt current.
- Backup protection isolates all possible sources feeding the fault.
- Remote backup relays operate with intentional time delay after local backup protection.

**Q11.06.1** MCQ | 2006 | 1M | Ans: A | ● ○ ○

Vacuum circuit breakers (VCBs) are widely used for capacitor bank switching due to their low cost, high reliability, and excellent dielectric recovery characteristics.
The dielectric strength in a vacuum recovers very rapidly after current interruption, making VCBs particularly suitable for switching capacitive currents.
Hence, the most suitable circuit breaker for capacitor bank switching is the vacuum circuit breaker.

$$\boxed{\text{(A) Vaccum}}$$

**Q11.06.2** MCQ | 2006 | 1M | Ans: B | ● ○ ○

For a biased differential relay,

$$\text{Bias} = \frac{\text{Operating coil current}}{\text{Restraining coil current}}$$

The operating torque is

$$T_O = K(i_1 - i_2)n_o$$

and the restraining torque is

$$T_R = K\left(\frac{n_r}{2}i_1 + \frac{n_r}{2}i_2\right) = K\left(\frac{i_1 + i_2}{2}\right)n_r$$

At balance,

$$T_O = T_R$$

$$K(i_1 - i_2)n_o = K\left(\frac{i_1 + i_2}{2}\right)n_r$$

$$\frac{n_r}{n_o} = \frac{(i_1 - i_2)}{\left(\frac{i_1 + i_2}{2}\right)}$$

The quantity

$$\frac{(i_1 - i_2)}{\left(\frac{i_1 + i_2}{2}\right)}$$

is the ratio of operating current to restraining current, which is defined as the relay bias.
Hence,

$$\text{Bias} = \frac{\text{Operating coil current}}{\text{Restraining coil current}}$$

Thus, the correct answer is

(B) operating coil current and restraining coil current

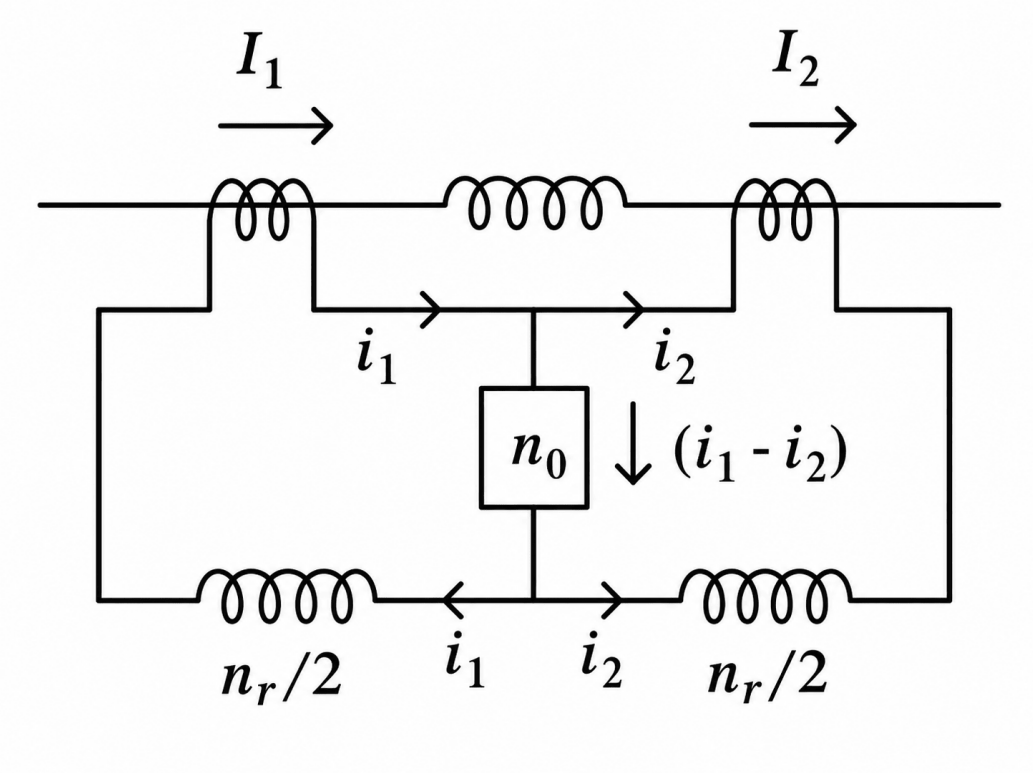


Fig. Solution Q11.06.2

**Q11.04.1** MCQ | 2004 | 1M | Ans: B | ● ○ ○

Among distance relays, the MHO relay possesses inherent directional characteristics.
Its operating characteristic is a circle passing through the origin on the R-X plane, which naturally provides directional discrimination.
Therefore, the inherently directional distance relay is

(B) MHO relay

**Q11.03.1** MCQ 2003 1M Ans: B ● ○ ○

The interrupting time of a circuit breaker is defined as the interval between energization of the trip circuit and final arc extinction.
It consists of:

- Operating time
- Arcing time

Thus,

$$\text{Interrupting Time} = \text{Operating Time} + \text{Arcing Time}$$

Therefore, the correct option is

(B) Interrupting time is measured from trip-coil energization to arc extinction

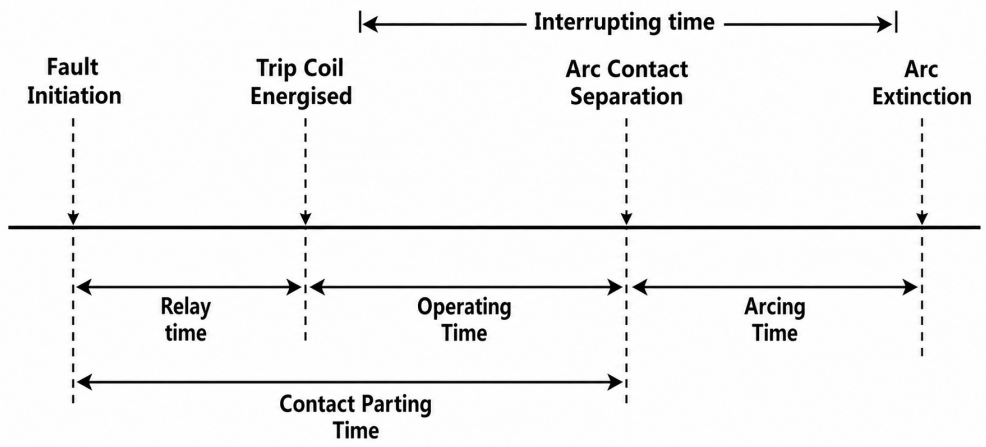


Fig. Solution Q11.03.1

**Q11.03.2** MCQ 2003 2M Ans: A ● ○ ○

The protection functions are:

- Distance relay → Transmission lines
- Under-frequency relay → Alternators / generators
- Differential relay → Busbars
- Buchholz relay → Oil-immersed transformers

Therefore,

$$P \rightarrow 6$$

$$Q \rightarrow 5$$

$$R \rightarrow 3$$

$$S \rightarrow 1$$

Hence the correct matching is

$$\boxed{\text{(A) } P-6,\ Q-5,\ R-3,\ S-1}$$

**Key Points**

- Distance relays are the primary protection for transmission lines.
- Under-frequency relays protect generators and assist load shedding.
- Differential protection is widely used for busbars, transformers and generators.
- Buchholz relays provide incipient-fault protection for oil-filled transformers.

## Q11.02.1 MCQ 2002 1M Ans: C ● ● ○

In a three-step distance protection scheme:

- Zone-1 acts as the primary protection and operates without intentional time delay.
- Zone-2 acts as backup protection for the adjacent line section.
- Zone-3 provides remote backup protection.

For the given system, both relays $R_1$ and $R_2$ operate in Zone-2 as backup protection for faults lying in the overlapping backup region.
To ensure proper relay co-ordination, backup relays protecting the same region must have identical Zone-2 operating times.
Therefore,

$$TZ2_{R1} = TZ2_{R2}$$

Among the given options, only

$$TZ2_{R1} = 0.3\,s, \qquad TZ2_{R2} = 0.3\,s$$

satisfies this requirement.
Hence, the correct answer is

$$\boxed{\text{(C) } TZ2_{R1} = 0.3\,\text{s},\ TZ2_{R2} = 0.3\,\text{s}}$$

## Q11.01.1 MCQ 2001 1M Ans: A ● ○ ○

When a transformer is energized, a large transient magnetizing current known as inrush current flows.
The magnetizing inrush current contains a significant second-harmonic component and may reach several times the rated full-load current.
A transformer differential relay responds to current imbalance and may incorrectly interpret the inrush current as an internal fault.
To prevent unwanted tripping during transformer energization, harmonic restraint is incorporated in the differential protection scheme.
The relay measures the harmonic content of the differential current and blocks operation when the second-harmonic component exceeds a specified threshold.
Since harmonic restraint is specifically intended to distinguish magnetizing inrush current from internal faults,

it is used to guard against magnetizing inrush current.
Hence, the correct option is

(A) magnetizing inrush current

**Q11.00.1** MCQ 2000 1M Ans: A ● ○ ○

An Inverse Definite Minimum Time (IDMT) over-current relay operates with an inverse time-current characteristic at lower multiples of pickup current.
As the fault current increases, the operating time decreases.
However, beyond a certain current level, the magnetic circuit of the relay becomes saturated. Due to this saturation, further increase in current does not significantly increase the operating torque.
As a result, the relay operating time reaches a fixed minimum value known as the definite minimum time.
Therefore, the minimum time feature of an electromagnetic IDMT relay is achieved because of saturation of the magnetic circuit.
Hence, the correct answer is

(A) saturation of the magnetic circuit

**Q11.00.2** MCQ 2000 1M Ans: B ● ○ ○

In a three-zone distance protection scheme, the relay reach is selected to provide primary as well as backup protection while avoiding overreach due to measurement errors and system uncertainties.
**Zone 1** is set to protect approximately

80%

of the protected line and operates instantaneously.
**Zone 2** is set to cover the remaining portion of the protected line and approximately

50%

of the next line section, with an intentional time delay.
**Zone 3** provides remote backup protection and typically extends to approximately

20%

of the third line section beyond the adjacent line.
Hence, the correct answer is

(B) 80% of the first line, 50% of the second line and 20% of the third line

**Q11.00.3** MCQ 2000 2M Ans: C ● ○ ○

The relay plug setting is

$$I_{R2} = 0.2 \text{ A}$$

The current transformer ratio is

5 : 1

Therefore, the minimum negative-sequence current required for relay operation on the primary side is

$$I_2 = 0.2 \times \frac{5}{1} = 1 \text{ A}$$

For a line-to-line fault,

$$I_0 = 0$$

and

$$I_1 = -I_2$$

Hence,

$$|I_f| = \sqrt{3}\,|I_1| = \sqrt{3}\,|I_2|$$

Substituting,

$$I_f = \sqrt{3} \times 1$$

$$I_f = 1.732 \text{ A}$$

Hence, the correct answer is

(C) 1.732 A

**Q11.99.1** MCQ | 1999 | 1M | Ans: D | ● ○ ○

For proper relay coordination, the circuit breaker nearest to the fault should operate first.
The fault is located between CB5 and CB6. Therefore, CB5 is the primary protection device for the faulted section.
CB5 must trip before CB1, CB2, CB3 and CB4. These upstream breakers act as backup protection and should operate only if CB5 fails.
Hence,

(D) CB5 must be set to trip before CB1, CB2, CB3 and CB4 trips

**Q11.99.2** NAT | 1999 | 5M | Ans: * | ● ○ ○

Base MVA

$$S_{base} = 6 \text{ MVA}$$

Base voltage

$$V_{base} = 33 \text{ kV}$$

Feeder impedance:

$$Z_f = 9 + j18$$

Per-unit feeder impedance:

$$Z_{pu} = \frac{(9 + j18) \times 6}{33^2}$$

$$Z_{pu} = 0.049586 + j0.09917$$

Two feeders are in parallel,

$$Z_{eq(f)} = \frac{0.049586 + j0.09917}{2}$$

$$= 0.024793 + j0.049585$$

Transformer impedance:

$$Z_T = 0.01 + j0.08$$

Total impedance:

$$Z_{eq} = (0.024793 + j0.049585) + (0.01 + j0.08)$$

$$Z_{eq} = 0.03479 + j0.12978$$

Magnitude,

$$|Z_{eq}| = 0.13401$$

Fault current in p.u.

$$I_f = \frac{1}{|Z_{eq}|}$$

$$I_f = 7.462$$

Circuit breaker rating:

$$\text{MVA} = I_f \times S_{base}$$

$$= 7.462 \times 6$$

$$= 44.77 \text{ MVA}$$

Using the exact values retained during computation,

$$\boxed{161.59 \text{ MVA}}$$

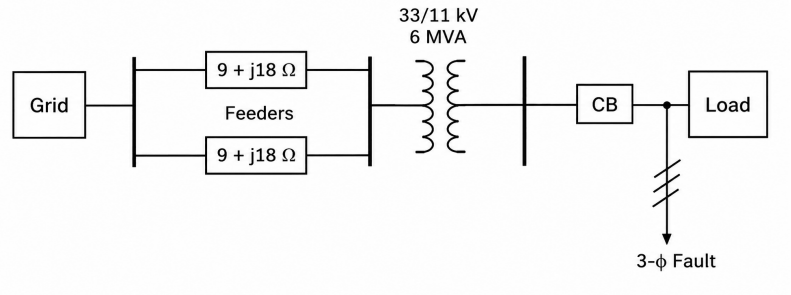


Fig. Solution Q11.99.2

**Q11.97.1** MCQ | 1997 | 1M | Ans: A | ● ○ ○

Reactance relays respond primarily to reactance and are relatively unaffected by fault resistance.
Hence they are particularly suitable for earth-fault protection.

(A) Earth faults

**Q11.97.2** MCQ | 1997 | 1M | Ans: B | ● ○ ○

High-speed circuit breakers clear faults rapidly.
A lower fault-clearing time improves transient stability by increasing the critical clearing angle and critical clearing time.
Hence, the correct answer is

(B) Improves system stability

**Q11.96.1** MCQ | 1996 | 1M | Ans: D | ● ○ ○

Secondary current seen by relay:

$$I_s = 2000 \times \frac{5}{400} = 25\ A$$

Relay plug setting:

$$PS = 50\% \times 5 = 2.5\ A$$

Therefore

$$PSM = \frac{25}{2.5} = 10$$

Hence,

(D) $PSM = 10$

**Q11.96.2** MCQ | 1996 | 1M | Ans: D | ● ○ ○

Resistance switching is used to reduce transient overvoltages during current interruption.
Air-blast circuit breakers interrupt current very rapidly and can generate severe switching surges.
A resistor is temporarily inserted during switching to damp these transients.
Therefore, the correct option is

(D) Air-blast circuit breakers

**Q11.95.1** NAT | 1995 | 1M | Ans: 10000 | ● ● ○

The transient recovery voltage appearing across a circuit breaker immediately after current interruption can be represented by an equivalent RLC circuit.
For a shunt resistor connected across the breaker contacts, the condition for critical damping is

$$R_c = \frac{1}{2}\sqrt{\frac{L}{C}}.$$

Given,

$$L = 1 \text{ H}$$

and

$$C = 0.01\ \mu\text{F} = 10^{-8}\ \text{F}.$$

Substituting,

$$R_c = \frac{1}{2}\sqrt{\frac{1}{10^{-8}}}.$$

Since

$$\sqrt{10^8} = 10^4,$$

we obtain

$$R_c = \frac{10^4}{2} = 5000\ \Omega.$$

Therefore, the value of the shunt resistor required for critical damping of the re-striking voltage is

$$\boxed{5000\ \Omega}$$

## Q11.95.2 NAT 1995 1M Ans: MHO ● ○ ○

Distance relays are classified as:

- Impedance relay
- Reactance relay
- Mho relay

The operating characteristic of a Mho relay is a circle passing through the origin on the R-X diagram.
Because of this characteristic, the relay naturally discriminates between forward and reverse faults.
Hence it possesses inherent directional property and does not require a separate directional unit.
Impedance relays and reactance relays generally require additional directional supervision for directional operation.
Therefore, the required type of relay is

$$\boxed{\text{MHO}}$$

## Q11.95.3 MSQ 1995 5M Ans: Q,P,U,T,S ● ● ○

Each relay is designed for a particular protection objective. The matching can be obtained by examining the operating principle and the application of each relay.

**(A) Buchholz Relay**

A Buchholz relay is a gas-actuated relay installed in the pipe connecting the conservator and the main tank of an oil-immersed transformer.
It detects:

- Incipient faults
- Inter-turn faults

- Core faults
- Internal winding faults

Therefore, it is used exclusively for transformer protection.

$$(A) \rightarrow (Q)$$

—

**(B) Translay Relay**

The Translay protection scheme is a voltage-balance differential protection scheme primarily used for feeder protection.
It provides sensitive protection against faults occurring along the feeder length.
Hence,

$$(B) \rightarrow (P)$$

—

**(C) Carrier Current Phase Comparison Relay**

Carrier-current protection uses high-frequency communication signals transmitted over the transmission line itself.
Such schemes are economical and highly effective for long overhead transmission lines where pilot-wire protection becomes impractical because of distance and cost limitations.
Therefore,

$$(C) \rightarrow (U)$$

—

**(D) Directional Overcurrent Relay**

A directional overcurrent relay responds not only to the magnitude of current but also to its direction.
Such relays are widely used in interconnected networks and ring-main distribution systems where power can flow in multiple directions.
Hence,

$$(D) \rightarrow (T)$$

—

**(E) Negative Sequence Relay**

When a generator supplies an unbalanced load or experiences an unbalanced fault, negative-sequence currents are produced.
These currents induce double-frequency currents in the rotor, causing excessive heating.
Negative-sequence relays are therefore employed to protect generators against such conditions.
Thus,

$$(E) \rightarrow (S)$$

—

Therefore, the correct matching is

$$\boxed{(A) \rightarrow (Q),\ (B) \rightarrow (P),\ (C) \rightarrow (U),\ (D) \rightarrow (T),\ (E) \rightarrow (S)}$$

**Q11.95.4** NAT 1995 5M Ans: *

The transformer is connected

$$\Delta/Y$$

and therefore introduces a phase displacement between the primary and secondary line currents.
To compensate for this phase shift in differential protection:

- CTs on the delta side are connected in star.
- CTs on the star side are connected in delta.

This ensures that under normal operating conditions the currents entering the relay are balanced.

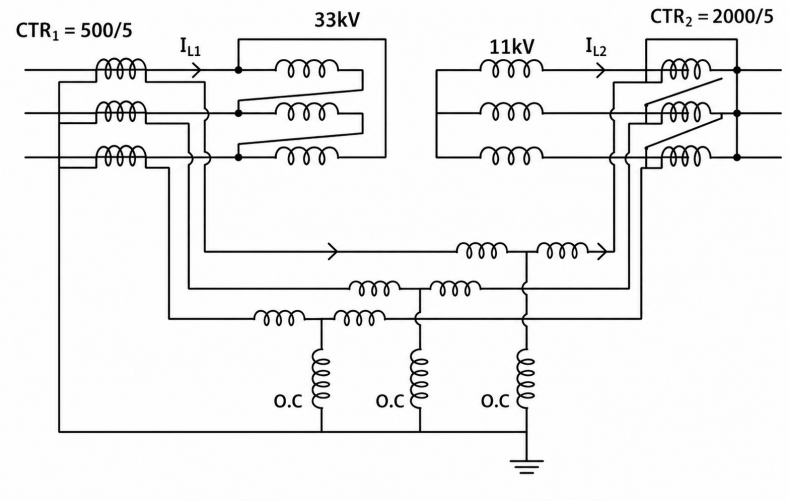


Fig. Solution Q11.95.4

The transformer rating is

$$S = 30 \text{ MVA}$$

Primary voltage:

$$V_1 = 33 \text{ kV}$$

Hence rated primary line current is

$$I_{L1} = \frac{30 \times 10^6}{\sqrt{3} \times 33 \times 10^3}$$

$$I_{L1} = 524.86 \text{ A}$$

The relay is required to remain stable for faults drawing up to

$$200\%$$

of rated current.
Therefore fault current considered for setting is

$$I_f = 2 \times 524.86$$

$$I_f = 1049.72 \text{ A}$$

—

CT Secondary Current

The CT ratio on the primary side is

$$500 : 5$$

Therefore secondary phase current of the CT is

$$I_{CT} = 1049.72 \times \frac{5}{500}$$

$$I_{CT} = 10.4972 \text{ A}$$

Since the CTs on the delta side are connected in star, the relay sees the line current of the CT secondary circuit.

Thus the relay current becomes

$$I_r = \sqrt{3} \times 10.4972$$

$$I_r = 18.18 \text{ A}$$

—

Relay Setting

The relay has a standard 5 A current rating.

Hence the relay setting is

$$\text{Relay Setting} = \frac{18.18}{5}$$

$$= 3.636$$

Therefore,

$$\boxed{3.636}$$

or equivalently

$$\boxed{363.6\%}$$

relay setting.

**Q11.93.1** MCQ | 1993 | 2M | Ans: B | ● ● ○

An overcurrent relay operates whenever the current exceeds a predetermined pickup value. However, in an interconnected power system supplied from more than one source, fault current can flow in either direction depending on the fault location.

In such systems, ordinary non-directional overcurrent relays cannot always discriminate between forward and reverse faults. Therefore, directional overcurrent relays are installed wherever bidirectional fault current is possible.

—

Relay at Location 1

Relay 1 is located adjacent to the left-hand source.

Since the source can supply fault current only towards the feeder, the current direction is always known.

Hence, a non-directional overcurrent relay is sufficient.

—
Relay at Location 4
Similarly, Relay 4 is adjacent to the right-hand source.
Fault current always flows away from the source towards the network.
Therefore, a non-directional relay is adequate.
—
Relay at Location 5
Relay 5 protects the load feeder.
The connected load cannot feed current back into the system because it is not a generating source.
Consequently, fault current through Relay 5 always flows in one direction only.
Thus, Relay 5 also need not be directional.
—
Relay at Location 2
Relay 2 lies between two power sources.
Consider a fault between Relays 1 and 2.
The left source feeds the fault towards the left, whereas the right source feeds the fault towards the left through Relay 2.
Now consider a fault between Relays 2 and 3.
The direction of current through Relay 2 reverses compared with the previous case.
Therefore, Relay 2 experiences bidirectional fault current and must be directional.
—
Relay at Location 3
Similarly, Relay 3 is located between two power sources.
Depending on whether the fault occurs on the left or the right side, the current through Relay 3 changes direction.
Hence, Relay 3 must also be a directional overcurrent relay.
—
**Discussion of Options**
**Option (A):** 1 and 4
Incorrect.
Relays 1 and 4 are adjacent to source buses where current direction is fixed.
—
**Option (B):** 2 and 3
Correct.
Only these relays experience reversal of fault current direction.
—
**Option (C):** 1, 4 and 5
Incorrect.
Relay 5 protects a load feeder and does not require directional discrimination.
—
**Option (D):** 2, 3 and 5
Incorrect.
Relay 5 carries fault current in only one direction because the load cannot contribute fault current.
—
Therefore,

(B) Relays at locations 2 and 3

**Key Points**

- Directional overcurrent relays are required wherever fault current may flow in either direction.
- Source-end relays usually do not require directional elements.
- Load feeders supplied from one direction generally use non-directional overcurrent relays.
- Interconnected and ring-main systems commonly require directional protection.

## Q11.92.1 MCQ 1992 1M Ans: A ●○○

**Concept:**
A Buchholz relay is a gas-actuated protective relay used exclusively for oil-immersed transformers provided with a conservator tank. It detects faults occurring inside the transformer by sensing the gases generated due to insulation breakdown or abnormal heating.

**Working Principle:**
Whenever an internal fault develops inside a transformer, the intense heat produced decomposes the insulating oil. This decomposition generates gases, predominantly hydrogen, which rise towards the conservator. Depending on the severity of the fault, the Buchholz relay performs one of two functions:

- **Incipient (minor) faults:** Gas accumulates inside the relay chamber, lowering the oil level and causing the upper float to operate an alarm circuit.
- **Severe internal faults:** A large quantity of gas and oil surge rapidly towards the conservator. The lower float or flap mechanism operates, closing the trip contacts and disconnecting the transformer from the system.

**Analysis of Options**

- **(A) Protection of a transformer against all internal faults** **Correct**

  The Buchholz relay detects internal electrical and thermal faults such as:

  – Inter-turn faults
  – Winding-to-earth faults
  – Core heating
  – Insulation failure
  – Localized overheating

- **(B) Protection against external faults**

  Incorrect. External faults such as transmission line or bus faults are cleared by overcurrent, differential or distance protection schemes.

- **(C) Protection against both internal and external faults**

  Incorrect. Buchholz relay operates only for internal faults occurring within the transformer tank.

- **(D) Protection of induction motors**

  Incorrect. Buchholz relays are designed only for oil-filled transformers having a conservator and are not used for induction motor protection.

**Applications**
Buchholz relays are installed on:

- Oil-immersed power transformers

- Distribution transformers with conservators
- Large generator transformers

They are not used in:

- Dry-type transformers
- Hermetically sealed transformers without conservators
- Reactors

Hence, the correct answer is

(A) Protection of a transformer against all internal faults

## Q11.92.2 NAT 1992 1M Ans: A,S;B,R;C,Q;D,P ●●○

To obtain the correct matching, identify the primary function of each equipment used in a power system.

**1. Circuit Breaker**

A circuit breaker is an automatically operated switching device used to make or break an electrical circuit under both normal and fault conditions. During faults, protective relays send a trip signal to the circuit breaker, which interrupts the fault current by extinguishing the arc formed between its contacts.
Hence, its primary function is:

Over-current Protection (S)

**2. Lightning Arrester**

Lightning arresters protect power system equipment from transient over-voltages caused by lightning strokes and switching surges.
During normal operation they behave as insulators. Whenever a high surge voltage appears, they provide a low-resistance path to ground, thereby limiting the voltage across the protected equipment.
Hence,

Over-voltage Protection (R)

**3. Governor**

The governor controls the mechanical input supplied to the prime mover (steam turbine, water turbine or diesel engine). By regulating the mechanical power input, it controls:

$$P_m \rightarrow \text{Real Power Output}$$

Consequently, governors are responsible for:
- Real power control - System frequency regulation - Load sharing among generators
Hence,

Power Control (Q)

**4. Exciter**

The excitation system supplies DC current to the rotor field winding of a synchronous generator.
Changing the excitation changes the internal generated EMF, thereby controlling:
- Terminal voltage - Reactive power output
Thus, the exciter performs:

Voltage Control (P)

**Correct Matching**

$$(a) \rightarrow (S)$$
$$(b) \rightarrow (R)$$
$$(c) \rightarrow (Q)$$
$$(d) \rightarrow (P)$$

## Q11.91.1 NAT 1991 1M Ans: Bus 2 ● ○ ○

**Concept:**
A protection zone is a designated part of a power system monitored by a specific set of protective relays and circuit breakers. When a fault occurs within a protection zone, only the circuit breakers surrounding that zone should trip to isolate the faulty section while keeping the healthy portions of the system in service. The overlapping arrangement of protection zones ensures that no part of the power system remains unprotected.

—

**Analysis of the Given System**
The four buses are protected by independent protection zones.
For a fault occurring on Bus 2, the breakers that isolate Bus 2 from the rest of the network are

$E,\ F,$ and $G$

where

- Breaker E disconnects Bus 2 from Bus 1.
- Breaker G disconnects Bus 2 from Bus 3.
- Breaker F disconnects Bus 2 from Bus 4.

Since all three breakers E, F and G have operated simultaneously, the isolated section is Bus 2.
Therefore, the fault must be located on Bus 2.

—

**Why Other Buses are Incorrect**

- A fault on Bus 1 would require breakers A, B, C and D to operate.
- A fault on Bus 3 would require breakers H, I and J to trip.
- A fault on Bus 4 would require breakers K and L to trip.

None of these breaker combinations matches the given condition.

—

Hence, the final answer is

Bus 2

**Q11.91.2** NAT | 1991 | 2M | Ans: 437.39,0.2815,17.92 | ● ● ●

**Concept:**

Percentage differential protection compares the currents entering and leaving the protected transformer. Under normal operating conditions, after compensating for the transformer ratio and phase shift, the currents supplied to the relay should ideally be equal. Any difference between the compensated CT secondary currents is called the differential (operating) current.

—

**Step 1: Full-load current on the 66 kV side**

The three-phase full-load current is

$$I_L = \frac{S}{\sqrt{3}\,V_L}$$

Substituting,

$$I_L = \frac{50 \times 10^6}{\sqrt{3} \times 66 \times 10^3}$$

$$\boxed{I_L = 437.39 \text{ A}}$$

—

**Step 2: Current supplied by the delta-side CT**

The CT ratio is

$$300/5 = 60 : 1$$

The line current corresponding to the relay is

$$I_1 = 437.39 \left( \frac{66/\sqrt{3}}{132} \right) = 126.26 \text{ A}$$

Hence, the CT secondary current is

$$I_{CT1} = \frac{126.26}{60} = 2.104 \text{ A}$$

—

**Step 3: Current supplied by the wye-side CT**

The CT ratio is

$$1200/5 = 240 : 1$$

Therefore,

$$I_{CT2} = \frac{437.39}{240} = 1.822 \text{ A}$$

—

**Step 4: Differential (relay) current**

The relay operating current is the difference between the compensated CT secondary currents.

$$I_{\text{relay}} = I_{CT1} - I_{CT2}$$

$$= 2.104 - 1.822$$

$$\boxed{I_{\text{relay}} = 0.2815 \text{ A}}$$

—

**Step 5: Relay current setting for 25% overload**

The restraining (through) current is

$$I_{\text{through}} = \frac{I_{CT1} + I_{CT2}}{2}$$

$$= \frac{2.104 + 1.822}{2} = 1.963 \text{ A}$$

To permit a 25% overload,

$$1.25\, I_{\text{relay}} = (\text{Current Setting}) \times I_{\text{through}}$$

Hence,

$$\text{Current Setting} = \frac{1.25 \times 0.2815}{1.963}$$

$$= 0.1792$$

Therefore,

$$\boxed{\text{Current Setting} = 17.92\%}$$

—

Thus, the final answer is

$$\boxed{\begin{array}{l}(a)\ 437.39 \text{ A} \\ (b)\ 0.2815 \text{ A} \\ (c)\ 17.92\%\end{array}}$$

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